

5th Applied Active Inference Symposium (Part 2, Nov 13, 2025) ~ (Full reupload)

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welcome back it's day two of the fifth applied active inference symposium

november 13th 2025 and the first session will be the future of active inference.jl

with peter jonathan and samuel so peter thank you to you thanks daniel all right it's state

please mute the youtube

youtube sorry i have some sound problems here thanks

one second

okay here we go again can you see the screen otherwise i'll just reshare it and and there

we are thank you daniel sorry for the the sound issues there i had another tab open that i

shouldn't have had and also thanks for organizing this symposium which is incredible the work

that's been put into it and worth being grateful for so just that's worth saying probably a lot

of times during this symposium again and again hi everyone so today i and sam and jonathan

will present on the ongoing and upcoming developments of active inference.jl which is a julia software

tool for running and applying active inference models to data or in simulation

and i'll start out with just a background and a conceptual thing and then jonathan will take

over and sam will take over and they will go through code and give a demonstration of how to use the

package in its newest version that's not yet released but and there's a link in the youtube channel if

someone wants to follow along the code to a github repo which can be cloned and used

with this new experimental version of the package

so um i'll briefly to start with background we're conceptualizing

conceptualizing active inference as a kind of cognitive model and what a cognitive model means

is just that it's a formal model that produces behavior of some kind so all of artificial intelligence

et cetera can be conceptualized as cognitive models and this is broader framework where you have some

agent and you have some environment and the environment produces observations to the agent the agent produces

actions that affect the environment and then the agent has a cognitive state called the theta here

that changes over time that could be its beliefs about the environment and um the agent has some cognitive

parameters that could be for example learning rates or its priors that govern how its cognitive states change

and govern how it produces actions and in general cognitive modeling and therefore also active inference has three

kinds of uses that i know of one is theoretical modeling so that's when you run simulations

to see and what a given agent would do given different parameters so there you know the theta here

and you simulate the environment and you want to see what happens and active inference is full of these so

all the simulation work where you're going to show what happens if active inference fish swim together

etc could be considered theoretical models and much of the backbone of the field is built on this

then there's applied cognitive modeling and that's where instead of having the parameters you have the

observations and the actions so you have the behavior for example because you recorded it from a human in an experiment and you want to reconstruct the behavior that govern the parameters that govern the behavior you want to find out what was the general model or what was the prior so what was the learning rate of a given participant that's what's used in computational psychiatry but also much beyond that and finally one uses artificial intelligence where again you have the parameters but you don't simulate the environment you have real environment for example you know burning house or whatever and then you want your

your artifact your tool to generate the right kinds of behavior for example a vacuum cleaner that's supposed to move in some ways and not other ways you can use active inference in all three ways our lab specializes in applied cognitive modeling so the second of these three and last year we focused quite a lot on how to use active inference for that but this year we will not because instead we will focus on new structure of the package and how that makes it more flexible for the kinds of active inference models you can implement whether you want to use them for theoretical or applied modeling or for artificial intelligence and i just want to briefly mention that there are many other modeling frameworks than active inference you know frameworks where you optimize for expected value frameworks where you run symbolic code operations frameworks where you think them as kind of a dynamic system in connectionism and later been renamed to deep learning in control theory etc and there's also a bigger family of Bayesian mind models where you think that the mind uses Bayesian inference so we should have models that use Bayesian inference to describe the human mind

an active inference which is the kind of variational Bayesian predictive processing model is a type of Bayesian

mind model that does three things so it commits to specifically using variational Bayes which is one way of approximating Bayesian inference it usually uses message passing schemes to do this and i'll get back to that

and notably it does not only model perception but also action as Bayesian inference just to place us in the larger field that we live in and you can combine models with these other fields these other frameworks if you want as well active inference can be combined with them

so as we know Bayesian inference you have some data some observations and you have some parameters or some

hidden states of the environment you have a model m and what you want to find so the general model is just how the data and the parameters usually co-occur could be just some joint probability distribution it can be quite complex

and as usual what we want is we want to find the parameters that are most likely given the data we get and we have Bayes theorem and we all know this and crucially we can't calculate the bottom part because we have to the bottom here to the bottom right and to calculate that you have to integrate across all possible parameters and if you have parameters that are continuous you have so full continuum for each parameter and if you have many

parameters hidden states that becomes impossible to do that means that usually you approximate Bayesian inference instead

there are generally three methods so on a sampling and i've mentioned these because we can actually use them with active

inference if we wanted to i will show that later so it's worth mentioning that they exist

in sampling you just pick values randomly stochastically with an algorithm that ensures that you're going to pick a value proportionate an amount of times proportionate to how probable it is in the posterior

and there are many ways to try and do this they can fail they can succeed but the main advantage is that they can get any shape of posterior in principle the main disadvantage is that they take a long time

and they're not so mathematically beautiful because you don't use equations to get it analytically

another one is simulation based inference where broadly you generate lots of behavior or lots of data

with a model and then you do something to get the posterior out of that and it's one of the classic and the

most used method here is amortization where you simulate data with your model and give them different

parameters and you train a deep neural network to revert that process and then further parameters given the data

and finally in variational inference you minimize a free energy to get the best to get the posterior

notably this is not anything new so it's been used in machine learning and in statistics for a long time

what's new in active inference is that it's used as a model of human cognition in general and it's combined

with action selection as well again i'll just briefly remind us that here you have the true posterior which we cannot calculate

we make up entirely arbitrary an approximate posterior cube and then we can calculate the free energy which is just

the kl divergence between the approximate posterior and the generative model on the right

we have both of those so we can calculate it

and then of course the magic is that if we find the approximate posterior with the lowest free energy

we also find the approximate posterior that's closest to the true posterior

we see this that you can rewrite the free energy

which is the difference between the approximate and the true posterior

and the evidence is the surprise

and notably if you change only your posterior

that only affects the left term here

and that means that you can only reduce the divergence

by reducing the

by choosing the approximate posterior that minimizes the free energy

that's all very exciting hallelujah

now notably that does not tell us how to actually find the approximate posterior with the lowest free energy

and there are different ways of doing this

that are all variational based and they're all free energy minimization based

and that's you know not necessarily an easy task especially if the

the parameters of q so if you have many

if you have high dimensional posterior because you have many states you're doing inference over it gets quite complex and the way that they relate to the free energy might not be very obvious either so in general I know of three ways that people find the posterior that has the lowest free energy one is just gradient descent so you just wiggle your posterior around until they find the one with the lowest and that requires calculating the gradient so how the free energy changes when you change your when you change your approximate posterior and you have a complex model that's usually difficult but there are slc methods to do that even for complex models in some cases when you're lucky you can also just analytically mathematically find in one equation the approximate posterior with the lowest free energy that's very fast that's very nice but usually not easy and finally there are message passing methods and those are the ones we know the best so there you do an additional approximation where you take your gentle model that's taken from the par in our paper you take a gentle model and you split it up so for each of the components each of the hidden states of parameters in your model you split it up and you update your beliefs about them one at a time usually you do what's called a mean field approximation where you assume that they're completely independent but you can do this in different ways so there's also these beta free energy for example approaches where you split them up into pairs instead of single so you can retain that you can see that two parts of your posterior might correlate but they're still assumed to be independent from the rest then you update them one at a time and because you split them up and because there's an approximation that won't always work so you keep doing that until they converge and you can see that there's a lot of different methods that are used to be independent of the other components that are used to be independent of the other components and that's the standard method in active inference currently and it bears just showing briefly the update equations for each of the components if your generative model specifically is a discrete state transition POMDP you can have other generative models but in that case you combine two things you combine a prediction which is your previous belief about the state of

environment plus or multiplied with the transition probabilities so you take what you thought the world was before you let it go through your model of how the world changes and you now have your prediction and then you add your evidence from the observation so in almost all message passing schemes you have this

prediction plus evidence approach and that's why we call it predictive processing because when you do message passing

you get something often you get something that looks like prediction out okay why spend all this time reminding us

that there are what message passing is but also that there are other alternatives i'll get to that

probabilistic program languages are ppls for short exists so that's a general technology where you have some system where you can specify all sorts of generative models and then they for free give you methods for

doing Bayesian inference and prediction and there exists a bunch of these so stan is famous infer.net for example is also widely used and they offer different kinds of methods so stan for example focuses on sampling

but can also do variational inference by gradient descent infer.net focuses specifically on message passing and the whole point here is that you're not limited to one type of general model you can just write depending on the ppl any kind of general model and and get your updates from that without having to hand write all of the update equations i spoke to some people from one ppl who said why not would you hand write your update equations all the time which we do so in julia there are multiple ppls

and they can all be used to do the bayesian inference that active inference.gi needs

the classic one is the active inference.discrete.pdp that's the in-house that we develop

and that's the one that's identical to the one used in pymdp and in the matlab code and so that's a ppl in the sense that you can specify many kinds of general models but it's also limited because you can only specify discrete ppls so where you think where you get discrete observations you think the world consists of discrete states that change in state transitions and that you have discrete actions that affect those state transitions.

We're also developing the hgf which i won't spend much time except to say that that's very very similar except that the world is not made of discrete states with discrete state transitions but Gaussian random ones. Here you also do message passing and that's different update equations but exactly the same logic and they cannot also be combined which we're working on. But there are also other more flexible ppls so rx infer is one which will do a message passing specifically as mentioned before so that's very fast but it's also an approximation of an approximation of Bayesian inference so that's why some people prefer others.

You can also use Turing or Gen which focus more on sampling or on gradient descent variation inference. So they are generally more flexible in the types of models you can write in them but are generally also much slower and they don't have as elegant mathematical single step updates that's why they're slower and are not they can be but they don't have to be free energy based.

And essentially what we want to show the rest of this presentation is that instead of being confined to have to use the discrete pumpdp method it's very nice as long as you are in a discrete environment

and as long as you think your actions can only change state transitions and that other stuff for example it's very nice but we want to allow active inference.jl to use any of these options as backend and therefore to allow people to use a wider range of generative models for a wider range of scenarios but still doing full-on active inference.

So the last part here I will just briefly remind us of two different ways that active inference-like models can be implemented.

One is let's call it planning as inference and that's the one where you just put in your possible actions or policies as random variables in your model and you do standard Bayesian inference.

Could be variational, could be anything, to find those sections and so this is a screenshot from the tutorial on RxInfer, very nice tutorial, where they have a discrete pumpdp so you can see this A matrix and B matrix as we know it and we have a previous state and then they write out that generative model.

So we have a current state which depends on the previous state and the B matrix and a previous action and we have our current observation which depends on our current state and our A matrix.

So this is specifying the same stuff we actually do in Pymdp normally.

And then we have a planning part where you go through each time step in your imagined future and you predict forward what would happen.

That's what happens here in this discrete decision from previous state down in the for loop, etc.

And then finally you have your goal prior or your preference prior in the end where you, in addition to predicting what will happen, you also have a fixed distribution.

Here it's only for the final state.

So here the agent actually only has preferences for the very final state.

So here you run this, you can use a sampler, a variational inference and you get out actions that are likely to make that goal prior come true.

And given your beliefs about the current state and your beliefs about A and B, etc.

Here there's no parameter learning but you can easily add that, etc.

And it's just to show you could make many changes to this.

It's very flexible, that's the whole point.

What this method does not give us though, is that it does not give us the information gain that is in the expected free energy in active inference.

So we get the finding actions that give us our goal prior, but we don't give the information gain.

And the action for people, if anyone were at IY, are working at ways to extend this.

So to have sneaky, clever ways of having priors about, for example, the ambiguity that will kind of get you the expected free energy anyways, while keeping everything local and using message passing.

You could do this with other things too.

I'll contrast this to, I'll call it modular planning for now, but this is the standard way that active inference is implemented in active inference.gl, PMDP and in MATLAB.

Where you really have three components to your model.

You have your generative model, the agent's general model of the environment.

Then you have a perceptual part where you take that general model and you form beliefs about the environment

and make predictions about the next time step in the environment.

And this is the coordinate descent variational inference that I described before.

This is the place where we minimize variation.

And then we have a third part where you do your planning.

And here in active inference.gi, etc. you do, for example, a tree search where you go through all possible, all, sometimes you go through all possible combinations of actions in the future.

Sometimes you do it in a more constrained way in order to find the set of actions that have the lowest expected free energy.

What's important here is that the planning part is separate from the perception part.

So in cognitive modeling, that's often called a perceptual model and a response model or an action model.

So it subsumes the perception part.

So when you are, especially if you're doing sophisticated inference, you're running your perceptual inference forward in time many times.

So you do your variational inference many times into the future as you plan.

The point here is that they are modular.

So you can have one generative model, say a discrete POMDP with ABC, AB matrices.

And then you can use many kinds of perceptual inference methods with that.

So you could use RxM1 message passing.

You could use RxM1 message passing.

You could use RxM1 message passing.

You could use Turing's sampler with the same generative model.

And similarly, given a generative model and a perceptual process, you can also

with different generative models and or different perceptual processes, you can combine that with the same planning method.

Because in the planning, you're just saying,

if I do that, what will I think in the future?

If it is doing sophisticated inference.

And you could, that step, what will I think in the future?

That can be done with sampling, with gradient descent-based variational inference, or with message passing.

And it can also work with other kinds of generative models than the discrete POMDP.

I'll note that standard for the planning is to do a tree search.

And tree search will explode if you go very far into the future.

So they get competition very costly.

So you have to find ways to prune, etc.

And there's a whole field of finding good ways to search through that large space of possible policies efficiently.

But one other option is actually to make the planning stage itself an inference call.

So that itself is a model, which we want to allow in the future.

So you have a generative model, inside of which there is Bayesian inference on a different generative model.

You have a generative model, which is your planning, and that contains the perceptual part.

And that, of course, requires us to nest PPLs.

And that's something we're working on.

But I just want to point out that that's usually what's called metacognition.

So you have a model of your model.

And sophisticated inference, that's exactly what you do.

You have a model of your future beliefs, of your own future beliefs in the future.

And I think that qualifies calling sophisticated inference metacognition.

Given that it's specifically used for planning, one might call it metacognitive planning.

It's just a different word for the same thing, but one that highlights what's happening.

You're planning your own beliefs in the future.

And to do this, you need to run inference within your inference.

And therefore, you need to nest PPLs.

And that's what we're building to allow.

So the excitement here, the hope is that you can plug in, for example, RxInfer or Turing as a backend, so that you can do other kinds of generative models.

And that we can allow for nesting them.

And we're building the infrastructure for that.

And in the following, we'll show you an example where we use two different inference methods with the same generative model and the same planning structure.

Yeah? Okay.

So that's it for me.

That's just the background, just to make the conceptual point that we'll build on.

From here, Jonatan will take over.

But control the PowerPoint.

And the second, we'll show you some code as well.

Super. Thanks, Peter.

Yes, so before we actually show you some code, I'm just going to go through conceptually how the new structure is to be understood.

And what kind of needs to be done if one wants to create a custom struct for either generative model, perceptual process or action process.

So, Peter, if you would be so kind to go to the next slide.

Yes.

So here we basically see the same figure as we saw before, where we divide this active inference model up into three main parts.

Where we have a generative model, our beliefs about hidden states and how they transition over time and how they generate observations.

We have the perceptual process.

How do we then infer these hidden states?

And then we have the action process and how we evaluate policies and then marginalize them into actions.

So the idea behind this is that sort of like Lego bricks, you can kind of interchange these parts.

And so you can exchange, as Peter said, the perceptual process.

And then it still works with the same generative model and the same action process.

So it's kind of becomes modular.

And if you go to the next slide, Peter.

Yes.

And then exactly what is required to make such an extension.

And so I'll just briefly introduce how it's kind of conceptually implemented in the code in the active inference.jl.

So we have the same model.

And here we have again, these three main parts where you can see the current implementations that we have is the discrete state space pound piece for the generative model.

We have the coordinate descent variational inference for the perceptual process.

And this is going to be the part that Sam is going to show how you can create a custom perceptual process.

So you can use other methods to do the state inference.

And then we have the action process where our default is just a policy tree search with evaluation over the expected free energies.

Yes.

You can go to the next one.

Yes.

And so here we can kind of see the overall structure of the package.

And it's basically based on an active inference core module, which kind of contains the template for the structs and functions.

So the functions and structs in the active inference core module are not really used, but they are made so that they can be extended into other modules.

So as you can see here, the module that we have currently implemented is the discrete POMDP.

And I'll show you briefly, shortly in the code, how you can actually extend these generative models and the functions like the perception planning and this kind of stuff into the discrete POMDP so that you can actually create an active inference loop

with the discrete POMDP generative model and with the perceptual process and action process that we have.

And then you can extend this by creating other modules as well, which we will in the future, and which is going to be where you would add the custom structs.

And if you go to the next one again.

Yes.

And so basically here, the idea is that we use abstract types, the model, where we are saying that for a generative model, if you want to add another generative model, then you have to make it a subtype of this abstract generative model.

And basically what we allow for here is that in the generative model, you can also specify what type of what sort of action or observational state.

If you go to the next one, Peter, just briefly just showing here that you can, for example, have a POMDP that can be both continue where you have continuous action or continuous observations and states, but you can also have a POMDP where you have discrete actions and discrete states.

And this way we can use,

we can use this to,

to for different functions to the word is escaping me, where we have different functions that multiple dispatch on the different types, even though it's the same,

the same, could be the same POMDP with the given different sort of actions or observations,

we can do multiple dispatches. And this way it becomes very modular and very easy

to work around. If you go to the next one,

is some, the same one here goes for the perceptual process. If you want to add a perceptual process,

then you have to make it a subtype of the abstract perceptual process. And if you go to the next one,

the same for the action process. And so if we combine all of these, if you go to the next one

again, then we have our active inference model, which is composed of these three parts of the

general to model, the perceptual process, and the action process. And here you can see that it's

interchangeable in the sense that you can just, and this is what Sam is also going to show you,

that you can have the same general to model and the same action process, but then you switch out the

perceptual process with a custom perceptual process instead of the Cavi perceptual process that we're implementing current. And just a brief note on the action models here, the action models is a sister

package to the active inference package, which is for doing model fitting to cognitive models,

of which active inference here is one of those models that we're planning to implement and to do

model fitting on or allow for model fitting. If you go to the next one.

Yes. And so there are four main functions that one needs to be aware of if one wants to implement

a custom struct. This is the perception, the planning selection and store beliefs. And this is,

I'm going to show the active inference loop and you'll see why one has to at least either modify

or make sure that these four functions will work with whatever customizable structure that you are

planning on implementing. So if you go to the next one here, we showcase also like the core module, how

you're supposed to extend from the core module into whatever module you're working on. So for example, here, when we're working in the discrete pump DP module, we're extending from the active inference core module, which has the perception function. This, the perception function from the active inference core module is never going to be used because it only contains or dispatches or abstract types where it will then give you an error and say, you need to create a concrete type. And this is what you need to do. You need to create concrete types for each of these parts. And then as you can see here in the PMDP module, you then extend it from the active inference where you then have the active inference model, where each the generative model, the CAVI, which is the perceptual process and action process are all prefixed with this discrete pump DP. And then you have the multiple dispatch that you have to do so much on the specific type of active inference model that you have. And if you go to the next one, and you can see this is the same for the planning. So the perception one creates the posterior state, the planning creates the posterior policies. And again, we have the multiple dispatch on the discrete pump DP and it's an extension from the active inference core module. If you go to the next, we have the selection, which basically marginalizes over the action. So it actually creates the act of the distribution over actions that we then sample from when we want a concrete action. And if you go to the next one again, yes, and this is a story, I believe this is just a like a utility function where we then take all the output from the previous three functions, and then we put it into the active inference model for use in the next time step. But basically all of these four functions need to work with whatever customizable or custom structure you're making and needs to be kept in mind. And yes, and the next one. Yes. And here you can see this is kind of like the higher level active inference loop that is in the the active inference core module. And basically, because we have the multiple dispatch, this loop can work with any active inference model as long as each of the functions, perception, planning, selection, and store beliefs have a multiple dispatch on the active inference model type and subtypes that you're using. And same will show how this is the case. And then the next one is just briefly showcasing here that in the perception action loop, you have the active inference functions in active inference function, which gives you probability over actions. And this probability, you can then use the Julia distributions page, to then sample an action that you then give to an environment, give you an observation. And in this sense, you have the perception action loop. So it should be high level, high level, it looks, it should be very clean. And then now Sam, he will show you how this is implemented with a walkthrough of a notebook. And in the next slide, you will see a link to a GitHub page where you can clone it, and should be able to follow along. All right, I give it over to you, Sam. The link is also in the YouTube chat. And I'll keep an eye on the chat. Meanwhile, in case anyone has questions. And finally, do note, I'll just add this, do note that creating your own alternative version of active inference models is maybe a bit of a power user thing that it's good to be able to do. But we also retain the old interface where you can do a classic active inference.GL models. Sorry for the introduction. No worries. Okay, so let me just share my screen quickly.

Can you at least confirm that you can see it? Well, yes, we can see it. Thank you. Awesome. Okay, so there's this little notebook that we have prepared for you to show you basically like the current

implementation. We'll be using the development branch because it has not been published yet as a new version. But basically, I don't know how familiar people are with Julia. But in the beginning, there are some formalities that we need to go through like activate the current environment. And we have read some utility functions, just for plotting and the environment. And then we bring our dependencies to the scope. And so in this example, we'll be using a simple multi-armed bandit task, which you can

imagine as a slot machine with three arms. And each arm is associated with either reward or loss with some probability. And so this example will initialize the environment with arm one being the best. So we'd have the probability of 0.9 of receiving reward. And the second best would be arm two and then arm three

would be the worst with only 0.1 probability of receiving reward, otherwise loss. And after each 25th trial, these probabilistic contingencies will switch. And we will make our active inference agents initially uninformed about what are the reward probabilities. And we'll let it play the game for 100 trials and we'll let it learn these contingencies while trying to stick to the preferred outcomes, which are the rewards in this case. This is just a visual of how it looks. Over 100 trials and the probabilistic switches between arms after each 25th trial. So now we are ready to set up our generative model. We'll just start with specifying some structural components. So we have three hidden states, where each hidden state corresponds to an arm in the bandit. We have two observations. So award our loss and controls. Those are actions. So we have three actions. Each action basically just means pulls arm one, pull arm two, pull arm three. And then we're going to use a policy length of one. So we're only planning one move ahead when making decisions. And we're just going to pass all this to create matrix templates. And we're going to tell it to populate it with uniform probability distributions.

And this is going to give us the A, B, C and D objects. And now we can populate it. So for the observation likelihood, this will give us a two times three matrix where each column corresponds to a hidden state or an arm in this case and rows correspond to observations. And we're just going to leave it populated with uniform probability distributions because we want the agent to learn these probabilistic contingencies. Now we specify transition likelihood. So we have three B matrices. In this case, each B matrix corresponds to one of the three actions. So basically telling the agent that when it's in the arm two, for example, and it pulls the arm one, it's most likely to actually end up in that state. And we make it a bit stochastic here just introduce some uncertainty to it. Right, so we have our B objects created, to be created above. And we just take like each slice and populate it with our matrices, each corresponding to one of the three actions. And the two last things we're going to need to specify are the C vector and the D vector. So the C vector is the preference. In this case, just the two element vector with the first element encodes the preference over rewards and the second one, the preference over losses. So here we're basically just saying that prefer words over losses.

And for the prior initial hidden states, we're going to tell it that it's going to start in the arm one.

All right, arm one. All right, so now we have the generative model. And we can move on to definition of the active inference model. And so I'm going to start with the default implementation.

So sort of the out of the box way of specifying the generative model or the active inference model, rather. So we are still keeping the old way of doing that, where we're just passing the individual components of the generative model to the AF model constructor. And then we don't really need to care about anything else. We're basically ready to use our model. However, the first thing we would like to focus on here is the new modular extensible approach.

And so here we need to specify the individual parts of the active inference model separately. So here we'll create the generative model object by passing it by passing the individual generative model components to it. And this is the default discrete pom2p generative model.

Then we'll need to specify our perceptual process. And as Peter and Jonathan mentioned in our package, the perceptual process which consists of two main parts, which is the algorithm to perform inference about hidden states. So here in the beginning, I'll just use a coordinate ascent variational inference from the discrete pom2p module, which is implemented by default. And then we need an algorithm to do learning. So basically means to update the parameters of the generative model itself. And here by default, we are just using the Dirsch-led capichortical model.

We're going to learn the A matrix. We're going to pass some parameters to it, like the concentration parameter or the forgetting rate, because there's some volatility in our environment to make the agent a little more flexible. And then lastly, we specify our action process. So the action process here is really just a planning algorithm used to evaluate our policies. And this can be standard inference, this can be sophisticated inference, tree-based methods. By default, in the discrete pom2p module here, we're using standard inference without graph. And in this case, we're telling it to use the policy length of one, which we have defined above. And then we're going to use all the three terms to calculate the expected free energy. So stays information gain, the parameter information gain, and the utility term. And once we have all these three parts ready, we can just collect them all three and give it to the AF model constructor. And this will generate the model object for us. And then we'll just need a couple of more things. We'll need to initialize our environment. So here we have some pre-made environment. And we're going to tell it that the best arm has the reward probability of 0.9. And then we're going to switch the contingencies after every 25th trial.

And some random seed. And then before we actually simulate, we're just going to set our initial observation, our initial actions, initial action, and we're going to simulate for 100 trials. And here I just have some empty vectors so we can store the beliefs of an agent during simulation for later for plotting. Now we get to the action perception loop. So the main function here is the active inference function which takes the model observation and action as arguments and it outputs the probability distribution or actions. Now because it's a categorical probability distribution, we sample an action from it. And then we pass it on to the environment which gives us the observation for the next time step. And it goes on like this for 100 trials. And then I'm also saving the first row of the of the A matrix and the posterior states later for plotting here. And so how does that look then in practice? So in this plot, the first part, the green dots are the reward observations and the red dots are the loss observations. In the second one, we have the first row of the A matrix, right? So this corresponds to the probability of

receiving a reward for each of the three arms as the agent was learning it throughout the course of 100 trials and in posterior overstates, which is a little noisy here because we introduced some stochasticity to it in the transition model. So this was the default out of the box way to do it. And now I'm going to show you how to create a custom perceptual process. So here it's just going to import two things from the package that we'll need. And that's the type of abstract perceptual process and the learn a type from this Qt pump to P module. And here we're going to create, we're just going to call it a container really because this mutable struct, like it's like its task is really just to store all the important variables that we will need for our perceptual process. And importantly, we're going to make it a subtype of abstract perceptual process so that our functions or the functions of our package can accept it. Right? So we have our inference function, which we are going to define later. We have our A learning algorithm. And then it stores, of course, posterior overstates, posterior overstates from the previous time step, empirical prior, current observation, expected states, expected observations, and then the B learning and D learning. We're just going to leave empty here. Right? And they're just going to create a constructor function. And we have our container for the custom perceptual process. So now what we need to do is we need to create the actually the the inference function that the agent is going to use to form beliefs about hidden states. And here, this is a very simple generative model. It's just a single state factor, which means that we can do exact Bayesian inference based on these two equations, which is basically just the soft max of the log prior plus log likelihood. And on the first time step, we just switch to the prior with the D vector. And otherwise, we're going to use the empirical prior, which is just calculated based on the transition likelihoods. Right? And to put it put this all together, we're just going to load the soft max function from the logs log x functions library. And we have our exact inference function here. So it takes just three arguments, which is the model itself, the prior and the observation at the time step. I'm going to take out the A matrix from the alternative model, we're going to get the index of the observation. And again, we're going to calculate the log likelihood and the log prior plus some small value for a numerical stability and bring it all together in a soft max function to get our approximate posterior. But actually, in this case, it's an exact posterior. Yeah, so now we have this ready and we can bring it all together again to the model constructor. So in this case, again, our generative model is just discrete pom to p with the components, which we have defined above. But now to the perceptual process, we are going to use our custom perceptual process, where to the inference function, we pass our exact inference function defined here. And we're going to tell it to use learning as implemented by discrete pom to p module. And the same goes for the action process. With the policy link of one, and we're going to use parameter information gain. And the rest is basically exactly the same configuration. Give it the first observation, first action, similar for 100 trials, some empty vectors to store the beliefs of an agent. And this one is basically exactly the same. But the only difference being here that we are using an exact patient inference to form our posterior overstates. And this is how it looks. So far, we've talked about using our default implementation and then using custom basically

handwritten perceptual processes. And now I want to move on to how to use PPLs as a part of an active inference model. So in this case, I'm going to use Turing, but this can also be an RxInfer or gen.jl, and can be really any Bayesian algorithm can be sampling, it can be a variational method, it could be other optimization methods. But in this case, we're going to use sampling via Turing. So the only thing we need to do is to create our Turing model, which is going to be our perceptual process function. And again, it takes three arguments. So our AIF model prior and the observation per time step. We're going to take out the A model from the generative model, get the index of the observation, we're going to tell the function that our likelihood is defined in our A matrix. And then we create a Turing model. And in the Turing model, we're just going to tell it that our hidden states are coming from our categorical distribution defined in our prior. And then our observation are coming from categorical distribution defined in our likelihood, dependent on hidden states.

And once we have the Turing model ready, we can sample from it. So in this case, I'm using a Metropolis Hastings algorithm, and I'm taking 5000 samples from the posterior. This will give us our chain. And then I can just average over samples for each category because it's a categorical distribution to get to our posterior states. And just return it as a factor. And that's it. And then the basically, the rest is exactly the same. We bring it all together in the model constructor, define a generative model. And then in the perceptual process, for the inference function, we set our Turing perception function here. And then the rest is basically the same. So we define observation, action, number of trials, define some empty vectors to store beliefs. And the action perceptual

Turing is exactly the same. And when we plot the result, we see that we have made three different models with different perceptual processes. Now. Yeah, and that's the end of the notebook. If you guys have anything else to say, or if there are any questions, we're ready for it. And otherwise, thank you.

Thanks, Sam. Amazing.

You know, so the, maybe the last thing to say to around the surface, just that we're trying to make this very flexible. So especially maybe some power users may be interested in adding, you know, new ways, custom ways, such as you use an exact inference method or a sampler, etc. But also make some of these methods ourselves so that they can kind of just be, you know, you can say, I want to use Turing instead. And specifically, when we use the discrete pumpdp as a generative model, we will rarely need to use a sampler because the message passing usually works quite well. But in some other generative models, we might need maybe not a sampler, but maybe a gradient descent variational inference, where we can use the

full flexibility of these peak counts. And another, I mean, the exact inference method you should say, should be faster than the standard message passing method, probably, because it doesn't have to converge over time and that kind of stuff. So, and yes, so we're trying to make it both flexible and able to rely on the huge power of some of these PPLs like Rx and Fer, but at the same time, keep an easy to use interface on the top, like Sam showed in the beginning. Okay, we have a few minutes left. If there are questions, I guess Daniel, you can decide how to spend those few minutes. And I've been responding a little bit in the chat. So I don't think any of the questions there.

Yes, up to you, Daniel.

Daniel Kahneman Okay, yeah, if anyone has a question, they can write in the live chat.

Because your defense is tomorrow, Peter, even if it's kind of related, what is the title of your dissertation? What are you defending tomorrow?

Daniel Kahneman Tomorrow I'm defending, it's called modeling the mind. And it's uncognitive modeling quite broadly. So we're including this software that we're showing here, but also the HGF software and the action model software. And there, so the whole point of action models is that when you do applied cognitive modeling, and you fit models to behavioral data, and you can either fit them one at a time to each participant separately, but there are much better ways to do that. So you can fit them in these hierarchical models where you assume that there's something common between people, you can even fit them at the same time as you relate them to, for example, questionnaire data, if you want to find out how people's learning rates relate to their schizophrenia, for example, fit that one or even have one model where you fit neural data and behavioral data together. If you're interested in how your active inference model relates to brain data, etc. So we're making software to do all of that. And since that's compatible with the active inference stuff we just showed you, and that all comes for free. And then the thesis otherwise is on there. So we use that to model hallucination, where you model hallucinations as false inferences. And in our results, they seem to come from having too strong priors relative to the observation. And the theory is that that's because of sensory noise, the noise in the sensory system, which makes you learn over time that the observations are noisy and then you trust your priors more, the priors to your predictions, same thing. And we do that with people in different modalities and find out that that's modality specific. So just because you hallucinate you hear stuff that's not there, that will correlate with having too strong priors in the experiment. But it won't have effect on visual necessarily. Yeah, that kind of stuff. And then I think last year, now that you asked Daniel, it's very pleasant, but also I'll talk for ages. So I'll just run off by saying that the last thing we also used is to not reconstruct the generative model of a person's underlying a person's behavior, but instead using it to reconstruct the generative model underlying a collective's behavior. So you simulate a bunch of little agents, you treat them as one agent together, you fit a model to it like this. You know, some of the stuff that we can do now that's shown us this, we can actually ask, we can use model comparison to test, do people use message passing or do they use gradient descent variation inference or might they even use a sampler? And it might be good theoretical reasons to not think that they use a sampler. There are people out there who do think people use samplers. But it just allows us to specify these active inference models more flexibly so that they can do, so that we can properly compare the full space of what they can play.

And is that enough response to that question, Daniel?

Daniel Kahlman, Cool. Thanks.

Daniel Kahlman, And then what will be next for the package or how do you see it going forward after this year?

Daniel Kahlman, So, and you can jump in Sam and you want to send, but I think

one thing is we want to make this interface as easy as possible, have many have a flexible. So that's what we've been talking about today, but make that make some good tutorials for that. So in case you want a continuous state space, continuous action, discrete observation for whatever purpose you have, that you can do that and still use full active inference stack on top of it. Apart from that one piece of excitement, I think is exactly this nested inference stuff where so theory of mind is an example of nested inference where I have a model of Sam's model of me. And that means that inside my model, when I do Bayesian inference, I have to do Bayesian inference, so it gets nested.

And the same with metacognition. So I'll have a model of my own model of stuff and the same with sophisticated inference also actually. We've tested this a little bit and it looks like you can literally put Rx and FUR inside Turing without any problems. And we haven't tested that much, but if we can get that to work, then we'll get basically theory of mind, full explicit theory of mind, full explicit metacognition, flexible, sophisticated inference for free. And that's quite exciting. But like I said, that's still under, let's say, vision space. And then there are other things like we want inductive inference in there. One of these, you know, where you have some goal states and you are slightly lost you for a second there, Peter.

Inductive inference, I think, try to see.

Inductive inference, I think, where you learn not the usually accounts, but you learn the precisions instead, that's faster. Like we want these bells and whistles to be there, but also be easy to add more of them. And that's kind of the hope within the package. And then we can rely on something like Rx and FUR for the heavy, fancy, you know, message passing and all sorts of weird types of models. Yeah. Lots of excitement. And if someone ends up being interested in collaborating or contributing, that kind of stuff, they should just get in touch, would be very happy to do this.

One point of making it as marginal as we can is to make it easy to contribute also. So someone might like to add this specific geronism model. And if they do, we'll add to the package so that other people can just press, I want that model and then it's there. I hope that makes sense.

Okay. Thank you, fellows.

Bye.

Thank you. Thanks for everyone who's listening.

Thank you.

Bye.

Bye.

Bye.

Bye.

Bye.

All right.

We're back with Rx and FUR in action with Bart van Erp and Fraser as special guest. So thank you, Bart. Go for it.

Thanks a lot, Daniel. Before I start, quick shout out to Peter for the great talk before this. Also,

Peter, best of luck tomorrow with your defense. Exciting time. So yeah, go for it, I would say.

Today's talk is Rx and FUR in action. So my name is Bart. I'm one of the core developers of Rx and FUR.

Also one of the co-founders of lazy dynamics.

So we have been basically at the cradle of Rx and FUR. And I thought the symposium would be an excellent moment to share our latest developments with the rest of the crowd, get some new inputs,

new opinions. If there are questions, feel free to hop in, right? I think I want to make it very

interactive, interactive as possible. So feel free to interrupt when needed. So what are we going to

talk about today? So for the people who've never heard about Rx and FUR, I want to give a short

intro on what Rx and FUR actually is. I'm aware that most of you already are familiar with this

technology, but just to get everyone on the same line, I want to make sure that everyone has the

same background information there. And from that moment on, I actually want to start building

applications. Some want to show some cool applications that we've been building over the

past couple months, starting with an actual drone demo. We'll also introduce gears.jl, a new package

that we've recently released. We'll talk about our RxInfer MCP server that we've been releasing.

And at the end, we'll have a sneak peek in stuff that we're currently working on. But I won't tell

you what it is. So if you stay long enough for the end of the video, I think you'll get some cool insights.

So let's get started. So what is RxInfer? So in short, RxInfer is an open source library for scalable

probabilistic programming. So what does this mean? Probabilistic programming is scalable.

Well, it has to do with active inference, Bayesian inference. We can do planning, control. Some people

call it explainable AI or better AI. But I think at the core of it is uncertainty quantification. We want

to create programs, algorithms, which are probabilistic in nature, which do not only propagate just values,

but also uncertainty margins, allowing for more robust programming or more robust algorithms.

And what RxInfer does very well is that it scales very well over dimensionality. So compared to

Turing, for example, which is mostly sampling based, RxInfer has a different technology under the hood,

which allows it to scale to problems with a lot more variables and a lot more dimensions.

So how do we do this? So at the core of RxInfer, we have factor graphs.

For the people unacquainted with factor graphs, basically, these are probabilistic graphical models,

which allow us to describe some model of the outside world. This can be the dynamics of a robot.

This can be some assumed dynamics of a signal that we're observing, for example, financial data.

And what we can do is we can create these probabilistic or generative models and decompose

them into smaller terms. Each term is here, for example, one of these nodes in this graph.

And the edges that connect these nodes, these represent variables. So you can think, for example,

that there is a state that progresses over time and each time it goes through some particular transition f.

And that's kind of these equations govern the process on how the state evolves over time.

And we also make observations. So the observations we plug in as these solid black nodes. Basically,

once we observe something, this information will be captured in the hidden state and will be used for,

let's say, making predictions or doing state inference. The benefit of using these factor graphs is that

they are explainable and white box. So we can basically dive into the underlying model

and figure out, okay, what's actually causing the behavior that we're observing.

And because these models are mostly generated by domain experts, by domain knowledge, we can actually figure out, okay, which piece of this graph, which small box is causing specific behavior to happen. And it's just like Lego, right? So you can tie together all kinds of different nodes with all kinds of different edges, which makes it very modular. But all the computations are performed locally. So although you create a huge model, the computations will be on a very local and small scale, compared to when you would do, let's say, all the operations at once in the entire large model.

So what we have developed in order to create these factor graphs in code is our domain-specific language called Graph PPL. And it looks roughly like this. So we introduce in Julia the `addModel` macro, and this `addModel` macro allows us to specify our model. So here we have, for example, a Gaussian state-based model,

where we have observations that we call Y . In this model, we specify two parameters, θ and τ .

And we say each of these parameters will get a gamma prior with rate and shape of one.

The hidden state, we say, okay, it's going to be from some submodel. And that submodel, we can also define in the model macro. So by doing so, we can kind of create this hierarchy of models, basically nested submodels. And this submodel is one, let's say, timestep in our Gaussian state-based model.

So we can see here that the next state, so s is a shorter state, will be distributed as a normal, whose mean is some function over our previous state, and the precision is governed by θ .

And our observations can then be extracted from this state through some function G , where we model the observation noise with the precision parameter τ . So this is one slice, let's say. And these function FNGs can be very different depending on the application. It can be drone dynamics, it can be robotics, modeling financial data. So it can be very diverse. And we can basically use this submodel macro to expand our Gaussian state-space model, basically putting the different slices after one another.

And this basically allows us to specify this vector growth in a couple of lines very, quite similar to the domain-specific language that, let's say, `turing.cl` also uses, but a lot more easier to use than some of its counterparts, right? There are also a lot of programming languages out there, which should not offer such a nice DSL. And so I think this is very nice, very great to interact with the tool. Makes it easier also for outsiders to actually get started building these models without having... So just some story. So all the way at the start, we had to manually create these nodes and then also manually these edges, and we had to connect everything together. Yeah, that turned out to be very, very annoying. So this is basically its evolved version. And in general, we're all very happy that this language actually supports these features. But basically, they allow us to specify a probabilistic model in which we want to do the computations, right? The way that we do the computations is through variational inference. So what variational inference basically does, although most of you are already familiar with this, is basically it allows us to approximate complex, often multimodal, and skewed distributions, which distribution that is from the exponential family. So basically, instead of doing all these exact computations, which are very difficult and tractable, what we do instead is we say, okay, we're gonna come up and say, okay, we're gonna approximate our result with, let's say, a normal

distribution. And with this normal distribution, we'll try to find the normal distribution whose mean and variance best approximate the distribution that we actually, the actual true distribution, let's say, that we can obtain through sampling, but sampling is oftentimes too slow. So here we use variational inference, basically turning this variation inference problem into an optimization problem, right? So what we minimize is this variational free energy functional, where we try to find, okay, what Q minimizes this specific functional for our probabilistic model P . And P is the one that we previously defined in the vector graph. This looks all complex, but in the end, everything is automated in Rxinfer. So there's no need to worry about these equations. What's nice about variational inference is that that scales very well, and it's very efficient in comparison to sampling-based methods, because we oftentimes have closed-form expressions, and we leverage those to basically prevent having to sample thousands and thousands of times. And this free energy functional inherently has a trade-off between complexity and accuracy and accuracy. That's also well-known in the active inference community. So basically, you can think of it as kind of an optimal approach for actually finding these posterior values that you're interested in.

Also for this, we have some domain-specific language. Basically, we allow with Rxinfer to give like an easy interface on changing these approximate distribution skew. So these distributions skew are the ones that we set. Let's say the simple normal with which we want to approximate a more complex distribution. And we can also apply different constraints on that. So here in the bottom, we say, okay, we want our posterior over S , over our hidden state, to be normally distributed, just using very common Julia syntax.

And we can also state as well these different states as we have them evolving over time. Modeling them all as one big Gaussian is not always desirable, because that grows, of course, pretty badly with the amount of states. So what we can say is we can say, okay, we want to actually impose a mean field constraint. So for each individual state in our model, we create a different queue, which is not correlated with the following one. So by that, we basically can reduce the computational complexity of this simplifying distribution in order to make inference run even faster.

So we can do that.

Then putting it all together. So we have vector graphs, we have aggregation of inference. What can we do with it? Well, RxInfer, as I hinted already on earlier, it doesn't do sampling. It's a bit, quite a bit faster than sampling based methods. Instead, what we do is message passing.

So basically what that means is we use the local structure of this vector graph. So these nodes, which are very sparsely connected to each other. And we perform these computations that basically optimize the free energy. We optimize this free energy on a local scale.

So each node basically tries to optimize or minimize this free energy on a local scale. And that allows it to do computations, approximations of our posterior distributions. So basically what we can do is we can compute these posterior distributions, Q , by passing information between the individual nodes. And this information flow, that's what we refer to as messages. Oftentimes, or almost always,

these messages are distributions themselves. So basically what we are doing is we're propagating distributions across the vector graph, across the network. What's special about the way that how RxInfer works on this is that it's fully reactive. So if there is no data coming in, also nothing will happen because it's lazy by nature. That's also one of the reasons why our company is called Lazy Dynamics. So we only really do stuff when it actually is very much necessary. And that's also the same principle that underlies RxInfer. So if there is no data coming in, yeah, the model doesn't have any information or needs to update its beliefs, right? So it then also doesn't do any computations. So this is quite adaptive, robust. So also when there are sensor failures, we can adapt to that. Once, let's say, one of the parts of the model breaks down, like in real world, then we can stop passing messages there. And because it's reactive, it also allows us to put to every computation in parallel, such that they don't wait for each other, but they can just pass the messages along each other in parallel, making it possibly also scalable to new chips that come available on the market.

So RxSpense, for the ones who haven't seen it, it's open source, like the core of RxSpense is fully there. On GitHub, you can basically find all the information to get you started. We have a bunch of examples to get started with, ranging from small drones flying around to discrete state space models, POMDPs, Gaussian processes. And basically, we're expanding this list of examples, like every couple months, there will be new, like a few new ones. I also recently saw that we have new examples on icing models, doing some very cool projection techniques for approximating posterior distributions. So there is a lot, basically, a lot of information in the example. So if you want to get started, I would definitely recommend to check it. Just go to rxspense.com and there you'll find the link to the examples. We also have discussions there, where we are very much open for feedback. If you have questions, want to know something, just post it there and one of us will definitely reply to you. Also, if you want to see, like, enhancements to the tool, we very much encourage you to post, like, your opinion, your wishes there. We are currently looking into the possibility of creating some kind of logging interface, where we actually are requesting people to also share their ideas, thoughts, such that we can basically have a look inside rxspense in real time to see what messages are being passed, and also how the computations actually happen. So basically, just running it in real time, debugging it in real time. So that's what we are currently requesting ideas for. So if you have them, please share. And of course, we also have issues. I wish we didn't have the issues page, but unfortunately we do. If you want to help us out, very much welcome to contribute. I think we have quite some nice ones or easy accessible ones out there. If you want to help out also reach, just send me a message or just post something on GitHub and we'll definitely get back to you. So good. That's rxspense. So rxspense, everyone now has kind of an idea of what's happening under the hood. And what we wanted to show to you is some drone demos that we've been developing over the past couple of months. So what rxspense basically allows is to create high fidelity control and planning systems. A couple, I think, yeah, a bit more than a year ago, we also presented a nice presentation here on the Active Inference Institute, where we build a drone with rxspense, a very simple two-dimensional drone, just to show everyone, okay, how can we actually put these complex dynamics of this drone into

our rxsper interface and how can we run it. So I would definitely, if you're interested in this kind of topic into robotics, definitely recommend to check this out because I think it's a very nice video. I won't play it here because I think it's rather long, but yeah, I think it's definitely nice for people to get started. What we've been doing for the past couple of months, basically we've been expanding upon this. So in this model stream, we worked on a two-dimensional drone example with simplified dynamics and we went all overboard, basically. We're now in three dimensions, six screens of freedom, and we're modeling the drones to a very, very precise... so we're modeling the dynamics in a very, very precise manner. We're incorporating drag coefficients of the body, not only translational, but also rotational. We are modeling the electrodynamics of the rotors. So basically all the dynamics that we could find of a drone are now basically inside a model. And I want to basically share with you the results that we obtained.

So we started off with dynamic obstacle avoidance. So we have this drone here in simulation, because it's a bit difficult to have these actually in these bulbs. And with this model, modeling turn the dynamics, the electrodynamics, everything there. And the idea of this drone is that it needs to follow this trajectory. So there is this blue kind of arrow indicating the target or the goal that the drone should follow. And this target moves along this trajectory, right? What's very cool about this particular demo is that all these spheres that you see flying around, the drone doesn't know about them. So while actually flying, it doesn't know about them. Only when the drone gets into a certain proximity, it sees them. And then of course it needs to adapt, right? So this drone, it will fly around. Once it sees one of these spheres, and if it's in its way of actually following this line, it will adapt to that actual obstacle and also plan around it. So it will try to reroute, find another path that at least follows the goal as good as possible in order to still kind of keep track of this target. So I think this is something that we are very happy with, with the results. Putting it onto hardware is of course a completely different challenge. But I think for the first experimentation, I think we're very far away or very, we made quite some progress. And this is for the simple one drone dynamic ride. What we've also been experimenting with is actually having multiple drones. Just to give you an idea, so the drones, they start around z is equal to zero. So on this ground plane, and basically we task all of these drones, like 20 in total, to fly upwards. And also in this scenario, there are these spheres which are like in the way of the actual reaching the target within a straight line, right? So each drone basically operates independently from each other. And they also don't know where the other drones are. So it follows a similar algorithm as on the left. But in this case, it actually, once it sees the sphere, it again tries to reroute around it. But it also isn't aware of the other drones. So at some point, some drones also encounter other drones. And here they will also adjust their trajectory in order to not collide. So you can see that for example here with the pink one. So you can see that the pink actually dodges the green one. By making a sharp turn. So just to show everyone what's possible with RxInfer, right? So RxInfer, I think the examples already are getting more advanced as we speak. But we can achieve very cool applications. Drones, I think, is already one that's very complicated. So we're very happy that we actually

managed to get these running also in real time. What we've also been working on is a new package. So RxInfer is fully reactive. And what we intend RxInfer to work with is basically sensors which operate on different frequencies. We can have, for example, a camera which runs at like 30 or 60 hertz. Whereas if we have a microphone, it has a sampling rate of 48 kilohertz. IMUs are also somewhere 300, 400 hertz. So we were thinking, okay, this is nice. RxInfer is kind of targeted to solve these particular problems. But how are we actually modeling them in Julia? Because it's not so easy to create a particular simulation environment where every sensor runs asynchronously in parallel. So that's quite complex. That's why we introduced gears.jl. So basically what gears.jl allows you is it enables fine-grained scheduling for simulations and real-time systems.

So just to give an idea what does it mean? It basically allows us to create simulations of systems where we have different observations coming from them at different sampling rates. So basically in the situations where we want to be in full control of when we observe what or when our environment gives us a new observation to process, that's something that gears.jl has been trying to solve, basically. So just to give you an idea of how it looks, once you import gears in Julia, you can create a variety of tasks. So in this case, we can say, for example, every 200 milliseconds, we want to print the following line that the task is executed at some time step. And then we run our scheduler. So we can say, run the scheduler for one second. So for the next one second, update. So basically the tasks that we previously defined, these will actually be executed during this one second duration. And we can make this more complex, right? So for the drone example, we can say, process IMU data every five milliseconds. Do this or perform the replanning or update the control sequence every 200 milliseconds or 100 milliseconds. If processing audio, maybe yeah, process the audio every 10 milliseconds. So we have very fine-grained control of the actual data sources coming in. And we can run this.

So we can do this. What's very nice about this is that the clock that basically runs these tasks and basically does the scheduling, we can also replace the machine clock that gets regularly used with some virtual clock. So basically, if you want to do simulations spanning over, let's say, years, like if you're modeling, I don't know, planets in their orbits, you want to do something with that, we can actually also add in a virtual clock, which allows us to speed up or slow down the simulations. If we want to model specific behavior on like a nanosecond scale, not sure what kind of applications go there, but I guess there are. Then we can also slow down the virtual clock, such that we're not modeling on the second level, as our hardware normally would, but on the nanosecond level. And we have a variety of schedulers to basically create our own simulation environments.

We'll have, we have the intervals that we just saw. So every 200 milliseconds, do this.

We also have something called an ASAP scheduler. So as soon as possible. So as soon as something happens,

perform a task. So this can be used for chaining tasks together and also event-based. So when the same observation comes in, I always make the example where we have, for example, an API request coming from another service. That's also something that you can work. And what's nice is that these

tasks, they can run asynchronously. So when you have a lot of different tasks at the same time, it actually figures that out and you can run them asynchronously in parallel.

So that's just a short introduction of gears.dl. So basically they allow us to create these complex simulation environments. And in general, I think it allows us to experiment with the core capabilities capabilities of RxInfer better than just native Julia code. So that was one release that we did a couple weeks ago.

Something that's a bit more recent and then that they're also very excited about is RxMCP.

So what we heard from the community is that although we have this nice domain-specific language, using RxInfer is not always the easiest, right? So it's still challenging.

So what we wanted to do with this project is basically what we wanted to give the capabilities of RxInfer a broader audience. We wanted to have a lot more people interact with the tool without knowing that they're actually interacting with the tool. So instead of downloading Julia, which, yeah, is another programming language, people are a bit hesitant if they want to go away from Python.

We just wanted to make it easier for people to interact, even like senior executives, right?

People don't do any programming anymore, right? Those are the people that we also want to target. But what these people often require is some form of analysis.

Some form of basically

like processing data with uncertainty because that's where it shines. So with RxMCP, I think it has never been easier to use some parts of RxInfer. So what it allows is basically grounding LLMs with RxInfer using our model context protocol, where MCP for short. So effectively, you can just chat with RxInfer. So if you have a data set that needs processing, you can throw it in, ask it a question, and it will basically figure it out. So together with the strength of LLMs, basically converting numbers into some natural language and using, let's say, the uncertainty processing of RxInfer, you can actually get the best of both words.

We already have integrations now with Mistral, with Cloud, Cursor, OpenAI. You can find the docs as well.

So we added in the QR code in case you're interested. At the moment, it's free to use. So we definitely recommend everyone else to check it, use it, give us feedback, and our goal is to improve it further along. To give you an idea of how it works, RxInfer, it starts with some context. So you can think of, oh, I have this huge data set. I'm going to throw it in one of my favorite LLMs, and I'm going to ask it to do something. So I'm going to query it. Well, RxInfer will then basically learn from the data that you have supplied. So it will try to figure out what are underlying relationships between variables.

And it will also predict, so it will give you suggestions, actually. It will compute the posterior distributions, and those posterior distributions will then hand back over to your favorite LLM, which can interpret these distributions. So basically, it can give you then recommendations, based on the query that you provided, and the uncertainty-aware results that RxInfer provided.

So basically, you get the best words, you get the best of both worlds. You get to use the analysis capabilities of RxInfer, whereas you have your LLM basically explain what it's returned, and why this is important. Just making the audience of RxInfer a little broader. So I think we're very happy. We're also

getting quite positive results here. A lot of positive feedback of people who otherwise would have never been able to use our tool. And we're very excited to also engage them. I also wanted to show a short video. It's two minutes of our CEO, Albert. He basically recorded the video demonstrating the capabilities of RxInfer. What he did, because it's a relatively fast video, he basically illustrates the use of RxMCP using a problem that a lot of companies, startups are actually struggling with. So suppose you have a thousand contacts and you want to figure out, okay, which contact should I actually pursue in order to make a deal? Which one is the most, which one has the highest return of investment? What is the most? Yeah, basically finding the person will get you the best deal in the shortest amount of time. He created the data file for this and I think I'll just play it and be silent for two minutes. I can let Albert do the talking for now.

Every startup has the same nightmare. You are making critical decisions with limited data. Which feature to build, what price to charge, who handles which lead. And every wrong decision burns runway. Using just LLMs for

this, well, those don't reason about cause and effect. You need grounded mathematical decision-making that actually learned from outcomes. Let's watch how RxInfer solve this. I'll use lead assignment as an example, but this works for any decision, pricing, feature, content, anything. Well, here's a bit of an irony. I'm using LLM to interface with RxInfer and even it struggled to parse with the, parse the unstructured data at first, but well, RxInfer underneath, pure Bayesian reasoning.

So we've got five new leads, RxInfer assigns them, Jake gets small companies, Lisa gets enterprise, and these aren't guesses. They are mathematically optimal given uncertainty. Now let's imagine we've got the results back after following these predictions. Well, Jake's sure bets lost 20k. Uh, Lisa enterprise deal made 40k. And here's the really cool thing about RxInfer. So I feedback these results and instantly RxInfer updates

its beliefs. Uh, and turns out that Jake isn't the right for a founder led deals. Try this instead. And this isn't just pattern matching or token prediction. It's a causal reasoning about actions and rewards. Next founder

lead arrives. Model says Lisa, not Jake. Uh, it learned the causal relationship from five outcomes.

Obviously you won't be typing these comments, uh, uh, yourself. Your agents will call RxInfer MCP directly. And this works for many things, right? For pricing, you can learn price elasticity per segment. Uh, for features, you can discover what drives conversion. And for content, you can understand the engagement causally. You need decisions that are grounded in reality, not just text generation.

Uh, RxInfer is open source. RxInfer MCP is free with integration docs available now.

We invite you to start learning from every decision today. Check out placedynamics.com.

So wonderful. So that was a very short demo. I, uh, was thinking about doing a live demo, but you always know

live demos never work. So, uh, that's how I wanted to let Albert show you. So what he also mentioned, and I think this is, uh, very, very interesting is basically RxInfer also really learns. So it's, it learns, um, based on new data. And once new observations come in, it will actually adjust its beliefs.

This is very different from the LLM chatbots that, uh, you have with a lot of companies. I was trying to book a flight over the weekend, actually. And I ended up chatting with this chatbot, which you of course know it's an LLM. And despite saying, well, just connect me through, connect me through, uh, it was quite stubborn. So I think this is also one of the places where RxInfer would shine basically update the beliefs of these LLM as data comes in instead of just appending it to the context, right? So that's currently what's happening.

And then a sneak peak latest developments. So this is very, very rough. And I want to kind of warn you that, uh, functionality is not fully there yet. Um, but we are still wanting to show it because we were very enthusiastic about it. Um, and we wanted to get as much feedback on this topic as possible before we actually further develop it.

I think it's very cool. So let's just get started.

So what we have been thinking about is doing uncertainty quantification in LLMs. So there is this huge problem of hallucinations, figuring out whether LLMs can actually work.

And on the other hand we have RxInfer, which is fully probabilistic. It can quantify uncertainty very well. So why not marry both? So we've been thinking about, can we actually combine LLMs with RxInfer? Not in the same way that we do with MCP. So with MCP, we say, okay, LLMs go use RxInfer.

But basically what we are thinking about is using RxInfer to upgrade these LLMs.

Um, basically that it doesn't just provide you with an answer, right? But that it also provides you with an uncertainty margin. Instead of saying, ah, you're right. You're fully right. Definitely.

Good question. This is the answer. Maybe it will be better if the LLMs become a bit more nuanced.

So that it says, I think this is right with, let's say, 50% confidence. Um, so we've been exploring ideas basically for RxInfer to integrate with these systems. Uh, we have a very short here on the right, yeah, visual representation of how something could look like. So next time you're using, uh, JetGPT, it asks, oh, do you want to use RxInfer for, let's say, verifying Q and A accuracy? You can say, yeah, I would like that. I would, I would want to know how certain the LLM actually is. Um, you confirm, and you can just then say, okay, RxInfer will, uh, will run on the backend and it will figure out how certain the LLM actually is in its answers. So we're still, this is still very much a work in progress, but we wanted to get some initial feedback, whether you think this is interesting, uh, what else would you like to see? If you have great ideas, let us know. And if you want to test it out, you're very much welcome to, and we highly appreciate it. Um, if you want to know how to get started, otherwise just send me a message or, uh, reach out somewhere, uh, and then we'll definitely give you, let's say the API keys to get started. So with that, I would actually like to conclude, uh, and I'm open to start a discussion, very interested in what you guys all have to say.

Thank you, Bart. Fraser, if you'd like, maybe say hello and check in, and then feel free to start with any questions that you have. Yeah. Hi, thanks Bart. Um, hopefully you can hear me. Okay.

Uh, so I'm a brazer. I'm a PhD student at the University of Glasgow. I also do a little, a little bit of open source work with the BIOS Lab. I'm very honored to, to do that. Um, yeah, I was excellent stuff. I'm particularly interested in this, you know, gears framework for, for scheduling

tasks and so on. Um, I'm trying to think about how to build certain active inference agents using our X and FUR that interface with a world that might itself be reactive and so on. Uh, which as you say is quite difficult right now. I'm, I'm just thinking, you know, gears is excellence. Um, in some sense, it's a way to, it is just a way to schedule tasks. You know, we had our environments before as a way to kind of talk about actual entities that are out there. Um, I'm thinking, well, if, you know, potentially that could integrate with gears, that'd be a really powerful way to, uh, afford those kinds of environments and, and things that we could then place agents in, um, to kind of give them a home. Is there anything you can talk to there about potential cross-pollination between, um, RX environments and gears, or if there's maybe some, some other replacement package for environments? I'm not sure I'd be interested in your thoughts there.

That is excellent. That's a very good remark. I don't think we have actually done any integration there ourselves. Um, for the very simple reason that, yeah, we didn't get around to it yet.

Uh, but I definitely foresee that like joining both with, yeah, we have this package already out there.

Um, so I think we can leverage both and use both strengths to, uh, to make it even easier to create these reactive environments. And I think gears allows us to, uh, not only make them reactive, right? But, uh, to also do the actual, let's say precise scheduling when it's needed.

So for example, for sensor data, it's yeah, in a sense, it's reactive, of course. Uh, but at the same time, it just samples every so many Hertz. Um, so I think

RX environments like, um, these are like fully reactive. And that's very nice. On the other hand, sometimes you do want to have this control over when actually sensor data comes in. And I think that's where this gears is really shiny.

Yeah. The combination of both seems like it'd be a real winning strategy.

Yeah. Okay. Maybe I'll think about that myself. What I can do there. So.

Excellent suggestion. Thanks.

I'll, I'll read a question from live chat and have some related questions. So Burt Berkens wrote, could RX MCP also work for POMDP models in the future?

Uh, yes, uh, definitely.

So these POMDP models, right?

Uh, in general, I think I'm not even sure whether we maybe already are supporting them.

Um, so just to give you a very quick overview of what RX infer MCP actually is. So we have RX infer, which is our software tool. We have RX infer server that basically runs RX infer in the cloud for some selected models. And then we have RX infer MCP, which connects the LLM to the server.

Um,

Um, so for RX infer server, which is also open source, it's possible to implement these POMDP models by themselves. I'm not entirely sure whether they then also directly become, let's say, connected to the LLM.

Uh, so that's something I would need to check, but at least we, you can send easy, a simple API icon, uh, connections or requests to RX infer server and have the POMDPs running there. Although I do foresee definitely that connecting this with MCP as well could be very valuable when you have indeed

some data that you just want to process. Uh, you don't want to write all these, uh, data loading functions and you just want to throw it in there. It's Friday afternoon. You want to finish this up before you leave, uh, leave to home. Let's say I think that would be excellent. In the end, the idea is for MCP to start supporting as many, uh, models as possible, uh, as relevant. Uh, but at the moment, they were prioritizing the ones that we have kind of open requests for, uh, for the very simple reason that we can, we can put everything there. Uh, but we're, yeah, trying to be a bit smart about our time and our, uh, yeah, time investment there. Yeah. On, on the RX MCP. So you said that it learns from the query and the data. So what structure of graphical models does it start with or sweep over? And do you get back at model blocks, which you can separately execute? I mean, is it doing an end to end RX infer for meta programming and then reporting outcomes or what is really happening underneath the call? Excellent question. So, um, what I mentioned, so we have this RX infer server where we have some models running for the demo that we just showed. We have a particular model in our for the one that created that already. Um, but it's still very generic. So basically everything that has to do with, uh, contextual decisions where you have some data and you want to make a decision. That's where basically MCP currently is targeted for or tailored for. Um, so the MCP is basically a connection going from your LLM to RX infer server in the cloud. Um, it does return quite a bit. I'm not sure whether it returns the full model macro, I don't think so to be honest, although this is something that if there is a real need to return this model macro, sure that can be made possible. Um, but what is actually returning is all the posterior inference. So basically all the raw results that you could normally obtain with RX infer itself, let's get sent back to LLM and then the LLM is used to interpret these results. Okay. Makes sense. So the MCP facilitates within the repertoire of what the server has access to. Exactly. Yeah. Okay. I'll ask a question from the chat and Bart, and then Fraser, you can add any thoughts too. Okay. Rob wrote broad question. Where is uncertainty valuable in applications? I see its benefits in fundamental science, but I struggle to find effective examples in applications. There are plenty, there are plenty, but I like, I like the question quite a lot because, yeah, we've been looking through this uncertainty lens already for a couple of years, right? So we can see where it's valuable in almost any application. Um, but yeah, I can, I can definitely give some examples here. Um, in robotics, we have sensors, for example, and these sensors, they make particular observations, but the sensors can also be noisy. So if you move, you say the arm of a robot or you drive a cheap robotic kind of around, it's, will give you some measurement of where its next location is, but at the same time, you know, well, the sensor readings aren't that accurate. Maybe it, it slips a bit, let's say, uh, on the ground. So yeah, there will be some deviation in the end result. So that's one place where you could use this uncertainty of raising these margins. Um, and you can use that to correct also for it at some point. Another one, which is, I think very, very important is everything concerning financials, like financial markets. I wish financial markets were deterministic, because if that was the case, we could just pour in money into one stock that we are aware that goes upward. But in reality, no one knows which stock goes up unless you have inside information, of course. But for most, it's just a guessing game, a game where you try to figure out, okay, which stocks are going

up, which stocks are going down. And oftentimes, what the term that they use in these financial markets is called volatility. So basically, it describes how volatile the stock is. So if you have a stock that's very non-volatile, its price will be very smooth over time, whereas the volatile stock will go all over the place. And that's something that you want to be aware of. You want to be aware of this volatility or actually uncertainty about how this stock is progressing over time. If you're going to use that investment fund for, let's say, throwing in your pension money, that's something you probably want to have stocks which are more secure, have the less limited volatility. Whereas if you're like private equity, that kind of stuff, you want to take a bit more risk. So that's also an application area where you can think of. And I think that makes uncertainty shine.

Yeah, like having the point estimate without the variability measure of some kind is just half the picture. Because if there's something that you expect it to be 1% improvement, if it's an engineering process or financial, but you don't know if that is a 1% plus or minus 10 or 1% plus or minus 0.01%, then you wouldn't really know if that was actionable. And then you would need to go another level. Okay, 1% plus or minus 2, but how confident am I about 2? And then that kind of dovetails back into why we want to have these composable, flexible Bayesian models where we can interpret those different uncertainty structures. Yeah, exactly. For some occasions, propagating no uncertainty itself is also fine, right? So for some applications, you don't need it. However, what you see is for like very critical applications, it's actually good to have some kind of quantification of margins, uncertainty. So yeah, definitely, definitely.

Pretty sure? Want to add anything?

Yeah, just maybe for a really speculative question now. I attended IY earlier in the year, the international workshop on active inference and laser dynamics and RxInFur team from the Bioslab you guys presented. There was some work on kind of recasting the EFE for planning in active implementations as the VFE. So that we just have one free energy marginal, well, one free energy functional that we're minimizing in order to do active inference. And you know, that'd be quite nice because we have two sort of canonically floating around in the active inference world. Maybe that's a little bit annoying. Obviously, this is the fruit of ongoing work at the Bioslab. So it's all very speculative. But is there anything you can comment on with respect to the way in which this might eventually land in RxInFur? You know, is there a way in which you can see how the death free energy might change in consequence? Or a way in which users might be able to plug in? Oh, you know, they want this kind of unified functional thing for message passing to happen here, as opposed to not. Is there is there anything you can say about that at all? It's very, very broad question. Sorry.

No, it's an excellent question. Now, indeed, so our colleague Wouter Wouter has been working on this particular topic, trying to frame the expected free energy as a variation of the energy minimization procedure by doing some smart tricks basically on this vector graph and in this message passing procedure. I think once, so the expected free energy has compared to, let's say, variation of free energy, some other perks, right? So you get, I think, one of the most important ones is that you get this information seeking behavior. At least that's one of the perks that variation of the energy by itself

might not always have. So for some applications, this is very critical.

You know, you're critical. It could help you a lot in actually trying to learn, let's say, the parameters of your robot or the parameters of the financial markets. Although for those, you don't really have an influence yourself. So I think robotics would be a better example. But indeed, it's ongoing work. How we see it is we see it kind of as a different flavor, different flavor to what we're already doing. It's also still very much under development. But I think, yeah, progress is very good. The results are very encouraging. So I can't wait basically for the results that will be popping up in half a year or a year from now. And once they do this, there's definitely incentive to integrate this in RxInfer as well, right? So it's not that we are thinking, oh, there's better free energy or variational free energy. It's kind of the holy grail. For some applications, XFRE is just more interesting, more nicer. It gives you properties that other targets wouldn't give you, let's say. So that's definitely something that once this research kind of matures a bit, once we have a cleaner idea of how we can integrate it in an improved manner on a larger scale, then that's definitely something that we are planning to work on. That's also why we're, of course, performing this research because, yeah, we want to keep improving the toolbox, not only for, yeah, but also for actual industry problems.

Excellent. Cool.

What are you most looking forward to or what do you think will be able to bring it into most people's hands in the coming year?

Oh, that's a tough one.

I think we're making very, very nice progress with

MCP. So as I mentioned, the motivation behind MCP was really to get RxInfer into the hands of as many users as possible. I think providing it through an LLM, basically just having something that everyone can access, even if you don't have university background or you don't know mathematics, you can still use it. You'll still understand that these uncertainty margins can be useful for specific cases.

I think getting it out in as many hands as possible is what we're really striving for.

And that's why this MCP, in our opinion, it's a very first step, right? It's kind of the first step in getting RxInfer out in the public.

And we are currently looking into applications where we might benefit most. So basically maybe specific application areas of this MCP where we can put it in as many hands as possible. So not just selecting these leads as we did, as always showed in the demo, but maybe there are also other application areas out there which are very much relevant, but that we currently are overlooking.

So yeah, if any, if there are any suggestions regarding this, feel free to reach out because we would be happy to extend the functionality of RxMP with these kind of new applications.

So that's basically one thing that we're looking very much out for.

Another thing is what I already mentioned is so we're kind of, and not reviving, but we always have this idea of creating this amazing debug tool for RxInfer. So from a developer point of view, I'm very excited about that as well. So what you can do now is basically it's very hard to debug because

everything's reactive. We do have some very deep developer options to actually figure out what's causing what. But we want to make that process simpler. So there is now an open discussion on GitHub, basically requesting ideas, announcements from everyone on how we can make this debugging or interface experience better for a lot more users, a lot more developers.

And from the laser dynamic side, we have already been experimenting with some of the tools, let's say. We haven't integrated them with RxInfer. But we have starting to get a better understanding of what's happening. What would be a nice interface, what would help. I don't think we want to go, let's say, all overboard and let's create our own full UI ourselves. So probably we want to leverage on existing technologies. But yeah, we're very much open for ideas, suggestions. If you think, oh, this would be cool, let us know. Definitely. Any feedback there is recommended or appreciated. Thank you. And it's been really fun with Fraser facilitating the RxInfer learning group and a lot of cross-pollination. And I think compared to where we were a year ago, you've really highlighted this bi-directional relationship with the LLMs. On one hand, there's the MCP direction, which is the LLM LLM using RxInfer as tool use. And then on the other direction, there's kind of two flavors. There's the uncertainty quantification of LLM outputs. So you can just ask the LLM repeatedly, what's the GPS coordinate of this location, and then use those estimates to get a posterior distribution. And then the next step there is to choose the prompt to the LLM with active inference to make informed sampling of those latent spaces. And so that is kind of building out that two-lane highway, going both ways between larger models and how they're increasing in power, and then all of the interpretability and benefits that graphical models and factor graphs bring. So that's very fruitful. And it's a very pluralistic, exciting time for building those kinds of hybrid models. Definitely. I think, like from our perspective, LLMs are very good with natural language, and we are all using them. Even people who claim that they're not using them, we're all using them, right? Drafting emails from time to time, and well, you know, right? But it does provide an easy interface. And I think that makes it very accessible for people to use it. So definitely, personally, I'm in favor of leveraging on the, let's say, the... Basically, using both tools, I think, combining them, and using best of both. I think that's what we really want to strive for.

I'll just ask one final short question from the chat. Jay Graves wrote, Several projects, including AI Knowledge Sidekick, are scanning selected information sources to create personal curated summaries. Could RxMCP be applied to monitor such a feed?

Excellent question.

I just need to think a bit on how this... So, in general, probabilistic models are very much useful for doing personalization. Just to give some context, it's a bit out of scope from the actual problem described here. But what we were designing RxInfer for from the very beginning is actually high-speed applications, and most specifically hearing aids, because that was the one the company was funding RxInfer from the start. What we were using RxInfer for, one of the parts, right? Not only the signal processing, but also for incorporating user feedback. So, when you do an interaction with the human being,

and I'm speaking for myself, right? So I had this... was struggling with this chatbot for booking a flight over the weekend. If I say, well, this answer is not listed, I want the model to kind of learn from this. I don't want to keep interacting. I don't want to say it 20 times, like, connect me through, connect me through. After two times, I think it should be very clear that I want to be connected through.

What we used RxInfer for personalization in hearing aids was basically figuring out, okay, what specific sound sources do we want to listen to? So normally, these algorithms, hearing aids or earbuds, they have this active noise cancellation, so they cancel everything. Or they have this hear-through mode, where you basically hear everything again. What if you only want to, let's say, listen to one specific sound? When you're maybe in a car, you just want to listen to sirens. Well, not that you don't, you want sirens to be all over the place. But once like an ambulance drives by, you want to listen to that, and you don't want to listen to the car engine. At least that doesn't make sense.

When you're out in the park, for example, maybe you want to listen to the birds, and you don't want to listen to the people who walk past you and have these conversations. So train station is also an excellent example, right? When you want to listen to basically all the announcements, but you want to zone out the actual sound of the trains. So figuring out what a human person would want to listen to is something that requires user feedback. And it's also context, very much context dependent, because if I want, if I'm at the bar, I want to have a different experience than when I'm working in the office.

So we have used RxInfer, it's a very long detour. But what I wanted to say is we have been using RxInfer for preference learning, where we have interactions with human beings. And creating these personalized reports, I think that definitely falls within the scope of what RxInfer can achieve.

I believe that having said that, I'm not sure whether it's directly applicable to...

Or that, sorry, that it's not applicable, that RxMCP directly supports this personalized summary generation. I think that was the original problem, right?

So, but definitely there are possibilities there. Definitely we can bridge this gap. It's not that we aren't able to do it. It's mostly we just didn't do it yet. I think that's, yeah, that's the current status of it.

Any final thoughts, Frasier?

I could stay for hours and hold the hostage for a long time. But no, I mean, I think that was really excellent. I enjoyed everything. I'm looking forward to everything that we can do together. There's a lot of leads there and what you presented. So yeah, thank you very much.

Yep. Thank you to you and the team and for engaging with the open source. And next year, we'll have another spiral of learning with RxInfer. So thank you guys. Bye.

Definitely. Thanks a lot.

Thanks.

Thank you.

Thank you.

All right.

Next is going to be a two minute and 30 second video by Michael Lennon.

Here it goes.

Michael Lennon Hello, I'm Michael Lennon, a member of the Active Inference Scientific Advisory Board and delighted to be presenting today on reflections on some economic myths that I've been encountering around AI over the last year.

And however, as I prepared this slide deck, I noticed that there was one point in particular that I wanted to draw attention to first.

So this is a rapid clip about a specific opportunity that I wanted to put in front of the community and speak about.

And if there's more time than to go into the formal Fuller presentation offered.

So the rapid headline is about the formation of a new consortium of major social impact investors in the United States who have pulled about half a billion dollars in funding, basically aiming to influence the present future of AI towards more pro-social social impact.

And so they are, as you can see, an impressive group.

They have three explicit goals that they've set to expand the group of people who are involved in the decisions of AI to strengthen the organizations that are bringing people together to shape AI in the future and supporting the creation of public goods.

So this is very well suited to what the active inference institute has been up to.

And so I wanted to start here drawing attention to two of the slides that I have later on in the second clip that speak about both bioregionalism, a form of local environmentalism that is organized around natural ecosystems rather than political or

or jurisdictions.

And how, in addition to having networks of allies that we have established methods and specialized tools that can make us really stand out in this space.

And we have been for some time, but I'm working on this, but that as a community would like to draw attention to how might we be building additional collaborations and how might we be building additional synergies to really make a collaborative difference in the world.

All right.

Next up is going to be a three minute video from Joel Dietz. Here we go.

I'm showing you a tool that I made originally for LevinLab that I've sort of been extending to Active Inference. But first I will show sort of how it works and the concept behind it. So the goal over here is to show the summary of our research and field and extending.

I'm showing you a tool that I made originally for LevinLab that I've sort of been extending to Active Inference. But first I will show sort of how it works and the concept behind it. So the goal over here is to show the summary of our research in field in a field on a time series to show how well various hypotheses are validated. And if you kind of roll back sort of clock within a field you'll be able to see you know over what time series you know you start validating various

hypotheses. This is ideally to cause people to make more and better hypotheses and to sort of automate this. It uses the probabilistic knowledge graphs. So the concept here is that everything is you know extracted sometimes from peer-reviewed papers that's mostly we're working off of into various evidence and assertion things. The weight of evidence is done by an NLM and so all these things are kind of bundled together and then published in an annual publications format which is on an immutable ledger. So it basically will give you you know a record and it also helps if the LLM you know you don't totally trust it or it's flubbing or whatever else obviously that's still an issue and it may be more an

issue for more contentious topics so that you can basically have a different LLM score or have multiple different versions or you know gradually improve it. So certainly this technology is technology that's in flux and so it's not always perfect. This is a sort of importing function so you can easily kind of import to various RDF files so you get these kind of nice sort of summaries and you know sort through the various aspects of it. So yeah originally again this was really just for some 11 things. This also had an economic calculator which I'm not going to show but it shows the sort of downstream economic impact of various things including projections and it has a bit of a comparison feature to see for people who are interested in investing or putting more money into these things what that might evolve with. So we kind of have a similar sort of concept here with active inference more or less replicate the same model but obviously the hypotheses are not the same as they come through. You can see the kind of evidence accumulation and sort of here what it thinks are sort of validated hypotheses including just a sort of variational free energy kind of concepts with the rest or with respect to sort of neural systems that are providing Braxian inference, decision making, control and reinforcement learning casts as inference down to the things that are like not entirely validated as it says to basically psychopathology, decision making and things like that. So we have some work to do I guess in a way because this has been a bit of effort to sort of build software in between but I just thought I would give it out as a sort of discussion point that could be useful way of doing new forms of meta-analysis and also for things that are maybe in the more contentious category to have sort of multiple different versions besides the fact that this can kind of be automated and potentially used to drive different types of experiments. Thanks for your time. If anyone wants to use this software, please get in touch with me. I'm very happy to make this available to other researchers.

I'm very happy to make this available to you. I'm very happy to make this available to you.

Okay, next up is going to be a five minute lightning talk by Alexander Siby.

Hi, my name is Alex and here are some ants. These ants are all mathematically driven.

I think of them as Markovian agents and the memory is actually stored in the actual environment itself.

So these are Markovian agents exploring their environment and you can see them starting to build little defense mechanisms here.

And you can also play with different variables and different parts of the script. You can add different layers to the network and so you can see what's going on.

And I've done this with lots of different biological and physical systems.

I'm really interested in what happens when variational free energy reduces and I see it as coherence increasing.

That is coherence as memory density through time. Here we have a mycelium network and you can see the coherence field there, growth activity, network formation. So all of the scripts are all mathematically driven using the same formalism, which I'll show you in a moment.

Here we can see mycelium network underground, glucose exchange and moss starting to exchange nutrients in the network there too. And some L-system trees up there as well with the photosynthesizing sun.

A hurricane. So on my talk you'll see that that has a higher omega value because it's more in an exploratory

campaign. So it's a sort of higher temperature. Here we have thunderstorm. And I'll just explain as coherence increases in the cloud as charged, eventually it reaches a rupture threshold.

And we get thunder and lightning. The sounds from these scripts where their sound is all diegetic from within the script itself. Here we have an atom and change the electrons and it will change the obviously the atom that we're looking at. And just to explain that CRR or Coherence Rupture Regeneration, which is the formalism I've been using, is of course grain formalism. So I'm not saying that this does anything microscopic within paradigms, but I find it a helpful way to think between paradigms.

So here we have the sun again, modeled with the CRR. And here again we're looking at how coherence builds through time. How we reach these rupture thresholds where exploitative campaigns can no longer operate and we need exploratory new generative model, which re-weights from the past states of coherence. Here we can see ice and change temperature, not ice forming. So I've done basic modeling as well.

This one has a lower omega value. It's sort of less exploratory, more rigid as a physical system.

And here we can see a hologram. So here we have perturbation of the coherence field.

And this is exactly modeled in the script. Here we can see a maze, a sort of rim solver. So the agent is a Markovian agent and the memory itself is stored in the field, in the environment.

So what is CRR or Coherence Rupture Regeneration? Well here we have coherence formalism. So coherence is building memory density through time. It's essentially all past states matter in a non-Markovian sense.

And we have this LX tau here, which is how memory density is built through time.

Coherence reaches critical thresholds where the environment provides new free energy spikes and can no longer, the exploitative campaign can no longer work. So we require an exploratory campaign or creative campaign. And this is where the rupture or the direct delta leads to a new kind of generative model. And you can think about this as scale invariant, moment to moment or coherence built through time. And then rupture being like ecosystem collapse or something in the body collapsing or a psychological shift or an ontological shift through time. Here we have regeneration. So here we have the historical states here, picking up from the coherence of memory density here. And we have a Boltzmann-like regeneration operator. So regeneration is exponentially building on what mattered most in those past states through time. So this is an exploratory playground. I think of it like a free energy playground, exploring coherence-based approaches. I've put various videos on here, explainers, other bits and pieces for people to play and explore. And I'm just looking for people that might be interested in this. So really happy to be part of this symposium. And thank you.

All right. Next, I'm going to play a video.

So

I'm going to play a video.

So

I'm going to play a video.

from Yana.

from Yana.

And Yana also requested that I read this github post, which I'm also going to put the text into the chat. Okay, here we go.

[illegible]

Here it is.

This is Rishu, PhD.

I'm a mom and a nerd running a business solo, which means that some days I feel like a CEO,

and other days a vending machine dispensing snacks and emotional support.

That's why I would be wearing a soft and warm mom nerd type shirt.

During this talk.

Hi, everyone.

So have you noticed that when we move a toddler's toy to a new place, it becomes a new toy for them.

The object doesn't change, but the context changes.

And yet, it feels completely new.

Because their brain learns through context and it's playful prediction.

And adults actually do the same thing, just with jokes.

For example, how to refresh old routines by setting them in new contexts.

Using the same punchline, but new premises.

And your relationship will thank you.

And by the way, punchline is a line that flips the normal expectation for the situation.

Premises, which is the same topic situation.

Example.

Gym culture, grad school, and parenting.

And somehow, it feels fresh every time.

And that's because humor, like play, is a little prediction game.

Your brain expects one thing gets another and rewards you with laughter.

In parenting, the premise changes every five minutes.

Say, my toddler is manipulating the system, aiming for something she wants.

But endlessly generating new premises.

That is, in grandma's room, in the bathroom, in the living room, in a car.

This is one of the best jokes I wrote the Chachipiki situation.

My toddler is basically running a multi-agent reinforcement learning experiment.

First, she goes to grandma for five minutes of Xiaohongshu's short videos.

Grandma says, you have to pee after, okay?

But it's prepaid screen time.

She ignores the contract.

And the second situation, then she comes to me.

No words.

Just takes off her pants, sits on the potty like, we both know the protocol.

Because in my lab, she only gets the phone if she starts by sitting on the potty.

I didn't teach her manners.

I taught her API integration.

Which means knowing the exact steps to unlock the system without understanding the system.

So it looks like she's obedient.

She pees, grabs my hand, and goes, let's go dance.

Which means, let's go watch dog dance video on YouTube.

Touchline.

At this point, we realize she's not learning politeness.

She's A-B testing our parenting models to maximize screen time.

And we're both failing QA.

This is Rishi, PhD.

I'm a science writer and AI strategist.

What I've learned during all my projects.

We don't have time to become an AI engineer.

But we need an AI as an appliance.

Like a microwave oven.

You push the button and something helpful comes out.

And you go back to life.

But the fun part is, we can use AI to make our everyday life efficient and everyday routine more playful.

We can use Chachapiti to think more deeply, not less.

And we can use it to understand our kids and ourselves with more softness.

Letting it expand what we can imagine, analyze, and express.

And I'm sharing these experiments through my new podcast.

That's called Struggling Nerdcast and The Chaotic Mom Nerdcast.

You can also find how I use ChatGPT to write this conference talk script that you are listening to with all the behind-the-scenes struggles.

I'm also working on how AI can support parenting psychology, why kids meltdown, how emotional regulation could develop, and how to respond with empathy instead of panic.

Okay.

Okay.

There will be a couple-minute break, and then we will be back with the next session.

Thank you.

Thank you.

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Thank you.

processes, P1 and P2 and so on, don't actually happen in brains. They happen only in our interpretation of the events in the brain. So without that interpretation, there could be completely different processes, Q1, Q2 and so on going on. So neural theories which say the physical things happening in the brain are the cause of consciousness, can't be right. Because those are theories of the brain have lots of encoding. We have lots of interpretation of what we think the brain is doing, but physically the brain is not doing that. The brain is just doing neuron firing. So how do we get out of that? And here's the positive proposal. Suppose there was an analog model of space inside the brain, a model that's not heavily encoded. And this is the physical picture here. Suppose we have neurons firing away in the brain and inside the brain there's actually a physical analog model of space. So there is a lot of decoding of the neurons to create that physical analog model of space. And the analog model of space is, as it says, very like real space. So in that idea, that analog model of space could be the source of consciousness. That is a theory of consciousness which I've developed and which I believe has a great deal to do with the free energy principle.

So you say, why does the brain have an analog model of space in it? What use would it be?

It could be very useful because it could help to store a Bayesian model of what's where in space, integrating what's sensitive. It could be a short term memory persisting that spatial information for short times. It could be used to classify things to know what things are around the animal. So what things need to be something to be done about them.

It could be used to simulate and plan physical actions. And it could be used to compare the present moment with previous situations.

So if you have the analog model in the brain, it could actually, in engineering terms, be very useful.

There are reasons for it to evolve. And there is a possible physical form for such a model in the brain.

And the pros for the analog model in the brain, which is in an archive paper here, is that there is a wave excitation in the thallus.

And as is well known, a wave excitation, a three-dimensional wave excitation, can hold 3D spatial information as a 3D Fourier transform in space, just as a hologram does.

A hologram is a physical example of an analog spatial model.

It goes through a Fourier transform, but that's not an extensive decoding operation.

It doesn't require a Fourier transform to get back from the hologram to the real world.

It doesn't require a huge amount of decoding information.

It just requires a scale for the Fourier transform.

Now, why does that fit the data?

How could there be a wave inside the thalamus?

The thalamus, as is well known, is a site of integration for lots of sense data of different modalities.

They all pass through the thalamus.

And the thalamus has a very well-preserved round shape, which is very suitable to hold a wave in three dimensions.

Now, the other piece of evidence that I won't go into is that if you don't have a wave, that the anatomy of the thalamus, a small round thing in the middle of the brain, doesn't make sense. It would do much better to disintegrate and move near the cortex.

And another piece of evidence is that lesions of the thalamus tend to interrupt consciousness, whereas lesions of the cortex, for instance, do not.

And so the hypothesis is that the wave in the thalamus is the source of consciousness.

So I'm doing quite well for time here.

We're going to undershoot time quite radically.

So how does that relate to the FEP?

Current FEP models look a bit like this, a very crude sketch.

There's a brain, which is a neural computer, and it has an outer Markov blanket.

Inside the brain's encoded information in neural pulse trains.

There's a decoding process, which converts trains into muscular action.

And that goes around to affect the world.

And through the senses, sense data comes in as encoded.

So there's an encoding and there's one particular kind of decoding, which is decoding spike trains into muscle actions.

But there's no decoding of spike trains into a model of space.

So there is no consciousness in these models.

How do the models have to change to accommodate the evolution?

To accommodate the proposal that there is an analogue model inside the brain, it looks roughly like this.

The same picture, but made a bit more complicated.

There's an outer Markov blanket between the brain and the external world.

There's an inner Markov blanket between the neurons in the brain and the analogue model in the thalamus.

So there's a lot more encoding and decoding goes on.

Sense data comes in, is encoded as neural spike trains, and is decoded into the analogue model.

And there's a little feedback loop here between the analogue model is encoded as spike trains again, and it's iterated until the analogue model is a good Bayesian model of the external world.

So there's an encoding and decoding cycle there.

Also for planning actions, there's encoding and decoding cycle here.

So the model is encoded, neurons work out possible planned actions,
and they are decoded to compute the impact of those actions on the world.

So there's more encoding and decoding goes on there.

And the proposal here is that consciousness arises from the analogue model.

So the key thing is that we carry on building FEP models.

We need this extra ingredient in them.

And it's a very necessary extra ingredient because without them,
the FEP cannot account for consciousness.

And consciousness is probably the key factor about the brain
that we really need to account for.

So I'm coming to my conclusion now well ahead of time.

The conclusion is we need to understand consciousness in the FEP.

But we can't do it with a neural theory because neural theories of consciousness can't be right.

Neural complication, computation is heavily encoded,
but consciousness itself, spatial consciousness is not encoded.

So we need something else in the brain.

The proposal is that the something else is an analogue model of 3D space.

Heldens thalamus.

And so we can extend the FEP in that way to have an analogue model,
as well as the neural model of the outside world inside it.

So there we are.

That's the proposal.

I would very much like to try and integrate this model in the FEP.

And if anybody else is interested in working on building FEP models,
which have analogue models of space inside, I would be very happy to work with them.

Okay. So I finished way ahead of time.

I hope there's a bit of discussion.

Thank you, Robert.

Ahead of time, but you can't be ahead of space.

Thanks.

Okay.

Okay.

People in the live chat can write any questions.

I'm just going to write down a question.

You said physical events don't have meaning without interpretation.

So then must interpretation be something other than physical?

Or is it that the act of interpretation is inaccessible, but it can hold the access consciousness of the

interpretant?

Well, some interpretation is done by us as people in our minds when we sit outside of the computer and we say the computer is computing this or that.

There is also no interpretation that isn't done by minds.

For instance, the main example I gave was where the brain puts out spike trains and muscles interpret those spike trains as physical commands to contract.

So that's an interpretation which is done outside the brain, but it is not done by any other mental entity.

It's just purely done by a physical thing, a muscle.

So then is interpretation the consequence of the stimuli or a sense of the relevance?

Because the interpretation is the consequence of the stimuli.

In other words, brain spike trains have a consequence that the animal moves.

What would make that?

What would make the possibly waveform implications of like visual stimuli be?

Be accessed in an aware sense as opposed to the ostensibly unconscious or at least not perceived by a central consciousness of a muscle contraction?

Well, the hypothesis is that for some reason which we don't understand, the analog model held in the wave is the thing that causes consciousness and consciousness exactly mirrors that analog model.

So that's, if you like, that's David Chalmers' hard problem, why something physical in the brain causes a mental thing of consciousness.

And we don't have an answer to that hard problem.

But I'm personally not worried about hard problems because in any science, all of physics, you come down to hard problems eventually.

So if you have a decent theory that explains most of the data and there's a hard problem somewhere, like the hard problem of the mind is how do physical events in the mind make consciousness?

And the hard problem of the hard problem is how do physical events in the mind make consciousness?

And the hard problem of physics is what's the fundamental particles?

You go down through atoms, molecules, protons, quarks, and so on.

And the hard problem is where is the fundamental particle?

I'm not bothered about the hard problem of physics.

I'm not really worried about the hard problem of consciousness.

There has to be laws which say physical events cause consciousness.

And we just have to postulate those laws and keep them as simple as possible.

The law here that's postulated is that when there's a certain kind of wave going on on the Thalamus, that is the cause of consciousness.

But it has to, you can't go down deeper than that.

What do you think the architecture of this model that you'd like to build would be?

And then what would it reveal and how it, what would be the implications of what we could learn from it?

Very good question.

I'll just go back to the previous slide.

The architecture of the model is essentially that picture.

So it is a fundamental addition to the FEP, to active inference models.

I think we have to explore its properties in detail to start asking ourselves what are the implications.

So it's a proposal that we build models along those lines with the inner Markov blanket as a boundary between the wave and neural firing.

I think we just have to explore the consequences.

They will be unpredictable until we've dug into them.

Sorry, that's not a very exciting answer, is it?

Yeah.

I'm just looking into the center of the decoded analog model.

And that is, I mean, does this just regress another layer into a sort of opaque nested onion model?

Is if, if, is the clay?

It's not onion at all.

It's not onion because onion, you would say, will go down to a deeper layer and a deeper layer and so on.

And so the, a lot of the ideas of the FEP, like sort of scale invariant, Markov blankets within Markov blankets, don't apply.

You have this Markov blanket, then you got the wave and that's the end.

No inside Markov blankets, no onions.

Is, is what's relevant, the statefulness of the center of the donut with the decoded kind of private key exposed?

Because if so, then why would the statefulness be the experiential elements?

Well, you're coming back to the hard problem again.

We don't know why.

I mean, it's just consciousness is there as a part of nature.

We know it's there.

And previously, we've had to invent laws of physics to relate parts of nature.

For instance, many years ago, we had matter, like hard things that you could touch.

And we had electricity.

And the two were very different.

And force and electricity were very different.

So we invented bridging law to bridge force and electric, matter and electricity, force and electricity.

Similarly, we have to invent a bridging law to bridge consciousness and a material wave, a physical wave.

And we just have to postulate that law and don't worry about it anymore, basically.

It's a hard problem.

Relating to completely different phenomena, like force and electricity or like waves and consciousness, that is, you have to just propose that there is a law.

Then the law has to succeed or fail by fitting experimental data.

You don't worry about the philosophical foundations of proposing such a law if it gives you a good theory that fits the data.

Yeah, it's a very interesting.

I don't know if meta science is the right take, but definitely.

Proposing a synthesis doesn't need to be constrained by repeatedly the fact of the matter that prior to the synthesis, the phenomena are fragmented.

That's right.

That's the starting point of the proposal, not the conclusion of the execution of the proposal.

And then where it becomes the research ahead in the development is what are those implementations?

Yeah, that's true.

What kinds of models do we need to sweep out here and what measurements and what empirical data sets to connect to, of which the data sets wouldn't necessarily be of a different kind than data sets that could even already exist.

It could be proprioceptive data or spatial awareness data.

It could be developmental information on when children or different species have awareness, like object impermanence.

That's right.

I think the test of the theory is the fit to data.

And as you say, the data is all kinds of data relate to consciousness.

The main piece of data that relates to consciousness, I'll go back to the slide, is this one.

The fact is that we do have this thing inside our heads that's just very like the real world.

That is a huge, important set of data points with lots of information.

At any moment of the day, we've got this whole conscious model of the world as a very tight one-to-one correspondence with the real world.

That correspondence, that matching, is a huge experimental fact.

So I think that's the main thing a theory of consciousness has to account for.

So we need to just build those theories and test them against that fact.

It's a kind of private fact, which makes it a bit awkward.

But nevertheless, I think it's a private fact that we can have.

What is the private fact?

The private fact is that I'm aware of the world right now as a three-dimensional model.

You're aware of the world right now as a three-dimensional model.

There are things we can't talk about.

Like, I can't talk about my green qualia in a way that persuades you that you're seeing green the same way as I see green.

So there's private parts of it.

But we can discuss public parts of it.

For instance, if we're in the same room, we can look at the same thing and say, that's a straight line or that's a flat surface or that's a bump.

And we can agree on that.

And those things are the correspondence between my private model of the world and the real world,

and the fact that they match up in this neat way is a big experimental fact.

Yes.

What are your...

Go ahead.

The main thing that people, I think, will find uncomfortable is that there are so many neural theories of consciousness,

and there continues to be...

There's a website with 350 theories of consciousness on it,

and most of them are neural theories, and all neural theories have to be wrong.

There is something they're missing out, this decoding step.

That's it.

And I'm saying the decoding happens physically in the brain to create the wave in the thalamus.

I'll read a question from the live chat.

So B. Sulpham wrote,

Fascinating and provocative theory.

Could you comment on the possible mutual illumination between this conception and of the semiotic thinking of Peirce and the connection with introspection?

Well, probably semiotic...

I can't actually comment much, but introspection, particularly, I will comment on.

I think introspection is the brain, is the mind thinking about its own thoughts.

And that, I think, is not necessary for consciousness.

In other words, plenty of animals have no introspection, as far as I know.

But in this theory, we would expect those animals to have spatial consciousness.

So the theory says spatial consciousness has nothing to do with introspection.

It's not a higher order theory of consciousness.

It's a base order theory of consciousness.

Yes.

Some people, to the question, some people have explored, Majid Beny and colleagues have explored the Peirce connection specifically.

And I think where it connects is this question of primary semiosis and higher order theories, metacognition, introspection, these kinds of inner semioses that don't deal with the primary redness of red, the onionness of onion, the at-handness of an affordance.

Yeah.

But rather the introspective awareness of that primary semiotic event.

Yeah.

I think primary semiosis is the way to put it.

Primary as opposed to higher.

Yeah.

So what are your next moves?

What would make it a good productive research holiday season and 2026?

Well, I'm not an expert on FEP implementation.

And so I would really like somebody to step forward and who does know how to do FEP models and say, let's build an FEP model.

Let's take the FEP toolkits we've got or new toolkits if we need them and let's build an FEP model complete with variational minimization and so on.

And let's apply it to apply this and incorporate the thalamus wave in it, see what the model looks like.

Yeah.

And as we've discussed also in the insect setting, it might be more accessible to integrate a biophysically based fly body or insect body simulator with a full brain emulation.

Yeah, absolutely.

I mean, I haven't talked about that at all, but this theory in the mammals have a thalamus, which could be a possible site for the wave.

Insects have the central complex and central body and other fan shape body and the lip side body, which are very well preserved across all insect species and seem to me to be a likely site for the sensory integration, the same sort of sensory integration, which could lead to the same wave exaltation and the same spatial consciousness.

And I think insects are a great place to look for it because their brains are so much simpler to think about.

Hmm.

Yes.

A lot, a lot to think about a lot of good motivation for, for tool development and for how we keep working on these questions.

Anything else you'd like to add?

Not really.

I'm sorry if undershot left you some fair time, but.

It's all good.

Okay.

Thank you, Robert.

Farewell.

Okay.

Thanks a lot, man.

Bye.

Thanks.

Bye.

Bye.

Bye.

Bye.

Bye.

Bye.

All right.

Next up is a nine minute lightning talk from John Graves.

Here we go.

Many automated systems take data and apply some kind of reasoning called an algorithm to help make a decision.

a decision. The more advanced ones can even plan and execute a sequence of actions to reach a specific goal. But the most sophisticated systems evaluate counterfactuals. Judea Pearl and Dana McKenzie called this the ladder of causation in the Book of Why. Seeing is the first rung, doing is the second rung, imagining is the third rung. The type of questions the system can address become increasingly complex as we climb this ladder. At the first rung, association, which is unfortunately about as far as traditional statistics can take us, we can ask how are the variables related? What if I see x? What then? At the second rung, intervention, there is prediction about the world as it exists and feedback from actions taken in the world with the possibility of the difference between the predictions and the outcomes driving a learning process. So doing enables learning. And here the questions become active. What if I do x? Does the action I take have the result I expect? At the third rung, counterfactuals, there is a prediction about the world as it might be or might have been? What could I do? What could I have done? So these second and third level systems don't just ask what is the answer. They ask what can we expect when we do or imagine doing things?

This is the essence of having a causal world model. The theory behind these systems is presented in the book Active Inference by Thomas Parr, Giovanni Pizzullo, and Carl Fristin. A learning system operating in the context of a world model can reason about hypothetical situations and potentially use that information to map out a sequence of actions which is optimal within the limits of the system's current knowledge. That means truth is subjective from the system's perspective, but empirically verified. Feedback from the actions taken can drive learning, expanding the knowledge available to the system, recursively improving future predictions, and driving more optimal actions. Alternatively, the book Everything is Predictable by Tom Shivers presents the

historical development of Bayesian logic. Shivers makes a number of very well-phrased observations. First, the great insight of probability theory is that we should look at the possible outcomes from a given situation, not what has gone before. The probability of an event is the number of ways we know of that that event can occur divided by the total number of things we know of that could occur. Probability thus gives a quantitative measure of the things we don't know for certain. It may require a subtle shift in mindset, but think of certainty as a continuous rather than a discrete binary value. One can represent complete

certainty and zero can represent complete impossibility, but the points in between can reflect levels of confidence and skepticism with 0.5, representing an equal possibility of true or false, a state of complete uncertainty. The key insight of the Bayesian method is that you must add any new information that you get to the

information you already have to make each subsequent prediction. Intuitively think of learning by trial and error.

When we learn from the errors, we don't make them again on the next trial. Sampling our statistical probability

is different from inferential probability. What can we predict about a sample, given that what we know or think we know

about the whole hypothesis testing? We don't really want to do that. With inferential probability,

we're really asking what can we conclude about the whole given a sample or samples we've taken?

Bayes showed how starting with an estimate of what is most likely to be true to your best knowledge, your model, you can make inferences and draw conclusions about what is likely to be true in the world.

Consider, as a practical example, the doctor who looks at a patient's data, particularly their symptoms, and has to reason to the most likely conclusion, a diagnosis. Diagnosis guides a decision about treatment, and treatment may or may not result in a cure. Since the world is messy and often ambiguous, data is never perfect. Systems must deal with uncertainty, changing circumstances, and incomplete knowledge. This is where genius agents can help. They're designed for decision-making under uncertainty. The following comes directly from the genius documentation.

The genius agents tackle ambiguity through probabilistic modeling and Bayesian inference.

This is very different from the approach of large language models, which are built on a foundation of neural networks. Genius agents use a probabilistic graphical model to represent the world, specifically a directed acyclic graph which links the model variables frequently from cause or prerequisite to effect.

We'll focus on two related types of graph. The first, discrete Bayesian networks are a great starting point for introducing the core logic and themes behind how these agents reason and make predictions.

Here in the sprinkler example model, the cloudy variable has arrows to both rain and sprinkler, while the wet grass

variable is the endpoint of the arrows coming from rain and sprinkler. The arrows make the graph directed, and the fact that there are no loops makes the graph acyclic.

The next step is to add probabilities to each variable.

Cloudy is shown as having a 50-50 split. Then rain on the left is shown as having an 80-20 split when it is cloudy, but a

20-80 split when it is not.

Sprinkler on the right is shown as off 9 times out of 10 when it is cloudy, but split 50-50 when it is not.

Finally, wet grass is shown with eight different probabilities, depending on the four states possible for the two variables sprinkler and rain. For example, in the first column, when the sprinkler is on and there is no rain, there is a 0.9 chance of wet grass. The notation for expressing these conditional probabilities uses a vertical bar to denote given. When a probability has multiple dependencies, they are listed separated

by commas.

A factor graph replaces the arrows connecting variables with these probability tables, which are then referred to as factors.

This sets up the system to make two types of inference. First, a forward inference, also called conditioning, where, for example, an observation of either the rain falling or the sprinkler running in green allows for the inference of wet grass in red.

Or the inference can go in reverse, so observations of wet grass with a sprinkler running in green can be used to infer the chance of it being cloudy in red.

The next lesson will cover how the variables, states, and factors come together in Genius.

The HIMO will be used with bois from bois from bois from bois from bois from bois from bois from bois from bois from
bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from
bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from
bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from
bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from
bois from bois from bois from bois from bois from bois from bois from bois from bois from bois from bois bois from bois
from bois bois from bois from bois from bois bois from bois from bois bois from bois from bois from bois bois
from bois from bois bois from bois from bois bois from bois from bois bois from bois from bois bois from bois
from bois bois from bois from bois bois from bois from bois bois from bois from bois bois from bois from bois
bois from bois from bois from bois bois from bois bois from bois bois from bois bois from bois bois

All right, Pablo, how are you doing?

Hello.

Hi, Daniel.

Doing good.

Are we live?

Not yet, right?

Not yet.

Not yet.

Okay.

So...

That sounds weird.

Thank you.

Thank you.

Thank you.

And then we're going to just run around in that space and you will share whatever you have prepared.

We're going to have 3D environments and with, of course, agents, hopefully agents with active inference systems.

And we have prepared this environment for the community and for the symposium.

Someone already, Paul.

Don't worry, you will be back.

You're okay.

And the idea of this space is explore the space and collect the four different symbols that we are around.

So we should be like exploring and sharing.

It should be very educative.

We have been involved with the community and the Active Inference Institute for, I think, already three or four years.

And it's an honor to be doing this test.

Take this as a first approach of 3D internet.

And hopefully next symposium, we will have more to share.

So we should be exploring the different islands.

This, as you can tell, this, as you can tell, this, as you can tell, is a kind of a brain and an interpretation of what could be a decentralized symposium.

So we can tell you how it goes, the different days and the different persons that are going to be talking.

So we can tell you how it goes, the different things that are going to be talking about.

Yeah, this is not working.

I'm going to try it any, anyways.

Oh.

So many levels there.

We're screen sharing.

Yeah.

You're screen sharing yourself.

Yeah.

Cool.

So the idea is that we can move around and, you know, we can do a lot of things.

So we can go for the ops and read here, the adventure.

Hmm.

Interesting.

Yeah.

It's fun how you pulled out those quotes that are a little bit like a doorway between the technical content in the textbook and just these bigger adventure and mystery questions.

Yes.

I'm not sure if it's behaving as expected.

Welcome to the adventure.

Here is a cognitive map of the active inference, but it's incomplete.

The four states of the system are missing.

So find the four ellipses, gems, and send them here to the correct state.

The books will give you clues about where the states of each gem.

And yeah, so we should be like looking for, wait, I'm going to fall.

I'm a little bit nervous because I think it's not behaving.

Yes.

It's working.

No, it's working.

Okay.

So here, here we have some friends that we can go all together to explore.

Yes.

You can join me and let's go first to this Island of the Drift.

The Island of the Drift.

Oh, here is the book.

So this state alters the environment and we see perception itself.

It does not arise from impulse, but from the need of reduced uncertainty.

Not only transform what exists, it transform what the system believes.

Key to move is just to this place.

It is to test a hypothesis with the body and reshape a world too much.

Your model.

Okay.

So.

So we're reading kind of a riddle and then we're picking which of the four states of the particular partition it corresponds to.

Yes.

So.

That's right.

Which one would you say it is?

Let's go with active states.

Yeah.

So for someone that knows active inference, it should be easy, but I think it could be a cool tool to explain newbies or people curious about the ecosystem.

how this works.

So let's go and let's find the next Island, which is a land of the Remota.

So one thing that we did here is that you can also infer which state it is without even reading,

which is not the perfect, but for kids and people that don't really get in this level of complexity, which is not the same.

So I think it's a good idea.

So.

In this case, it's not part of the system, nor it is under its control.

Even so, it saves every signal that could be interpreted.

It's existence must be inferred.

It can never be observed.

And some causes can be touched, but without them, your perception would be noise.

What will happen, will be perceived even when unseen.

Sorry for my English.

So yeah, I think it's the external state.

But okay, we got two out of four. Whoops. You can see already there

that we are completing the riddle or the treasure hunt. Oh, let's go for that one on there.

Maybe we need to teleport to the island of Vestiga. There we go. Yes, we use that teleport and here is the fourth state. This state is not directly perceived, yet it guides every decision the system makes. It is a probabilistic representation of the world built on experience and expectations. It does not respond to the present, but to the model the system manifests in the future of reality. Key. Predictions is not what you see, it's what you believe in you are going to see and that already changes what you see. Okay?

So as we can infer is the blue, so we already know that this is going to be sent to the internal state.

Okay, awesome. Awesome. We need to go back to the main square and it's only one more to go. It's already...

it's out there, Iceland of Haze. So let's find... yes, this is the teleport. Whoops, here we are.

Last one is... this is the only state directly accessible. It is a discontinuous changing, arriving like a rust.

It can inform, but also the safe if taken as truth. It's the first thing that happens and also the easiest to break.

It's only useful if you knew what you expect to feel. Okay?

Okay. Sensory state. There we go.

So now we have complete the adventure. We have some Easter eggs and you can also download the active inference book from here. It opens the link straight to the download and here is a 3D image of something that could be familiar for most.

And then because we have discovered everything, we can get our seed of knowledge.

So...

Um... now...

We have managed to do this small adventure.

And this is going to lead us to another adventure that we did last year.

I'm gonna... I'm gonna enter into the...

into this world.

We have to build three different spaces for the Active Inference Institute and having a blast the building of them.

And trying to explain as easy as we could for new persons that would join the Active Inference Institute. We are not scientists. Our team is mostly...

mostly...

...study tellers and...

...tech lovers.

And this is the adventure of curiosity. And this is another riddle that tried to explain how the Active Inference...

...system works. So we have the internal states...

...the road. And here we have a small...

...instructions.

And here we have also...

...an outside. So we have our brain.

And this is a representation of...

...what we already have been...

...seeing as the main...

...explanation of the complex maths.

This is the...

...represent the Markov blanket.

And this is the sensory...

...state...

...where some of the nodes are missing...

...on each section. So we should take notes of what are the missing nodes.

...and...

...with...

...this information...

...we can...

...work...

...walk around...

...and try to figure out how...

...how all this...

...goes.

...here is...

...the basic music lesson.

...everything.

Okay. So these are the missing nodes that we...

...were...

...already inferring.

We can talk with the teacher.

It could...

Is it...

...is it intentional or possible that the missing notes...

...are the matrices that we need...

...for active inference?

Yes.

I mean it's intentional that you need to infer...

...to...

...finish the...

...adventure.

I'm not sure if I understand...

...properly.

But...

...the idea is that you have...

...here...

...like...

...to walk through the different nodes.

And...

...let's say...

...this one is the first that was missing.

But if I...

...fall...

I have not...

...taking notes of the...

...second node.

So...

...I'm...

...gonna fall.

...um...

...yeah...

...oops...

...yes...

...here...

...so the idea is to fill the gaps...

...and reconstruct the...

...the piece...

...of the...

...that is missing the paralysis.

And...

...yeah...

...we keep improving this...

...Proof of Concept...

...that...

...or...

...very early...

...alpha.

Here it starts.

Okay.

...

...playing the song.

Yes...

...and...

...checking the...

...the last ones...

...do on the second...

...re...

...me...

...so...

...we go back...

...I'm not gonna finish the adventure...

...I'm gonna...

...ah...

...the...

...the other part...

...so...

...to...

...go to...

...do...

...which is...

...si...

...re...

...e...

...e...

...e...

...e...

...e...

...and here...

...oh...

...I don't recall...

...I think it's last...

...no...

...

...so...

...yes...

...we just need one more...

...and...

...it should be...

...it should be done...

...and then you can get...

...another...

...piece of...

...seed off...

...of...

...knowledge...

...so I'm gonna...

...go back to the...

...symposium space...

...and...

...keep a little bit...

...with the exploration...

...because it's another...

...few things...

...that's around here...

...so...

...if you're new...

...also...

...and you want to...

...so...

...what's been...

...how we build...

...things on the Active Inference Institute...

...you can...

...you can come here...

...and probably...

...through a more visual way...

...get...

...the links...

...where...

...the lessons...

...and the courses...

...and...

...the different...

...information...

...that we are...

...building...

...would be...

...would be available...

...the idea...

...is that we already have...

...five different...

...years...

...of the symposium...

...and it would be...

...have like...

...one island for each year...

...or...

...something similar...

...so you can...

...you can go through the...

...uh...

...different information...

...that is...

...uh...

...is being...

...uh...
...served...
...and I have...
...thought...
...something like...
...the Active Inference...
...Institute...
...ecosystem...
...so...
...go to the ecosystem...
...this is...
...a...
...part of the...
...ecosystem...
...that...
...most...
...uh...
...that...
...we have...
...so...
...we have built...
...and the other...
...space is...
...inspired...
...on...
...this...
...uh...
...and it should serve...
...to...
...um...
...explore...
...on...
...a...
...3D...
...oh...
...this link is...
...broken...
...no...

...it's working...

...um...

...the...

...the third space...

...is that it should...

...help...

...the community...

...to...

...to join...

...these 3D spaces...

...the idea is that...

...on the near future...

...we could...

...speak...

...share screens...

...here...

...have...

...meetings...

...meetups...

...and...

...and...

...pretty much...

...interact on...

...a...

...3D...

...environment...

...which we believe...

...is part of the future...

...of the internet...

...um...

...so here...

...it should be...

...like the information...

...to be a...

...volunteer...

...um...

...if you are more about...

...learning stuff...

...you would...

...uh...

...have there...

...the different links...

...if you want...

...to become an intern...

...or...

...you...

...here would be like...

...the...

...the...

...teleport to the...

...interns...

...space...

...and...

...it's...

...it's just...

...um...

...much...

...funnier...

...gammified...

...way of understanding...

...the internet...

...and the learning...

...and...

...we...

...we're pretty much...

...working on...

...on all this...

...so...

...yeah...

...I don't know...

...if we have any questions...

...or...

...someone has some...
...curiosity...
...it's something that we're gonna keep building...
...so...
...and it's...
...evolving...
...and hopefully...
...we...
...we...
...we do something...
...very fun for...
...for the Institute...
...and...
...we believe that the education is...
...way better...
...with a gamification...
...gamified...
...and storytelling approach...
...so...
...here...
...it should be the space for the...
...board of directors...
...and if you look...
...this...
...from the top...
...which I cannot fly yet...
...but...
...on the future...
...we will be able to fly...
...and what we...
...what we...
...will see...
...is the...
...pretty much the same structure...
...as we...
...we saw on the previous picture...
...yeah...
...very cool...

...yeah...

...it's...

...it's...

...it's a different...

...mode of...

...of learning and playing...

...it's really fun...

...to see it...

...when...

...and...

...and just...

...it's...

...as we kind of break...

...into the digital...

...and...

...these kinds of spaces...

...um...

...we can have all of these...

...linked spaces...

...different...

...rooms...

...and adventures...

...and...

...right now...

...are those characters...

...just...

...procedural dialogues...

...or they could have...

...LLMs...

...or they could be other kinds of things...

...or what?

...yeah...

...they can have...

...LLMs...

...we have already tried...

...uh...

...that...

...in the past...

...and...
...is very...
...cool...
...because then...
...you can speak to them...
...in any language...
...they can be...
...um...
...very...
...notchable about...
...one specific topic...
...and...
...if we have...
...one LLM...
...that only is for...
...onboarding...
...it would...
...help people to...
...get a better...
...experience...
...and they can explore...
...and...
...be helped by these...
...LLMs...
...and...
...they do guys...
...yes...
...during this year...
...start...
...uh...
...having...
...not...
...whenever...
...because I think it's useful...
...to have like a...
...writing...
...or...
...or...

...previews...
...um...
...uh...
...NPC...
...speaking...
...and telling you...
...what's...
...what's going to happen...
...and at one point...
...it can become...
...an LLM...
...and...
...and...
...it could be also...
...very fun...
...and...
...and...
...cool...
...to...
...to have them...
...moving around...
...these spaces...
...and...
...serving...
...uh...
...to the...
...to the institute...
...and to the...
...to the ones that have...
...curiosity about...
...how everything works...
...and...
...and all that...
...uh...
...anyone that...
...want to participate...
...on this project...
...is an open source project...

...uh...
...it's all about...
...experimentation...
...learning...
...and having fun...
...and...
...imagine...
...a better digital future...
...uh...
...we really believe that...
...and have hope on...
...building something...
...that is not very...
...dystopical...
...so...
...um...
...we...
...we...
...try to do our best...
...and...
...put our grain of...
...of...
...of...
...arena...
...to...
...to build a...
...a better future...
...for the ones to...
...to come behind...
...so...
...anyone that's curious...
...or can...
...add value...
...or...
...have any ideas...
...or...
...or...
...like or dislike anything...

...any feedback...
...is welcome...
...and...
...yeah...
...we are around...
...the Discord channel...
...mostly...
...and...
...yeah...
...very passionate about...
...education...
...and...
...what Active Influence...
...is bringing to the world...
...and...
...how we are...
...uh...
...on Open Science...
...uh...
...learning...
...and...
...yeah...
...it's been a blast...
...being around this...
...this...
...so...
...yeah...
...thank you Pablo...
...definitely...
...and I'll...
...ask some questions...
...from the live chat...
...so these will be some fun...
...things to discuss...
...okay...
...B Solfum...
...wrote...
...will the link...

...work after the symposium?

Yes...

...uh...

...the idea is...

...uh...

...uh...

...internet...

...uh...

...3D Internet...

...is gonna be live...

...and we could...

...be improving...

...and using these spaces...

...to...

...uh...

...uh...

...share...

...maybe the...

...youtube...

...uh...

...recordings...

...or...

...streams...

...that we do...

...we'll try to keep that...

...link up...

...but then...

...more...

...more generally...

...whether it's...

...um...

...adventure...

...dot...

...active inference...

...dot institute...

...or some other...

...URL that we'll be developing...

...over the coming years...

...we'll...

...we'll...

...we will have these spaces...

...be very...

...uh...

...clear...

...um...

...um...

...okay...

...next question...

...is there a separate URL...

...for whatever...

...the PI logo...

...refers to...

...what does the PI logo...

...refer to?

...hmm...

...hmm...

...I don't know...

...Daniel...

...you know...

...uh...

...we...

...we have...

...we have this logo...

...on 3D...

...we will...

...we will give these assets to...

...uh...

...we will put them...

...so anyone can use them...

...uh...

...but yes...

...please let us know...

...oh yeah...

...oh yeah...

...definitely...

...a lot of things it could be...

...but that's the Institute...
...logo...
...um...
...where's the sign-up?
...well...
...for this...
...there's no sign-up needed...
...this just runs in the browser...
...and for other...
...Institute programs...
...we...
...always try to communicate it clearly...
...do our best...
...so that it's available...
...without any...
...barrier...
...and you can just...
...email blanket...
...at active inference dot institute...
...with questions...
...and as...
...you enter other...
...kind of programs...
...or courses...
...or experience...
...then those are...
...signaled clearly...
...yeah...
...yeah...
...the idea is that...
...to keep it...
...open...
...on as low barrier...
...as possible...
...and then of course...
...in the future...
...is going to be a sign-up...
...mode...

...but it's not going to be...

...necessary for...

...most...

...of the content...

...in the future...

...in the future...

...in the future...

...in the future...

...it's...

...it's fun...

...it really does feel like...

...a new...

...kind of journey...

...and chapter...

...in what...

...even just technical...

...education...

...can look like...

...and also...

...these broader...

...work and professionalism...

...questions...

...um...

...I mean...

...it wasn't so long ago...

...I was in grad school...

...and the idea that...

...science...

...would be...

...carried out...

...in a...

...3D space...

...with chatbots...

...and all these kinds of things...

...I...

...I...

...just...

...didn't even have it on my radar...
...and now...
...it...
...is a delight...
...and it's just...
...we have all of these...
...options...
...for...
...chat...
...and text...
...and code...
...we can kind of generate...
...the personalized textbooks...
...for...
...for...
...broad...
...fractions...
...of material...
...maybe...
...at some...
...cutting-edge level...
...there's some...
...um...
...poor performance...
...but for...
...core background topics...
...where...
...many of us will spend...
...many years...
...the materials are there...
...and so I think it's really...
...important and forward-looking...
...about what is the...
...experience...
...of those...
...and how do we stay...
...in this for...
...the community...

...and the long run...
...and the marathon...
...and...
...and the balance...
...and health...
...not just...
...um...
...propagating...
...a sort of narrow...
...view...
...of what it...
...looks like to learn...
...in just kind of one style...
...or just one type of material...
...when...
...the...
...real technical material...
...are available...
...in these...
...fun formats too.
Yeah...
...and it's a...
...layer...
...much easier...
...to...
...bring...
...more people in...
...and then...
...you can go deeper...
...in different layers...
...of the knowledge...
...and...
...yeah...
...for...
...for...
...for us...
...it was like...
...the...

...the main idea...
...to...
...make a...
...top-down...
...layer...
...of...
...learning...
...gamification...
...and...
...exploring...
...and...
...is still a...
...proof of concept...

...yeah...
...we believe that...
...it's...
...it's...
...going to be something...
...in the future...
...already...
...we have...

...like Minecraft...
...and...
...Roblox...

...a lot of experiences...
...that...
...are very cool...
...yeah...
...it's...
...an interesting connection...
...with the earlier talk...
...of...
...Robert Warden...
...and the spatial...
...consciousness...

...because...
...in these digital spaces...
...it's often...
...hard to have awareness...
...of like...
...how many other people...
...are in this channel...
...or in this live stream...
...or like...
...what...
...resources are there...
...and...
...so...
...we have...
...folders...
...of files...
...and all of these...
...infinite branching...
...coda documents...
...and then...
...once you have that...
...inner map...
...exploring them...
...is very...
...streamlined...
...but there can be...
...a big learning curve...
...with just getting...
...the first situational awareness...
...of what is even there...
...it's like a big...
...library...
...with kind of...
...branching...
...adventure...
...chambers...
...and so having...
...a...

...browser-based...

...top-level...

...spatial...

...navigation...

...visual...

...form...

...helps...

...put it...

...all in front of you...

...with what...

...ways that you can move...

...and then that's the...

...kind of learning by doing...

...and ultimately...

...the active inference itself...

...is like...

...what is off to the right...

...it's like...

...I do have to look...

...oh...

...that's epistemic value...

...or...

...this is not what I expected...

...for that...

...and so just...

...even in the gameplay...

...and in the awareness...

...basically...

...there's...

...all of the touch points...

...that...

...we could want...

...for going into...

...these technical...

...topics...

...just from...

...this...

...gameplay...

...and we could talk about...

...perception...

...and cognition...

...action...

...just with what's happening here...

...so...

...it's...

...it's a great...

...combo...

...okay...

...one more question in the chat...

...do you plan voice interaction...

...with nearby neighbors...

...in the 3D space?

...yes...

...it's...

...we...

...did...

...test it...

...last week...

...and...

...sharing the screen...

...the idea...

...for this year...

...at least...

...during this day...

...sharing...

...the screen...

...on the...

...on the main stage...

...on...

...and...

...audio...
...it would be...
...available...
...so...
...if you are...
...watching together...
...all together...
...this movie...
...we could...
...you know...
...stop...
...and discuss...
...and talk...
...on...
...on...
...on...
...what's going on...
...work...
...for...
...this year...
...but we are...
...super close...
...of...
...of...
...doing that...
...and...
...in other spaces...
...it's already working...
...also...
...it's an Easter Egg...
...anyone...
...interested on...
...po-ops...
...or...
...3D...
...objects...
...sorry...
...NFT objects...

...or...
...digital goods...
...it's...
...it's...
...this...
...this...
...this thing...
...I will share...
...the link...
...later...
...and...
...if you have...
...if you write down...
...this word...
...you can have...
...a collectible...
...and...
...that's also...
...a fun part...
...that...
...I think next year...
...we will share more about...
...digital goods...
...and the future of digital goods...
...and how we could...
...gamify better...
...the learning experience...
...through...
...these objects that...
...ensure...
...that you at least...
...have...
...a degree...
...of...
...familiarity...
...with...
...the concepts...
...that are being...

...talk...

...or told...

...in the different areas...

...so...

...this is...

...yeah...

...very cool...

...so...

...awesome...

...so...

...thank you Pablo...

...to...

...you...

...and to the...

...rest of your team...

...thank you...

...Daniel...

...okay...

...till next time...

...see ya...

...Ciao...

...bye...

...bye...

...

...

...thank you Pablo...

...farewell...

...good luck with everything...

...thank you...

...man...

...

...so...

...I'm gonna go ahead and...

...talk about making sense of uncertainty...

...designing experiences for adaptive interface...

...and I actually wanna start by taking a moment to explain...

...what this title is...
...because we're gonna be bridging two...
...two...
...historically sort of separate worlds...
...one of...
...really sort of technical space of what's called active inference...
...and free energy...
...and the other is...
...we're gonna be looking at...

...what does that mean...
...for those who might not know those exact same words...
...um...
...or might not be intimately familiar with them...
...but are either practitioners or individuals...
...looking to better understand themselves...
...and how to make changes...
...um...
...within themselves...
...or create content...
...um...
...or applications for others...
...that are net positive...
...um...
...so...
...I'm Anna Pereira...
...um...
...I'm a research fellow here at the Active Inference Institute...
...I also have a set of projects that you can see...
...at actincycle.com...
...um...
...I'm a research scientist...
...and also have a recent book...
...Navigating Uncertainty...
...Simple Scientific Insights for Everyday Life...
...so let's go ahead and jump into it...
...so...

...by way of introduction...

...you can talk about...

...okay...

...why does this matter...

...so we have these things...

...um...

...are called Active Inference and Free Energy...

...and they tend to look like the image...

...at the top of the slide here...

...but they're largely inaccessible...

...so if you haven't heard about it...

...that's pretty common...

...um...

...it's been around for 20-ish...

...some years...

...usually takes about 40...

...or...

...to 100...

...for something like this...

...to become...

...common vernacular...

...but it eventually does...

...right...

...like a lot of us didn't know what DNA was...

...but now we conceptually understand it...

...um...

...so that's some of the hope...

...behind what's happening here...

...um...

...and why I think it's especially important...

...is that this helps explain how humans...

...can overcome uncertainty...

...and we have a high degree of that...

...in society right now...

...um...

...and design thinking...

...so more than just colors or fonts...

...um...

...but the overall experience and understanding...
...offers a bridge between this abstract theory...
...and our lived experience...
...um...
...and so that's what we're gonna explore...
...in our time together...
...okay...
...so in this context...
...and we're gonna slow down a little bit...
...to make sure we're bringing everyone along...
...so...
...if you're technically seeped...
...um...
...in the space...
...perhaps thinking about it from a different perspective...
...is of interest to you...
...if you don't yet know what active inference is...
...don't worry...
...um...
...we're gonna talk about that as we go...
...so...
...active inference...
...is like the science...
...the very physics of us...
...so...
...who we are...
...we're constantly inferring...
...what's happening within us...
...and around us...
...so you're doing that right now...
...and listening to me...
...um...
...processing...
...with the felt senses that are coming up...
...and what you're doing...
...is you're working to reduce the uncertainty...
...and the dissonance...
...so...

...every thought...

...feeling...

...and action...

...ties back to this fundamental process...

...SO...

...through understanding the science...

...we gain the insight to more effectively...

...manifest our intentions...

...right...

...because it's not my place to tell you...

...what to do with your life...

...um...

...it's really up to you...

...how you want to do that...

...but part of my design thinking...

...might be to encourage you to remember...

...that you...

...this is your single precious...

...um...

...sort of remembered...

...lived experience of life...

...uh...

...and then to put this in a bit of context...

...SO...

...over the past...

...year and a half...

...there's been a series of presentations...

...from...

...preliminary research...

...to a book proposal...

...around how this work could be done...

...to create...

...accelerate active inference...

...and the free energy principle...

...in a way that was mutualistic...

...for the larger community...

...um...

...leading to a preliminary book release...

...it ends up some formats...
...like audio...
...actually take a really long time...
...but hopefully by the time you're seeing this...
...it's actually available...
...and then finally...
...we have a deeper discussion...
...around some of the science and principles...
...that are in the book...
...which is happening here...
...uh...

...thank you to the Active Inference Institute...
...in hosting this symposium...
...uh...
...so we can have these conversations...
...okay...
...so...
...for outline...
...we're gonna talk about design strategy...
...and then we're gonna go into active inference...
...and free energy...
...and conceptualize them...
...make them easy to understand...
...then we're gonna talk sometime about...
...um...
...one...
...small sort of...
...um...
...activity around it...
...right...
...and...
...and...
...inferences...
...and...
...free energy is a really large concept...
...but we're gonna narrow in...
...on a potential activity...

...of strategy and design...

...then have a discussion...

...and...

...and...

...finally close...

...so...

...design...

...strategy...

...um...

...again...

...thinking about more than just fonts and colors...

...um...

...um...

...but thinking about translating...

...complexity...

...into clarity...

...so...

...complex ideas like active inference...

...and free energy...

...they're...

...um...

...the downside of doing this...

...in this case...

...the cultural and contextual factors...

...um...

...but it...

...it really depends on what we're trying to do...

...um...

...in this place...

...in this case...

...the cultural and contextual factors...

...um...

...but it...

...it really depends on what your...

...your design goals are...

...so...

...the particular design goal for this...

...was to distill into clear...

...relatable ideas...
...that were meaningful to contemporary...
...Western audiences...
...and so that means a lot of...
...simplication...
...simplification...
...right...

...look at some of the stats...
...about 28% of American adults...
...can understand the science section...
...of the New York Times...
...so...
...effective communication requires...
...a very high degree of...
...of...
...simplification...
...and one of my favorite places...
...to...
...draw inspiration from...
...is...
...Kriska's out...
...they do a really great job of this...
...um...
...particular media and form...
...um...
...an additional goal is to think about...
...breaking down our perceived reality...
...right...
...um...
...sometimes I almost think of it...
...like a...
...neutral...
...obnipotent perspective...
...of what we're thinking...
...and experiencing...
...as a way to help us...
...kind of break down...

...some of the biases...
...that...
...that we might have...
...uh...
...so...
...as part of this work...
...we kicked off with a preliminary...
...survey...
...it was a sample of...
...representative US population...
...we saw a substantial interest...
...in the intersection of this...
...larger wellness space...
...and science...
...it's very interesting to...
...um...
...note that people were...
...especially interested...
...when it was giftable...
...or under 20 US dollars...
...um...
...on the right...
...I have an image of...
...what it looks like...
...at the best case...
...uh...
...to get...
...the sort of content out...
...so this would be...
...a book...
...with printed illustrations...
...that you could have at home...
...that you could refer to over time...
...you could mark up...
...you could come back to...
...um...
...so not something...
...that's in a digital format...

...it's away from your devices...

...um...

...but it's also not at the library...

...because it's more personal...

...and more accessible...

...and marked up...

...in the best case scenario...

...you're more than double that...

...this is not funneling money...

...in my direction...

...this is simply the cost of printing...

...shipping...

...and...

...um...

...the third parties involved in that...

...um...

...um...

...and I want to highlight this...

...because it's a very interesting...

...design constraint...

...um...

...um...

...it's really interesting to reflect...

...what does this mean from a societal perspective?

...we're not going to get into that now...

...it has interesting implications...

...implications...

...so...

...um...

...about the design strategy...

...people are frequently more receptive...

...to information...

...that is visually inviting...

...and emotionally resonant...

...so...

...thus...
...attention...
...is applied to aesthetics...
...as a central construct...
...um...
...because it helps build trust...
...and engagement...
...so...
...it's a tool that we use...
...in that regard...
...um...
...it's not something personally...
...um...
...that I...
...um...
...devote...
...a lot of time to...
...outside of a strategic sort of lens...
...um...
...um...
...yeah...

...like a design lever...
...um...
...to help do this work...
...so...
...let's think about conceptualizing...
...active inference...
...so...
...historically...
...if you're familiar with it...
...um...
...it looks something like this...
...if you wanted to know...
...what active inference was...
...you put it into Google right now...
...you'd probably...
...um...

...land on the Wikipedia page...

...and this is the image you would see...

...um...

...um...

...it's complex...

...it's abstract...

...it's technical...

...but then again...

...is sort of behind us...

...uh...

...so...

...one way that we can break it down...

...is to start to think about domains...

...right...

...and we're remembering our particular...

...design objectives...

...um...

...so...

...we considered...

...what it unfolds across...

...the historical representation of...

...threes...

...in normative...

...cognitive capacities...

...right...

...you see a lot of threes in math...

...XYZ...

...you see it...

...um...

...in social contracts...

...constructs like religion...

...at trinity...

...um...

...and so the goal was to...

...concept...

...contextualize these key principles...

...in a way that help people...

...see new possibilities...

...so we ended up landing in...
...environment...
...mind...
...and body...
...um...
...and...
...you know...
...I think after they're learned...
...they're great to start reminding people...
...well...
...you know...
...these...
...sort of like the Markov blanket...
...it exists...
...but it doesn't...
...it depends on how you're looking at it...
...uh...
...and then the next thing you can do...
...is start to play with color motifs...
...um...
...and trying to think about...
...okay...
...well...
...what's distinguishable...
...and what's memorable...
...so for environment...
...you end up with green...
...invoking this sort of nature...
...with body...
...you end up with red-orange...
...sort of balanced...
...and inclusivity...
...um...
...and...
...things like blood...
...and internal warmth...
...and then with mind...
...you often...

...um...

...we found ourselves with blue...

...thinking about calm...

...and reflection...

...so the result is this visual language...

...that is grounded in science...

...and it's structured to be accessible...

...designed to be repeated and elaborated...

...so here you see...

...um...

...a description of the active inference...

...an early description we use in the book...

...kind of breaking down these processes...

...but because we're thinking about...

...petological methods...

...and slowly introducing people to more complex methods...

...at this point we only allude to the predictive brain...

...um...

...so you can see arrows moving in the opposite direction...

...for those who are less familiar with what the...

...the predictive brain is...

...it's the idea that...

...what you're probably aware is that you can see the world...

...you can understand it...

...you can make actions...

...so that's potentially happening in this direction...

...but at the same time...

...your brain has already learned about the world...

...and is making predictions about what it's going to see...

...so what we're experiencing is always these two things...

...happening...

...and it has a lot of interesting implications...

...um...

...for the human experience...

...uh...

...that we end up getting to...

...getting into in later chapters...

...so if we think about...

...conceptualizing free energy now...
...so we talked about active inference...
...which is the how...
...the ongoing cycle...
...the predicting...
...the sensing...
...the acting...
...the updating...
...we start to think about the why...
...like why is the cycling happening in the first place...
...and how do we reduce this uncertainty...
...so we end up illustrating free energy...
...um...
...by adding gray for the unknown...
...so building on our existing...

...visual motif...

...

...so this is just one example...

...pulled from one...

...um...

...chapter...

...where you see active inference start to be laid...

...on some summaries from the social sciences...

...especially psychology...

...so with fragmented cycles...

...or cycles with more free energy...

...you see things like rigid beliefs...

...resistance to new insights...

...sort of closed dialogue...

...you see the body having increased stress...

...and disconnection...

...limited regulating practices...

...the environment...

...there's perhaps more conflict between science and religion...

...increased social division...

...and then in cohesive cycles...

...you see things like more like reflective beliefs...

...openness to new insights...
...open dialogue...
...increased coherence in the body...
...connection...
...regulating rituals...
...um...
...and in environment...
...and in environment we mean more than just...

...say nature or pollution...
...we're talking about anything that isn't you...
...so this could be communities...
...for example...
...this moment I'm in my home in Switzerland...
...um...
...so my current environment is in my house...
...um...
...so in the larger environmental context...
...in a cohesive cycle with less free energy...
...you might see things like respect for science...
...and personal beliefs...
...increased social sort of cohesion...
...so...
...activity strategy...
...yeah...
...so now that we understand the principles...
...you could imagine we've been sitting at a...
...whiteboard sketching out what it means to ride a bike...
...and how you ride a bike...
...we can start to think about...
...okay...
...well what does it mean to actually...
...learn to ride the bike...
...um...
...so we can start to think about some of that strategy...
...and so one of the things we...
...we're thinking of doing...
...is how do we increase people's capacity...

...um...

...and again it could be towards whatever...

...intentions they have...

...it could be...

...how to adapt to rapidly changing technology...

...it could be...

...changing behavior patterns they've been stuck in...

...um...

...you know...

...in...

...um...

...in more extreme cases you might have...

...people who are really entrenched in things like...

...addiction...

...who might be...

...um...

...might be looking to break free...

...in...

...in assistance from others...

...um...

...and so some of the ways that we can start to think about this...

...and obviously these activities would change...

...kind of depending on what your intentions were...

...and where you were looking...

...were in larger...

...in much more of the larger like wellness space...

...um...

...um...

...and...

...yeah...

...so you can then think of things like...

...prediction...

...action...

...and learning are all deeply interconnected...

...right...

...and so we have these suboptimal inference loops...

...um...

...like maybe some...

...beha...

...change behavior patterns...

...I know in my past I would find myself...

...arriving at the same destination...

...even after I made a series of changes...

...and it confused me as to why...

...um...

...and then how intentional practices...

...and again this is a subset of the larger active inference space...

...but how intentional practices can support...

...flexible updating greater coherence...

...essentially breaking out of some of those...

...worn in behavior paths...

...uh...

...that we might have...

...or the ability to learn a new skill set...

...um...

...um...

...that we would like...

...um...

...and so...

...um...

...this work...

...um...

...kind of the more in-depth exploration of what chapter 27 is...

...in the book...

...uh...

...Navigating Uncertainty...

...um...

...and...

...what we're gonna do is we're gonna take a look at...

...well what does it mean to update ourselves through activities...

...and we're gonna compare that to algorithm training...

...so we're gonna borrow from machine algorithm training...

...combined with active inference...

...what's interesting here is that humans train through...
...embodied both mind and body experience...
...so you'll see how we kind of...
...um...
...expand what we do for normal algorithm training...
...um...
...to understand the science of humans...
...and what that means for the human experience in updating ourselves...
...SO...
...ground truth for human systems...
...perhaps unsurprisingly...
...we landed in a space of mind, body and environment...

...one possibility...
...um...
...and so for...
...for mind...
...really landing in the intention...
...you know...
...is it like reflection, regulation or focus...
...you know...
...with the body...
...what are you doing with it...
...are you walking...
...are you sitting...
...and environment...
...like what's the context...
...um...
...is it social...
...and the larger environmental conditions...
...um...
...and when aligned...
...it reduces free energy...
...or it creates a sort of ecological coherence...
...um...
...promoting adaptive clarity...
...effective learning...
...and potentially meaningful experiences...

...uh...

...so we could take a running example...

...um...

...you could almost put this to any sort of fitness activity...

...um...

...and so...

...if...

...this again would depend on the larger context...

...of what's happening...

...but in this case...

...we could consider it as an opportunity...

...to...

...think about...

...um...

...increasing adaptive capacity...

...through...

...practicing...

...um...

...reduced free energy...

...so...

...increased free energy...

...more of this like...

...lack of coherence...

...and would be like a busy city landscape...

...songs with words...

...wandering thoughts...

...but if you were to use the same time...

...and same activity...

...to decrease free energy...

...and increase adaptive capacity...

...you might run loops in nature...

...with reflective soundscapes...

...um...

...borrowing...

...from musical playlists...

...used for...

...uh...

...psychedelics can be really great here...

...I know John Hopkins has a really great playlist...

...um...

...that's just really great for...

...interoceptive sort of...

...reflection and awareness...

...and then the focus might be...

...somatic awareness...

...instead of wandering thoughts...

...um...

...but again...

...really pulling in that...

...individualization...

...really matters...

...right...

...and it needs to be flexible...

...to meet the goals and realities...

...of the individual...

...um...

...that might mean age...

...we've already talked about...

...what economics might mean...

...for some...

...cognitive capacity...

...live context...

...these are just a couple examples...

...of things that might play a role...

...um...

...and really emphasizing...

...supporting this sort of...

...self-generated exploration...

...with enough structure...

...to reduce uncertainty...

...so it's very delicate...

...sort of balance...

...so essentially...

...the goal is to...
...use the science...
...to help humans today...
...um...
...and this kind of communication...
...can help people...
...begin to shift their experiences now...
...I also hope...
...and think that the science...
...can help others...
...create...
...um...
...for others...
...through a lot of different means...
...um...
...and that's...

...one of the reasons why...
...and choosing to communicate...
...in this way...
...because we're at a point in time...
...where there are...
...there is more work to be done...
...than there are people to do it...
...and that's...
...so one of the hopes from this work...
...is that this can spill out...
...through different avenues...
...okay...
...so let's think about...
...actually designing the activity...
...uh...
...so the additional details...
...especially those that are more...
...technical...
...and extend the sort of...
...algorithm metaphors...
...are available at the paper...

...on actincycle.com...

...slash paper...

...um...

...just in order to streamline...

...and make this condensed...

...um...

...we went ahead and just focused...

...um...

...a little bit more on the human side...

...and...

...uh...

...we landed in...

...again...

...for this particular example...

...in this particular chapter...

...um...

...a framework that looked at...

...a four-quadrant model...

...defined...

...defined by the two dimensions...

...of body...

...low intensity...

...high intensity...

...and mind...

...quieted...

...and highly stimulated...

...um...

...um...

...and this offers opportunity...

...for increased...

...psychological flexibility...

...and adaptive learning...

...um...

...um...

...but I'm gonna keep saying this...

...over and over again...

...right...

...um...

...um...

...I just like to remind myself...

...how important that is to...

...to remember...

...and...

...uh...

...people get to choose...

...where they wanna focus...

...and some...

...may choose to...

...narrow their focus...

...depending...

...on their goals...

...their context...

...and their current capacity...

...um...

...we focus on broad spectrum updating...

...um...

...because...

...that is supported by...

...active inference...

...algorithm development...

...in the broader...

...psychological literature...

...uh...

...I can't help but wonder...

...those who do choose...

...to...

...to sit in one quadrant...

...how many might...

...um...

...choose to change...

...and kind of diversify...

...a bit more...

...if they understood...

...what they were...

...what they perhaps...

...had to gain...
...by perhaps...
...visiting some of the other...
...quadrants...
...and what that might mean...
...for their main quadrant...
...uh...
...since...
...since...
...since...
...of...
...one that...
...that has been...
...made less...
...consciously...
...um...
...and what people might decide...
...if they have the opportunity to...
...uh...
...okay...
...so let's start with...
...quiet mind and body...
...so...
...mechanistically...
...this is supports...
...the...
...uh...
...sort of...
...interoceptive awareness...
...paraceptic...
...regulation...
...can function as a form...
...of...
...denoising and...
...detect subtle...
...persistent signals...
...masked by everyday noise...
...um...

...so for this particular example...
...we have restorative yoga...
...uh...
...to nature sounds...
...and meditation to know music...
...um...
...these are just two examples...
...of many...
...um...
...selective because they fit well...
...in the presentation...
...and they're well supported...
...in the literature...
...but really...
...there's a lot to choose from...
...if we step into...
...active mind...
...quiet body...
...you see things like...
...uh...
...cognitive simulation...
...without physical exertion...
...this is a link to...
...improved memory...
...consolidation...
...and cognitive flexibility...
...so things like...
...journaling and reading...
...then you have things like...
...active mind and body...
...uh...
...so you have embodied...
...practices that require...
...both movement...
...and mental strategy...
...so it supports...
...sensor-motor integration...
...social coordination...

...rapid prediction updates...
...um...
...the examples we have here...
...are martial arts...
...in team sports...
...and then finally...
...quiet mind and active body...
...so this supports...
...down regulation of mental activity...
...while maintaining...
...somatic engagement...
...this is associated with things...
...like reduced rumination...
...improved mood...
...and enhanced default mode...
...network regulation...
...uh...
...so we end up...
...with this framework...
...that...
...um...
...a number of people could play with...
...either the individual...
...or...

...a practitioner...
...um...
...who is...
...perhaps working with somebody...

...to explore...
...okay...
...well...
...how...
...how might a person...
...use activities...
...as one of the ways...
...to...

...create a more...
...adaptive...
...sort of...
...um...
...set of capacities...
...within themselves...
...um...
...and again...
...this would...
...potentially be one tool...
...among many...
...um...
...as the human system is...
...is complex...
...with a lot of layers...
...so...
...some additional...
...considerations...
...um...
...you have this one...
...I find pretty interesting...
...that it might be beneficial...
...to intentionally...
...include activities...
...with higher levels...
...of unpredictability...
...or...
...informational load...
...um...

...especially...

...in a time...
...of so much...
...uncertainty...
...really training...
...um...
...this part...

...can help cultivate...
...resilience...
...by reinforcing...
...strategies...
...for self-regulations...
...um...
...as a sort of form...
...of adaptive training...

...since I know...
...there's a lot of...
...um...
...academics...
...on this particular forum...
...I'll mention...
...one of my favorites...
...is actually to go to...
...un-conferences...
...so imagine...
...instead of showing up...
...in a sort of...
...domain-specific conference...
...an...un-conferences...
...across domains...
...and you show up...
...and there's a large whiteboard...
...and essentially...
...you're gonna fill in your talk...
...and...
...um...
...so you're gonna attend talks...
...across...
...um...
...and then in the evening...
...spend time together as a community...
...say like...
...by doing a massive cooking project...
...um...

...with complex social dynamics...
...and loud music...
...um...
...yeah...
...and I just wanted to leave that here...
...as...
...um...
...I imagine...
...if some of you are looking for ways...
...to sort of stretch those quadrants...
...uh...
...given this is an active...
...um...
...an academic community...

...I wanted to share that...
...that potential one...
...with you all...
...okay...
...and then it's also interesting...
...to think about the balance...
...of variation...
...you know...
...which activities might want...
...to stay consistent...
...but also...
...you know...
...does it make sense to bring in...
...some variation as a...
...way to extend the sort of...
...adaptive training...
...and to explore more nuanced...
...patterns of engagement...
...and build resilience...
...within particular...
...cognitive or physical modes...
...great...
...so let's step into the discussion...

...um...

...so this body of work...

...is meant to be this...

...initial step...

...that really...

...uh...

...contributes to...

...translating active inference...

...and free energy...

...into forms that are...

...conceptually...

...rigorous...

...and acceptable...

...accessible...

...and the framework...

...induced here...

...is intentionally...

...adaptive...

...designed to involve...

...and dialogue...

...real world...

...use...

...and ongoing research...

...its value...

...is also intended...

...to lie...

...in the conversations...

...applications...

...and refinements...

...its inspires...

...um...

...if you have any...

...feedback...

...um...

...you can leave that...

...at actincycle.com...

...slash feedback...

...and I'd be...

...um...

...delighted to receive it...

...so this work...

...it relies on...

...multiple sources of...

...rigor...

...um...

...uh...

...perhaps the biggest...

...is the existing...

...research...

...in foundational...

...literature...

...um...

...it's been...

...truly remarkable...

...to step into...

...um...

...that...

...body...

...how robust...

...it is...

...um...

...across different...

...domains and methods...

...um...

...dialogue...

...with leaders...

...and active inference...

...in the free energy...

...communities...

...as well as...

...collaboration...

...with experts...

...across other domains...

...pulling in other...

...types of knowledge...

...and wisdom...

...and feedback...
...from users...
...and peers...
...it's also...
...interesting to...
...really think about...
...the fact that...
...humans often...
...update through...
...discomfort...
...but many avoid...
...one of the greatest...
...challenges...
...for implementation...
...so helping...
...people update...
...more effectively...
...with less suffering...
...may be a really...
...crucial lever...
...conclusion...
...okay...
...so this is...
...essentially a starting point...
...for bridging...
...complex theory...
...and practical application...
...um...
...accessible...
...visuals...
...and experiential...
...framework strategy...
...uh...
...based on...
...robust science...
...aim to foster...
...learning...
...engagement...

...and adaptive action...

...you know...

...how can we make...

...these changes...

...we might be looking...

...in ourselves...

...whether it's...

...desired to...

...learn a new skill set...

...or...

...break an existing...

...pattern of behavior...

...increase capacity...

...be a better leader...

...develop a new tool...

...for somebody else...

...like...

...these are all...

...all intertwined...

...the same science...

...stance...

...um...

...through feedback...

...continued dialogue...

...and interdisciplinary...

...exploration...

...um...

...the references...

...they're listed in...

...the paper...

...actoncycle.com...

...slash...

...paper...

...and I want to take a...

...quick moment...

...for the acknowledgements...

...I'm super grateful...

...for the group that came...
...together around this...
...um...
...Carl...
...Frinton...
...the public advisor...
...and...
...taking the time...
...to really engage...
...with the community...
...even...
...when seeped in...
...the more...
...technical world...
...I'll...
...Ludo...
...who...
...just...
...was such a great designer...
...to work with...
...and iterate on some of these...
...illustrations...
...and motifs...
...to really make sure...
...we got those things right...
...um...
...and the colors...
...and just really built such a...
...help build such a...
...beautiful design strategy...
...for this...
...um...
...Zach...
...who was the editor...
...um...

...uh...

...used AI...

...as a complementary tool...

...um...

...everything...

...originated with humans...

...and was the final...

...stamped by humans...

...and...

...created by humans...

...in the middle...

...but...

...really helped make sure...

...that that communication...

...um...

...was very human...

...and then...

...Sylvia...

...which I...

...imagine you'll hear from...

...um...

...um...

...um...

...um...

...um...

...um...

...um...

...um...

...in the book...

...um...

...I personally had a chance...

...and again...

...we'll just...

...um...

...I know many of you...

...watching this...

...have probably...

...contributed to that space...

...um...

...um...

...it's truly a remarkable...

...body of work...

...um...

...of which to...

...to get to intersect...

...and work with...

...so...

...um...

...thank you for that...

...and...

...um...

...I hope it's not too much...

...for me to...

...to sneak in this last thing...

...in the next...

...of this work...

...so...

...please consider supporting the work...

...um...

...either through...

...um...

...reviews...

...recommendations...

...acquiring a book...

...or...

...providing feedback...

...um...

...um...

...any of the above...

...are very much appreciated...

...so...

...um...

...thank you so much for listening...

...and I look forward to continuing the dialogue!

...as a question from...

...as a question from...

...as a question from...

...as a question from...

...as a question from...

...as a question from...

...as a question from...

...as a question from...

Thank you.

Thank you.

Thank you.

Thank you for joining again, Matt, and to you for presentation.

Thank you so much, Daniel, and glad to be back.

I hope I'm not giving anybody whiplash.

That was a great talk by Anna, and this is kind of a big shift in topic and probably a bit of a provocative title.

So I'm Matt Brown.

Great to be here.

I'm with a company called ThoughtForge, and we're doing something a little unusual.

ThoughtForge began as a research project back in 2007, exploring alternate biologically plausible approaches to machine learning, and we're applying it to adaptive real-time dexterous control for robotics.

Just a little background on myself.

I've been working in AI for almost 25 years now at this point, including time in the defense industry, video games on franchises like Halo, Bioshock, Splinter Cell, stuff like that.

And then about 10 years ago, I was an early pioneer in deep reinforcement learning at a company called Bonsai, which was acquired by Microsoft.

And the last five years, I've been really kind of focused on taking this interesting research technology and applying it to real-world applications.

So just a brief agenda about what I'm going to talk about.

Last year, I gave a talk around the low-level mechanisms about what we're doing and kind of the approach from the bottom up.

And this year, I'm starting from the top down.

I'm talking more about the current motivations and goals for the technology, how it's being applied, and a little bit about the methods.

And I'm trying to keep it to more interesting aspects, more of the philosophical stuff.

So I'm going to start out by talking about robots and dexterity, why it's hard, and why I think a new approach is needed.

Then kind of contrasting current AI approaches with deep learning.

And then kind of just talking briefly about the techniques that we're using and our implementation.

And then I'll show off some videos and applications that we've been working on over the last couple years.

So why are people trying to apply AI to robotics?

And I apologize for anybody in the audience here who's very familiar with robotics because a lot of this is going to be super obvious.

Traditional robots operate in controlled environments doing repeatable things over and over and over again.

And that's kind of what they've been really being good at for the last half century, roughly.

And they haven't really been applied outside of that because the real world is messy, unpredictable, variable.

You know, everyone's aware of these challenges, I think.

What AI is looking to solve or what people are using AI to try to solve are three main things.

Perception, understanding the world, you know, understanding when things are not in their controlled state.

State adaptive manipulation.

People really would like robots to be able to do a lot of the tedious manipulation work that's done by humans around the world on a daily basis.

And then generalization, being able to learn and adapt without having to be explicitly reprogrammed for every situation that they're in.

And at least right now, AI seems to be the only viable path to unlocking robotics in these kinds of challenges.

So let me just make sure I didn't accept this slide.

OK, so these are the kind of cutting edge approaches in AI that are currently being explored and where

there's tons of money.

You probably hear about, you know, bunches of companies getting funding to go after AI and robotics.

These are generally the techniques that they're using.

Visual language models basically use LLMs to kind of ground vision and actions through language.

And there's some examples here with RT2, Aurobogen, OpenVLA.

And none of these solutions have yet been proven to solve all the problems or gotten to the point where they can be used in factories reliably or in homes.

I should probably start out by saying that.

And so, you know, each of these have their own challenges.

With visual language action models, they have poor action fidelity, so they're very kind of poor quality motion.

And also, language is not necessarily the best way to understand the physical world.

And they're also very non-deterministic.

And so it's very hard to know with confidence that they're going to be able to accomplish the task reliably.

Diffusion-based control and planning, this is basically using techniques from image generation or video generation, where they use kind of diffusion from noise to generate images.

They're doing the same thing for robotic trajectories, for generating motion.

And sometimes this is combined with visual language action models and with foundational models, so people try to mix and match.

Right now, this is very, very ripe for a lot of experimentation.

The problem with diffusion is that you need really, really high-quality data, and that's really hard to get.

The simulations usually are not good enough, so people are teleoperating their robots.

You'll see a lot of that to generate the data.

But that's also a very tedious way of getting data.

It's also very expensive to actually use in the real world.

And it's brittle to distribution shift, so anything changes, anything about the task changes or the objects change, and then the policy stops functioning well.

And the last one is kind of a broader goal, which is the foundation models for embodied intelligence.

And this is generally people trying to create a chat GPT moment for robotics, where if they get enough data about physical manipulation in the physical world, you'll get to some threshold.

And these AI models will suddenly understand how the physical world works, and then you can kind of use that as a base model to go after individual tasks.

And the problems here are somewhat obvious.

The amount of data that you need is enormous.

The amount of compute that you need is enormous.

And I don't know if I talk about it in the next slide here.

Yeah, I was going to talk about how challenging the data problem is.

Actually, I'll just move on to the next slide.

So these are the main problems with the traditional approaches in AI applying to robotics.

And for the most part, data is the biggest problem.

There's just not enough.

There's not an internet of data, you know, of robotics data to train on the way that chat GPT might have.

And there's a lot of different embodiments, you know, different numbers of actuators, different kinds of actuators, different kinds of sensory input, vision, tactile force, proprioceptive.

And every robot is different.

And so getting enough data that can generalize across all different kinds of robots in the physical world, we just haven't gotten there yet.

And the example I like to give is self-driving cars.

You know, self-driving cars are finally getting to the point where they're usable and they're functional.

But it's taken over a decade of data collection, billions of miles of road driven.

And that's just to handle one use case, driving a car, as opposed to the expectations for robotics are that they're going to be able to do everything, you know, clean your house, work in factories.

And so the mismatch here between the data set sizes and the expectations are really, really quite large.

And there's, you know, same problems that you have with self-driving, with the long tail variability.

It's very hard to simulate reality accurately.

There's just anything can happen.

And similar to self-driving cars, in fact, when one of these models fails in the real world, it has very expensive and often dangerous consequences.

The other problem is compute.

The amount of compute to actually run one of these new age models is absolutely astronomical.

And robotics have real-time constraints.

They're latency sensitive.

So if ChatGPT takes five seconds to give you an answer, that's not a problem.

But if the robot's in the middle of manipulating something and it takes, you know, more than a fraction of a second to make a decision, you probably have broken something or dropped something.

And so the performance constraints are much, much more challenging than in language.

And this is one of the reasons why AI in robotics is so challenging.

This also means that your AI model can't live in the cloud.

That means that your AI has to be small enough to be on a computer right next to the robot because you just can't deal with the latency that you get from working in the crowd.

So the biggest challenges here for AI and robotics still exist in that, you know, they don't really adapt to real-world variability in the way that they need to.

And so what we've been working here at ThoughtForge is kind of a new way of looking at the AI problem in general, using cybernetics as kind of the original body of theory that we're working on.

And obviously there's a lot of connections between cybernetics and active inference.

And I'll go into that a little bit later in a couple of slides, a little bit more thoroughly explaining that relationship.

But, you know, when I started kind of looking at these problems and, in fact, the learning problem, I was very curious about how nature solves these problems.

Because obviously we exist in the world, animals exist in the world.

How do animals kind of deal with these problems?

And they don't do it with huge data sets and huge computers training on that data over a long period of time.

And they don't just train once and then hope that they can generalize.

And in particular, animals don't separate learning from acting.

You know, it's kind of concomitant and it happens at the same time.

And so this is just kind of contrasting here.

The kind of paradigms for what people are looking at with AI currently and what we should be looking at and what we're kind of at ThoughtForge are kind of attacking.

Continuous feedback loops for perception and action simultaneously.

Adaptation at the point of contact.

And learning shaped by physics and, you know, active inference and the free energy principle are kind of foundational principles and theories around that.

And the idea here is to leverage learning in the moment versus learning in the past.

So just to kind of dig into that a little bit deeper, you know, most AI systems today are focused on prediction and are giant prediction systems.

And so the approach we've taken is a huge shift over these old fashioned deep learning approaches.

And I'm terrible at naming.

So deep learning is you can look at most deep learning networks as universal function approximation.

So what you're doing is you're taking huge piles of data and then approximating that into a function, an input output function.

I've called what we're working on inactive synchronization.

I'm terrible at naming.

I guess there's a quote, you know, there's two hard things in computer science, caching validation and naming things.

And we may have mostly solved caching validation, but I haven't really quite figured out naming things.

This is my current favorite name for our approach, inactive synchronization, because I think it's detailed enough and specific enough to be useful.

Other things I've thought about calling it is, but are related, but maybe too generic and don't really give much insight is, you know, generalized synchrony, cybernetic learning, metastable adaptation, things like that.

Homeostatic intelligence or manifold synchronization learning.

These are all accurate, but perhaps not as specific.

And what I mean here by inactive synchronization is that it's a synchronization process with kind of an inactive tint to it.

Inactivism views cognition as a dynamic interaction between environment and organism.

And the world model that's created through this process, through the synchronization process, is co-created through interaction.

Model and control are kind of intertwined.

And so the world model itself is constructed of sensory expectations and motor control percepts, essentially.

And so these are kind of intertwined.

And so this world model is created through interaction, created by the control dynamics themselves, or they emerge from the control dynamics themselves.

So other ways to look at this is universal function approximation, you know, current deep learning systems are retrospective.

You know, they basically take past data and to approximate functions.

And what we're doing is prospective equilibration, basically continuously adjusting to maintain predictive coherence.

And maybe this will make more sense once we get a little bit more into the details.

But traditional approaches are very data dependent, time agnostic, and feed forward, meaning they're input-output functions.

So once you have an input, it computes and generates an output.

And that output is directly related to the input.

And there are ways of making it with recurrent neural networks and things like that where it has some history.

But for the most part, every time there's an output, it's a direct response to an input.

This is very, very different.

These are time-aware, which means input signals are coming in, output signals are going out.

But the output signals, the motor controls, are not directly a computation on the inputs.

They are, you can think of the model both generating, you know, predictions about their sensory experience and motor controls at the same time.

And then where the sensory expectations are incorrect, kind of filter through the network and modify future outputs of the network.

So very different way of looking at the learning problem.

And the goal of what the inactive synchronization here is to build up this model of the world that can kind of predict how the world is going to react to your motor actions.

And then the goal of the motor actions here are really to get the, change the world to become easier to predict or to put you into a preferred state.

And a lot of this aligns very, very well with active inference, obviously.

But this kind of gives kind of a high-level philosophical view of what we're doing, the top-down approach.

So this is just a very, very quick slide to connect the kind of cybernetic approach with active inference.

And obviously these are very closely tied together.

The free energy principle is, you know, I probably don't need to go into it too much.

But it talks about how all living systems act to minimize their free energy, basically pulling from physics a lot of these really great ideas about how the natural world works.

And then active inference is a framework within free energy principle explaining how organisms actively gather information to reduce uncertainty and maintain internal stability.

This is kind of the kind of connective tissue into cybernetics because cybernetics was a field, you know, that's kind of disappeared in recent history, but built upon the same ideas.

Where it provides kind of the theoretical foundation for how feedback and self-regulation leads to this minimization of predictive error and, you know, this kind of self-regulation, self-maintenance process.

Digging a little deeper, our specific techniques are modeled around a cyberneticist known as Ross Ashby.

And he had some really kind of foundational ideas.

He's most known for the good regulator theorem and the law of requisite variety, which are both related to each other in some sense.

And the good regulator theorem, you know, basically says that if one system wants to regulate or control another system, that system has to, you know, have or be a model of that other system.

And the law of requisite variety is somewhat similar.

If you want one system to kind of control another system, it has to be at least as expressive or it has to have at least as much variety as the system is trying to regulate your control.

And he saw homeostasis as kind of the goal of stability, as kind of the universal mechanism for adaptation in the natural world.

And so he built a device in the 1940s called the homeostatic, very, very small, and I'll talk about it in the next slide a little bit more, where he kind of demonstrated adaptation as stability or stability as adaptation.

And it uses feedback between the units to kind of generate a distributed homeostasis through a local property called auto-stability.

So basically he had these nodes that in a network, these were built out of World War II bomb parts.

So he was, this is a pre-digital computer, he was using, you know, doing pure analog computing at this point, using recycled bomb parts.

And each node is basically trying, is performing this property called ultra-stability and when networked together, generate distributed homeostasis amongst the group.

And you can kind of look at the network as a self-model, where each node is trying, this network is trying to predict its neighbor's nodes.

And if all the nodes in the network are doing this, you kind of reach this equilibrium, this homeostatic equilibrium over time through this ultra-stability process.

So just to talk a little bit more about the homeostat, because it is incredibly important to our techniques and what we're doing.

It was the first device kind of capable of adapting itself to the environment, I guess, you know, depending on how you want to look at it.

It exhibited behaviors such as habituation, reinforcement, and learning through its ability to maintain homeostasis.

This, he kind of viewed homeostasis as, you know, the fundamental mechanism for adaptation and complex systems.

And he basically built this device to kind of show it off, kind of what he was thinking.

And it was very kind of primitive at the time.

But, and I think for the most part, people didn't really, really understand what's going on here.

But at the low level, in terms of what he was trying to solve, you know, there's a lot of, a lot of what they're trying to solve with reinforcement learning right now was kind of exemplified in the system.

And, and I, the contrast this with going back to kind of the 1940s, you know, models of the mind and things like that.

Deep reinforcement learning now, a lot of people don't know this, is based on, loosely based on BF Skinner operant conditioning.

And so, really kind of old-fashioned ideas kind of put into computer science is what we're dealing with now. I would like to think this is a little bit more forward looking in terms of concepts for, for how learning works. This was also described as an isomorphism generating machine, which, which is obviously useful when you're building a world model and you wanted to kind of abstract concepts in the world as groups and things like that.

And I think I've already talked about how ultra stability generates homeostasis. Here's just a more in-depth look at what it was.

You can see a picture of it right there.

By the way, this is, this is Ashby himself posing in front of his homeostat in case anybody was curious who that person was.

That's the picture of the homeostat there.

You know, these big clunky things made out of recycled bomb parts.

On the upper right there, you can see just kind of a network diagram of what was going on.

So basically each node in the homeostat was connected to all the other ones and they also had a self connection.

And the diagram on the lower right there is meant to describe the mechanism of what's going on with ultra stability and how these, these, these nodes update.

So in deep learning, the nodes in the network are perceptrons kind of actually probably five years, five or six years before the homeostat.

You know, it's kind of funny that all these chaotic attractor definitions came out of the same decade in human history.

But so this was just, you know, a few, half a decade later than the perceptron was this kind of model.

And, and how they differ is that chaotic attractor definition.

And so, you know, it's very similar to the perceptron in the sense that you have incoming signals or incoming connections from other nodes with weights.

And, but how we actually update the network is very, very different.

These obviously don't use back propagation.

Instead, they use this ultra stability.

And so another way to contrast deep learning with this kind of homeostatic network is that in deep learning, this back propagation process is a global update.

You have to freeze the network, you know, not use it, do a global update of all the nodes through back propagation.

With ultra stability, the update step is local.

So you don't have to freeze the whole network.

The, the nodes are continuously performing this ultra stability process in parallel.

And so this is, this, this works a lot better for large networks.

If you, especially if you're trying to implement this on computing hardware, you can kind of distribute your network across computing resources and you can kind of update them independently of each other.

Because of the way that the connections are all, you know, recurrent, you can kind of think of this network as trying to predict itself.

If each node is trying to predict its neighbors, the, the kind of group itself is building a kind of a self prediction model.

Aspie never actually connected this to external control systems or sensors.

And so the, the base homeostat was really just trying to show how this equally equilibrium, this homeostatic equilibrium could be reached in kind of a mechanistic way.

So I just like to include this letter in here in my talks just because it's, it's a fun little piece of history.

At the time, Ashby and Turing were, were, I wouldn't call them rivals, but Ashby was very much, you know, pushing for analog computing, this homeostasis way of building computers and computing machines.

And Turing was very much about digital computation.

And, and this is a letter from Turing to Ashby basically saying that, you know, his universal computing machine is more general than the homeostat because you could simulate a homeostat on a universal computing machine.

And, and, and while this is true, I think they're missing the point.

This is kind of the original analog versus digital argument.

And at the time, you know, analog systems didn't really, you know, complexity theory hadn't been developed.

And so stabilizing analog systems hadn't really been figured out in terms of how you can use negative feedback to make them reliable and things like that.

And so I think they're kind of conflating two things that I think are still being conflated here where computation is not the same as for lack of a better word, cognition or universal adaptation is the different than pure semantic computation.

And, and we've kind of, I feel like this paradigm that we're seeing in AI with, you know, all these different approaches are really just pushing further, further down that semantic computation direction, as opposed to understanding what's actually going on in intelligent systems or complex systems.

And so I feel like this is the beginning of this, this, this divergence in beliefs about how we should build a compute, a computing future.

Obviously, Ashby lost this argument at the time.

Turing, everybody knows who Alan Turing is.

But it's, it's a fun, it's a fun little tidbit for, I like to include.

So how do you actually build one?

You know, the, there, there was, there was one physical object built.

I think it's still in existence somewhere.

I think maybe the Ashby family has it.

I'm not sure.

But I basically took Turing's advice in the last letter here and simulating.

So I'm using a modern Turing machine and simulating it on, and simulating the homeostat as an analog system in there.

This is based on techniques.

Before I got into all this stuff, I was an early pioneer in what's called software radio, which basically was, I don't know if I can call it a science, but a field around how do you, how do you properly simulate analog systems and digital hardware in kind of accurate, in physically accurate ways.

And so we kind of used, I used a lot of those techniques to build the first prototype.

And one thing that's interesting here is because of the chaotic interactions between these, these nodes in this, in this, in these systems and the way that the homeostasis generates is that it's, you can't, you can't, you can't analytically calculate the network.

And so you kind of have to simulate it step by step and let it evolve over time.

And that's what I mean here by you can't, you can't account for the interactions algebraically.

You, you, the circularities and the recurrence really have to be kind of play out iteratively over time.

And there's a great quote here about, from Frigogaine about complex systems.

Don't forget their initial conditions.

They carry their history on their backs here.

And what I'm trying to say here is that the structure of how the system evolved itself stores within it a memory of the history of how it got there.

It's a lot like, you know, how a snowflake itself is structurally a history of the, the temperatures and pressures that it went through during its formation.

In the same way, these systems are also very complex and unique.

Every time you run one of these, the way that it stabilizes is always unique, kind of like a snowflake.

But the, the end result that you get there kind of contains the history in a way, in a very kind of compressed way.

And that's kind of a property that we want in a world model, our dynamically generated world model through interaction.

So I started with kind of Ashby's, in terms of how to actually build this.

I started with Ashby's kind of physical descriptions and analog hardware here.

Some of this came from his technical journals, which were digitized almost 20 years ago.

Some really great stuff.

It's all handwritten.

So be prepared to read his handwriting.

He's got some amazing illustrations in there, but it's about 7,000 pages of technical journals.

And so there's some gems in there if you're patient.

So I started by pouring through stuff and building kind of a very accurate simulation of the homeostat.

And then over a long period of time, abstracted it down and simplified it down to something that looks a lot more like modern AI networks, you know, where you can kind of build these into abstract shapes and topologies.

And each node is doing the same thing.

It's basically trying to predict its neighbors as the system as a whole reaches equilibrium.

But you can think of this as kind of a loosely coupled set of dynamical systems.

You know, each one of these is a homeostat, an abstract homeostat.

And, you know, I'm trying to think how much I want to dig into the dirty details on this slide here.

You know, you can kind of imagine that this is still disconnected from an environment.

So we haven't yet, at least at this point in the slide, connect this to an environment with sensors and motors

and anything like that.

So really all it's doing is when it's reaching equilibrium is finding a set of positive and negative feedback loops.

You can imagine these connections in the network here forming a large set of overlapping feedback loops.

And depending on what the value of the weight, some of these are positive, some of these are negative.

And what's going on through this local alterability process is that they're searching a state space here of positive and negative feedback loops such that they all are self-balanced.

So all the, you know, once you're reaching equilibrium, it's not a quiet equilibrium in the sense that it's very much active in that all these feedback loops are very perfectly balanced so that they're stable.

And this, you can kind of think of this as a critical state.

And this is really important because it means that the slightest change or the slightest new piece of data can completely destabilize one of these feedback loops and cause cascading changes throughout the network.

And as long as you can kind of restabilize the network quickly, this is great.

That means that one small piece of data can drastically change the dynamics of the model.

And that's what you want in a system.

That's the opposite to backpropagation-based systems where you need a lot of data in order to change the behavior of a model.

Instead, you want one surprising piece of data to drastically change how the model behaves.

That's much more data efficient.

You need a lot less data to get to a useful world model this way.

Let's see if there's anything else that I want to talk about here.

In terms of how this relates to active inference, when it's reaching equilibrium here, what it's trying to do is it's trying to minimize the prediction error of its data.

It's prediction of its neighbors.

And so once you're at equilibrium here, you're basically at zero prediction error and all the networks are successfully predicting each other.

So just at a high level, I have to spend a couple of slides on these next few topics here.

First is how I actually connect these to an environment through essentially motor coupling.

Sensors and motors kind of define that interface between the environment and the model.

Then I'll talk a little bit about embodiment and why this is important.

And in fact, I think to me, the most important aspect of embodiment is the existence of time in the system.

The time awareness of it.

For me, I have a hard time looking at traditional deep learning AI models, even in robotics, where you have a body being embodied because they're considering them to have embodiment because they're time agnostic.

You know, you have these input output functions and they have no concept of being situated in not just in a place, but in a time where their last action that they took is relevant to the current one, for example.

And there are ways to trick this by feeding, you know, especially in reinforcement learning, this is often what you'll do is you'll feed in your previous inference results as inputs to the function.

But I think of that more as a hack.

And then I'll talk about a little bit about synchronization as a learning process itself in case that's not super obvious.

So once you've built a homeostat, even a large one with a large topology, it's still kind of boring.

All it's doing is kind of learning itself and learning to get to equilibrium.

So, you know, one of the first things you have to do is so it's self-synchronizing.

You can kind of imagine it as a self-synchronization process.

So the next thing you have to do is, you know, if you want this to do something useful, you have to connect it to an environment.

And so sensors kind of define how the agent is embedded in the environment.

And the topology itself defines what the adaptive capacity of the homeostatic agent is, you know, what it can do and what it can learn.

You know, obviously, the larger and more complex the network, the more expressive and more complex behaviors you'll get out of it.

And then the sensor definitions, we've kind of, I call them sensors, but maybe they're inputs.

I'm not great at semantics here.

Naming things are hard.

The sensors here are actually really important because they also act as teleodynamic priors.

In terms of getting this to do something useful, you basically can define what the homeostatic set points are in the sensors.

So the sensors aren't just data.

They're not just, you know, pixels coming in, but they have goals implicit with them.

And an example might be instead of having a temperature sensor that's just saying 68 degrees or 70 degrees, instead it's saying, you know, I am two degrees away from optimal or I'm five degrees away from optimal.

So the idea here is they have kind of implicit goals in them.

And those can be seen as like input attracting sets, which you can kind of, from externally, you can modify those in real time to get more interesting behaviors and to kind of change what the model is doing at any point in time.

The motor definitions here are basically just coupling points that go outward.

So they're inside the network.

There's a lot of recurrence, a lot of bidirectional connections.

The sensors are almost by definition input only.

So they're unidirectional and the motors are also unidirectional outward.

And they basically provide affordances to the network for modifying the environment to, you know, and it provides the basis for future sensory predictions.

You can imagine the network here is basically trying to generate motor outputs that help satisfy the input goals, you know.

And so the preferred states of the sensors become the preferred outcomes of the motors.

And so you kind of have this synchronization loop that goes outside of the network.

So the network is doing self-synchronization, and then once you connect it to an environment with sensors and motors, that synchronization manifold basically extends outward, you know, and it basically is performing a kind of homeostasis through the environment, outward through the motors and through the sensors.

You know, the model kind of imagines what it wants and what its preferred state is from the sensors and then generates motor actions to kind of drive it towards that state over time.

So, like I was saying, I think embodiment is a critical feature of what we're building.

And I think the missing piece here is time.

So in these networks here, there's no forward or backwards propagation because there's so much recurrence.

You can't really define a front or a back or a top or a bottom.

You know, signals just propagate in all directions.

The closest thing you can get to is distance to the environment.

So how deep into the network you are and how far away you are from a sensor or a motor.

You know, this is not like the sandwich model, I guess, is what people like to say, where you take input, you compute it, and you generate output.

Instead, you know, information is flowing into the network from the sensors and actions are kind of, you know, flowing outward from the network through the motors.

But once the information hits the network, it just starts bouncing around in all directions depending on the network topology.

You update the network and information flows in all directions.

And through this kind of, when you're reaching equilibrium and you're hitting this state, you know, that's when you can consider that the network is successfully predicting the sensory inputs.

And, you know, whether or not the motors are contributing to that depends on the specifics of the motors and the sensors themselves.

But the expectations of the sensory information here becomes embedded with the motor actions themselves.

So the idea here is the network, while it's generating sensory predictions towards the front of the network, it's also generating, you know, motor connections to the motors.

And these are kind of happening in parallel.

So as it's updating its model to better predict the sensory information, it's also updating the motor actions that are being generated.

And so the kind of the motor actions and the sensory expectations kind of get intertwined in the network here.

And this is why I like to call it the inactive synchronization process, because, you know, these are kind of being co-created as part of the model, the world model that's being directed.

But this this is, you know, if those familiar with predictive processing or predictive control, sorry, perceptual control theory, this will look very kind of familiar in terms of how this actually works.

Finally, I figure I'll talk a little bit about synchronization and think about homeostasis and prediction and where the prediction errors are.

Right. If you if you kind of imagine the drive towards homeostasis as kind of a goal, the distance you are from that equilibrium could be seen as like a prediction error.

And so you're basically trying to drive this prediction error down.

And this becomes more obvious if you actually look at the chaotic attractor definition in terms of the way that you're taking a weighted multiply of your input signals.

And you're trying to get that to be zero. You're trying to you're trying to basically predict that, you know, what your neighbors are doing and a successful prediction will result in a local prediction error that is near zero.

And when when the network and when this happens, you can kind of think of it as a synchronization process when when you're you can when you can act when you can successfully predict your neighbors, you're kind of synchronized with them.

You know, you know what they're going to do before they do it. And so that's how you can have the sensory expectations meet the sensory inputs through a synchronization process.

And synchronization is kind of magical. Once once you have two systems that have synchronized or let's say you have multiple systems synchronized and you can imagine each node in here synchronized to its neighbors.

And so you kind of have this synchronization over a distance.

that's a way for two systems that are not in direct communication to coordinate their behaviors.

And so, you know, you kind of get very interesting collective behavior out of these networks, collective here of the nodes through the synchronization process where the each individual node behavior is going to be consistent with or

or

or

coordinated with distant nodes in the network.

And so this is a very important aspect of of how these functions work, because if you had to if if two distant nodes that are representing different aspects of the environment had to be in direct communication all the time in order to maintain their their correct modeling of the world,

this whole system would become incredibly computation expensive and impossible and intractable.

So, so the synchronization process is very important.

I think I talked about previously how this is maintained at a critical state.

And so the smallest prediction error that's coming in through the sensor can have cascading effects through the network that cause very, very fast destabilization.

And so the trick here is to synchronize faster than the environment can destabilize it.

So if you're updating your network too slowly and the environment is changing very, very quickly, you're going to be constantly in a state of chaos.

But if you can update your network much, much faster than the the changes that are being experienced in the environment, you can imagine that this network is able to re synchronize and re re stabilize continuously over time faster than the environment can.

And this, this goes back to kind of Ashby's law of requisite variety.

And the idea here is that the variety that's being generated is very much time dependent and depends on the rate of change of the environment versus the rate of change of the network.

But the trick to this is ultra stability is very, very efficient, and then we can run it really, really quickly.

And in most of our robotic applications, for example, these networks are operating usually around the update state, the update rate of these individual nodes.

And so for robotic applications, usually between five kilohertz and 50 kilohertz.

So we do this very, very quickly.

And depending on the application and what's being done, that's generally fast enough to maintain the synchronization manifold between the model and the environment.

You know, it's a prediction error minimizing synchronization manifold.

But if you if you run the network that fast, you're able to basically maintain the synchronization manifold over time.

And that's how we kind of get useful behavior.

So I think I've talked enough about synchronization.

Now I'll get to the fun stuff that I started talking about the beginning here where the killer application for this kind of learning approach to this adaptive approach really is robotics.

You know, early on, we we kind of considered different applications of this technology for like anomaly detection or even analyzing, you know, model drift in AI or deep learning systems, for example.

But the challenges in robotics that AI is facing right now are really insurmountable with the amount of data that they need.

And at least in my opinion, I don't much of a going to get there in the next decade.

And the they call it a simple real is the other problem.

So if you train a robot in simulation because simulations are not entirely physically accurate, transferring it to the real world is a big problem.

But doing the learning post post post deployment in the robot solves a lot of that.

A lot of those challenges, not by it doesn't solve it more sidesteps it because you're doing the learning in the real world as opposed to in simulation.

So this is just a quick look at some of the industrial applications we've been doing with this technique.

These models are entirely using homeostats to do the robotic control.

The these are some of the more recent ones that we've been working on on the left here is an insertion process for automotive manufacturing, where we're kind of inserting small rubber parts into this case, a door latch assembly.

We've also been looking at connector insertions for cables and cable manipulation.

And on the right here is an older model that we built for turning valves and bolts.

This was actually I'll talk about each of these in the coming slides here.

This is the one on the right here.

We take the conferences.

It's a small little robot and we let people kind of move it around and let the robot figure out how to turn the bolts as people are kind of playing around with it.

So this is one of our first models.

We built this several years ago, and this was originally designed for doing inspections on a moving vehicle.

And so the vehicle could be bouncing around and they needed to kind of point the robot at something that's being inspected as it's being bounced around in unpredictable ways.

This was originally designed for solar farms, but can be used.

And we use this as kind of a base model for all sorts of stuff.

If you just need a robot that points at something.

Next, we started working on more manipulation use cases here.

This was also originally done for the energy industry.

They wanted to see if we could automate the turning of valves.

You know, you never know what size or shape a valve is going to be in a facility.

And sometimes there are injuries that these are high pressure gas valves.

For example, they have, you know, people get injured and killed very frequently.

So, you know, that's a great, great thing to do for a robot to do is something that's dangerous.

And so we would turn this into a demo that we bring around to conferences sometimes.

And we would let people like move around these bolts while the robot tries to tries to turn them in.

It's kind of fun.

People like to play around with the robot and torture it a little bit.

And there are also not very many interactive robotic demos at conferences.

This is another video of this conference demo.

And I included it because it has this video here on the right.

So that's our actual homeostatic network debugger.

But you can actually see the network updating in real time here as people are jerking around with the robot here.

Trying to, torturing the poor robot while it's trying to turn a bolt.

And people like to watch the, you know, it's an internal debugger.

And so it's, we built it as a tool for us, but it's people like looking at it while they're actually playing with the robot.

These are, this is some, some more recent work.

And I'll show what the final result of this ended up looking like.

This was for a project with an automotive manufacturing where they wanted to basically insert, you know, tight fitting rubber parts.

And the problem is you can't really do this with vision inputs alone because they're tight fitting.

And, you know, humans are really good at this.

You can kind of feel your way through it and kind of wiggle it in really easily.

And so that's essentially what we're doing.

There's a force torque sensor in the wrist.

And in this, in these videos here, there's, we're actually not using vision to guide the process at all.

So we basically tell it, and we're actually telling it the wrong location.

And we were telling the wrong location, usually between two and five millimeters of error in like a random direction.

And it's using the force sensor in the wrist to kind of feel its way through the insertion process.

And the one in the bottom left there, you can see where it kind of has to, you know, it catches an edge and then it figures out that they can insert it and it kind of pushes it in.

But we're basically using the same model for all these parts.

So one of the big goals of this project was to show that we could build one model for insertion and have it work across a large variety of parts.

With traditional AI, in a lot of cases, you'll have to build and train with separate data.

Each of these models, each of these would be a separate model for each part.

And that's a problem in manufacturing because these parts change every year or two.

And what they don't want to have to do is generate a huge data set every year or two at one of the parts change.

And there's also about 500 of these parts.

So they'd have to build a lot of parts specific models to do this.

And this is just a video showing what this actually ended up looking like when we finished it out.

And so this, you can see that this, in this case here, we've integrated the gripper.

We are using vision to figure out where the insertion housing is, but it's then it's using the force torque sensor in the wrist to kind of drive the rest of this process.

And in case I hadn't mentioned it, so essentially we're using the homeostatic networks to kind of do the control here.

So as it's doing this wiggling in, it's basically looking at the forces in the wrist.

And we've basically given it the goal of trying to minimize the lateral forces and things like that and maintain that the insertion,

the force vector is perpendicular to the direction of insertion, for example.

And it just kind of, you know, as it's trying to do the insertion, the network is constantly updating its understanding of the, you know, what it's doing and modifying its insertion,

modifying its behavior to perform the insertion task here.

I'll give it one last second here.

There's one more part.

This is showing all four parts being inserted back to back using the same model.

So we're not switching models here.

Like I said before, I'll just let this finish real quick.

Hooray.

All right.

We inserted all the parts.

So, and then this is some more work that we've done recently along similar lines.

This time inserting, I'm not sure why, let me see.

Here we go.

I just hit the button twice.

Doing insertions of various cables.

So we're using a single model here to insert cables.

In this case here, like a 15 pin, you know, parallel cable and then an ethernet cable.

And the idea here is this should generalize to different kinds of cable insertions.

Again, this is using force to kind of drive the insertion process here.

So even if the vision system is not entirely accurate and it's giving a location to insert that's incorrect, you know, a couple millimeters off here and there, it's able to basically figure out how to actually insert the cable in the right spot.

And we're continuing this work right now in more interesting cable manipulation cases here.

So the insertion is one part of this.

The next step is being able to manipulate the cable around, you know, untangle a little bit.

You know, cables are very, very challenging to work with.

This is why we're working on it is because it's very hard to automate with traditional robotics.

Anything can happen with cables.

They can get caught, they can get tangled, they can be in places you can't reach.

So you might have to pull another part of it to get it into a place you can reach because there are obstacles and things like that.

So that's kind of where we are with applications of this stuff.

And super excited about next year, what we're going to be doing in the future.

Oops.

That's the end of my talk.

Thanks.

Thanks.

Thanks for having me.

It's been great.

Thank you.

And feel free to reach out if anybody has questions or anything.

Happy to hear from you.

Very cool.

Okay.

I'll just ask any of the questions that are written in the live chat.

All right.

Oh yeah.

I should have been paying attention to that.

I'm sorry.

Oh, no, no worries.

I'll, I'll, I'll read it.

Okay.

So ready player Sid wrote, are we headed into a decade of tele operation, tele robotics?

What is the next decade of robotics going to look like with respect to embodied AI?

I mean, a great question.

I think we are, I think just in the last year, I feel like I've seen at least 50 to a hundred startups that are doing

tele operation or selling tele operated data or selling systems to do better tele operation.

One of the reasons why we went after soft body insertions, for example, is because tactile data and force data is very, very hard to get.

You can't really get accurate data in simulation.

And, and it's, if you need to train a deep reinforcement learning model, you really need accurate data for that.

And, you know, there's obviously doesn't exist out there.

That's why a lot of vision models don't, the visual language models don't have any kind of force data in there.

And so, and so for example, I've seen just like 10 startups in the last year that are providing tactile data collection for tele operating.

So, so there's still do using tele operation, but they're not adding force sensors and tactile sensors to the, to the robot so they can collect the data during tele operation.

But I think, I think if the money keeps going in that direction, that's what you're going to see.

And I think that's what you have been seeing in a lot of cases here.

There are a lot of startups that are just hiring people to do a task over and over and over again.

And then they'll train a model to do that task.

And often what you'll see is you'll, you'll get a successful model that can fold a napkin or fold a towel, or I'm sure those are the ones that are coming to mind.

But whether or not that, that model can then be translated to a different shaped towel or a shirt is questionable, mostly because of the science.

And so you'll have to tell a rate, you know, tele operate to collect data to, for that, that, you know, for the shirt instead.

But you know, there's so much money involved.

That's the direction that people seem to be going.

Like people don't seem to have a problem with spending a hundred million dollars paying people to tele operate robots day and night.

I've even seen a startup where what they're, they're not even selling AI.

What they're doing is they're, they're basically you rent a robot for like \$10 an hour to like operate your shop or I don't know.

You know, let's say have a corner store and have a robot operating.

And then it's literally, they're just paying somebody in another country, you know, \$2 an hour to operate the robot.

And that's literally their business model.

So I think you are going to see a lot of this and there's going to be a lot of smoke and mirrors.

I think, you know, I'm a little bit biased.

I obviously think that there are better approaches.

And I think, I think eventually the hype will die down and people will realize that we'll need more, more efficient and better world models.

And that's, that's what I'd like to think that we're building, even though it's, it's kind of a, in an active world model.

And that the model is comprised of sensory inputs and motor controls kind of intertwined with each other.

So it's not the same kind of world model that most people kind of envision in their minds.

But I do think it is, it is one that is relevant to a robotic, you know, task load.

Anyway, I kind of got off topic there, but I like to think that there are going to be better approaches, kind of like what we're doing here, that will take over at some point here.

When the, when the economics pan out, I, I think with self-driving cars, for example, the, the value of having a self-driving car, the ROI there is, is high enough that it can justify spending a decade of collecting data.

But for things like inserting rubber parts for automotive manufacturing, the ROI doesn't exist.

Like you can't spend a decade collecting insertion and tele-operated examples and have it be worthwhile in the end.

So I think once you want robots to go after kind of lower value problems like that, and it's still highly valuable, but just not as much as a car that can drive itself.

You're going to need different techniques. That's my opinion.

Cool. Okay. I'll read another question in the chat and people can write if they have more.

Question two from Sid. Why do DeepMind Robotics and other companies such as Figure take other approaches when it's so obvious adaptive models are better in so many ways?

That's a good question. That's a good question.

I think it's because it's, it's a clear straight line to something that they can imagine working and it's something that it's that it's an approach that they're familiar with.

Back propagation is like perceptrons have been around forever, but back propagation has been around for 30, 40 years, which doesn't sound like I mean, in the history of man, it's not very long.

But like in the AI world, that's, that's a long time.

In which case, what I mean by that, I guess, is you have like multiple generations of people trained in that paradigm and people understanding how to that paradigm.

So it's very comfortable and people can just, you know, hand wave around. Oh, if we get enough data, we'll get there.

I think the reasons why other like approaches with other implementations of active inference, for example, that are, that are adaptive, don't have the scalability that they are looking for.

And yet, and, and, and so I think it's easy to dismiss.

I think the, I think the tide is turning though. I think people are becoming much more aware of these other approaches.

Our approach using homeostasis and Ashby's techniques is very unique. Like I don't, I, as far as I'm aware, I don't think anybody else is doing this.

And so, you know, the reason why nobody else is doing is because they don't know it exists and they don't know how to do it.

And if I'm gonna be honest, building a, getting that KIDR contractor, that homeostat, that ultra stability definition is very, very hard.

Like I'm a, I'm an engineer by background. And so that's what I built my career about around taking theory and turning it into real engineering.

And it's challenging. It's hard. I want to say there's probably been about a half dozen attempts in the last 80

years to build a useful homeostat.

And I, having looked at them, I disagree with the implementations. I think there are problems with each of them, but this, I mean, it's all subjective at this point.

Nobody, there's no definitive homeostat that you really just have to kind of look at the properties and argue that, you know, it has this property or it doesn't have that property.

So I like to think we have something unique here and special, but, and that, that would explain why nobody else has it.

Okay. And Sid with the hat trick, what kinds of adversarial cyber physical attacks are possible on thought forge robots demonstrated in the video that use adaptive models?

You know, I hadn't really thought about that.

And I would like to think that, you know, yeah, it's something I haven't really thought about too much.

The models themselves are robust to like sensory noise, like, you know, this kind of inherent in how the algorithms work.

And so if you have noisy sensors and things like that, but if you can like straight lie to the sensors and you tell it that the world exists, the world as it exists is different, you know, that would, I mean, that, that would certainly work.

And I guess, you know, cyber attacks and cybersecurity, there's, there's lots of different ways of looking at that, you know, and if you, if you get access to the code, it depends on what you want to do.

But it would be pretty easy, for example, or not easy.

It would be possible if, for example, like, let's say we have an insertion model that, or that goes and picks up a part and does an insertion.

If you're somehow able to trick the, the perception system into thinking that a person was the part to be inserted, it would very happily go and try to grab that person and insert them.

And obviously this would be very dangerous for the person.

So there's no, like, you know, if you can alter the perception of reality for any of these models, then you can get them to do whatever you want.

And that's kind of how we're getting to do something useful in the first place.

If I'm going to be honest, the, the way that we've defined the sensory inputs, where we're kind of implicitly putting to the dynamic priors, we're putting our goals in those sensory inputs specifically so that we can control them to do something useful.

Because otherwise a homeostat will just kind of sit there.

If you disconnect it from an environment, it will reach equilibrium and do nothing.

And so the way that we're actually getting these things to do something interesting in environment is by manipulating its perceptual experience, which, you know, I, I, I guess the moral and ethical consequences of that is probably for a larger time.

Probably for a larger talk.

Yeah.

Very interesting.

And, and, uh, connects with this tension of the expectations for like a sensory prediction to be veridical, but also actionable because just veridically predicting is being a spectator.

It's passive inference, but then that is the question of how much to, to have that kind of biased counterfactual homeostasis that takes you into the step that you need to go towards.

Yeah.

And maybe there are ways to protect against that by having, um, higher order.

Tealy dynamic priors where like goal, instead of inserting, uh, inserting a part is to construct the part or construct the car.

And so maybe then it can evaluate the effectiveness of its subtask of like, Oh, is this thing I'm inserting actually going to get me closer to building a car?

You know, that might be one way out of it.

We're, we're not there yet, uh, with our technology, but, uh, I think from like a theoretical perspective, I think that's, that's one way to approach it.

Cool.

And then sort of in, in closing, any last thoughts, of course, and just what do you see for robotics or active inference?

What are some of the constraints or limiting steps right now?

And where do you see things updating or changing in 2026?

Oh man, that's a hard prediction.

Robotics is super exciting right now.

There's tons of stuff going on.

Um, I'm super focused on like practical applications, building the business, trying to make sure that this is an economically viable approach.

Um, and so that's, that's where most of my energy intends to get spent.

I'm super excited by all the traction that, that active inference as a field has kind of made in the last year, just in terms of, um, public awareness.

And, and I, and I think if the AI bubble bursts, I mean, I don't know if it's ever going to burst or maybe just deflate a little bit.

Uh, I definitely think there's a hunger out there for new approaches.

I think right now people are starting to see that even, even LLMs with their huge capabilities and what they can do is super useful, super interesting.

There are limits, you know, at some point here, language will not get you everywhere and will not, is not what's going to get you to, you know, truly, you know, intelligent software.

If you want to, if you want to put that way.

Um, I am much more lower.

I, I'm not trying to create Rosie the robot for your home.

I'm trying to, if, if I, if I have a lofty goal, I'd much rather make ants, you know, um, because they are, are, they, they're, they're capable.

They can build stuff.

They are loyal to their queen.

And like, you don't have to worry about, um, does the queen of an ant colony worry about the ethical consequences for its workers?

Um, maybe, maybe not, but, but I feel like that's a much more lofty goal than trying to create sentient, you know, AI and AGI, which is a lot of what you hear on, you know, on, from Silicon Valley these days.

And to me, you know, I, I, I, I don't think we need that.

I think we need, we, what we want is, you know, machines that can take away the dangerous, uh, dreary work that nobody wants to do.

You know, that's, that's, that's what I'm looking forward to.

Very nice.

Thank you, Matt.

So thank you for joining the symposium again and for helping on the scientific advisory board as well.

Thanks.

I'm looking forward to next year.

All right.

Peace.

Cheers.

All right.

Next is going to be a 33 minute recorded talk by Maria Luisa Iannaca.

Here it goes.

Hello, everybody.

Uh, thanks for having me in this wonderful symposium.

My name is Maria Luisa Iannaca, but you can call me Maria.

I am a PhD candidate in philosophy at the university of São Paulo and the university of Porto.

Uh, well, my talk will be, I think a bit different from, from the rest because, um, I come from philosophy and I will try to discuss the ambiguous status of attention in, uh, predictive models of cognition.

So, um, the content, uh, will be divided into like a brief description of precision modulation and salience, followed by a description of attention across disciplines.

And then I will finish with this discussion that I've been, uh, labeling as, uh, having an ambiguous status.

And, you know, as I am a philosopher, again, this is a philosophical analysis.

And so my focus will not be, uh, on any specific models or any specific formalisms, but on the general framework of both hierarchical predictive coding or predictive processing and active inference, which I believe you all know very well.

Okay.

Okay.

So, uh, attention in these two frameworks, uh, is usually related to two things, precision modulation and salience.

Precision modulation or, you know, precision weighting or precision optimization is, uh, formally the expected inverse variance of prediction errors.

In other words, uh, in the predictive brain, the precision modulation happens within second order predictions that control the confidence of the first order predictions.

So, if the brain is highly confident in a prediction, the modulation will reduce the gain on error signals, choosing to trust its prior beliefs over the unreliable new data.

However, if the context is uncertain or noisy, uh, an error will be given, uh, significant weight, you know, prompting a major update to its internal model.

And it is believed, uh, to be implemented via lateral or self-inhibition in the error units with the biological substrates being, uh, like inhibitory synapses, neuromodulatory adjustments, and also fast gamma synchrony.

And attending here, uh, would be the modulation of, uh, uh, the synaptic gain afforded to top-down predictions and bottom-up, uh, prediction errors based on the precision they are estimated to have.

And this is pretty much how the endogenous and exogenous attention would work.

Saliency, on the other hand, has to do with action selection.

So, saliency has been, uh, characterized, uh, as intrinsic motivation and denotes actions promising maximum information ratio gain.

Uh, that is, uh, highly unpredictable and yet interpretable observations after which sampling is suppressed in a manner akin to inhibition of return.

And in neurobiology, this would be tied to action planning, uh, specifically, uh, in vision, we would say, uh, the action of basal ganglia, uh, where the saliency ranks potential saccadic targets and thus corresponds to competing saccade, uh, evaluations.

So, attention then comprises actions selected to elicit information that cannot be referred passively.

And saliency indexes the epistemic value of such actions as quantified by, uh, expected free energy.

And, of course, uh, the predictive attention has been around for a while across, you know, various disciplines.

And here is just an example of three tasks that were simulated using predictive models in visual neuroscience.

But, you know, uh, there are tons, uh, of different other examples in the literature.

So, this first, uh, on your left is the famous Posner queuing task, which revolutionized, uh, the neuroscience of attention.

And it was successfully simulated by Feldman and Freeston in 2010.

And this task basically, uh, makes participants fixate, uh, centrally while cues indicate where a target might appear.

If the cues are valid, it will speed up the reaction times when the target appears at the queued location.

If the cue is invalid, slow responses will happen, uh, when the target appears as well.

And it's, you know, uh, uh, uh, people say that it's reviews how attention can be, uh, covertly shifted without eye movements.

And the example in the middle is an active inference model of selective attention developed by Mirza and colleagues in 2019.

Uh, they focused on contextual epistemic foraging.

That is how agents seek out task relevant information while ignoring irrelevant stimuli.

And for that, they simulated typical attentional behaviors like this one in Yavuz task, uh, where you basically, uh, have a picture and you, and you do, and you have to look to this picture.

And then, uh, your, uh, your saccades will be, uh, recorded.

And so on the left, uh, side of the figure, uh, there is the human saccades.

And on the right side, there, there are the simulated ones.

And they framed attention, uh, not just as the, uh, reactive process, but as a proactive part of planning that will be tightly linked to the saccadic eye movements and also believe updating.

And the third example on the right is more focused on the movements of the eye itself and also the neuromodulation that happens.

Uh, and this paper was done by Paul Freeston and they used an active inference model to simulate how different, uh, neuromodulators, specifically the cholinergic and the gabagic modulators affect, uh, ocular motor behavior during, uh, delayed saccade task.

And this task is about a cue that indicates a target location followed by a delayed period requiring fencing session.

And then, uh, saccades to the remembered location.

And so by manipulating precision parameters associated with each neuromodulator, they, uh, were able to model how pharmacological interventions could alter both the decision to initiate a saccade and the trajectory of the eye.

And so predictive models have also been used in computational psychiatry.

That is, uh, a new and progressively important area that aims to, you know, understand the psychiatric disorders using both real data and simulations and models.

And the majority of these studies work with predictive coding schemes instead of active inference.

So they kind of focus on the precision modulation parts instead of salience.

And I'm not going, uh, in details about these papers that I put, uh, on my slide, but, you know, I, I left them here just for you in case you want to, uh, check, uh, later and read if you want.

But the idea that I wanted to say here is this many, if not all the psychiatric disorders in the predictive context happen because of the malfunctioning of precision modulation.

And even though the malfunctioning is not the only or the main reason as everything, you know, about our psyche happens because of multiple things, it is a current and a core reason.

And this means that if we understand attention as precision modulation, then all psychiatric disorders could be, in fact, attentional disorders, at least in principle.

So can we, you know, really say that attention is precision modulation?

Well, attention, like, uh, any other phenomenon of the mind has been studied since forever, uh, across cultures and fields.

But one definition that was kind of foundational to, uh, the science of attention was the one made by William James.

That is, uh, the taking possession by the mind in a clear and vivid form of one out of what seems several simultaneously possible objects or trains of thought.

Focalization and concentration of consciousness are of its essence.

And it implies withdrawal from some things in order to deal effectively with others.

And is a condition which has a real opposite in the confused, dazed and scattered brain state.

This is like the classical definition.

But more recently, uh, neuroscientists Sabine Kastner and Kianobri coined a more scientific or more

operationalizationable, uh, definition.

That is the prioritization of processing information that is relevant to current task goals.

And a couple of years ago, philosopher Wayne Wu made an even shorter description and operational, operationalizable definition of attention.

That is, uh, simply attention being the selection of a task to guide behavior.

But of course, uh, as the literature is huge, there are many different effects, mechanisms, and interpretations of the behavior.

And interpretations of what attention might be.

So if you take a look in your right, you see, uh, this impressive taxonomy made by Caldas.

And you can see that selection fraction is just one sub kind of the many other sub kinds and kinds of attention that coexists.

Nevertheless, uh, I'll stick with, uh, Wu's idea or, uh, or these definitions of selection, uh, because they shaped the visual neuroscience.

And of course, for the sake of time and scope of this analysis.

Okay.

So, um, there are three main neural networks that people commonly attribute to attention.

If you look at these pictures I borrowed from Ross, uh, you can see in detail, uh, the activations.

So, um, the first network is the alerting system, which keeps the person awake and not asleep.

And this neural network is composed of the thalamus, locus coeruleus, frontal and parietal cortex, and has the noradrenaline as the modulator of the network.

The second is the orienting network, which drives attention overtly and covertly.

And it is composed of the pulvinar, uh, superior colliculus, superior parietal lobe, and parietal temporal junction, and also frontal eye fields, having a cytokine as the modulator.

The third is the conflict network, which, you know, monitors conflicts, problem solving, problem solving, or, uh, you know, basically, decision-making.

And it's composed of the ventral lateral prefrontal cortex, basal ganglia, and the anterior cingulate cortex.

And it has dopamine as its modulator.

And additionally, I included here a figure from Parm-Friston that illustrates the network of eye movement, which, you know, is fundamental in attention studies in visual neuroscience.

And this network is composed of the superior colliculus, uh, basal ganglia, frontal cortex, cerebellum, and also the vestibular system.

And it also has dopamine as the modulator.

Very well.

So, from the eighties to the nineties, there was, you know, a significant progress in understanding the effects of attention.

Um, and with that, many theories and models were developed.

Here I brought, uh, three theories that are well consolidated in the science of attention,

at least that I am aware of.

And the first one is feature integration theory that was developed by Anne Treisman and Gary Gallaud in 1980.

And this theory is about attention being a kind of glue that binds the perceptive features together in a single percept.

And one core item in this, uh, in this theory is the cognitive maps, which have been extensively used in modeling neural networks.

Uh, the second theory is the primoto theory of attention that were developed by Lila, Craig Hero, and Giacomo Ricciolati in 1987.

And this theory is very important for spatial attention because it kind of proved that attention shares, uh, the same activations as eye movements and also action representations in the brain.

So they basically described the dynamics of overt and covert attention.

And the third, uh, theory is the bias competition theory, which was proposed by Robert Desi Desimon and John Dukan in 1995.

And this theory is highly influential because it basically describes, uh, the dynamics of top-down and bottom-up attention, uh, specifically how visual competence would compete for cognitive resources.

So yeah, pretty important.

And when it comes to personal level of attention or clinical attentional effects, there are many, uh, mainly three disciplines that kind of investigated.

Psychology, psychiatry, and philosophy.

Of course that, you know, before technology and tools, neuroscience was psychology.

Uh, today's cognitive psychology.

And you, you, you, you, you, you find attention literature before the 19th mailing in psychology, psychology textbooks.

So anyway, the personal level effects of attention are basically to focus, right?

When you kind of pay attention to something, uh, and how well you perform it, the, you know, this will depend, uh, in many things like moods, interests, environmental distractions.

And so on.

And we also have the internal agency because not only do we pay attention, but we choose what we pay attention to.

So even though, uh, our attention can be grabbed sometimes, like, uh, you're trying to pay attention in my presentation, then your phone rings.

Even though, uh, it will distract you for a moment.

You kind of, you can choose between giving the attention to your phone and answering the person or whatever, or getting back to the presentation.

So, uh, you choose basically, uh, you, you, you're in personal level, you choose.

You choose.

And when it comes to attentional disorders, when these things do not work well or properly, uh, science and, you know, psychology, psychiatry, uh, worked a lot with lesions, uh, developmental problems, uh, that make

people kind of neglect one side of the visual field.

And this was, uh, uh, sometime later, uh, labeled as hemispatial, hemispatial, uh, neglect.

And, of course, we have the attentional disorders, ADHD.

That is pretty common nowadays and people are being diagnosed even, uh, when adults.

So, uh, yeah, uh, it's like, it's important to highlight here that, uh, not only ADHD, but all psychiatric disorders have a common symptom.

That is the decreasing attentional task performance.

And so, of course, when you talk about lesions and development, uh, it's kind of a different thing, but when we talk about psychiatric disorders, uh, this attention performance is a common and current symptom.

And lastly, real quick, I would like to present to you some of the philosophical theories of attention.

So, the first here is Wade's selection for action.

He proposes that attention is best understood through the role, uh, in guiding behavior.

So, in his view, attention would be an active process of selecting information for, uh, uh, that is relevant for action.

So, for instance, if an agent is preparing to grasp a cup, the visual and the motor systems will prioritize or select cup-related features.

And this selection, according to him, is what constitutes attention.

Sebastian Wald's priority structure depicts attention as the structuring of consciousness itself.

So, it, it would be a way of prioritizing mental content across perception, uh, thought and emotion.

And for him, attention is not tied to a specific modality or task, but it reflects how the mind allocates its limited resources.

And so, this prioritization shapes what becomes salient, accessible, and integrated into our stream of experience.

Christopher Moe's cognitive unison theory links attention to, uh, something related, like, uh, something like an orchestra.

So, attention would not be a single process, but the harmonious functioning of multiple cognitive systems.

And for, because of that, attention for him would be an adverbial, adverbial phenomenon, uh, that is not something that the brain does in one place,

but how various process operates together.

Now, uh, some theories of consciousness have attention at their core, uh, as it happens with at least these two here.

So, Michael's Graziano's attention schema theory will argue that the brain builds a simplified model of its own attention,

that would be the attention schema, that is similar to how it tracks the body.

And this internal model helps the brain monitor and control attention.

And it's this model that gives rise to the feeling of awareness.

So, consciousness then would be, uh, the brain's interpretation of its attentional state.

Lastly, uh, Jesse brings attendant intermediate representation theory.

Uh, that is to say that consciousness happens when attention targets intermediate level perceptual

representations.

Those reach enough to be meaningful, but still tied to sensory input.

So, only attendant representations at this level will become conscious, making then attention a gatekeeper of awareness.

So, uh, now that I've talked a bit about attention, let's see why I think, uh, its status in predictive models of cognition is a bit ambiguous.

So, first of all, uh, what are the differences in attention addressed in hierarchical predictive coding and active inference models?

You know, I'm aware that this question may be more because I'm not entirely familiar with the formalities and, you know, uh, I don't know how to reproduce these models.

I don't model anything, but, you know, by reading the papers, I could not spot real differences in the way attention is treated.

And I would be happy to learn.

I mean, if you know, if you know to answer how to answer this question, please email me.

I mean, formally, uh, precision modulation and salience are different things in the models, of course, but the way attention is related to them is what is not clear to me.

Uh, you know, for instance, uh, precision modulation happens every time in all scales, but does it mean that we are attentive at all the time in all scales?

I don't know.

I don't know.

I don't think so.

Or even if we are, maybe it means that attention would be more like the foveal spot of our experience.

I don't know.

Uh, and even like if it is the case, then what's the difference between this and conscious perception?

And I don't think that attention disorders mess with perception in this manner.

Of course, there are some lesions in the brain that could, uh, that could be related with, you know, holes in a vision or black shadows.

I don't know.

Things that messes with visual perception, but I don't think that attention disorders would mess with perception in this specific manner, you know?

And what about salience?

Is salience the same as the dynamics of overt and covert attention?

It kind of makes sense to put it that way, but I don't see the relation of this dynamics and what salience means in the predictive models.

Specifically the action choosing thing.

I don't know.

But anyway, secondly, uh, can these models address, uh, the various, if not all, uh, types of attention beyond spatial attention?

You know, uh, this idea of precision modulation and salience, uh, seem to work well if related to bias competition and the promoter theory of attention, but these two are about spatial visual attention.

And so I, I don't know.

I was just wondering if we could, uh, if it will be ever possible to address the other kinds of attention.

For instance, there is this work by, um, by Ransom and colleagues that argue that, uh, the effective biased attention cannot be simulated by predictive models.

So I don't know.

It's just, it could be just a matter of times, uh, until all the kinds of attention will be addressed, but I don't know.

Anyway, uh, my third question is, uh, my, also because, uh, of my naiveness towards the formalism.

And again, I'll be happy to learn more, but what is the original contribution of these models to attention literature or to attention on neuroscience?

The way I see it, uh, there have been no new predictions made by these models, only reinterpretations or simulations of non-scientific schemes and more.

Uh, many of these schemes are also simulated by other computational models, like reinforcement learning, normalization, and so on.

So what's the novelty here?

You know, and lastly, where is the external world?

You know, exogenous attention is famously known for its dependence on the external world and the external stimuli, but it looks like here, uh, it's all about top down modulation, even to externally triggered attention.

I know, I know that prediction errors come externally, but, uh, if, whether they are going to be salient or not will depend on this modulation and not on, say, the intensity of the stimulus or, or something like that.

Uh, and honestly, it, it kind of makes me feel a bit down because, you know, I'm studying active difference precisely because of the narrative of the organism environment coupling and stuff.

And, uh, as I understand it now, uh, it looks like the external world is really, uh, minor detail in the predictive biomachinery.

So, I don't know, I would be wrong.

But, okay, uh, if we are to accept that attention is precision modulation, maybe it means that attention is something more fundamental than a simple cognitive feature, right?

I mean, it can be what Molz described as an adverb in a cognition, meaning that it is a mode that things happen.

But I, I think it still wouldn't explain the integral functioning of, of things, as sometimes we do not perform an attentive behavior at all, but it might be an interesting start.

And also, uh, if we understand attention as a more fundamental thing, maybe there won't be as many types of attention as we currently have, because once we reinterpret these types, they will kind of disappear into, uh, the model updating scheme, I guess.

And another important implication of this take is specifically for psychiatry or computational psychiatry.

So, it is known that bad performance in attention tasks is common to psychiatric disorders, but what if they are, at their core, different expressions of attention disorders themselves?

You know, for instance, uh, people say that too much gain in prediction, in predictions leads to psychosis, whereas too much gain in errors leads to autistic behavior.

So, could that be the case? I mean, is everything attention, attentional?

Um, you know, this idea is not entirely new, okay?

There are some works, uh, in the literature suggesting that psychiatric disorders are attentional disorders, and so this might be a fruitful spot to investigate further.

And, of course, um, would we be closer to a unifying theory of attention if we continue this reinterpretation of models and theories of attention into the predictive account?

I don't know. I don't know. It looks like, at least, reinterpreting biased computation and premotor theory within, uh, active inference framework, I think it worked well.

So, uh, we could keep doing this and see what happens. So, yeah.

Well, my last slide is what I think it kind of should be addressed by the community of active inference.

Um, first of all, what are the advantages of saying that attention is precision modulation and salience?

You know, could we say that, uh, what I said a couple of slides ago is an advantage or is it more confusing?

Like, if we have a theory that explains everything, then this theory explains nothing, right?

So, I don't know. What, what are the advantages, the real advantages of saying that attention is precision modulation?

And how to acknowledge that sometimes modulating weights has nothing to do with attention?

Because it may be the case, right? I mean, we don't need to, of course, if we say that attention is more fundamental than other cognitive features, then we could, uh, say that at least, uh, uh, a primitive or a minimal, uh, attention, uh, function is working all the time.

But then, I don't know. I mean, is it the case really? Um, well, and besides, uh, the kinds of attention that could be dissolved into, uh, the model updating scheme, can the predictive attention say anything about the personal level of attention?

That is the act of paying attention or, uh, uh, the internal agency or, you know, anything like that, that requires personal level interaction with the word.

And lastly, this is more like a request that are a suggestion than anything, but, uh, so I understand that active inference and predictive coding frameworks are still young in the literature and people are applying them to everything, which is cool.

But specifically for philosophers, it feels like philosophers got away from the literature, you know, uh, well, scientists have always been skeptical and still are, but philosophers were once super attracted by, uh, the predictive idea.

And then I think they are kind of getting away from this, uh, if we see philosophers talking about it, they either have a very specific idea that may even not correspond to the formalisms at all of the frameworks, or they are critiqued, they are critics of these frameworks.

Also usually having a very specific and maybe even unrealistic idea of the frameworks.

So my request as a philosopher is still trying to deepen and learn the models and trying to work on their implications to real life is that the community, uh, should stop scaring philosophers away.

And I don't know, we should try to, uh, uh, make a second bridge for philosophers again, to be interested in the, in the active inference models and see what's new, uh, and to continue working on the implications of these frameworks to real life.

And so to help modelers to not be so abstract and lost in their maths.

I dunno, it's just something that I feel like saying here.

Anyway, that's it.

Thank you for your attention.

If you need to contact me, uh, you can scan this QR code and see my website.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Okay.

We are back with Samuel Montañez et al.

Agentic Finance.

So Samuel, looking forward to your presentation and anyone in the live chat can write any questions.

Thanks.

Go for it.

Thanks, Dan.

Hi everyone.

My name is Samuel Montañez.

I am, uh, an artificial intelligence PhD student based in Mexico city.

And I'm here with my coauthors, John Klippinger and Matthew Moroni.

We're basically, uh, a working group and a research group, uh, working mainly on the intersection of finance, economics, uh, neuroscience, and in this case, bridging active inference for financial applications.

So that's pretty much what we're going to share today with you.

So what we have for today is Igentic Finance, active inference applied to dynamic portfolio management.

So any questions, please put in, put them in the chat.

And also Matt here has some practical industry oriented questions for that.

That should be interesting that we're going to do, do in the end as well as the public's questions.

So let's start first.

I like to share with you the finding a very, a very important finding in economics that pretty much launched the behavioral finance revolution in economics.

And it's called the, the accessible utility puzzle.

So in standard finance and economics, we, we think about the prices of securities or financial assets pretty much depending on their fundamentals.

When we say about fundamentals, it's their economic fundamentals.

So for example, for stocks or other type of securities, we usually value them as taking the present value of the expected dividends discounted by some risk rate.

So that's pretty much the standard formulation that you will learn in any introductory finance course, but that's also the, the most, the standard way that most people do it, do valuation, both in industry, in corporations.

And that's pretty much what you would be taught in an MBA.

So what's behind that formulation in, in terms of the theory of what we are assuming about the economic agent is that we are assuming a theory that economists called rational expectations.

We assume that economic agents are rational.

Rational, rational agents in the sense of having complete information, having transitive and static preferences, and a bunch of other assumptions that makes, makes the model easier, makes just the model easier to work for.

And in general, most financial models have this assumption behind it.

So in this case, if we assume that we, we assume that we have this rational expectations model, and we say that we expect that a certain security would be correctly valued, depending on the expected value of their dividends discounted back to the present, we're going to check if that's true.

We use the standard and the market index that represents the 500 largest companies in the US.

And if we assume a rational expectations models of the dividends I just shared with you, we would have from the 1980s to the

225, pretty much the red line that we're seeing here.

That would be the expected correct price under that rational expectations model.

That would be what we should be looking at or something very close to that.

But what we're looking at really is a blue line.

That's the actual real backtest on the S&P 500.

And pretty much Robert Shiller, one of the founding fathers of behavioral finance at Yale, pretty much measured the difference in volatility between the rational expectations model and the real market price.

And he found an astonishing finding, which was that real S&P or market returns were 25x higher than what would be predicted by a rational expectation model.

And later on, subsequent studies tried to find the answer of these difference in volatility.

And they pretty much analyze everything.

Difference, for example, in costs of trading, differences in taxes, because of course the index is just like a, well, it's just that, it's just an index.

So it has no transaction costs, stuff like that.

So they accounted for all of those things and none of those things could explain a 20x difference in volatility.

So basically, this led us to found to, I mean, this led the economics profession to start importing

methodologies and knowledge from other fields.

And in this case, psychology to try to answer why is that markets are behaving this way, that prices are the version of what would be expected on the rational model.

And essentially what Shiller claimed and what is like the tenet of behavioral finance, behavioral economics, is that markets are driven by belief dynamics and not only by their economic or cash flow fundamentals.

So given that we're speaking now about beliefs, we need to import things or knowledge or techniques for, from psychology and later on from neuroscience.

And pretty much what we're proposing here, our framing is, let's say the integration of different fields coming together.

On the one hand, we have Robert Lucas as one of the, I mean, probably like the last mainstream economist who is a strong proponent of the rational expectations model, especially in macroeconomics.

We can also mention others like Eugene Fama with the efficient market hypothesis.

We can also mention Paul Samuelson at MIT, but all of those are the mainstream economies who have pretty much established that rational expectations is a way to do economics.

And it's still the ruling school in economics.

Now, later on with the, with this, this, this, this work from Shiller here, famously, Kahneman and Tversky, two psychologists who were exploring decision-making under uncertainty, published their famous prospect theory.

You've probably seen it or heard it or read it in the, in their famous book, Thinking Fast and Slow.

And they basically found that, that individuals consistently performed bad economic decisions and, and came to several decision errors, especially many errors concerning calculations, probabilities, and money in general.

And later on, and later on, with this gentleman, Richard Thaler here, a recent Nobel laureate in economics in 2017, actually received a Nobel laureate in economics because he was able to integrate those findings in, in psychology and import them into economics.

And that, that gave rise or formalized the field that we know today as behavioral economics.

However, as with, as with, as with rational expectations, economists usually think about behavior.

For example, we can remember from Paul Samuelson's, revealed preferences, see behavior, and we infer preferences from that observed behavior.

And that standard lens to look at economics was pretty much prevalent also in the behavioral economics school.

So it didn't change much the experimental methods or the techniques that they were trying to use or that they were using to try to, to understand economic behavior.

So later on, especially from neuroscientists like Paul Gleishmer at NYU and Colin Kaminer at Caltech, there was a neuroeconomics revolution, which was basically saying, why aren't we using the latest neuro scientific findings to try and inform our understanding of economic decision making?

So basically, these two gentlemen are probably the leading academics or intellectuals following this trend.

But interestingly, I mean, they have made ways in economics, but, but, but it is not nearly the most popular field in economics.

But basically, what we are claiming here is that we can come together into a more general theory that accounts for a principle based scientific theory, which is the active inference theory or framework and

integrate all.

I mean, both of these schools of thought into a single school of thought into a single school of thought, which we're going to call cognitive economics, where we're basically going to unify the behavioral finance, behavioral economics, and even some, some aspects of rational expectations as well with neuroeconomics findings that come naturally in the active inference framing.

And that's pretty much what we are proposing, using active inference as a general framework or a general umbrella to make these fields come together into what we call now cognitive economics.

So essentially, we are taking, I mean, more, I mean, this is not limitative, but we are specifically focusing on bridging these aspects from each field.

From neuroscience, bounded rationality, also the neural process theory in active inference, and also other aspects of decision theory in neuroscience.

From economics, mainly behavioral economics, behavioral finance, but we can also reconcile some aspects of rational expectations as well, because it has been tremendously useful in economics.

From finance, from finance, from finance, from finance, asset pricing theory, and from active inference, we're taking pretty much everything to integrate this into the framework.

And actually, cognitive economics is not per se, let's say, our own creation.

Other major economists or mainstream economists like Gabay at Harvard or Bordalo in Italy and Schleifer at Harvard have been actually pursuing this line of thought in the last five to 10 years, let's say.

And, for example, Gabay is studying the effects of attention and limited processing capacity that naturally for almost anybody in the active inference community.

I mean, we are well acquainted with Landauer's principle, precisely having thermodynamic constraints in decision making.

So that comes obvious to people in the active inference community, but it's actually sort of novel in economics.

Also, for example, Bordalo from Bocconi and Schleifer at Harvard have been pursuing what they call diagnostic expectations, which is basically integrating findings from cognitive psychology, some from neuroscience, and bridging it or importing them to economics.

And they actually propose laws of motion for understanding belief dynamics.

And naturally, I mean, that naturally resembles basic mechanics in active inference.

And finally, loss aversion, which is the famous finding from Kahneman Tversky in behavioral finance, can also be tracked like one of the, let's say, pioneering works on trying to understand decision making from a narrow grounded theory.

So basically, active inference would allow us to bridge all of these field in one part from a normative standpoint.

On the other part, from a descriptive standpoint, because it gives us much richer tools to understand economic decision making.

And the other is a mechanistic model to understand how those belief dynamics are forming and are being updated.

And for this, we are proposing a change in paradigm in finance and in financial valuation.

So we're proposing an agenting finance type of model.

So instead of using static models as, for example, taking the present value of the expected dividends, which

is basically a standard discounted cash flow type of model, and many other financial models, we're saying instead of doing that, let's use agents-based modeling.

Let's assume that our economic agents are actual agents and that they perform inference as planning.

Because in a sense, all type of economic decision making has a reference to goals.

I mean, almost mandatorily has to do with economic financial goals, and it has to do with planning.

So when we integrate those two key aspects, that makes the agenting framing very natural, very, very natural.

And we are proposing that this agenting finance framework of financial valuation would allow us, first, that the agents can infer distributions on their uncertainty and actually update those inferences about the distributions that they are observing.

Where we're speaking about distributions of asset prices, returns, et cetera.

Because in traditional financial models, most of them assume, for example, a fixed distribution with known parameters, most often the normal distribution.

But what if the distribution is not normal?

What if we don't know the parameters?

And that's what we often observe in financial markets, where we see changing distributions.

We also see distributions with fatter tails than what would be predicted by a normal distribution.

So that's very, very common.

And finally, traditional financial models completely ignore the agent.

Completely ignore what the agent believes.

Pretty much always assumes a type of heterogeneity in preferences.

So all agents are the same.

All individuals make the same type of economic decisions.

They are fixed.

Whereas in agentic finance, we can always tune our graphical model to the specific type of agent with a specific type of investment goals of risk preferences, of prior distributions, financial knowledge, et cetera.

So it allows us to really personalize valuations depending on the type of priors that you have.

Whereas in traditional finance, that's fixed.

So basically, we are proposing using active inference to frame financial valuation as planning through inference, essentially financial planning in this case.

Now, the PNDP architecture that I just showed you in this graph, in this visual graph, is essentially the standard way to frame or to apply active inference.

So the partially observable mark of decision process is partial.

I mean, you might be acquainted with a mark of decision process in standard reinforcement learning, for example.

But it's partial because the agent only partially knows the external states in this case.

It has partial information.

And it has to infer those external states.

Now, a key aspect that we are framing here is that we can get into great detail in what financial aspects we

want to take into our valuation model.

Here is just an example of how you could frame it.

But it's really not limitative to public assets.

It can also be used in private assets.

It can be much more detailed in the actions that you can take or the many observation variables that you can define.

And this is just for visual clarity and that we can understand it.

So first, we have the state spaces.

And we're going to define two state spaces or two factors.

So one is going to be the market regimes.

When we speak about the market regime is what we usually assume naturally.

The market is going up, it's going down, it's flat, or we're in a crisis like financial recession.

So those are our four market regimes.

And those are the ones that the agent, the trader, wants to infer.

And then we have portfolio positions.

So portfolio positions means what I'm going to do or, I mean, where do I stand in my portfolio in the sense of going long, going short, selling, or, I mean, or expecting a decrease in price, or just holding.

I am holding my position flat.

And when we observe, what we observe and what the agent is observing from the external world to try to learn are two things in our model.

One is a return observations for the asset that we are going to analyze.

It can be a basket of assets, an ETF.

It can be whatever we want.

And second, we have the P&L observations.

So when we say P&L, it's the profit and loss of our own portfolio, of the portfolio of the agent.

And we're going to divide that portfolio into quartiles.

And we're going to divide it into quartiles so the agent can analyze if I am on the top quartile or in the bottom quartile.

So we can have some sort of ranking of order of preferences.

And finally, we have the action space.

So the agent is going to be able to act on its sensory inputs to try to change their positions and get closer to its expected preferences.

So those trading actions are going to be the three standard trading actions called buy-sell.

But we added a fourth one, which is reduce.

Because in some cases, you might not want to sell your entire position, but you might want to decrease your exposure.

So that's like a fourth detail that we added.

Now, if you want to think about how would this framework would evolve, how would it work while being active, basically we say that the state spaces are the hidden states.

So we have two hidden states.

One, again, the market region.

And naturally, as investors, we would want to learn if the market is going up, if it's going down, or if it's just like staying flat as it is.

And the interesting thing is that we have no control over that.

We have no control over market returns.

And nobody really knows what the market is going to do.

For example, economists all over the world, but especially in the United States, have been forecasting a financial recession or an economic recession in the U.S.

for, I mean, almost since 2020.

So we have been five years waiting for a recession that never comes in the U.S.

And also, for example, Jamie Dimon, the CEO of Goldman, of JP Morgan, recently said that he expects that perhaps in 2026 we're going to get to the recession this time.

So if not even the best economies, the world financial leaders can guess in which market regime we are, that's pretty much the best example to see how they are completely unpredictable, uncontrollable.

But what we can control is our portfolio position, because we can decide to go long, go short, I mean, just hold our position as it is.

So we want to change that.

And then we have the observations that the agent is going to learn from.

And we divide the observations in this financial application into exogenous observation and endogenous observations.

And why?

Because, again, in the external states, we have the market.

And there's the financial securities, and they generate returns.

So we can observe those returns.

We can observe them, especially for public securities like equities, public equities.

Let's say, for example, for an Amazon stock, we can observe its price at all points in time.

But we cannot have any influence on that price.

And then we have the P&L of our own portfolio, of our portfolio.

And we can observe that.

So that's endogenous to the agent.

And that distinction is going to be very relevant to what's coming next.

Also, I mean, a key difference between a standard financial model, and perhaps not that different from an econometric model, but much cleaner in its implementation, at least for an agentic implementation, is that we can factor these different types of information that the agent receives and clearly classify it.

So that's key, at least for financial information.

And return attribution, which is a key aspect of applying different artificial intelligence techniques for financial markets.

And also, we can add action-dependent transitions, which is that we can act upon our portfolio to alter the P&L,

but we cannot act on the market.

So we can make that modality distinction, which we often cannot make in a standard model where we just put all of the variables into a black box.

And finally, in the C matrix, which is the preference matrix,

that would be a quasi-version of a reward in reinforcement learning, for example.

We can also embed into that risk aversion parameter, which is also standard in finance.

And we are using that 2.25 standard, I mean, risk aversion factor, which was pioneered by Keneman and Tversky,

which basically says that humans compute equal losses to gains in twice the magnitude.

So that's what we are also putting into the model.

So we can actually embed this type of neuroscientific or decision theory findings into the model to have theory-based parameters.

So I just mentioned that the C matrix basically is a matrix where we can put the traders, in this case, the economic agent's preferences.

Now, how it is done in the standard active difference is that we work with lock preferences of, well, we work with lock preferences of defined as lock of prior probabilities.

Basically, so that's the standard way of framing it, as we can see in the upper box.

Now, instead of using that type of implementation, we're changing it a little bit,

and we are defining our C matrix as it is done in risk-sensitive control or KL control, where we just define preferences in terms of utilities.

And one question would be, but why would you define the C matrix like this and not use the standard active inference implementation?

And our reasoning to do this is mainly because of knowledge transfer and, let's say, industrial applications.

So if we want to convince or persuade real investors or financial institutions to use this model, it's much easier to explain to them how preferences are encoded in just expected utility.

So if the C matrix is defined as expected utility, we can say the model is maximizing expected utility subject to a specific risk preference.

So we can actually frame it just like that.

Whereas if we frame it as the lock of prior distributions,

we will have to say something like our market returns are expected or are modeled as a uniform distribution, let's say.

And that's much muddier, let's say, than just saying it out loud as any economist or financier would say it.

And this is not new or anything new.

This is the standard way to do, let's say, maximum entropy reinforcement learning or KL control, as mentioned.

And actually, KL control has been used in financial management and portfolio management before.

So that might be a method that also industry, I mean, general industry practitioners are acquainted with.

So how are we defining this?

Essentially, remember that we have market returns and the P&L.

We just said that market returns are uncontrollable.

I mean, the market is just there.

We cannot control it.

So we're going to define that C matrix as zero.

So that means that we are neutral to market returns because we cannot influence them.

Whereas the P&L is controllable because we can act upon selling, buying, et cetera.

So it's position dependent.

That's what we mean.

So it's not going to be zero.

And essentially, an additional benefit of framing the C matrix or the preferences as utilities is that scaling them by the risk aversion, scaling the losses by the 225 Kahneman and the risk aversion is transparent.

So it can be done in the active inference framing as well.

But in this case, it's super transparent.

As you can see, for example, from the utility vector, the risk scaling parameter is only applied to losses because we say that gains are just, I mean, factor by one.

So for the 0.6 and the 1.0 in losses, we can scale them, as you can see here, by 2.25 and then that would yield the reward preferences for the P&L, the updated reward preferences.

So as you can see, it's pretty much transparent.

Now, how does this link to standard financial theory?

So when we take an introductory corporate finance or investment management course, the first pricing model that we usually learn is called the CAPM, the capital asset pricing model.

So the CAPM right here is essentially a regression model, a one-factor regression model that essentially says all expected returns on a security, let's say on an equity, on a stock, depend on just one factor, which is the market risk premium.

So essentially, the model says you are strictly and only being compensated by the amount of market risk that you are taking.

So we can decompose the CAPM into two types of essential risks in financial investments.

One is called systematic risk.

So that systematic risk is pretty much what I just told you, the market risk.

How much market risk are you taking?

And that's the only risk, according to this financial theory, that you get compensated for.

Whereas the other term is that idiosyncratic risk.

So that idiosyncratic risk, we can refer to it as the firmest specific risk.

So for example, if I'm investing in a restaurant, I am exposed to specific risks that the restaurant has, like having a fire or then employees don't show up or a flood or whatever.

But that's not systematic risk.

When we speak about systematic risk, we refer to it as general market risks, like inflation risks, like interest

rate risk.

So COVID as well, those are systematic risks that we cannot control in any way.

So we cannot diversify it away in any way in standard portfolio management theory.

Now, what are the disadvantages of this model?

Number one, it's static, meaning that the betas in the model, which is often called the market beta, is not dynamic.

So in the standard framing, it's not dynamic.

Now, there are more recent proposals where you have a dynamic beta.

But interestingly, for I think that most of the viewers, if you go today to Harvard Business School and pursue an MBA,

and you get taught a corporate finance course, you're going to get taught this model, not a dynamic beta model.

This is the model that you're going to be taught.

And most investment banks and consultants still use this standard model and not more sophisticated models that do exist in the literature.

So that's very important because perhaps there's updated models, but those are not the models that most people use in industry.

And additionally, it's a passive model in the sense that there's no agency.

Because again, economic agents are just static in a sense.

So it's just, I mean, it's in a sense just saying you can buy an index fund and just stay there for, I mean, as long as you can.

In a sense, it's saying that.

Because you cannot really adopt much more return unless you're willing to take firm specific risk, idiosyncratic risk.

But that's up to you.

And in theory, this theory says you shouldn't because you're not being compensated for that.

So it's not even logical.

Now, in our active inference framing, I just explained that we cannot control market returns.

We can learn from them, but we cannot control them.

So those are neutral preferences in the sense that we cannot act upon them.

But we can act upon our P&L.

And our P&L is a function of two things, of the weight that we want to have exposure to in a specific asset times the market return on that asset.

So that's what we can act upon.

That's the controllable modality that we can act upon in our model.

So how's different from the CAPM?

It pretty much changes the whole framing because, number one, that double view the weight is adaptive.

It's not fixed.

It's not static.

So that's one main difference.

Second, it learns.

Because even though we cannot act directly upon markets, we can learn from that market information to change our weights in the position.

So we can learn from that information and we can act upon our portfolio position to modify our P&L.

So as you can see, it completely changes the way to see finance as just measuring temperature to an agentic vision of financial markets.

So how does this link to standard framing in economics?

So as I was saying, even though the rational expectations model comes across sometimes as very antiquated, let's say, for many people,

because they say, how do economists actually believe?

I mean, it's not that they believe, but how do they take those axioms which are very extreme into their models?

And they just think that the models are going to be very successful as those axioms are extremely radical, let's say.

But Robert Lucas is probably the, I mean, one of the most famous living economists.

He won the Nobel laureate because of his famous Lucas critique.

And basically what he claimed was the following.

In econometric models, those econometric models, again, implied a certain stability of financial agents.

So policymakers just assumed that, again, individual ones in their societies were fully rational, respond to information.

And the most important aspect is that preferences were stable, that people don't change, don't change on their preferences.

So preferences are stable.

And essentially what Robert Lucas says is that parameters of econometric models are not structured.

So what that means is that they are agent dependent.

So agents, I mean, humans, we respond to information.

So we respond to actions that the government takes.

So when the government implements a certain policy,

they should not expect that people are just going to be fixed and acting as they always did.

I mean, people are going to react and probably, or in some cases, likely are going to modify their behavior.

And as surprising as that may look, that wasn't accounted for in standard macroeconomic models in the 80s, 90s.

And again, you can see that here's another place where we start thinking about beliefs in economics.

Now, how does this critique is relevant for our model?

Because if we frame our active inference agent as a standard active inference implementation,

where we say we have different modalities where one observation is going to be, let's say, the different senses.

One is going to be visual information.

The other is going to be auditory information.

Naturally, that information is coming into the agent.

We can weight its importance, but it's all coming into the agent.

And you can put on glasses, you can put on headphones or whatever, so you can act upon those observations.

But in this case, if we were to conflate that you can act upon the P&L and to the markets as well, that would cause a problem into the model because we just explained why they are not different.

They are actually different.

So what's our approach?

We want to break the potential issue in this type of implementation of policy endogeneity.

So we are proposing an application where we have selective Bayesian learning instead of full Bayesian learning as it would be done in the standard active inference.

So, for example, we have in the standard active inference model, we just mentioned that we have two modalities.

So we would have two learning equations as they are posed here.

And in that case of standard application, we would be just using the two modalities entirely to update our beliefs.

But as I just mentioned, we don't want to use the two modalities fully.

We want to learn from the market, but we don't want to learn from our portfolio.

And why do we don't want to learn from our portfolio?

That's like a weird question.

But imagine this.

I am looking at my portfolio and my portfolio performed well during the last six months.

And given that it performed well, I am going to update my internal model to act and start trading with that information from my portfolio.

So my portfolio, if I am not Vanguard or BlackRock, is not going to be nearly representative at all of the general financial market.

So if I am learning from that information, I'm going to contaminate my internal model to actually predict, to try to predict what's going to be the next market regime.

Because again, if Samuel owns 0.00001% of Tesla, it has literally no, my position has literally no effect upon Tesla or the robotics market or the electric cars market at all.

So what we are doing here is frozen that part of the learning equations in our model.

So that's what we call it, selective based on learning.

And we are only updating our internal model based on external market information, which is what we can learn from and which actually can provide relevant information.

Additionally, we have adaptive confidence gating.

So we're implementing ad hoc confidence gates, where we are saying that when there's a certain level of policy entropy, the agent is not going to act.

So in standard portfolio management, we have different types of risk management tools.

But there are some standard ways to frame a trading strategy is to implement some systematic trading rules.

So for example, some of the rules can be, you can never invest more than 5% in a specific asset.

Or you can never invest less than 1% in a specific asset.

Or you cannot have an over concentration to a specific sector of 20%.

So those are standard risk management trading rules.

And in the same sense, we're imposing, let's say, an active inference flavor trading rule, which is having a confidence gate that where if the policy entropy goes beyond a certain threshold, even if the agent wants to act, it's not going to act.

And that could be naturally adjusted depending on how conservative we want our trading agent to be.

Additionally, we are also using an exponential forgetting factor as well.

And also that can be adapted depending on the type of agent that we want.

But it is also recognizing that people don't remember market history perfectly.

I mean, not nearly.

So it also recognizes that economic agents forget, naturally forget.

Now, if you want to frame this financial agent or this financial market in terms of the Markov blanket, like the standard framing in active inference, we could say that we have an external world, an external market.

And in this case, we're specifically going to focus on trying to predict the market regimes.

So that's going to be the external state, the market regime that we want to predict.

Again, if the market is going to go up, it's going to go down.

What's going to happen with the market?

But we cannot have direct access into that market.

And those are uncontrollable dynamics.

On the other hand, we have in the right hand side, the internal model of the agent.

So that internal model of the agent is going to be the internal model about what's going to be the market regime.

I'm going to try to predict as a trader what's going to be the next market regime.

And how this model actually works is between the sensory and active states.

So the external market is going to yield sensory states.

And again, we just mentioned that two sensory states are going to be, or the two observations, are going to be market returns from the assets and the P&L from my portfolio.

We just explained, I can learn from, I want to learn only from the returns of the assets in the market.

And I'm going to freeze the information on my P&L because I don't want to contaminate or I don't want to have an endogeneity issue in my learning.

And finally, I can act upon that information.

And in this case, I'm going to act in two ways.

One is going to be performing trading.

And the other is going to be looking for information, the epistemic component into the active inference framework.

So as you can see, this type of framing, I think, more naturally accounts for standard trading and economic

decision making in financial markets than, for example, the model we showed.

So essentially how we can look at it now, I mean, when it's actually updating in the action perception cycle in active inference is depending on the expected free energy that can be decomposed on a systemic value, the intrinsic value, and the extrinsic value, which is going to be the pragmatic component, the expected value in economic terms.

Essentially, essentially, one of the key advantages, I mean, well known to everyone in the active inference community from the active inference framing, at least in this case, against economic type of approaches, is that it allows us to embed in a single functional in the optimization process, both the risk or the epistemic component or the intrinsic value,

and the expected utility into a single optimization process.

So is balancing these two components, or another way of framing it, is balancing the exploration exploitation dilemma automatically into the model.

So how is this different from standard models in economics, in finance?

Well, in economics and finance, we almost never, ever use epistemic value whatsoever, or intrinsic value, we only care about expected utility, expected extrinsic value.

Now, in other type of approaches, for example, in reinforcement learning, you can add an ad hoc parameter to add an exploratory term, for example, in eGRED methods, but that's an ad hoc addition to the model.

In this case, an advantage is that it's embedded into the optimization model of the model, in the optimization model itself.

Now, for just to provide an example, we're going to use the ARC Innovation ETF.

So for those who don't know it, ARC is an investment management firm, one of the most popular today in the US.

It's owned by a famous investor called Katie Wood.

And she became very popular, at least in financial markets and, let's say, investment community, because she has been a strong proponent of thematic investment, and especially focused on what we could define as transformative technology, highly growth stocks.

And, yeah, like transformative technologies, ranging from currencies to bio-digital health, green technology, many of those megatrends.

And her asset management firm issues and manages several exchange-traded funds.

And we are portraying here the last 10 years of performance of her most popular fund, the ARC Innovation ETF.

That pretty much is an investment fund that gives exposure to investors to the major trends, again, in innovation.

Now, as you can see, the fund performed extraordinarily well since 2015 up to 2021.

So it went up 700% in five to six years, 700%.

So at that time, Katie Wood was appearing in TV all of the time.

She was claimed to be the next Warren Buffett.

And her fund draws, I mean, drew investments for more than \$20 billion.

So it became extremely popular.

Now, as of now, I mean, after that peak, it went down 80% in its lowest point.

And even after almost six years, people who were most of the investors who entered into the peak have not recovered even their initial capital.

Now, I mean, why we like it is because this type of popular, trendy, momentum-chasing type of investment is precisely the type of asset that allows us to measure regime shift.

Because almost visually, we can see that that would be a bull, a bear regime shift, and then more or less a flat regime shift.

So we are using ARC as an example.

And how we're going to train our active inference agent, taking almost like a standard machine learning type of training methodology, where we're using 70% of the historical information.

And we're performing a backtest on the last three years from 2022 to 2025 to see how our agent would perform.

Now, what's the advantage? Because one may say, why not use just deep learning and NLM?

And the advantage is that we don't require retraining to just keep following the performance of the ETF.

Because by definition, this type of active inference implementation is adaptive.

It's adaptive learning. It's full online based on learning.

So that's a completely different way to do machine learning.

And I would claim it's a much more efficient way.

And also, I think for financial applications, this adaptivity makes you much more prepared to adapt to changing market circumstances,

especially with this type of volatile assets, which are one of many in the financial markets.

Now, just a brief theoretical bridge as well with standard financial and economic theory.

Using the Helmholtz decomposition, which is a standard part of the active inference and free energy principle, we know from Helmholtz that we can decompose any vector field into two components or two parameters.

One is the gradient, we can say, almost a deterministic component.

And the other is the solenoidal flow, which is the cycle that is never changing into the model.

Now, how can we build a link between this theory behind active inference and financial economics?

One model, which is one of the most important models in valuation, is the Black-Scholes-Merton option pricing formula.

So this model was created in the 70s and very quickly became the standard to price financial insurance, financial options in markets in general.

But originally, the model was created to value corporate liabilities, actually.

So another extension of the Black-Scholes model is what we call the real options Black-Scholes-Merton model,

which can be actually applied to value private projects, private equity projects,

and mostly innovative projects where there's a contingent aspect into that type of investment, where there's huge changes.

For example, like all of the innovation trends embedded in a fund like ARK.

Now, what I mean, the Black-Scholes-Merton option pricing formula defines that stock prices move as a geometric Brownian motion, where you have a drift, the first term here, defined by μ ,

and a stochastic component where you have the volatility here and the Wiener process, which does the stochastic part.

But what is the probability evolution of this model?

And the probability evolution pretty much will yield this equation here, which basically just left you with the gradient,

meaning that by equilibrium economics, that dissipative aspect, I mean, comes to an equilibrium. So there's no selenoidal flow.

I mean, put into economic terms or into physical terms in this case, because again, the model is an equilibrium model.

So it will always reach equilibrium.

So if it always reaches equilibrium, there will be no exploration, no epistemic value into the model.

So what we're claiming here is that this agentic active inference framing

is a non-equilibrium dynamics applied to financial markets,

and it can be applied to other type of markets,

where we know that even though at certain very quick points in time,

the financial market or one specific market might reach a certain equilibrium,

that equilibrium would rapidly dissipate and start moving towards another equilibrium.

And the reason why, again, is because standard equilibrium economics assumed that agents were sort of statics, sort of robots.

But when you have changing belief dynamics,

it's always going to be non-equilibrium dynamics.

And in this sense, we say that the synergetics theory,

also behind a lot of the active inference,

can naturally be applied in the self-organization and market regimes.

And interestingly, in the econophysics literature,

which is like a, what we could say,

like a niche group within the economics field,

has basically been studying this,

being, I mean, applying synergetics

and applying other type of phase transition physics literature

to try to understand market regime transition

and also the emergence of prices

and different market phenomena from the microscopic.

And we define microscopic as, for example,

the acting of an individual like Samuel in his own portfolio

and how that accumulates between millions of investors.

And that gives rise to the broad macroeconomic behavior

that we see in markets
and eventually even in world financial markets.
So we can see this emergence of economic phenomena
coming from essentially microeconomic foundations.
And naturally, that framing as well can be embedded
into trying to understand the theory of market regimes
from an active inference perspective as well.

So what we're just saying here
is that this framing of gradient flows,
synodoidal flows, synergetics,
gives you a very natural technology
to see this type of macroeconomic phenomena
and macroeconomic questions
and address them with this type of technology
and in an agentic fashion
with the applied active inference agent.

So what are the advantages?

I have discussed some against both,
for example, LLMs,
and in this case,
a closer application would be deep reinforcement learning
and also against traditional finance.

So against deep reinforcement learning,
number one is the sample efficiency.

Instead of requiring cillions of data points
to train the LLM,
in active inference,

you have an adaptive agent
that is training from real data or coming data.

So it's much more efficient.

Now, in deep reinforcement learning,
we can have ad hoc exploration parameters,
whereas in active inference,
it's inbuilt within the optimization function.

Deep RL, or in general deep learning,
the most important aspect
is explainability and transparency aspect
because by definition, it's a black box.

And in investment management
and also from coming regulation in future years
is going to be extremely important
when applying AI methods
to financial decision making
that you can actually explain
how the agent got to a certain recommendation.
It's going to be key for regulation purposes.
And the active inference framework
with a graphical model
allows you to clearly understand
how and why the agent
recommended a certain decision or why not.
very clearly, very transparently.
And finally,
you have implicit uncertainty
in deep reinforcement learning,
whereas, again, in active inference,
the entropic term is explicit into the model.
And you can also add confidence gates
regulated by entropy,
policy entropy,
like we did in our implementation.
Now, again, traditional finance,
I mean, not to joke,
but it's sort of a low-hanging fruit
because, again,
most of standard finance
has stayed in models
that were built in the 60s and 70s,
and they haven't been updated much.
Now, naturally,
academics have updated them,
but in industry practice,
I mean,
besides some small quant shops
in very few places,
most financial departments

in most corporations
are just using models
that were built literally
in the 70s and 80s,
which are static,
which have, I mean,
have no way to adapt themselves
to changing circumstances,
have no account
for information gathering
or epistemic affordance,
none of that, pretty much.

So, as I mentioned,
epistemic value is completely ignored
in traditional finance.

Most standard applied finance
doesn't use findings
from behavioral economics.

I mean,
more often is now included
in some aspects
in marketing,
but not that much
in trading.

And in terms of bridging adaptation,
at least in the applied sense,
is almost completely ignored
in applications
of real investments.

And we can see that,
again,
standard discounted cash flow models,
the blackshed models,
which are real models
that most people learn
and most people use,
is completely,
I mean,

completely ignored
all of these points.
And,
right,
I mean,
fine,
I mean,
we're about to conclude,
but just before,
the standard metric
to measure the performance
of portfolios in finance
is what we call
the Sharpey ratio.
So,
the Sharpey ratio
is essentially this,
measuring the expected
excess return
of my portfolio
against the risk-free rate.
So,
that's what we call
the excess return
because the risk-free rate
would be like that,
the baseline of money
or investments,
divided by
the standard deviation,
which in finance
we call the volatility
of that portfolio.
So,
in plain English,
it's basically measuring
a ratio
for risk-adjusted returns.

So,
what we care
in investments
is not
the absolute return,
is the risk-adjusted return
because
if I have
a very large return
but taking
huge amounts of risk,
that's not interesting
because probably
if I follow on
on that strategy,
I'm going to lose money
eventually.

So,
what I want
is a high return
with low risk
or controlled risk,
let's say.

That's what I care for.

Now,
what we're proposing here
is that
given that
our agent
is
adaptively learning,
has online learning,
we actually
want
to measure
that information
gathering efficiency.

So,

we're proposing
including
an additional metric
to investments
called the
entropic Sharpe ratio.

So,
we're doing
the same thing
that we do
in the Sharpe ratio,
taking the excess return
of our portfolio
against our risk-free rate,
but we are dividing it
not by the
portfolio volatility
but by the
policy entropy.

So,
that's what
we are dividing it for
and is essentially
measuring
the excess return
per bit
of informational
work performed.

So,
it's measuring
the ratio
of how
information
gathering efficient
or how efficient
the exploration
component
is happening

within
our active
inference agent.
So,
for example,
how could we
read it
if we have
an expected Sharpe ratio
above 50%,
we can say
that the model
is well calibrated,
is being very efficient
on using
the information
it is gathering
to trade
and to generate profits.
If it's low,
we can say
that that information
seeking aspect
is not being efficient.
Now,
why this is important
is because
increasingly,
investment management
firms
are implementing
LLMs,
natural process,
natural language
processing strategies
to gather
lots and millions
of bits of data

of market information
to trade.

and it would be
useful to measure
how efficient,
energy efficient
and cost efficient
that information
gathering
is being
for generating
profits
into the portfolio.

And if you
can actually
show
that
your model
is much more
efficient into that,
that can be
also accounted
for in the
economic,
let's say,
valuation
of how
effective
the agent
is.

And
almost last slide,
the explainability.

As I just
mentioned,
a key aspect
that is coming
in,

I mean,
very,
very,
very few years
is going to be
AI regulation.
And especially
in fields like
finance,
in fields like
healthcare,
given,
I mean,
just having
recommendation
agents
that you
cannot
actually
see
how they
are coming
into that
recommendations,
it's out,
that's probably
not going to be
acceptable
for regulators.
Because if people
lose money,
if people get
bad medical
recommendations,
somebody needs
to be
accountable for.
And if

deep learning
or LLMs
cannot be
actually interpreted
in that sense,
in how they came
into the output,
then
this type of
machine learning,
we believe,
is going to be
the standard way
to do
financial machine
learning because
first,
I mean,
first and foremost,
you can
decompose
the entire
model to see
exactly
what caused
the model
to come
to a specific
decision,
to a specific
recommendation.
So we believe
that that's
a key
advantage
against
other standard
deep learning

methods.

So to

conclude,

what we reviewed

today was

essentially

that

active

inference

is not

per se

a different

theory.

It's a much

more general

theory that

can actually

account for

most of

what has

been done

in economics

as of now,

but done

right,

let's say,

or done

much more

robustly.

It also

integrates

other aspects

that have

been missing

into economics,

for example,

adaptive learning

or online

learning,
and it's
framing,
again,
valuation
or asset
pricing as
planning
plus
control.
And also
we can
show that
standard
financial
models like
discounted
cash flows
emerge as
special cases
where you
only focus
on the
pragmatic
aspect of
expected free
energy,
whereas this
model is
much more
robust.
It's also
including the
epistemic value
component,
which again,
we just
mentioned,

is becoming increasingly relevant in the world where AI agents are going to be all over the place trying to gather data and to distinguish from fake data as well.

That's also going to be a key value that we want to have in agents performing any sort of task.

And finally, more specifically to the market regime theory, this is a type of agent that will allow you to

understand
more
macroscopic
market
phenomena
or more
macro
finance
type of
phenomena,
but at the
same time
being embedded
into a
single
individual
agent.

So it
allows you
to,
let's say,
reconcile
some
pending
phenomena
in
macroeconomic
theory
with
microeconomics
and finance
in this
case.

So we
believe that
our core
contribution
is that

active
inference
provides
a principal
framework
for
portfolio
management
under
uncertainty
and as we
just showed,
we've
demonstrated
risk
management
advantages
because
if you
define
that the
agent is
risk
averse
as naturally
people are
risk
averse,
it will
usually engage
in much
better risk
management
than standard
traditional
methods
or also
reinforcement

learning methods

where it's

very much

focused on

the reward.

So thank

you very

much.

So we're

going to go

with some

prepared

questions from

Matt and

then we're

going to

address

some

questions

from the

audience

as well.

Or we're

running out

of time.

Awesome.

Thanks,

Daniel and

Sam.

Excellent

stuff.

I want to

take the

perspective

of someone

who would

be a

practitioner

in the
industry
and there's
a lot
to be said
about risk
reward
management.

There's a
lot to be
said about
portfolio
construction
and the
separation
of trading
and compliance
in most
firms.

But I
really want
to focus
in on
this
contribution
of the
entropic
sharp
ratio
because to
me this
fundamentally
reframes how
we think
about data
economics.

An asset
manager might

spend tens
of millions
of dollars
on Bloomberg
terminals
and satellite
imagery
without any
idea of
whether each
new data
source
actually improves
their returns.

This actually
suggests that we
can measure
the value
of each
information
source by
policy
entropy.

So I
wanted just
to make
it super
practical.

Could you
outline how
you would
do real-time
scoring of
different
information
signals like
tweets or
data releases

or earnings
reports that
could help
readjust a
portfolio based
on the
quality of
those different
information
feeds over
time?

Yeah,
sure.

Basically,
we would say
that, I
mean, pretty
much the
higher the
ratio is,
that means
that whatever
information you're
feeding into
the model is
being much
more efficient
in terms of
the signal
to noise
ratio.

So it's
being much
more efficient
on signaling
the correct
information or
the valuable

information to
maximize its
model evidence
or minimize
its surprise,
let's say.

Whereas if
you're feeding
whatever
information,
other information,
if the
policy entropy
is low,
that means
that it's
doing the
opposite effect.

It's not
helping the
agent much
on maximizing
its model
evidence or
minimizing
its surprise.

So that
would be
like the
clear implementation.

And what's
valuable,
as you just
said,
is that later
on,
firms could
literally

quantify it
in monetary
terms,
I mean,
on weighting
how much
money they're
spending on
feeding data
into a
specific strategy
and how much
that data
is providing
returns.

So that also
gives market
opportunities
for people
willing to
create
methodologies,
data sets,
et cetera,
to make
that process
much more
energy efficient
and data
efficient as
well.

That's really
important.

I have a
quick question,
sort of a
big picture
question.

If you really
look at what
you're saying,
you're providing
a really new
paradigm for
identifying value.

and it
fuses
economics and
finance,
and it
actually
combines
monetary
policy and
fiscal policy
in so
form.

And when
you look at
the current
economic
system,
where it
is using
an arbitrage
of different
opportunities
and bid
and ask
risk spreads,
this actually
is a
replacement
mechanism
viewing the
market as

a Markov
blanket that
aligns different
agents around
specific outcomes.

So this
becomes,
it's far
bigger than
sort of me
being in
simple technical
trades.

It's a very
different way
of envisioning
the market.

And yes,
you,
and I think
that is going
to have really
significant
implications
at multiple
scales.

Would you agree
with that or
is that
overreach?

No,
totally,
totally.
as you
just mentioned,
right,
I mean,
we are

literally
reframing
how we
would
conceptualize
macroeconomic
structures
or market
structures
within
Markov
blankets
and how
the agents
as active
agents,
not as
passive,
like just
static agents
in standard
economics,
are actually,
I mean,
taking,
I mean,
are accounted
for into
that theory.
So,
as I mentioned
in the beginning,
it's a way
of framing
in general
finance,
and in this
case,

asset pricing,
but in
general,
framing
economics
into an
agent-based
theory.

Yes.

Which I think
is the right
way,
because it's
how people
behave in
their economic
decisions.

So I think
it's the
natural way
to frame
the theory.

great.

Well,
yeah,
just as
we're out
of time,
I'll just
for,
for,
um,
completeness
and just
for your
all's
future work,
Active

Bloomology

wrote,

I hope

they discuss

how they

propose

integrating

this model

into

underwriting.

Bert

wrote,

what is

epistemic

value when

actions do

not influence

the market?

And then

also Active

Bloomology

wrote about

the parameters

of defined

efficiency

and financial

modeling.

So just,

just a few

notes for

possibly a

future symposium,

but thank you

all for the

awesome work.

Thank you.

Thanks a lot.

And for those

who send
questions,
you can send
them to me
through this
email and we'll
get back to
you.

Thanks a lot
for your time.

Farewell.

Bye.

Thank you.

Bye-bye.

Bye-bye.

Bye-bye.

Hello.

Welcome back.

We are with
Denise Holt
for
Coordination
to Cognition,
Rethinking
Agent
Communication
Protocols
for Active
Inference
in the
Spatial
Web.

So Denise,
thank you
for joining
again.

Anyone in
the live

chat can
ask any
questions and
thank you
for the
presentation.

Thank you
so much,
Daniel.

Thank you
for having
me.

Okay,
I'll just
jump right
in.

So today
I'm going
to be
talking about
current
agent
communication
protocols
that we're
using with
current AI
versus what's
coming with
the spatial
web and
active
inference.

So with
the rise of
intelligent
agents,
agentic AI,

comes
essential need
for multi-agent
coordination.
So as a step
beyond reactive
chatbots or
standalone systems,
LLM agents
are now being
designed to
collaborate
across tasks,
domains,
and organizations.
But how do we
get these agents,
these LLMs,
to talk to each
other, share
context,
access tools,
perform tasks,
and make
decisions in a
way that is
secure,
scalable,
and interoperable?
The response
has been a
new class of
communication
protocols.
You have
MCP, which
is model
context

protocol.

It's from

Anthropic,

and this

gives agents

access to

essential data

sources.

ACP,

agent

communication

protocol from

IBM and

Cisco,

enables

structured

collaboration

communication

between agents.

A2A,

agent-to-agent

protocol from

Google,

standardizes

agent tasks

across vendor

systems,

and then there's

ANP, which

is an agent

networking

protocol.

And this is

open source,

and it supports

decentralized

agent discovery

and secure

identity

management

over the web.

Now,

why do

agent

communication

protocols

matter?

Each of

these

approaches

has unique

strengths

and limitations.

They're all

new, and

they're beginning

to be

implemented,

piloted,

and standardized

today.

They just

came on

scene this

last spring.

And they

help developers

solve urgent

problems,

enabling the

first wave

of autonomous

systems.

So they're

not just

valuable.

They're an
essential part
at this
moment in
time, and
it's agent
communication
for the
worldwide
web.

But a
larger shift
is underway.

Parallel to
these efforts,
the spatial
web is
rising.

We have
HSTP,
hyperspace
transaction
protocol,
and HSML,
hyperspace
modeling
language.

And this
forms a
radically
unified
infrastructure
to
contextualize
and connect
everything
from people
and places

to devices
and data.
It enables
a globally
consistent
framework
for describing,
discovering,
and interacting
with entities
across physical
and digital
realms.

And this
framework was
ratified by
the IEEE,
one of the
largest core
standards bodies
in the world,
on May 29th
of this year
as the third
official
foundational
internet
protocol,
layering on
top of
HTTP and
HTML.

And this
protocol doesn't
just accommodate
agents,
it redefines
the terrain

that they
operate in.

So this
raises a
pressing
question.

As the
spatial web's
universal domain
graph begins
to take shape,
will today's
agent communication
protocols evolve
to align with
it, or will
they eventually
become redundant?

These new
agent communication
protocols reflect
a legacy of
centralized
computing and
static data
models.

Active
inference, on
the other
hand, requires
something more,
real-time
action perception
loops, causal
modeling, and
dynamic adaptation
to unfolding
environments.

The current agent protocols are solving today's problems, but they may not be structurally suited for the agents of tomorrow.

So agent protocols at a glance, bridging the gaps in today's intelligence systems.

So let's go one by one just so we can understand what these protocols are doing currently.

MCP by Anthropic, the purpose, it provides a standardized way for LLMs or agent frameworks to retrieve structured context like files, APIs, databases, system tools, and it acts as a

content feed that simplifies how a model pulls in relevant data.

The strengths, it's simple standardized integration of tools and knowledge for LLMs like GPT or Claude.

It enables straightforward integration, reducing development time and cost, and it enables developers to plug in live information to otherwise static models.

But the limitations, MCP agents, they don't reason about the world.

They pull in data, they process it, and output a result based on prior training and

prompted
instructions.

You're still
dealing with
LLMs.

The agent's
understanding is
contextually
injected.

It's not
self-modeled.

Active
inference doesn't
just fetch
context.

It
continuously
infers it,
updates it,
and acts
upon it in
real time.

The MCP
model isn't
designed to
support that
cycle.

ACP, agent
communication
protocol by
IBM and
Cisco.

This enables
structured
multi-agent
collaboration
between agents
in enterprise

environments.

Its strengths,

it supports

stateful threads

of communication

and allows

agents to

negotiate and

reason together

across tasks,

facilitates task

coordination and

collaborative

workflows,

uses standard

APIs and

JSON schemas,

helps AI-driven

enterprise

applications

coordinate

complex tasks

like order

fulfillment,

and it

establishes

clear workflows

and roles

for agents.

But the

limitations

are it

requires a

centralized

directory for

discovery and

coordination,

and context

is carried
in messages.

It's not
modeled
internally in
the agents.

In an
active inference
architecture,
agents don't
just cooperate,
they co-regulate.

ACP enables
interagent chat.

Active inference
agents exchange
beliefs, align
internal models,
and adapt
as they act.

Now,
A2A by
Google,
its purpose
is cross-vendor
interoperability
for agent
tasks between
tools like
CRMs,
HR systems,
et cetera,
from various
vendors like
Salesforce,
SAP,
Oracle,
and others.

The strengths,
it solves
real interoperability
pain between
tools like
CRMs or
HR systems
or productivity
agents by
reducing
operational silos
within your
tech infrastructure.

It's well
supported by
major enterprise
vendors,
and it's built
on familiar
standards like
HTTP,
JSON,
and OAuth.

Now,
the limitations,
it's optimized
for task
exchange,
but not
cognitive
computing.

So there's
no real-world
grounding
or embodied
understanding.
Agents operate
like modular

services in a
cloud-native
workflow.

Active inference
agents perform
perception,
planning,
and action
within a
unified framework.

A2A enables
interoperability,
but it doesn't
enable inference.
agents.

And then you
have ANP,
and this one
is open
source.

It enables
decentralized
discovery and
semantic
identity for
agents,
and it comes
closest to
aligning with
what's needed
for active
inference in
the spatial
web.

And the
strengths are
that it
utilizes

decentralized
identifiers
and linked
data,
potential
foundation for
semantic
interoperability,
discovery via
open registries
and search
indexed agent
descriptions,
and it empowers
secure agent
interaction without
a single point
of failure.

But the
limitations are
that it lacks
shared global
context and
transactional
knowledge graph
that the
spatial web
provides,
challenges in
scalability and
real-time
context
synchronization.
connection.

It connects
agents, but
it doesn't
connect them

to their
environments.

And it's
difficult to
scale enterprise
wide without
global context
management.

So to sum
it up,
you've got
MCP, which
is easy
context access,
but there's no
internal inference,
and it's static
and non-adaptive.

ACP is for
task coordination,
but it's
centralized and
limited semantics,
and it's not
flexible.

A2A is
cross-vendor
integration, but
there's no
adaptive context
modeling, and
it's not
context-aware.

And then
ANP is
decentralized
security, but
it has

limited global
synchronization,
and it's
complex to
scale.

So
introduction to
active
inference AI,
and this is a
complete paradigm
shift.

The principles
of active
inference,
agents
continuously
update
internal
models to
reduce
uncertainty.

It's based
on the
free energy
principle.

It
employs
generative
models for
decision-making
and predictions,
and it
moves beyond
reactive
machine learning
towards
adaptive

predictive

behavior.

And you

minimize

uncertainty.

These agents

adapt in

real time.

So why does

active inference

matter to a

business?

Active

advanced AI

approach.

It's an

advanced AI

approach that

helps agents

predict and

adapt dynamically,

enables proactive

responses to

emerging business

challenges, and

it significantly

reduces

uncertainty and

improves

operational

resilience.

Now,

generative world

models and

causal

reasoning.

Active

inference

agents proactively
model
environments.
They are
continually
creating
generative
models of
understanding
about these
environments.
They're
simulating
multiple
business
scenarios and
outcomes.
And this
allows agents
to understand
why something
happened, not
just what
happened.
And it
improves
strategic
decision-making
and risk
management.
management.
Now, one
type of
active
inference
model is
renormalizing
generative

models.

And these

are an

advanced form

of active

inference

modeling.

They're

hierarchical

and scale-free

and they

efficiently

manage

complexity.

They enable

real-time

modeling and

adaptive

predictions

across

domains,

efficiently

handle

complex

operational

scenarios,

and significantly

reduce

data

requirements

and

computational

overhead.

So, while

generative

models like

RGMs

reveal how

agents learn
and predict,
the next
evolution,
including
multi-agent
systems,
brings
physics
itself into
the inference
process,
where belief
and behavior
evolve inside
a shared
field for
multi-agent
coherence
and belief
propagation.
So,
when you're
talking about
this field
of belief
for multi-agent
belief
propagation,
there is
an architecture
called
Field of
Belief
Hamiltonian
Monte Carlo.
And this
takes us

from models
to fields.
It moves
beyond
hierarchical
layers to
a unified
field of
beliefs
shared across
agents.
It has
Hamiltonian
dynamics.
It applies
energy-based
equilibrium
for efficient
inference and
real-time
adaptation.
It includes
isohedral
symmetry.
Each agent
is seeded
from the
same
geometric
program,
enabling
instantaneous
alignment
and a
silent
consensus
among the
agents.

And
decentralized
coherence.
Agents act
locally,
yet remain
globally
synchronized
through the
shared field.
and this
establishes a
physics-grounded
architecture for
scalable,
self-organizing
intelligent
systems.
So,
given all
of that,
what kind
of communication
protocols are
needed for
active
inference?
Active
inference
requires
continuous,
dynamic
updating
of context.
It needs
protocols
supporting
real-time

semantic
synchronization.

And it
pushes beyond
task-oriented
protocols
towards
holistic
environmental
modeling.

So,
why businesses
need active
inference
compatible
protocols?

They need
them for
tradition,
because
traditional
protocols
lack the
dynamic
adaptability
that
businesses
require.

Active
inference
demands
protocols
capable
of managing
real-time
changing
conditions.

And this

is crucial
for
responsive
operations,
for
customer
experiences,
and for
innovation.
communication.

So,
the spatial
web,
building the
world agents
can think
inside of.

So,
when we
look at
the spatial
web
infrastructure,
and we
consider
the space
between the
physical and
digital worlds,
right, you
have connectivity,
and then you
have this
spatial
management
area that
includes
things like

identity,
and activity,
and location.
interaction.
And it's
in this
space where
trust and
privacy and
security and
interoperability
lie.

So,
this is a
new
internet
infrastructure
layer that
unifies physical
and digital
worlds.

And it
provides
unified
context,
semantic,
shared
meaning,
spatial,
and temporal
interoperability.

And it's a
foundational layer
that enables
complex,
adaptive agent
interactions.

So,

it's the
next-generation
internet that
integrates
digital and
physical business
operations with
intelligence layering
over all complex
systems.

And it allows for
the coordination
of processes,
people,
entities,
and places
seamlessly.

So,
what is
hyperspace
modeling language,
HSML?

It's an
ontology-based
semantic modeling
that provides
universal,
consistent
meaning,
and definitions.

It describes
entities,
relationships,
permissions,
locations,
and states.

And it
ensures clear,

consistent
communication and
interoperability
across domains.

So,
when you think
of a spatial
domain and
a web of
spaces,
you need to
know who
or what is
authorized to
access the
domains,
what content
or data is
available to
view,
who can
publish or
modify content,
who can
transact or
interact within
it.

So,
this
standardizes
business
semantics.
It provides
a universal
business language
for digital
and physical
entities,

ensuring
consistent
meaning across
all business
units and
partnerships.

So,
hyperspace
transaction
protocol.

This
protocol is,
it provides
secure,
context-aware
queries and
transactions.

It supports
decentralized,
multi-dimensional
data exchanges.

It enables
real-time
consistency and
context coherence
across the
spatial web.

So,
it offers
secure and
context-rich
transactions,
a secure,
trusted method
to coordinate
complex,
multi-party
interactions.

It enables
enterprises to
respond immediately
and precisely
in complex
environments,
and it
enables
decentralized,
multi-dimensional
interactions
across the
enterprise.

And what it
results in is
this universal
domain graph,
or UDG,
a decentralized,
permissioned,
and constantly
evolving model
of everything,
not just
information,
but real-time,
spatially-anchored
representations of
the physical
world.

With global
decentralized
knowledge graph
of all
entities and
states,
it stores
spatial,

semantic,
and temporal
data.

It's a common
source of
context for
agents and
systems,
and it enables
real-time
queries and
context updates
and synchronization
across domains.

So,
context is not
fetched,
it's embedded.

Entities don't
just reference
data,
they're part of
the global
graph.

Every agent,
location,
object,
and event
can be
described,
queried,
and updated
in a
universally
consistent
way.

So,
agents can

locate
themselves,
interpret
relationships,
and respond
to unfolding
changes without
relying on
private databases
or hard-coded
integrations.
So,
the UDG
creates this
shared business
reality,
a common
real-time
operational view
for your entire
enterprise ecosystem,
and it
dramatically
improves
visibility,
responsiveness,
and decision-making
accuracy.
So,
for active
inference,
the spatial
web offers
something profound.
Active
inference agents
operate on the
principle of

minimizing
uncertainty about
the world by
continuously
updating
internal
generative
models of
understanding
based on
incoming data.
data.

They don't
merely perform
tasks.

They infer
the hidden
causes of
their observations
and act
in the world
to fulfill
expectations
while adjusting
their beliefs.

And this
requires a
deeply contextual
understanding
of space,
time,
causality,
and intent.

The spatial
web delivers
what active
inference needs.

It delivers

spatio-temporal

grounding.

Every entity

is located

in time

and space,

enabling agents

to align

their models

with real-world

dynamics.

It offers

semantic

consistency.

HSML

ensures that

room,

temperature,

or flight

path mean

the same thing

across domains.

It offers

contextual

granularity.

Agents can

navigate the

universal domain

graph to

retrieve just

the data they

need at the

right level

of abstraction

and scale.

And it offers

permissioned access

and trusted

frameworks.

And this is
crucial for
real-world
applications.

Agents can
access content
securely and
act within
regulated
boundaries.

So the
spatial web
provides the
shared external
memory,
sensory input,
and environmental
structure that
active inference
agents need
to operate
effectively at
scale.

So contrasting
current agent
protocols with
the spatial
web.

Excuse me.

So this
takes us from
API calls to
adaptive worlds.

The spatial
web is not
just a medium
of interaction.

It's a model
of the world
itself accessible
in real-time.

It isn't
static.

It evolves
as agents
act.

The feedback
loop between
agent and
environment is
central to
active inference
and it's
what makes
the spatial
web a
uniquely
compatible
architecture
for the
next era
of intelligent
systems.

Where legacy
agents process
tasks in
isolation,
active inference
agents embedded
in the spatial
web can act
as contextual
participants,
constantly sensing,
reasoning,

and responding
to a dynamic
multi-scale
world model.

There are
architectural
differences
with current
agent protocols.

They're point
to point
versus shared
context.

Agent protocols
like MCP,
ACP,
etc.

They're point
to point,
isolated context,
task driven,
and centralized.

Spatial web
is graph-based,
unified global
context,
inference driven,
integrated,
decentralized,
and globally
synchronized context.

So you have
centralized
versus integrated.

Current protocols
are narrow
task-based
communication,

and the spatial
web offers
unified real-time
operational
visibility
and context.

So the spatial
web supports
active inference
agents by
maintaining
a persistent,
structured
reality that
they can pull
from,
update,
and adapt
to.

Discovery
and identity
management,
isolated registries
versus global
semantics.

Current agent
protocols,
they're localized
states,
centralized registries,
or basic
decentralized
discovery.

Spatial web
is real-time
global context
management,
semantic global

entity identification
and universal
discovery via
spatial web
identifiers,
which are
kind of like
URLs,
but within
the universal
domain graph.
So in the
spatial web,
entities are
discoverable by
meaning,
location,
time,
function,
or any
combination.
And if you look
at this graphic
that I inserted
in here,
it's one of
my slides that
I used often
when explaining
this,
because HSTP
and domains
in this
universal domain
graph,
these new
spatial web
domains,

they're very
different than
what we know
as a
worldwide web
domain.

And if you
look at
current AI,
it's built
for the
worldwide web.

So a
worldwide web
domain
includes
websites
and apps,
and these
serve up
content,
and generate
data.

And they
enable you
to do
things with
that content
and data,
right?

So current AI
is built
for this
system.

Now,
a spatial
web domain
is an

entity with
a persistent
identity
through time
with rights
and credentials,
defining people,
places,
things,
and rules
in machine
readable
context and
spatial grounding.

So active
inference AI
is built
for this
new framework.

So for
high-level
causal
reasoning
of active
inference,
agents need
more than
just who
to talk
to.

They need
to know
what exists,
where,
when,
and why
it matters.

State

management,
local
versus global.
Agent
protocols,
these current
agent protocols,
their local
state,
tasks,
messages,
context,
and ACP
and A2A
allow for
message threads
or long-lived
tasks between
agents,
but the state
is local
to the
conversation.
And MCP
supports
context
injection,
but not
persistent
world-state
synchronization.
Whereas the
spatial web,
the universal
domain graph,
becomes the
single source
of truth

for all
agents.

Every entity,
event,
and state
change is
part of
the global
context,
accessible
and queryable
by permissioned
agents at
any time.

To minimize
free energy
and make
accurate
predictions,
active
inference
agents
must align
their internal
models with
an externally
observable
state.

A common
source of
synchronized
reality,
like the
universal
domain graph,
allows agents
to coordinate
without needing

custom logic
to reconcile
individual
perspectives.
semantic
interoperability.
Current agent
protocols,
their message-based
formats without
shared meaning,
whereas the
spatial web
is ontology-driven,
semantic
interoperability
via HSML.
So clear
semantic
standards
ensure
consistent meaning
so that a
term like
location
or battery
status
refers to
the same
concept
across
systems.
causal
reasoning
and adaptation.
Fixed
workflows
versus

embodied
understanding.

Today's agent
protocols treat
agents as
isolated
services,
reacting to
tasks and
triggering
responses.

Active
inference,
by contrast,
enables agents
to simulate
multiple futures,
plan under
uncertainty,
and infer
hidden variables
based on
context and
change.

The spatial web
enhances this
by providing
real-time
structured context
that agents
can reason
about causally,
bridging the
gap between
perception
and action
across a
shared world

model.

This enables
enterprise-wide
responsive,
intelligent
decision-making.

So,
active
inference
agents
within
the
spatial
web.

This
ushers in
a new
era of
autonomous
intelligence.

Agents
don't just
cooperate,
they
co-regulate.

They
learn,
infer,
and adapt
in concert
with their
environments.

In
FOBHMC
terms,
each agent
is a
localized

excitation
within a
shared belief
field,
acting
independently
yet
maintaining
global
coherence.

Agents
don't just
follow
commands,
they
understand
situations,
simulate
consequences,
and act
accordingly.

And most
critically,
active
inference
agents
don't need
to be
tethered
to a
centralized
database
or cloud
tools.

They can
operate
on the
edge,

inferring,
adapting,
and acting
autonomously.

But to
thrive,
they need
an environment
where context
is available
as structure,
not just
as data.

And that's
where the
spatial web
becomes
indispensable.

The future
of agent
protocols
in the
spatial web
era.

So navigating
the road
ahead.

The agent
protocols
emerging
today,

MCP,

ACP,

A2A,

ANP,

they seem

to be

solving

different
problems
than the
spatial web.

They manage
task

execution,
data retrieval,

peer
messaging,

and the
spatial web

aims to
model the
entire digital
and physical
universe.

But when
we examine
how these
systems interact

and where
technology is
heading,

a pattern
begins to
emerge.

These protocols
are not
competitors to
the spatial
web.

They're early
bridges,
temporary
scaffolding
across a
digital landscape

that's still
taking form.

They're a
necessary layer
for now.

There's no
doubt that
today's agent
protocols are
doing important
work.

These protocols
are gaining
traction not
just because
they're useful
but because
there's no
shared substrate
yet.

Each protocol
is a patch
for a
fractured
internet,
stitching
together siloed
APIs,
services,
and data
under a
common
interface.

They're vital
in a world
where agents
are still
grounded in

machine learning
architectures
that lack
self-modeling,
physical
grounding,
or native
contextual
awareness.

the spatial
web from
protocol stack
to contextual
substrate.

The spatial
web is not
simply a
next-generation
protocol stack.

It's a
paradigm shift
in how agents
interact,
not just with
each other,
but with their
entire environment.

Active
inference provides
the internal
generative
model.

The spatial
web provides
the external
semantic
and spatial
context.

Together,
they enable
real-time,
scale-free,
causally
grounded
intelligence
across
distributed
networks.

In this
convergence,
the universal
domain graph
becomes the
ambient
knowledge base
for all
agents,
no matter
who built
them.

HSML
becomes the
language agents
use to
interpret
their world.

HSTP
becomes the
protocol they
use to
act upon
it.

So what
happens to
these current
agent

protocols?

Well,
in the
short term,
there's
collaboration.

Today's
agent
protocols
continue to
address
urgent
interoperability
needs.

They form
essential
bridges
toward
broader
spatial
web
adoption.

Midterm,
we're looking
at
convergence.

Protocols
will begin
to integrate
or evolve
to align
with the
spatial
web
standards.

Integration
pathways,
HSML

for agent
descriptions,
HSTP
for transactions.

businesses
should start
aligning
protocol
strategies
with the
spatial
web
principles
and
early
adaptation
will
ensure
competitive
advantage
and future
readiness.

Now,
in the
long term,
there's a
risk of
protocol
inaction.

Many
protocol
functions
like discovery,
negotiation,
task
management,
they're likely
to be

natively
supported
in the
spatial
web.
interoperability
challenges
will
disappear
and
failure
to
adapt
may
lead
to
technological
redundancy.

And if
you look
at,
you know,
the evolution
of the
internet
historically,
universal
and open
standards
inevitably
replace
fragmented
systems.

Early
web
protocols
were
displaced

by
HTTP
and
HTML.
So,
with my
company,
with AIX
Global
Innovations,
we're
working on
an
ecosystem
of
adoption
that
includes
education
and
guided
activation
and
tools
and
solutions,
a
continuous
path
from
learning
to
deployment.
And if
you'd like
to learn
more,
feel free

to scan

this QR

code.

It'll

take you

to

Learning

Lab

Central.

It is

a

global

education

hub

that

we

have

built

that

is

hyper

focused

on

active

inference

AI

and

spatial

web

technology.

I host

monthly

learning

labs

and

we

have

courses

and
certifications
available
there.
And it's
just a
great
place to
come
together
as a
community
and find
other
people
who
are
interested
in
building
in this
field of
technology
as well.
So,
I will
stop sharing
now.
Thank
you,
Denise.
Awesome.
Thank you,
Daniel.
Okay.
some fun
questions
in the

live
chat.
So,
I'm
just
going to
read
them
off
and
we'll
go
with
it.
Okay.
Bert
Berkers
wrote,
when and
how can
people get
started
building on
the
spatial
web?
Yeah,
that's a
great
question.
And
I don't
have an
exact
date.
I know
that the
standard

was
approved
and
ratified
by the
IEEE
in
May.
and
I
know
that
the
consensus
is that
towards
the
end
of
this
year,
I'm
assuming
end
of
this
year,
beginning
of
next
year,
we'll
start
to see
some
tools
that
will

make
it
really
easy
to
start
building
registering
domains
in the
universal
domain
graph.
I know
the
spatial
web
foundation
dot
org
is
the
authority
that
will
be
handling
the
universal
domain
graph.
And
so
I'm
assuming
there's
going
to

be
tools
that
can
instantly
start
to
register
and
I've
seen
tools
where
you
can
put
in
documents
you
can
put
in
information
it'll
register
the
domains
it'll
spit
out
the
html
for
you
and
I
know
those

tools
are
on
their
way
I
just
don't
have
a
date
for
those
yet
so
I
feel
like
as
soon
as
start
you
know
kind
of
going
to
town
on
this
and
exploring
and
figuring
out
how
this

looks
in
the
wild
Awesome
okay
Scott
David
wrote
I'll
put it
in
the
chat
just
in
case
you
want
to
see
it
he
wrote
what
are
the
steps
through
which
the
spatial
web
and
context
rich
AI
interactions

can
be
onboarded
to
existing
organizations
companies
governments
etc
cost
savings
risk
reducing
yeah
and
that's
similar
to
the
answer
you're
going
to
get
that
I
just
gave
you
I
don't
have
all
those
details
right
now
I

know
that
they're
coming
and
yeah
I mean
I
would
keep
an eye
on
the
spatial
web
foundation
because
they
will
be
the
entity
that
starts
to
onboard
people
and
give
information
on
how
to
onboard
and
how
to
start

building
out
in
this
space
yeah
I
think
it
was
a
great
framing
and
very
helpful
to
get
that
review
of
these
protocols
and
I
just
got
the
sense
of
all
these
different
pairwise
or
hub and
spoke
type

interaction
architectures

they
but
they're
still
very
generic
and
they're
not
necessarily
about
any
real
physical
or
embodied
space
so
some
domain
ontology
is still
required
to have
the
kind
of
content
of that
agent
to
agent
channel
be
meaningful

beyond
just being
like a
handshake
right
right
I mean
we're really
good at
message
passing
and
injecting
data
but
you're
still
dealing
with
the
LLMs
you know
and
they're
you know
it's
they have
hard
limitations
right
you know
and
when we
talk
about
active
inference
and

we
talk
about
the
capabilities
of
active
inference
we
really
want
to
enable
all
of
those
within
a
framework
that
enables
this
embodied
AI
right
that
can
tie
these
agents
to
the
physical
world
and
this
kind
of

digital
twinning
of
everything
all
right
I'm
gonna
read
a
question
in
the
chat
and
then
I'll
just
read
a few
more
things
that
were
followed
from
there
so
Alex
Sabine
wrote
there
are
some
quite
profound
ontological
implications

for humans
who share
this field
of belief
with agents
how are
we
managing
the
ethical
discussions
around
this
and
then
Scott
David
added
to
that
plus
one
ethics
is
about
human
ideas
about
human
behavior
so
maybe
we need
new
approaches
to
ethics
more

akin
to
an
ethics
of
treatment
of
other
species
ethics
of
environment
interactions
with
nonhumans
so
I know
that's
a few
different
pieces
but how
do you
engage
with this
at that
level
so
what's
really
interesting
to
me
about
that
is
when
you

try
to
think
of
the
approach
we've
tried
to
take
with
ethical
use
of
AI
and
bias
mitigation
and
different
things
like
that
it
doesn't
really
work
because
you
can't
there's
no
amalgamation
of that
globally
if you
look at
our

world
you
have
all
kinds
of
belief
systems
all
kinds
of
different
cultural
differences
all
kinds
of
different
governing
styles
so
you
have
to
have
a
way
that
preserves
all
of
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For example, one way to tackle this is to investigate neuro-robotics, which is a subfield of robot

learning.

Basically, it's modeled a robot's learning process, very similar to that of a human.

So a biologically inspired auto-modus learning system often investigates human perception, memory, attention, sensory motor control, and so on.

So this neuro-robotics field also features robot learning from real-world interactions or online learning.

So with online learning, a robot can adapt to new environments with less data collection costs and then can be learned to be set around humans.

So first of all, we need to, in order to do like neuro-robotics, we need to investigate cognitive architecture.

So the cognitive architecture is very much studied in the active inference process theory.

So basically, it investigates different modules inside the brain that can be used to learn a perception system.

So first of all, there's the generative model, which is basically the perception system itself.

So it's basically trying to, through the sensors, making observations in the environments and then learn the actual interactions between the environment and the agent.

The second one is the motor, which is the action controller.

It's actual, the action will be taken by the agent and then executed inside the environment to return a new observation.

And then there's like the memory module, which stores the interactions between the agent and the environment.

So that memory is kind of memories will be stored in here in order for future use of like training or, you know, just maybe remembering of knowledge and knowledge purifications, for example, or distillations.

So basically, the prior preference distributions trying to predict the desired representation or the desired goal that the agents want to achieve in the future.

So generally, in active inference, a prior preference distribution is provided in advance.

But in our work, we are trying to have like a separate neural network to predict the prior preference in the future dynamically.

So the last one is sensors, right?

Like for example, the robot has like eyes and arms and so on.

So before going to the main topic, I want to touch on a little bit about deep active inference and the background around it.

So basically, the interactions between the agent and the environment can be described as the interactions between the generative process and the generative model.

So first of all, is the generative model process.

Basically, it's the real world environment itself, right?

And inside the real world environment, there's like a hidden state.

So this is like the true probability distributions of the environment and we cannot actually see it.

And in here, we can see that there's an arrow between the true state and the observations, which means that the generative process actually generates the observation that can be observed, observable.

So the agents can actually see this observation.

And this, when compared to like, it is very similar to a partially observable Markov decision process that is often formulated in the reinforcement learning literature.

So now when we have the generative process, we have the generative model, which is the actual agent itself.

So the agents also have a state distributions here.

So it is actually the agents best approximations of this generative process.

Basically, we can call it like a dynamic states or a agents belief state or a world model states, however you want to call it.

So this distribution will be best estimates of the hidden states inside the environments.

So the hidden states here can be like the world dynamics, for example, or the physics or the logic of the world, for example, something like that.

And in active inference process theory, the agents basically continuously trying to make the observation through time and also continuously modify its belief after making the observation.

So how do we further formulate this?

First of all, we have the agent states here.

It is like the initial belief or representation or the world dynamic states.

It's going to be called the prior over states, which is $P(s)$.

And then we also have $P(O)$, which is the model evidence, which is the observation distribution.

And then in active inference, we are trying to have the agents both observing and predicting.

So first of all, we predict the observations, which is basically the likelihood of observations.

It is the agent's best estimate of the observation distribution given the prior or the posterior information that it has.

And then finally, the $P(s|O)$ is posterior function.

Basically, it's the agent's best estimate of the state after making the observation.

So it's going to be after making the observations, the agents actually making or changing the posterior belief.

So now we have all these definitions of all these functions.

How do we learn this generative model?

So we have to learn, first of all, the likelihood functions and then the posterior approximation.

And then we have to best estimate the model evidence distribution here as well.

And then we estimate the posterior and then the prior.

So how do we learn this?

So active inference process theory is proposed that we are trying to minimize the variation of free energy or the marginal free energy.

So in order to do that, we first have to predict the likelihood of the observation or the modal evidence.

So in deep active inference generally, this is actually called the decoder and it is often parameterized by your neural network.

Next is the posterior functions.

So at the same time when we are trying to predict the observation, we also have to make an observation.

And then we produce the approximate posterior at the current time step.

So in deep learning literature, this is often called an encoder.

And it is also often parameterized by a neural network as well.

So in here, active inference state stats, we have to try to continuously correcting the belief after making new observations.

So we are trying to minimize the belief after we have made observations and the belief states prior to making observations.

So we are trying to minimize the surprise between those two kind of distributions.

That also means that we are trying to minimize the posterior and the prior distribution.

That also means that we are trying to minimize the Kullback-Leibler divergence between the posterior Q of S and the prior Q of S .

So this is also means that we are trying to correct the belief after making any observations.

And then next is in parallel to minimizing the uncertainty between the posterior and the prior.

We also have to maximize the prediction accuracy of the observation as well.

So in this case, we are trying to maximize the similarity between the true observations and the predicted observations.

And this is also called maximizing modal evidence or maximizing observation prediction accuracy.

And we do this by minimizing negative log likelihood of the predictions.

So that is for like one step.

Right. And now we want to extend it to multiple steps.

And then we have to integrate actions here as well.

And so at the time step T , there is several different, there's hidden states, right, that produce the O of T , which is the observation that the agent can see.

And then the S of T is like the actual initial, it can be prior or posterior of the agent.

And then there's the actions of the agent that the agents plan to take.

And so when the agents take the actions in the environment, it is executed and it's produced a new state, a new hidden state that we cannot see.

And then produce a new observations, right.

At the same time, the agents can estimate the next state given the previous state, S_T and the action S_T .

It can estimate the next state S_{T+1} here.

And then we do the same process, right.

So S_{T+1} is actually the prior of the next state.

And we can do the same process by predicting the observations of the next state.

And at the same time, we observe the observation of the next state and then encode it to an approximate posterior.

And then we minimize the variational free energy by minimizing the KL divergence between the posterior and the prior,

and then maximizing the prediction accuracy by minimizing negative log likelihood of the prediction, O_{T+1}

plus one.

So here's the formula for the variational or marginal free energy.

You can see that in here, there's like a complexity term where it's minimizing the KL divergence between the Q of S given O

and P of S given S^T plus one minus A^T plus one and A^T plus one.

It's basically the previous state and previous actions that's caused the current state, right.

Leads to the current states in the current prior.

And then this complexity term is like measuring the amount of surprise that we want to minimize.

And then finally, we have like the accuracy term where we are trying to maximize the prediction accuracy of the observation.

And so we're trying to minimize the negative log likelihood of the prediction observation distribution.

So now when we have done the learning for the world model or the generative model, we have to learn the policy as well.

Right. So we have to know a way to like decide what kind of actions should be executed in order to optimize or solving the task.

So there's so many actions that can be taken in one single time step.

And each of those actions can lead to like different possibilities.

So the agents can like estimate like each possibilities based on like previous actions in the current belief.

And so on. Right. So there's like as the time steps or the horizons spans out the number of possibilities of the belief states actually goes up a lot.

Right. So how do we choose the optimal trajectories of actions that we want to take?

So that's exactly like this.

That brings us to the expected free energy.

Right. So we are trying to say that whatever kind of trajectories in the future.

That yields the lowest expected free energy.

Then we choose that as a optimal trajectories of actions that we want to choose.

And then we learn the policy on that.

That's exactly like this.

The expected free energy is eventually decomposed into an epistemic term and instrumental terms.

So in this topic, I only focus on the instrumental signals.

So the θ here is the neural networks actually parameterized by θ .

The π is planned action.

So the instrumental signal is decomposing here.

It is we can look at all these three terms.

First of all is the estimation of future observations here.

And then q of s given π is basically the approximate posterior in the future, like given the policy in the future.

And then there's $\log$ of p of o given $\log$ of p of t .

It's basically the prior preference distribution at time t .

So generally in active inference, right, we provide the agents with the prior preference distribution in advance.

But in, yeah, I've mentioned before, in our work, we will predict this kind of distributions dynamically, right?

So that the agents can follow its own wanted or preferred trajectories and learn the policies that can follow the preferred trajectories.

So yeah, that brings us to the main parts, active neural robotics and goal-guided preference learning.

So before going to the main main topic, it's basically I want to reiterate a lot of problems in current model based robot learning nowadays.

So popular robot or model based robot learning algorithms often deploy the offline policy onto a real-time robot.

What does it mean? So it's basically it's learning online, an offline policy, and then it's evaluated in the real time.

So there's no little to no online learning capabilities.

Second of all is they do not leverage the concept of prior preference learning.

So no goal predictions.

So the agents might not want like might not know what it wants, right?

So most of the algorithms actually only maps from the vision or the language or both directly to the actions.

That means it is only learning, only treating all these experiences as identically distributed, independent and identically distributed.

And then next is depend depends largely on expert demonstrations.

So there's no if there's little to no expert demonstrations, then the policy will suffer from suboptimal policy.

So another problem here is nowadays we can see that a lot of big or large organizations doing like all these vision language action models.

There's foundation models and fancy and so on.

So all of that is just come down to training a very large model that have a very large language model and large vision model.

And then encode that with transformers and then map it directly to the action tokens.

And so this is basically a schematic of model behavior cloning and implementation learning.

And so in this case, expert demonstrations are treated as identically and independent distributed.

distributed, and every single interaction

system is put into a single data set,

and then it's trained, and then it's

going to be evaluated in the real world or in simulations.

So this is more like a supervised learning setting

where they could use a lot of problems.

The classic problems here is mentioned by also Sergey

Levine and a lot of other popular people.

It's a cascading problem.

So cascading errors actually accumulate

over time in actual trajectories because of the domain data

on the domains that the expert administration is trained on,

is used to train on the robots.

It's different from the actual real world domain.

So it's kind of similar to a data set distribution shift or domain shift.

And therefore, the robots actually deviates from the demonstrated examples.

And then this leads to policy collapse in real time or online executions.

So in here, we are trying to develop a very intuitive way of overcoming all these problems.

So behavior cloning and imitation learning, they actually can learn to match the expert performance at best.

And the expert policy is potentially suboptimal.

As you can see in the figure on the right, we can see that behavior cloning in red.

For example, it's going to be solving all these, all these, it has like the reward trend.

It's very low.

For example, this is like the suboptimal policy.

It's learning, it's learned through expert demonstrations.

For prior preference learning, right?

The agent first learns to predict the expert trajectories as its prior preference.

And then throughout several interactions or further interactions between the agents and the environment, the prior preference learning model can try to dynamically produce more optimized goal trajectories or more optimized prior preference trajectories.

And so the prior preference, the predicted prior preference trajectories will actually

eventually be better than the expert demonstrations.

And then we can see that the prior preference, the dynamic prior preference is independent of potentially suboptimal expert demonstrations.

So that is the intuition of the method.

So in our work, we also have a, generative world model.

It is also very similar to what I've talked about in the deep active inference background section.

But in here, there's a little bit difference.

We have another model.

So in this case, the similar thing is we still build our generative models on the concept of a world model, which is basically get back to David Ha.

And yeah, so we still have like observation.

We have like encoder posterior that encodes the current observations.

And then we have like the transition model that have like, that predicts the next state prior given the previous state and the previous actions.

So this prior will be, which serves as the prior information prior to making observations of the next state, of the next time stack.

And at the same time, we also have like the observation at OT, right?

And then we encode it to an approximate posterior.

So at the same time, we have like two kind of states.

One is prior to making observation.

One is after making the observation.

And then we also have the decoder reconstruction as well, where we have the posterior and then we reconstruct it in order to predict the observation.

And then we also have the complexity loss, which is the Kullback-Leibler divergence between the posterior and the prior.

We also have the accuracy loss, which is the negative log likelihood of the predicted observation.

There's a difference here, right?

We have the RSS, which is basically called recurrent states, recurrent state space model.

So this actually is very simple.

And this is introduced in the Dreamer series from Danny Jarhaver.

So the RSSM is, here's the inside of the RSSM.

So we still have like the encoder posterior, encoder posterior in orange here.

And then we have an additional recurrent neural network.

Normally it's a gated recurrent unit.

It takes in the recurrent state and then it takes in the distributions of states, post-area.

And then it's predicting the prior of the next state itself.

And then it's also predicting the recurrent state, h_t , or the deterministic state.

The prior, the next prior over state is often be called stochastic state.

Right, and this h_t is also served as an input into predicting the next state posterior, in addition to the next observation, l_t .

Right.

So now we have trained our world generative world model.

That is RSSM.

We can now learn the prior preference model.

Right.

So learning the prior preference model requires, I'd say, requires the states of the world model itself.

So in this case, we are trying to propose a dynamic contrastive learning system where CRSP is contrastive recurrent states prior preference model.

So basically, we're trying to keep the RSSM the same or fixed when we are trying to learn the CRSP or the prior preference model.

So the point or the goal of learning the prior preference model is trying to have that model predicting the optimal trajectories, right?

The optimal trajectories of the state in the future such that those state trajectories represent the goal or the preferred state or however you want to call it.

So in order to do that, we have to collapse the state of the prior preference model to the state of the RSSM whenever the trajectory is successful.

And in contrast, we have to pull away or push away from the RSSM state when the trajectory is not successful.

So in order to do that, we have to introduce a term called role or prior preference rate.

So we're literally saying whenever the trajectory is successful, the role is positive.

This positive term will collapse all these two states together, right?

And then whenever the role of prior preference rate is negative, then these two states are pushed away from each other.

So when we keep the state of the RSSM fixed or stop the gradients of the RSSM, then only the state that gets pushed and pulled is the state of the prior preference model down here.

So in the end, the CRSP or the prior preference model will learn to only predicting the optimal state of the RSSM without predicting the failing states.

So you can see that in here there's a little bit difference, right?

So the RSSM is not conditioned or the RSSM is conditioned on the actions, right?

But the CRSP or the prior preference model is not conditioned on the actions, right?

This basically trying to say that, give me an observation.

I will predict future trajectories that is needed or future goal trajectories that is needed to be in, or future optimal state that is needed to be, that the agent needed to be in.

So eventually, we will use this goal trajectories to guide the policy learning, right?

So now there's, here's an example of how we calculate the prior preference read row.

So basically, we are trying to say, whatever the immediate step that leads to successfulness of the trajectories, then that one will have the most strength of collapsing two states to each other.

So that's the two states of the RSSM and the states of the prior preference model.

Right?

So we can say that whenever the trajectory is successful, for example, the car is in, like in the flap right here is successful, then that's prior preference rate at that point is one.

And then we decayed backward.

Similar here, the push button, we also decayed backward.

So on the other hand, when the agent is in an unsuccessful state, or trajectories, for example, then the prior preference rate will be minus one near the unsuccessfulness state.

And then we also decayed backward.

So here's an example of positive row that is used in the successful trajectories.

So in here, when the positive row is used,
these two RSSM states and the CRSP states
is collapsed to each other.

And the CRSP has been able to predict all the state
of the RSSM that is successful.

So here's an example representation
of the CRSP state, which is trying
to predict the successful trajectories.

And for example, in here, there is negative row.

When the row is negative or prior preference rate
is negative, these two states from the RSSM
and the CRSP is pushed away from each other
so that the CRSP will not predict any failing goal or failing
states from the SSM.

So there's no complexity objectives here,
no chiral divergence between the posterior of the CRSP
and the prior of the CRSP.

So throughout time steps or future horizon, which
is in the x-axis, right?

So we can see we have two environments and benchmark here.

First of all is the observation, which is the first line, first
row.

Second row is the prior preference rollout.

So we plot it in order for visualization.

So we can see that throughout all these observations, the car has
to go to the left in order to build enough momentum to go
toward the right hill in order to reach the flag.

Otherwise, it cannot.

So we can see for every single observation in the mountain car environment in the top row,
we can see that's the prior preference rollout.

So we can see that the prior preference model is trying to predict the goal that the car needs
to be in.

For example, in here, the first observation, the car is in the middle or the bottom of the hill.

Then the prior preference is trying to predict the car, trying to go in to the left in order to build momentum.

And then until like the frame number five and six and seven, the prior preference model is trying to predict the goal of the car is fully moving toward the right.

Right.

And in the next benchmark environments, it is also happening in the same case, even though there's a little bit of noise here.

But we can see that the prior preference rollout wants the agents to push the button.

Right.

So it's trying to predict the preference rollout.

So we are halfway there.

Now we need to integrate the language part.

Right.

So when we want to compete with like all these vision language action models, very popular, for example, like RT1, RT2.

Right.

We also need to have like a form, some form of like multi-modal encoder.

Right.

So it is because language instruction or dynamically provided goal can be considered as a form of observation.

Right.

So in this case, we're going to use like pre-trained large language models as frozen feature extractors, for example, GPT, LAMA, T5, and so on.

Right.

And then there's vision encoders.

Basically, it's trying to extract real-time observation.

And we can just train it as normal.

So generally, I often saw people do trainable text encoder as well.

But generally, it should not keep the same embeddings and integrity of the language encoder.

So there's a lot to be experimented.

That's not what I'm trying to say.

And then finally, we have the language embedding and the image embedding.

And then potentially, we can have like several other modalities as well.

But here's like I demonstrate the two case, most common case.

That sounds weird with the language and the vision.

Then we concatenate them together as embedding, like a 1D vector.

And then it's going to be further also known as modality fusion technically, Meet Fusion.

And then this can serve as the observation embedding that can be fed into the posterior network.

So in here, you can see that the multimodal encoders are producing all these, for example, embeddings.

These embeddings fit into the posterior network as normal in our system that we have described.

This is the RSSM. And then we can see that the posterior joint feed in the recurrent neural network and then it's producing the deterministic state.

At the same time, the recurrent network also predicting the next state prior distribution or the stochastic state.

And then the deterministic state also affects the predictions of the next posterior distribution as well.

So we can see that this is kind of like a multimodal world model system where it can execute actions by the language instruction.

So now that we learned the world model, we also need to learn the prior preference model as well.

So notice that here is the CRSP without the language input.

Right now we need to expand the CRSP modules so that it can be guided by the language as well.

So that's the prior preference model can predict the optimal trajectories that is conditioned on the language.

So in here is the actual way that we did it.

The CRSP now can be conditioned on the language model.

The language here basically can be encoded using all those models, the live language models, for example, GPT, Llama.

Right.

And then we do the same process as well as I described in the previous slides.

So now we need to learn the policy as well.

Right.

So after we have like the generative model, we have the prior preference model.

Now we need to find a way to like determine the robot's actions.

Right.

So when we are trying to, there's a lot of popular actions, determination models.

Right.

So when we're trying to do that, we have to be very mindful of like, like what kind of policy learning that is, that yields the best performance as well as efficiency.

So there's two competing methods.

So first of all, it's cross entropy methods where trying to say for every single step.

We sampled and Gaussian distributions of actions.

And so usually N is 1000.

And then we're trying to prune all these distributions using like the world model and the preference model to estimate the scores of all these possible trajectories that is yielded by all these 1000 Gaussian distribution of actions.

Right.

And then after that, we tried to take the top K.

So maybe K is 30 to 50, normally 30 to 50, and we choose the best K performance trajectories.

Right.

And then we take all those actions to make a new meaning and the standard deviations of the action distributions.

Again, we initialize, we sampled and Gaussian distributions again.

Right.

So we trying to iterate that those like five times, for example, normally is two to five times and optimize the actions populations.

So generally, this process is very slow.

We have to do it for every single time step.

That will use the computational time complexity of like O of n by k .

Basically, n is like the populations and k is the iteration numbers.

Right.

So imagine we have to forward the world model 5000 times for every single step that we want to take the actions.

Right.

So it's very slow.

So other methods here can be active critic.

It is very classic in a reinforcement learning literature.

And it is faster because we only need to make one single forward pass in the world model and then make one single forward pass in the policy model.

So.

So.

In order to learn this model, we trying to have this state dynamics model here and then it's produced a representation and the preference states throughout all these trajectories rollout.

Right.

And then we can use the representation and the preference to compute the instrumental signal and the epistemic signal.

For example, the representation and the.

preference can be used in here is basically, for example, instrumental signal can be computed by the difference or the divergence between the representations, actual representations and the preferred states that the agents want to be.

Right.

So if it's far away from each other, then the instrumental signal is low.

Basically, it's the score of like rewards is as low.

And on the other hand.

Right.

So if the actions that use the represents the representation that match our preference or goal or predicted goal.

Right.

Then the score internal score should be like higher.

Right.

So it's.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

agent or uncertainty in the environment itself.

So to reduce the uncertainty, we have

to take appropriate actions that explores the agent,
explores the environment enough.

And there's also other epistemic signal
that is investigated by Carl Friston as well.

So it is a beautiful literature.

So yeah, we use all those scores, compute all those scores.

And then finally, we accumulate them
to compute the advantage value and return value.

This will serve as the label for the value models
to predict the return.

And then finally, we learn the policy learning
by prop through the value model or value network.

This is also called policy gradient.

So yeah, after learning the world model, the prior preference model,
and then the policy model, now we are
able to actually perform online learning in simulations
and the real world.

There's several different benchmarks to it.

So there's first of all, the front car kitchen, which
employs front car kitchen robot arm.

There's a lot of tasks, eight tasks.

For example, the robot has to open the microwave,
or turn the switch, or move the kettle, and so on.

And then there's also MetaWorld, where the Sire robot arm is featured
with over 30 manipulation tasks.

And then there's RoboSuit, which is a very classic and popular benchmark
for testing robots algorithms.

So it features so many robots.

But in here, we actually use Panda robot arm with a total of 20-plus manipulation tasks.

And in the future, we also may want to integrate real-world robot learning as well.

So here's our preliminary results.

We have compared our agents on the top left here.

We have compared our agents with a lot of other state-of-the-art benchmarks

or baselines on different benchmarks.

So we have Dreamer v3 and demactive inference variational policy gradients.

And then some other active inference baselines as well,

as well as model-based enforcement learning baselines.

In general, we have our agents actually perform,

has the best performance in most tasks.

And it usually converts faster and learn to solve the task faster,

as well as can solve complex tasks where other agents cannot.

For more complex tasks, for example,

where the tasks requires completing multiple subtasks,

and then it's going to have the language instruction here as well.

For example, open the microwave and then move the kettle

and then finally open the slider.

You can see that the robots actually do according

to the language instruction.

Yeah.

And then after that,

we also test on the robot's sheet as well.

For example, in this case,

open the drawer, pick the cup,

place the cup into the drawer,

and close the drawer.

We can see that the robots actually have a little bit of struggle

to put the cup into the drawer.

But eventually it did it.

Yeah.

Just while you're on this slide,

how does the model come to understand the semantics of what a cup is?

Is that the vision model connecting it to the embedding?

Or how does that,

how does it know what a cup is?

So it doesn't know by a concept per se,

but I would say like all these actions is encoded

based on the visual information that is received.

So it's basically the robots has received the information

of both views,

like from this view and from this view.

And it's kind of encode,

like use the convolutional networks
to encode the perception information
into some form of a state
and then use the states
to decide on actions using the policy network.

Did I answer your question?

Yes.

So it doesn't necessarily have a,
like a knowledge graph world model,
but it uses the embedding.

So then if,
if the model could represent a chair,
that would go from the visual input
to the embedding space of chair
that gets hooked up with the,
how does it know how,
how to,
if you said put it to the left of there,
how would it know that?

Right.

So in this case,
when we have like expert demonstration data,
right,
it kind of have like the initial understanding
of like,
of like each pixels is on which pixel,
right?

So there's like different set of pixels
inside the visual image
that's kind of activates the representation
that's can use the policy activations.

And it's trying to solve,
trying to like actual do,
solving the task.

But if you actually move the cup
to the left a little bit,
it will still struggle,
right?

Because the distribution of pixels
now is different.

Right.

So it's,

I think it's going to be totally dependent
on like the distribution of pixels
as very susceptible to like the image quality
and stuff.

If it is like different from the image quality
of the expert demonstration,
then it will struggle.

So to answer your question,
there is no knowledge graph
or I'd say causal model.

That's the robot learn.

So there is no causal model
and no concept models.

It's just the representation
or like the embedding
that is learned
from the distribution of pixels
throughout the encoder or CNN.

And that,

by that we called
all those representation,
the approximate posterior
and the prior itself.

Yeah.

Yeah.

That's very interesting
how even without
the formal causal model,
you could still in language
say something like,
move the cup to a place
that will look better.
but then that implicitly entails
a causal model

of how actions would lead
to something looking better
or looking different.

So there is a sort of way
in which the causality
is at least actionable,
even if it's not represented
in that declarative way.

Yeah.

And then,

yeah,

I'd say there's also
failure case as well, right?

So this is
because like,

there's

like the,

you know,

pixel distributions

kind of different

from trajectories

to trajectories.

So,

for example,

with the same problems

right here

and the distributions

of like,

action that the robots

is different as well,

then the robot

will struggle

to,

to interact

with the environment

and use

the optimal trajectories.

So,

yeah,
it's because,
for example,
the robot like,
moved the kettle,
but now the kettle
is in different position
as the learned one,
right?

So it's going to be
struggle to like
having all those
information encoded
into the policy
and the policy making,
you know,
appropriate actions
to solve the task.

So,
yeah,
that is one challenge.

There's also other
challenges as well.

For example,
I'm going to see
that's like the,
in the figure
on the right here,
right?

We can see
like,
this is an example
of like,
the inability
to predict
different blocks
in the language
table benchmark.

So this,
the figure on the left
is the true,
actual observation
that the agents saw.

But in here,
like the agents
actually predicts
the observation
like has no blocks
at all,
right?

So you can see
that like the subtle
changes in pixels
cause the agents
to struggle
to predict,
especially in the
environment
that have like
so many different
blocks
that is like
in subtle pixel,
right?

There's small pixels
inside the environment.

And then there's
like the other
strange interactions
and observations
as the subtle pixels
varies,
right?

And all these blocks
like varies
in positions,

in size,
and in colors,
right?
So this actually
caused the agents
to predict
all these frames
with disappeared block.

and then,
you know,
there's also
other several problems
as well.

For example,
like decoder learning
is computationally expensive
and then research
well-tests
investigates
encoder-only
active inference model.

For example,
in the very popular
paper from NERIPS,
it's contrastive
active inference
is also that
paper that investigates
encoder-free
active inference.

And then
experimental design,
finally,
yeah,
right?

It's also challenging
because,
you know,

experimental design
for real-world
online robot learning
is something
that's not
usually possible
because we have
to design
the reward function,
right?
We have to design
the reset function,
for example,
for every single episode,
right?
We want to reset
the state
of the environment.
We cannot,
like,
use the human
to, like,
manually,
like,
reset the state,
per se,
because the world,
the real world,
is kind of like a,
you know,
single-life process
where the,
it's like,
one episode,
one trajectory
is, like,
long forever.
It doesn't reset,

right?

And,

for example,

in here,

for example,

pick and place the block.

The robots

can play around

with the block

inside one episode

but when

it can put the blocks,

push the blocks away,

then technically,

the environment

needs to be reset.

But there is

kind of

to build a system

that's trying to

pick and place the block

and put it back

to the initial position

so that the robot

can learn.

It's either

chicken and egg problem

or maybe

we need,

we need,

like,

a larger system

to be in,

it's very tedious

to set up,

for example.

And if we can use,

like,

human to,
like,
manually pick and place
the block,
it will cost
a lot of effort
and time
and money
for data collections
and stuff.

So,
that is also
very challenging
to design
a real world
online robot learning
experiment.

So,
that concludes
my presentation
about our topic.

Thank you very much
for having me.

If you have any questions,
feel free to,
feel free to mention.

Thank you.

Awesome.

Very comprehensive
presentation.

I'll just see
if anyone in the live chat
wrote a question.

Meanwhile,
though,
what stage
are you at
in your PhD

and what,
what are you looking
to kind of
get towards
with this phase?

Yeah.

So,
I'm currently
a fourth year PhD now
and I am
very active
in developing
different
robotic learning
algorithms
using
biologically
inspired
algorithms
and active
inference.

So,
I focus
on building
all these
generative
model
and making
use of
all these
concepts
that active
inference
literature
propose
and all
these
mathematical

concepts
as well
in active
inference.
So,
I think
it's very
interesting
to further
extend this
to real
world robotics
humanoid.
can
automatically
or
autonomously
explore
the environment
in order
to do
save and
rescue
operations
and stuff,
for example,
right?
or can
do
autonomous
explorations
of deep
sea
missions,
for example,
or we
can send
humanoid

robots
on Mars,
for example.

It's
going to
be a long
way,
but it's
inspiring me
so much
about that.

Yeah.

Cool.

Okay,
from the
live chat,
Jason Fox
wrote,
what number
of expert
demonstrations
did each
task require
to train
a good
prior
preference
model?

So,
in previous
research of
R,
we used
five
trajectories.

Generally,
in,
you know,

RT1,
RT2,
they use
like several
thousands
in order
to do
like a
single
task.

But,
in our
task,
we use
five
trajectories.

Yeah.

Yeah,
that was
an interesting
part with
the
behavior
cloning,
but then
there was
a second
plateau
of
performance
with the
secondary
prior
shaping.

So,
what is
the lift
or the

improvement
from the
expert
trajectory?
What is
that secondary
improvement?
And then,
do you
think that
that could
be replayed
back to
a human
expert
to kind
of update
their
improvement?
or is
it
learning
a
technique
that
doesn't
really,
for some
reason,
it's
more
optimal,
but not
for a
human or
something?
So,
yeah,

when we
say it's
going to
be learning
the
suboptimal
policies and
then be
better than
the suboptimal
policies,
we are
trying to
like,
we're
trying to
prove that
the,
like,
our algorithms
kind of
use less
expert
demonstrations
than,
like,
normal
behavior
cloning.
And
eventually,
right,
the prior
preference
model
predicts
the expert
demonstrations

accurately,
right,
and then
eventually
through the
process of
self-imitation
learning,
right,
so when
we're trying
to collect
all these
interactions
from,
between the
agents and
the
environment,
eventually,
the agents
also
explore the
environment,
right,
when the
agent is
currently
having,
like,
suboptimal
policies
equal to
the expert
demonstrations,
it also
explores
the environment

and using
epistemic
terms,
right,
and then
it can
try to
explore the
more optimal
policy,
and then now
when we have,
like,
further
interactions
between the
agents and the
environment,
that is more
optimal than the
suboptimal policy,
we save that
into the memories,
right,
and then the
prior preference
model will
look at those
memories and
then see that,
oh,
this one is a
better interaction,
this one is a
more optimal than
the suboptimal
trajectories from the
experts,

then the prior
preference model will
try to learn that
and try to predict
the more optimal one
instead of now
predicting the
suboptimal
expert trajectories,
right,
so with that,
when the
prior preference
model predicting
the more optimal
trajectories now,
it can guide and
influence the
policy learning,
right,
so that is the
reason why
the agents
can outperform
the experts
in case of,
in case here we
leverage, like,
self-imitation
learning and
prior preference
learning,
right,
in case to
answer your
question about
can we go back
and improve the

expert,
I would say yes,
that would,
there's several
different ways to
achieve this,
right,
so we can
treat the
prior preference
model as another
expert, for
example,
and then we
have the
expert model,
right,
we have the
expert model,
and then now
the role is
changed,
right,
the main
agents,
the active
inference agents
now is an
expert,
and then it's
trying to
interact inside
the environment,
and then produce
all these sort
of interactions
between the
agent and the

environment,
and then all
these interactions
are saved into
the memories,
and then the
memories used to
train the
experts using
normal supervised
learning,
so that would
increase the
amount of,
increase the
optimality of
the expert
itself,
so yeah,
so that
answer the
questions.

Yes,
cool.

Anything else
you want to
mention or
point towards?

I think for
now it's
good,
yeah.

Yeah,
that was an
awesome presentation,
really cool to
see how going
from that

basic state
estimation,
this project is
really taking it
all the way
through the
vision language
action models,
and cool to
see.

So thank you
again,
Viet,
for joining.

See you next
time.

Thank you.

Bye.

Okay.

Next,

I will play
a few
pre-recorded
videos.

We're over
with the
live presentations
for day two,
and those will
return tomorrow.

I'm going
to play
several
pre-recorded
videos,
and we'll
see how far
we get on

this stream.

First video

is going to

be a 57

minute

by J.F.

Kotcher,

research fellow

at the

Institute.

So,

here it

goes.

Here it

is going to

be a

research fellow

at the

Active

Influence

Institute,

where I carry

out a research

project in

artificial agency.

technology.

The title of

my talk is

The Phenomenal

Dash Logical

Robot.

My research

project asks

the question,

what does it

take at a

minimum for an

autonomous robot

to learn to
survive in a
world it knows
initially nothing
about?

So, in the
hope of finding
answers, I'm
programming a
mortal robot to
have felt
experiences, to
make sense of
them, and to
grow the
agency it
needs to
survive.

The robot in
question is a
rover made
of Lego
EV3 parts
that is
equipped with
a variety of
sensors and
motors.

So, what
this project is
not, first
of all?

Well, this is
not an input
output simulation
of human
level cognition,
which is what

LLMs do.

It is also
not meant to
be a general
model of
cognition.

It is, at
best, an
attempt at
implementing one
form of
basal cognition
for a
simple robot.

This is not
a robot solving
free energy
principle
equations.
in the same
way that
bacteria do
not do
math.

And this is
not a robot as
an extension of
its programmer.

Very importantly,
the robot's
experiences are
not going to be
grounded by its
observer.

It has to
make its own
sense of its
own reality.

So, what is
this?

What is this
project?

Well, I want
to achieve
basal cognition,
not simulate
the more
complex human
cognition.

I am building
an artificial
agent, not
proposing a
general model
of cognition,
though this
project is
principled.

Free energy
principle is
applied here as
an analytical
framework, not
in implementation.

It will be used,
but a
posteriori.

And finally,
experiences are
to be intrinsically
meaningful to the
robot, and not
necessarily to me,
its programmer.

Well, this
project is a

learning by
building
exploration,
which is at the
intersection of
phenomenology,
active inference,
and agency.

Phenomenology
seeks to
determine the
essential properties
and structures
of experience
itself, and to
provide an
objective account
of subjective
experience.

So what is
experience, and
not is this
experience correct
in terms of
some objective
reality?

Active
inference.

Well, according
to active
inference, an
enduring self
actively infers
the causes of
its sensory
input and
shapes its
niche to

minimize

prediction

errors.

And agency.

Well, agency

is the ability

of a self to

act purposefully,

to achieve its

own goals, in

an uncertain

world it

experiences.

So my project

is to program a

robot so it

learns to

survive by

experiencing,

engaging, and

making sense of

its world.

world.

And the

robot is to

do so all

on its own,

unassisted by

its programmer

or some

observer.

Okay, so

first let's

look at the

principles guiding

this project.

Well, to have

agency, a

self must be
mortal.

We take that
position.

I take that
position.

agency implies
a self that
is distinct
from its
environment.

I think that's
pretty straightforward.

Though the
self is not
disconnected from
its environment.

So distinct
but connected.

A self is
not a thing
but processes
that are
organized to
endure.

In other words,
a self is a
dynamic
organization of
processes that
constrain and
sustain one
another.

The
organization is
not fixed
but can
adapt to

avoid

dissipation.

What matters

is the

lack of

discontinuity.

A self's

overarching goal

is to

survive.

And survive

in a

dissipative

environment

it depends

on.

The

processes

that produce

the self

must adapt

their

organization

to preserve

continuity

of being.

A self

strives to

remain a

self.

but not

necessarily

to stay

the same.

Mortality

is critical

because it

introduces

value.

What helps

the self

persist

is good

for it.

What

threatens

to

terminate

it

is bad

for it.

Value

is

subjective.

Here

we see

a small

bacterium

that's

breaching

a larger

one to

feed on

its

insides

or

clearly

what's

good for

one

is bad

for the

other.

So for

our purposes

let's define

an agent

as a

mortal

self

choosing

how it

survives.

So to

survive

an agent

must make

enough sense

of itself

in its

environment

to predict

well enough

the consequences

of its

actions

or inaction.

To

endure

a self

must have

felt

experiences.

That's

very important.

So

what are

experiences

and where

do they

come from?

First of

all,

well,
the agent's
environment
impinges
on its
senses
experiences
and so
do the
agent's
own
actions.

The agent
must
constantly
process
streams
of
multimodal
sensations.

The
agent
integrates
these
sensory
streams
in
space-time
in order
to create
its
experiences,
actively
create
its
experiences.

So,
for example,

when I
biked
into
an
apple,
I
experienced
its
taste,
crunchiness,
color,
weight
at once
in
unified
moments
of
experiences.

It's
worth
noting
that an
agent
did not
be
aware
it
is
having
experiences
in
order
to
have
them.

But
that's
the

word
in
different
topics.
I think
if we
go deeper
into
this,
we
start
entering
the
domain
of
consciousness.

Let's
not do
that.
Not
yet.
So,
the
agent
keeps
track
of
risks
to
survival
by
constantly
measuring
its
well-being.
It
is
mortal,

and
measures
of
well-being
indicate
how
its
survival
is at
risk.
Let's
posit that
the
well-being
of an
agent
is a
function
of
three
things.
Its
fullness,
how
energetic
it is,
its
integrity,
how
healthy
it is,
and
its
engagement,
how
involved
it is.
And let's

assume that
the functions
calculating
well-being
are
predefined
and
unchanging.
They are
given to
the
agent
at birth,
so to
speak.
So,
the
agent
will
correlate
its
experiences
with
fluctuations
in its
well-being,
detected
fluctuations
in its
well-being.
And this
is how
it will
have
felt
experiences.
An
agent

can thus
have
good
and
bad
experiences.
So,
sense-making.

An
agent
must
make sense
of its
world
if it
is to
survive
in it.

It
cannot
just
act
blindly.

An
agent
makes
sense
of its
experiences
by
modeling
what
causes
its
sensation.

It
builds
a

causal
theory.

The
causal
theory
explains
what
causes
the
sensations.

The
causal
theory
is
used
to
make
predictions.

It
is a
generative
model.

It
generates
incoming
sensations.

Now,
actual
sensations
that
contradict
the
predicted
sensations
become
prediction
errors.

This

happens
when the
causal
theory
is
inaccurate
or
complete
or
when
the
world
has
changed
since
the
causal
theory
was
formulated.

This
is
basic
predictive
coding,
predictive
processing
principles.

What an
agent
senses
will be
the
uncontested
predictions
it makes
plus the
prediction errors

it receives.

so the

perceived

sensations

are a

combination

of

predictions

and

prediction

errors.

What the

agent

experiences

then is

limited

to what

it can

predict

but what

it can

predict

changes

over time.

Let's not

forget that

the

causal

theory

that generates

these

predictions

is going

to be

changing

and evolving

over time.

So

we
remember
that an
experience
is the
integration
of streams
of
sensations
and
so an
experience
will be
surprising
when it
incorporates
sensations
that are
prediction
errors.

So the
more
prediction
errors
are
integrated
into an
experience
the more
surprising
the
experience
will be.

So
at first
an
agent
will

hold
a
trivial
causal
theory
one
that
simply
predicts
that
nothing
changes
so it
predicts
unchanging
sensations
but
clearly
expecting
a world
that never
changes
leads
quickly
to
many
prediction
errors.
The
agent
updates
its
causal
theory
when
too
many
experiences

are
surprising.
As
we've
discussed
the
agent
uses
its
causal
theory
to
predict
incoming
sensations
but
also
to
predict
what
actions
favor
good
experiences
over
bad
one.
So
the
causal
theory
is
involved
in
planning.
So
not
all

experiences

are

worthy

of being

acted

on.

So

we need

to

look at

the

concepts

of

attention

and

goals.

The

agent

pays

attention

to

the

experiences

that

matter

most

to

its

survival

those

it

feels

most

intensely.

Those

will

be

primarily

the
ones
that
feel
worse

but
also
the
ones
that
feel
best.

The
agent
will
assign
itself
the
goals
of
impacting
experiences
that
it
pays
attention
to.

Experiences
it
doesn't
pay
attention
to
will
be
neglected.

The
confidence

an agent
has
in
individual
experiences
will
modulate
how it
responds
to
them.

So
an agent
seeks
to gain
confidence
in its
experiences
before
it
attempts
to persist
good
ones
or
attempts
to
terminate
bad
ones.

An
agent
has
confidence
in
experiences
that are
not

surprising
which is
experiences
that are
not
made up
of
many
prediction
errors
or
if
they
are
surprising
it
will
have
confidence
in
experiences
that
are
enduring.
So
even
if it
is
surprising
the
longer
an
experience
persists
the
more
confident
the

agent

will

be

about

it.

An

agent

will

prioritize

existential

goals

it

will

prioritize

goals

so that

they

maximize

its

well-being.

But

when

survival

is not

at stake

the

agent

can

choose

to

act

just

for

the

sake

of

new

experiences.

It
will
pursue
exploration
goals.
We
say
that
it
babbles.
Okay
so
having
reviewed
at a
very
top
level
the
principles
guiding
this
project
let's
now
look
at
its
implementation.
Implementing
a robot
that
has
agency
does
present
significant
challenges.

Well
first
somehow
the
robot
must
be
made
mortal.
So
it
has
good
versus
bad
experiences.
Somehow
well
the
robot
must
experience
its
world.
it
must
make
sense
of
its
experiences
and
act
to
promote
good
one
and

do
that
entirely
on
its
own.
Which
means
that
the
robot's
experiences
its
sense
making
and
its
goals
must
be
entirely
its
own
not
the
programmers.
Now
what role
does
active
inference
play
in
the
implementation
of
this
project?

Well
we have
to
recognize
that
active
inference
does
not
tell
us
how
to
program
a
mortal
agent
especially
one
as
we'll
see
that
has
an
expanded
cognition.
That's
fine
because
that's
not
its
purpose
or
intent.
what
active

inference
does
is
formalize
what
is
required
informationally
for
itself
to
persist
in
a
dissipative
world.
So
active
inference
provides
us
the
requirements
on
the
implementation
of
our
autonomous
agent.
If
I'm
to be
successful
the
observable
states
of

the
robot
will
conform
a
posteriori
to
the
requirements
set
forth
by
the
free
energy
principle.
Now
let's
talk about
building
a
robot's
mind.
So
how
am I
programming
a
mortal
experiential
and
agentic
robot?
Well
my
implementation
draws
inspiration

mainly
from
cognitive
biology
biology
biosemiotics
and
philosophy.
I'll
find
inspiration
in
concepts
from
cognitive
biology
such as
organism
being
mutually
sustaining
and
constraining
processes
concept of
multi-scale
competency
stress
as
cognitive
glue
problem
solving
in
morphospace
and
cognitive
light

cones.

Those

of you

who

follow

the

work

of

Michael

Levin

will

recognize

these

as

some

of

the

key

concepts

underlying

his

work

and the

work

of

his

lab.

I

also

take

inspiration

quite a

lot

from

biosemiotics

with

concepts

such as

umwelt,
aboutness,
signal
processes
and
sense
making.
And my
two
principal
sources
of
inspiration
are
Dr.
Terence
Deacon
and
Dr.
Jesper
Hofmeyer.
I
also
lean
pretty
heavy
on
ideas
that
are
coming
out
from
philosophy
the
application
of
Kant's

metaphysics
its
synthetic
unity
of
apperception
plays a
central role
in my
implementation
and we'll
go into
a little
bit more
details
and more
specifically
I take
very
keen
inspiration
from
the work
of
Richard
Evans
and his
team
in the
development
of the
apperception
engine
again
we'll
talk
about
this

a
little
bit
later
for
you
programmers
watching
this
the
software
is
built
using
the
elixir
and
prologue
programming
languages
and
an
extension
to
prologue
called
constraint
handling
rules
CHR
so
elixir
is a
process
oriented
and
functional
programming

language
prologue
is a
logic
programming
language
and
CHR
is a
constraint
solving
extension
to
prologue
these
are my
programming
tools
the
hardware
used
in
this
project
is
a
combination
of
the
robot's
devices
plus
a
high-end
server
on
the
robot

we
find
a
set
of
Lego
EV3
sensors
and
motors
a
Raspberry
Pi 3
for
computing
aspects
and
a
Dexter
Industries
brick
pi
board
that
interfaces
the
EV3
sensors
and
motors
with
Raspberry
Pi
so
they
can
be
controlled

through
the
Raspberry
Pi
and
the
Raspberry
Pi
is
connected
through
Wi-Fi
to
the
network
making
all
of
these
devices
remotely
accessible
On
the
back
end
I
have
a
128
core
server
that
I
actually
do
need
to

do
the
compute
intensive
processing
of
various
cognitive
functions
robots
that
that
is
a
necessary
piece
of
hardware
because
the
computing
power
of the
Raspberry
Pi
is
far
insufficient
for
what I'm
trying
to do
ok
let's
talk about
mortal
code
can

mortality
be
programmed
can
a
robot
be
made
mortal
well
obviously
the
robot's
physical
structure
is
immortal
however
the
processes
that
animate
it
can
come
in
and
out
of
existence
and
if
these
processes
implement
let's
call it
the

mind
of
the
robot
well
the
robot's
mind
can
be
made
mortal
so
as it
operates
as it
explores
its world
the robot
will grow
a
mortal
mind
this
mortal
mind
is
what
animates
its
immortal
body
so that
it can
survivably
explore
and exploit
the world

it's in
and
the
goal is
for the
robot
to learn
to maintain
sufficient
well-being
or else
its mind
will die
I
built
a
virtual
grid
world
to provide
a virtual
robot
with an
environment
that can
both
nourish
provide food
and damage
the robot
through obstacles
and poison
patches
the reason
why I need
a virtual world
and virtual
robots

is just to
accelerate
development
if I were to
do all the
testing
debugging
in
with
physical
hardware
and in
a physical
world
this would
be
largely
unfeasible
but however
at some
point
a physical
environment
will be
built
that recreates
the conditions
of this
virtual world
but add
all the
unpredictability
messiness
and chaos
of an
analog world
which we
can't do

without
to really
fully test
this
implementation
all right
so
we said
that the
robot
will grow
its mind
as it
explore
exploit
its world
in order
to capture
what it
learns
so let's
talk about
growing
the robot's
mind
more
specifically
so what
are the
architecture
and dynamics
of this
robot's
mind
well
first of
all
I truly

believe that
all intelligence
is collective
intelligence
which means
that the
robot
is not
a monolithic
agent
its mind
contains
multitudes
so the
robot's
mind is
implemented
as a
dynamic
collective
of
cognition
actors
so you
should think
of a
cognition
actor
as a
process
that's
part of
a community
of mind
processes
it is
a rich
interaction

between
cognition
actors
that
instantiates
the agent's
society of
mind
its
collective
intelligence
each
cognition
actor
is
implemented
as a
process
that
makes
observations
which is a
combination of
predictions and
prediction errors
unifies these
observations
into
experiences
makes sense
of these
experiences
assigns value
to them
are they good
or bad
experiences
and then

acts
to have
better
experiences
each
cognition
actor
in a
collective
of
cognition
actors
has its
own
umwelt
umwelt
you can
think of it
as the
self-centered
world
of a
cognitive
process
in this
case
the
cognition
actors
so
each
cognition
actor
has its
own
umwelt
that
it

observes
make sense
of
and
act
on
a
cognition
actor
is made
up of
other
cognition
actors
except
for the
more
low-level
cognition
actors
who
connect
directly
to
sensors
and
effectors
so
this
collective
of
cognition
actors
that
grows
during
the
life

of
the
robot
is
organized
by
their
own
belts
and
organized
into
abstraction
layers
the
bottom
layer
is
populated
a
priori
so
before
the
robot
is
turned
on
by
static
cognition
actors
these
static
cognition
actors
interface
directly

with
the
robot
sensors
I call
them
sensor
cognition
actors
and
with
motors
I call
them
effector
cognition
actors
their
on

belts
is
the
robot
immediate
external
world
it's
the
distance
perceived
the
ambient
light
perceived
it's
the
contact

perceived

so

but the

static

cognition

actors

connect

the

robot's

minds

with

its

body's

sensors

and

motors

so

that's

the

bottom

layer

all

other

layers

come

in

and

out

of

existence

during

the

lifetime

of

the

robot's

mind

they

capture
as I
said
what
the
robot
learns
from
striving
to
survive
so
if the
robot's
mind
is a
collective
how
does
the
collective
learn
to
survive
well
clearly
the
apriori
bottom
layer
composed
of
the
static
cognition
actors
is
not

enough
by
itself
for
the
robot
to
find
nourishment
and
avoid
hazards
additional
layers
providing
abstractions
are needed
to create
the
big
picture
experiences
and
affordances
that are
needed
for
competency
and
survival
pure
low-level
perception
and action
are
simply
not
enough

so
as we
said
the
first row
is
pre-populated
by the
static
cognition
actors
but
all
of
the
layers
are
populated
by
what
I
call
dynamic
cognition
actors
as
the
robot
experiences
its
world
and
learns
how
to
survive
in
it

we
have
to
imagine
that
the
society
of
mind
grows
shrinks
reshapes
itself
by adding
and removing
dynamic
CA
as
cognition
actors
and
for
example
a
dynamic
cognition
actors
with
insufficient
well-being
can trigger
the birth
of another
to feel a gap
in
umwelt
coverage
so

obviously
we have
a bunch
of
static
cognition
actors
that
are
not
in
the
umwelt
of
a
higher
level
one
so
coverage
is
incomplete
and
so
we
fill in
this
layer
and
the
layer
must
fill up
before
a
higher
level
is

started
there's
a
logic
to
the
growth
of
this
hierarchy
of
cognition
actors
so
as
dynamic
cognition
actors
are
added
into
new
layers
well
clearly
they
form
an
abstraction
hierarchy
but
it's not
only
growth
a
dynamic
cognition
actor

will
remove
itself
if it
persistently
serves
no
purpose
if
a
dynamic
cognition
actor
does
not
participate
in
another
umwelt
or
if
it
if
it's
never
called
upon
to
share
its
experiences
or to
execute
its
affordances
if it's
useless
it will

eventually
remove
itself
and
this
will
allow
maybe
new
and
different
cognition
actors
to take
its
place
and
maybe
be
more
successful
so
this
adding
and
removing
of
cognition
actors
and
layers
of
cognition
actors
in fact
explores
the space
of possible

organizations
of the
mind
so
let's say
that this
particular
collective
illustrated
here
is
struggling
to
maintain
the
robot's
overall
well-being
well-being
is going
down
this is
not
working
where
death
is
impending
so
we are
searching
for a
competent
collective
and this
is
not
it

so
the
top
cognition
actor
might
remove
itself
and it
can
because
no
other
cognition
actor
depends
on
it
this
allows
for
other
underperforming
cognition
actors
to also
remove
themselves
and so
on
reducing
the
energy
costs
of the
entire
collective
and thus

the loss
of
fullness
by the
collective
so
we've
essentially
stopped
the
bleeding
of
fullness
of
well-being
and
we're
now
in a
position
to
grow
again
so
the
collective
having
now having
a lower
energy
draw
can afford
to
regrow
itself
and this
time
differently

this is
a lot
of
randomness
in
how
things
grow
so
the
odds
are
that
as
it
regrows
itself
it
will
be
different
from
the
collective
that
existed
before
and
was
found
to
be
insufficient
so
the
robot's
mind
searches

for
an
organization
of
conscient
actors
that
keeps
it
nourished
and
safe
the
robot's
mind
is
doing
problem
solving
in
morphospace
!
so
what
about
experiencing
how
is
that
implemented
well
a
cognition
actor
assembles
unified
experiences
from

its
observations
sensor
cognition
actors
so
that
these
would
be
the
static
cognition
actors
in
the
bottom
layer
have
very
simple
experiences
from
observing
distance
color
luminance
touch
etc
their
observations
of sensor
readings
are
their
experiences
so
while a

static
cognition
actor
observes
and
experiences
the
readings
of its
associated
sensor
a
dynamic
cognition
actor
observes
the
experiences
of its
umwelt
cognition
actor
so
when
a
dynamic
cognition
actor
is
born
it
is
born
with
an
umwelt
a
subset

of
the
cognition
actors
from
the
layer
below
and
that's
a
static
umwelt
and
this
is
what
the
cognition
actor
will
be
observing
it
will
be
observing
the
experiences
of
its
umwelt
but
in order
to observe
it
it needs
to

predict
so
the
cognition
actor
in
an
umwelt
will
communicate
its
experiential
domain
to
the
parents
cognition
actors
so
that
these
can
make
meaningful
predictions
experiences
and
experiential
domain
describes
the
kinds
of
experiences
a
cognition
actor
has

so
if
I'm
a
cognition
actor
I'm
part
of
an
umwelt
my
first
task
is
to
communicate
to
my
parent
these
are
the
kinds
of
experiences
I
have
so
you
can
make
predictions
about
the
specific
experiences
I'm

having
and
that's
what the
parent
cognition
actor
will
do
it
will
observe
its
umwelt
the
cognition
actor
is in
its
umwelt
by
predicting
their
experiences
and then
receiving
or not
prediction
errors
and
the
observations
of the
parent
cognition
actor
are
the

summation
of
predictions
and
prediction
errors
but
predictions
of
experiences
of
its
umwelt
cognition
actors
so
there's a
there's
a sense
where
the
experiences
of your
children
become
your
observation
but
the
parent
cognition
actor
then
unifies
its
observations
into
its

own
experiences
so
multiple
observations
of
its
child
connection
actor
experiences
will be
combined
into
single
um
experiences
so
multiple
an observation
multiple observations
of child
experiences
become
a parent's
experience
so
for
example
having
observed
in
its
umwelt
that
distance
keeps
shrinking

a
dynamic
cognition
actor
would
experience
the
more
abstract
concept
of
approaching
so
the
experience
of a
cognition
actor
becomes
the
raw
material
that
are
unified
into
the
experiences
of
more
abstract
cognition
actors
and so
forth
all the
way up
from

static
cognition
actors
so
clearly
higher
up
dynamic
cognition
actors
have
more
abstract
experiences
than
the
lower
down
cognition
actors
for
example
the
the
experience
of
I
stopped
approaching
is
more
abstract
than
the
experience
of
approaching
which

is
more
abstract
than
the
experience
of
distance
it's
worth
noting
that
experiences
are
constructed
and
named
by the
dynamic
cognition
actors
and
they're
not
constructed
nor
named
by the
programmer
in fact
they need
not
make
sense
to me
as an
external
observer

so
we
talked
about
self
experiences
where
does
the
robot
get
its
feelings
from
the
feelings
it
attaches
to
experiences
so
that
the
experiences
are
either
good
or
bad
or
indifferent
well
everything
a
cognition
actor
does
increases

or
decreases
its
feeling
of
well-being
so
well-being
is
determined
as we
saw
by
fullness
integrity
and
engagement
measures
any
internal
or
external
work
by
a
cognition
actor
reduces
fullness
in
predefined
ways
for
example
spending
time
over
a

food
patch
will be
captured
by a
cognition
actor
as
increasing
fullness
or
hitting
an
obstacle
will
be
perceived
by a
cognition
actor
and it
will
reduce
integrity
!
A
cognition
actor
participating
in
a
new
belt
will
increase
its
engagement
so

throughout
its
life
each
cognition
actor
comprising
the
collective
that makes
up the
mind
of the
robot
will
have
fluctuations
in
its
own
well-being
and
this
fluctuating
well-being
determines
whether
a
cognition
actor's
concurrent
experiences
feel good
or bad
It might
be good
to note
that

well-being
measures
are not
entirely
separate
For example
activating
motors
is work
that causes
loss of
fullness
it's an
energy cost
but also
a loss of
integrity
we're going
to be
simulating
wear and
tear
while also
increasing
engagement
we are
moving into
our
environment
so we
are
engaging
Loss
integrity
is regained
over time
think of
healing

for example
but that
drains
fullness
and on
and on
Now
if a
cognition
actor
senses
touch
senses
a collision
it will
have a
loss
of its
integrity
well-being
measure
it will
experience
a loss
but
it's important
to note
that the
loss or
gain
in well-being
of an
individual
cognition
actor
will diffuse
throughout
the collective

we can
think of
losing
well-being
as
stress
so losing
well-being
causes
stress
stress
and
stress
is
contagious
stress
experienced
by a
cognition
actor
and in
others
umwelt
will become
will be
will dissipate
will diffuse
to the
parent
cognition
actor
which will
then
diffuse
downward
to the
children
and upward

and eventually
through the
connections
the
connections
of
cognition
actors
stress
experienced
by one
will diffuse
and be
experienced
by everyone
and because
a
cognition
actor
cannot
distinguish
between
the
self-induced
change
to its
well-being
and a
change
caused by
the
diffusion
of
well-being
from other
cognition
actors
well

one
cognition
actor's
problem
eventually
through
diffusion
becomes
every
cognition
actor's
problem
now
when
a
cognition
actor
feels
a
a
reduction
in
well-being
it feels
stress
and stress
alters
its
behavior
a
stressed
cognition
actor
is more
likely
to try
something
different

from what
has worked
in the
past
obviously
things
need
something
different
needs to
be tried
but a
comfortable
cognition
actor
one that
does not
feel
stressed
is more
likely to
settle
into
habits
why
change
what
has
worked
before
now
let's
take a
bit of
a dive
into
acting
so

a
dynamic
cognition
actor
again
it's
part of
a
collective
of
dynamic
cognition
actors
acts
for
different
reasons
one is
to
confirm
its
experiences
and
then
to
terminate
its
worst
one
or
and or
persist
the
better
ones
so
to
confirm

persist
or
terminate
an
experience
a
dynamic
cognition
actor
must
first
identify
the
observed
umwelt
experiences
that it
unified
into its
own
experience
that it
now
seeks
to
impact
so
the
experience
of a
cognition
actor
is
a
unification
of
observed
experiences

by
cognition
actors
in its
umwelt
so
if I
want to
change
as a
parent
cognition
actor
if I
want to
change
an
experience
I
have
I
have
to
understand
its
composition
in
terms
of
the
observed
experiences
of
my
umwelt
cognition
actors
now

the
cognition
actor
plans
will
plan
how
to
alter
these
umwelt
experiences
so as
to
impact
its
own
experiences
as
intended
the
dynamic
contractor
each
one
has
a
causal
theory
and
we'll
see
more
in
details
how
it
acquires

it
that
explains
the
observations
it makes
in its
umwelt
and
thus
it will
use
this
causal
theory
to
anticipate
what
would
happen
if
experiences
in
umwelt
were
to
change
this
way
or
that
way
would
this
achieve
its
goal
of

persisting
or
terminating
one of
its
best or
worst
experience
so
knowing
how
its
own
experiences
are
made
up
of
umwelt
experiences
having
a
causal
theory
that
explains
how
the
observed
umwelt
experiences
change
over
time
then
it
can
formulate

a
plan
to
change
these
umwelt
experiences
in order
to
alter
its
own
experiences
so
question
number
one
as we
said
which
observed
umwelt
experiences
need to
change
to
impact
my
experience
says
a
cognition
actor
and
then
cognition
actor
would

say
okay
these
are
the
changes
that
need
to
happen
to
the
experiences
in my
umwelt
that
I've
observed
and
unified
into
my
own
experiences
that
I
want
to
change
and
it
will
formulate
a
plan
as
a
set

of
directives
and
what's
a
directive
it's
simply
a
to
modify
impact
and
experience
in
its
umwelt
so
it's
basically
a
goal
a
plan
is
a
set
of
goals
that
are
sent
to
the
umwelt
for
the
umwelt

to
accomplish
so
the
parent
dynamic
cognition
actor
sends
goals
to
its
umwelt
goals
to
persist
or
terminate
experiences
that
are
being
had
in
the
umwelt
but
it
doesn't
tell
its
umwelt
exactly
how
to
change
these
experiences

the
umwelt
cognition
actors
will
interpret
these
directives
they receive
and then
make
their
own
plans
because
their
experiences
are also
made
of
observations
of
their
own
umwelt
so
this
this
this
way of
at a
high level
saying
i want
you to
accomplish
this
and then

that the
lower level
say okay
fine
and i'm going
to decide
on my
own
how to
accomplish
it
and then
ask the
lower level
to do the
same
this
implements
a multi-scale
competency
again
for each
goal
it takes
on
a
cognition
actor
interprets
it
as
directives
planned
directives
that it
then
sends
to

its
own
umwelt
for
them
to
interpret
and so
on
so
plan
action
is
goes
all the
way
down
from
the
top
to
the
bottom
all
the
way
down
to
effector
cognition
actors
where
a
directive
might be
simply
to
spin

a
wheel
now
a
dynamic
cognition
actor
keeps
track
of
directives
that
are
sent
and
the
goal
they
are
meant
to
achieve
directives
that
appear
to
have
achieved
their
goal
are
remembered
as
affordances
for
later
reuse
effector

cognition
actors
at
the
very
bottom
have
atomic
preset
affordances
such
as
spin
or
reverse
spin
whereas
the
cognition
actors
higher
up
the
dynamic
cognition
actors
discover
abstract
affordances
such as
go forward
or even
go up
the
luminosity
gradient
so
as you

go up
the
abstraction
layer
the
affordances
become
more
abstract
more
high
level
affordances
are
discovered
and named
by the
dynamic
cognition
actors
not
by the
programmer
that's
important
in fact
they don't
even need
to make
sense
to an
external
observer
such as
myself
the
programmer
now

let's
look at
how
cognition
actors
build
the
causal
theories
they
use
to
predict
incoming
observations
and plan
their
actions
!
so
a
cognition
actor
observes
its
umwelt
one
time
frame
after
another
and a
time
frame
is
thick
now
and the

more
abstract
the
cognition
actor
the
longer
its
time
frame
so we
could
say
that
a
more
abstract
cognition
actor
has
a
bigger
cognitive
light
cone
if you
wish
and
observation
made
within
a
time
frame
are
considered
synchronous
!

So even
though they
may be
made
sequentially
within a
time frame
of say
half a
second
two
seconds
five
seconds
they are
considered
to have
to be
concurrent
to be
happening
at once
in the
real
world!

So a
dynamic
function
actor
constructs
its
causal
theory
to make
sense
of
what
it

observes
within
each
individual
time
frame
and
this
causal
theory
is a
form
of
induction
that
is
the
production
of
general
rules
from
specific
examples
examples
being
the
observations

A
causal
theory
here
is a
set
of
if
then
rules

in
the
program
it's
a
set
of
if
then
rules
that
are
constructed
and
live
about
three
things
three
kinds
of
things
what
observations
ought
to
occur
at the
same
time
in
the
same
time
frame
what
observations
cannot

coexist
in the
same
time
frame
they're
mutually
exclusive
and
what
kind
of
observations
would
follow
from
previous
observations
for
example
we
might
have
this
causal
theory
that
says
if
the
distance
is
getting
shorter
then
nothing
is
being

touched
so
if
you
are
observing
the
distance
is
getting
shorter
you
should
also
be
observing
that
nothing
is
being
touched
and
if
you
don't
you
can
infer
that
nothing
is
being
touched
another
constraint
another
rule
which is

a
constraint
here
that
says
the
distance
cannot
be
getting
both
smaller
and
larger
at
once
so
you
cannot
have
these
two
observations
coexisting
and
a
third
rule
which is
more
of a
causality
rule
proper
is
if
both
wheels

spin
forward
then
the
distance
will
get
smaller
so
this
rule
predicts
what
happens
in
the
next
time
frame
so
these
are
examples
of
a
causal
theory
an
interesting
side
note
is
a
causal
theory
can
posit
hidden

that
is
latent
facts
that
are
needed
to
make
sense
of
actual
observations
so
a
causal
theory
can be
partly
an act
of
imagination
and
this
positing
of
hidden
facts
is a
form
of
abduction
that
is
the
inference
of
likely

causes

from

effects

given

general

rules

for

those

who

enjoy

logic

!

So

the

discovered

rules

that

make up

the

causal

theory

are

expressed

in the

implementation

as a

logic

program

the

logic

program

embodies

a

causal

theory

and it

is

constructed

without
human
intervention
learning
is achieved
entirely
from
experiencing
there's
no
prior
instructions
or
feeding
of
examples
before
the
robot
is
turned
on
and
this
type
of
learning
happens
repeatedly
over
the
lifetime
of a
cognition
actor
and it
happens
for

all
cognition
actors
making
up
the
collective
the
mind
so
the
logic
program
serves
as a
generative
model
you
can
feed
it
the
current
observations
and it
will
produce
the
predictions
of the
next
observations
each
dynamic
cognition
actor
constructs
and runs

its
own
logic
program
though
at first
they run
the same
trivial
program
so
the
whole
thing
bootstrapped
with
every
cognition
actor
have
a
simple
program
that
says
it
is
as
it
is
and
nothing
ever
changes
and
so
the
dynamic

cognition
actor
will
run
its
logic
program
to
predict
the
next
observations
but
also
to
do
what
if
reasoning
before
acting
okay
so
when
predictions
made
from a
causal
theory
result
in
too
many
prediction
errors
that's
because
the

causal
theory
is
inadequate
then
the
dynamic
cognition
actor
will
replace
it
with
a
better
one
if
one
can
be
found
so
how
are
these
logic
programs
that
embody
causal
theories
generated
well
by
an
app
perception
engine

the
app
perception
engine
is a
computer
process
that
inputs
a
conscient
actor's
observations
it
outputs
a
causal
theory
that
explains
them
so
you
feed
in
a
history
of
observations
turn
the
crank
out
comes
a
little
program
logic

program
that
if
you
feed
the
latest
observations
will
produce
the
likely
next
set
of
observations
in
the
next
time
frame
the
predicted
observation
so
when
they
need
a
causal
theory
and
they
will
need
it
pretty
quickly

especially
in
the
beginning
dynamic
cognition
actors
call
upon
the
app
perception
engine
as
a
service
when
they
need
to
replace
their
current
causal
theories
again
they
do
so
because
they
experience
too
many
surprises
when
using
their

causal
theories
or
they
fail
consistently
to
predict
the
consequences
of
acting
upon
their
umwelt
now
finding
a
logic
program
from
remembered
observations
one
that makes
accurate
predictions
about
future
observations
is a
difficult
search
problem
it's
a hard
knot
to

crack
because
the
space
of
all
candidate
logic
programs
the
haystack
if you
wish
is
ridiculously
large
so
to
find
a
good
enough
program
in
what
I
call
a
useful
time
while
it's
still
useful
to
have
that
logic

program
the
search
needs
to
not
look
over
the
entire
haystack
it
needs
to be
biased
so
it
productively
ignores
much
of
the
search
space
that's
where
the
work
of
Immanuel
Kant
comes
in
is
synthetic
unity
of
apperception

it's
a
part
of
its
method
physics
focuses
the
search
on
causal
theories
that are
coherent
and possess
explanatory
power
so
Kant's
theory
says
these
are
the
properties
of
causal
theories
that
are
worthy
they
are
coherent
they
have
explanatory

power
and
you
can
ignore
any
causal
theory
that
does
not
have
these
properties
so
the
apperception
engine
is a
program
that
applies
Kant's
synthetic
unity
of
apperception
to
find
logic
programs
that
encode
coherent
and
useful
causal
theory

so
the
the
original
apperception
engine
was
built
by
Richard
Evans
and
all
at
Google
Deep
Mind
and
I
simply
quote
unquote
re-implemented
and
optimized
it
and it
was a
subject
for my
presentation
last
year
and
there's
a
paper
that you

can
read
to
find
more
of
the
details
okay
so
all
in
all
this
project
aims
to
express
and
connect
all
of
these
concepts
into
working
code
and
this
code
will
animate
a
mortal
and
adaptive
robot
that

is
capable
of
agency
that
is
the
task
so
where's
that
code
well
this
it's
the
open
source
project
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So we'll begin with motivation and background,
and then we'll go into the four main corners of our project,
then implementation, how we did it,
and then how we evaluated how well the model did,

and then just the last conclusions and whatnot.

So first, let's start with motivation and background.

So the first challenge that motivated this project is defining action spaces.

When you consider action spaces, you have to look at continuous spaces and discrete spaces.

Each space has its own advantages, but continuous spaces.

You have to read about computational efficiency and suboptimal solutions.

But defining discrete spaces that are super flexible is really tough, especially when you consider that you have infinitely possible discrete actions, but you need to define a space somehow.

Our next challenge is abstractly representing the environment.

So for planning, an agent needs to understand what its environment actually is.

For humans, we can just say there's a tree, or that's a computer, but an agent can't just do that.

So how do we turn these raw sensor data that we can give an agent into meaningful ideas?

There are some issues with deep learning, like it doesn't have an internal world model, and it learns correlations.

And while deep learning scales by avoiding the structure, active inference actually builds up a structure from the ground up.

And we want to combine deep learning scalability with active inference's structured understanding for a good environment representation.

The last challenge that motivated this project is considering computational efficiency.

In robotics, offloading computation to GPUs or cloud servers is often not possible when you consider issues like latency and bandwidth limits.

This requires us to consider edge processing or on-device processing.

Both deep learning and active inference are computationally intensive for various reasons.

And our goal with this project is to maintain the active inference framework's interpretability and adaptability while maintaining real-time inference and control on edge devices.

To address these challenges, we combine four main concepts.

First, hierarchical collective inference.

As from the diagram on the left, the brain works hierarchically with higher level prediction and planning and lower level perception modules.

Each layer updates beliefs to reduce prediction error.

Hierarchical active inference models this top-down, bottom-up like blue by minimizing uncertainty at each level to guide adaptive behavior.

Information from lower levels is compressed into higher levels.

And you still retain these key concepts, which is the main thing which will address a lot of our challenges.

And it also allows, in practical use, computation can be distributed among multiple levels.

The next concept is a foreign theory.

A foreign theory was originally described in the context of ecological psychology as abstract ideas offered by the environment that can restrict or assist possible actions.

However, this is very adjacent and can be directly used with the active inference framework to enhance policy selection while contextualizing the environment.

We can take the concept of a foreign theory, which can replace traditional states with abstract ideas and reducing the need for compute for inference.

And this serves as a complementary role as behavior emerges from interactions between the agent and the environment,

which bridges the abstract cognition with real-world context.

Another approach we took is the idea of scalable policy selection.

To scale policies, you need to have some sort of structure that can allow you to maintain low computation while still scaling up the amount of policies you consider.

In our case, we use a latent suitability state in an affordance map.

What that essentially means is we have an affordance called suitability, which encodes numerical or categorical evaluations of some region or path that the drone is about to take. So as you can see in the diagram below, low suitability indicates high obstacle counts, and it's not very desirable.

But high suitability means you have lower obstacle counts and you're closer to the target location and the agent wants to be there.

This contextualization allows you to scale up policy selection because you can filter out those low suitability actions.

You don't need to consider them because they're just not suitable.

And this reduced policy set can be efficiently evaluated through EFE without the need for some advanced hardware or off-sourcing compute.

The last area we need to consider is autonomous drone navigation.

So notoriously, autonomous drone navigation has a lot of different issues that we need to consider.

So just quickly going over some of them.

The first is that sensor data is often noisy, which makes perception and localization very tough challenges.

The next is that environmental factors like low light, glare, or clutter can additionally reduce our sensing's

reliability.

After that, you have to consider that real-time adaptation to dynamic and uncertain environments is very difficult.

If you have a drone autonomously navigated in an unknown environment, it needs to be able to move and think very fast to adapt to brand new situations and issues that's never come across before.

After that, we have to think about the fact that there's limited interpretability and robustness in existing navigation algorithms.

We already talked about some of the issues with deep learning, but in the field of autonomous drone navigation, these issues are more and more important.

Then high computational load restricts deployment on low-power edge hardware.

Because in autonomous drone navigation, we're navigating environments where issues like latency are amplified.

Almost everything has to be on edge hardware, but this isn't always possible with the most frontier models.

Finally, there's often difficulty infusing multimodal data from LiDAR, camera, and IMU consistently.

We need to have all these different data sources to enable all the different elements of our navigation.

But diffuse them and provide them to the model is a challenge.

So now let's go into the actual architectural structure.

So we have a three-tiered architecture.

We have high-level inference and planning.

And these lower levels, all of this builds up and sets up the high-level planning to be as efficient and streamlined as possible.

We have our mid-level perception and encoding, which builds our world model.

It builds it from observations and its suitability and then creates a world model and beliefs about the world model, beliefs about the environment.

Finally, we have low-level control, which right now is very simple.

It just executes movement and records change in position to maintain egocentric navigation.

So let's go over these areas a bit more in-depth.

So first is perception, where we take environmental data and process it into a point cloud.

So you can see that we have the Aerosens simulation and we collect data from the LiDAR and camera, which are two primary methods of perception.

We turn these into structured observations, specifically target distance, azimuth elevation, and obstacle distance, obstacle density.

And we use these to update our world models' state base, with distance, azimuth elevation, and suitability here.

Then we take our affordances, which calculate expected suitability scores for the region around us, based on the same perception data we got from Microsoft AirSim.

Then we move to high-low planning, where we sample optimal policies from a filtered action space that's created based on our affordances.

And this high-low planning works to minimize expected free energy with its pragmatic and efficient component.

Finally, we do our final execution, setting it to a very simple low-level module for physical execution in Microsoft AirSim.

So now, let's talk about a factor graph of our architecture.

Our architecture is currently very simple, so there's not much special about our factor graph.

It's a typical message-passing structure between t and $t + 1$, but there's a very key difference.

This S_4 , or suitability, actually feeds directly into the observation, and it dynamically weights that preference to then move into EFE.

We'll definitely talk about this more in detail once we get to the planning calculations, but this is a very key difference, and it highlights a key problem with our architecture.

Because it directly feeds in, it's actually completely deterministic, and there is no probability involved.

It's just, okay, I look at my information, I see this amount of suitability, that's my observation.

We need to make this probabilistic so that the drone doesn't have 100% trust in its sensors, because sensors are noisy, as we mentioned earlier.

Yeah, so that's definitely a feature improvement that we need to do.

Now let's go more in-depth onto the perception and world model area.

So the first thing to consider is the multimodal data acquisition.

As we mentioned earlier, our two primary methods are from the LiDAR and depth camera.

So a lot of it gives us distance data. The effective range we chose is from 0.1 to 50 meters, which is kind of arbitrary, but we found this based on testing in Microsoft AirSIM.

This is the effective range that we need for our purposes.

Then is the depth camera, which is only from 0.1 to 20 meters, and this gives us very accurate depth data.

This is an image you can see of what this some kind of sensor data would look like.

For us, what we decided to do was use a db scan algorithm to cluster all of the fuse points we get.

And we have an epsilon of 1.0 meters, which basically means that the max cluster size is 1.0 meters.

So in each area of 1.0 meters, you're going to cluster into one point and the min samples 2, meaning for one cluster, you need at least two points.

This process then abstractly represents the raw points into a discrete set of obstacles, and this internally performs denoising.

For example, if there was a random point somewhere in an individual point, then the db scan algorithm would ignore that because it's below min samples 2.

Then going more into this for preprocessing and fusion, we take our raw sensor data and filter it for some additional noise reduction and remove outer range points within the effective ranges we talked about.

And there's an optional subsampling in case computational efficiency means that we can't handle large amounts of data.

For fusion, we again just take both water data and camera data and transform it into a single unified coordinate frame,

which creates one large point cloud, which is more similar to the image you see here.

Then we do our data packaging and output for the technical part of it.

So, so far everything's been done on Python and we're calculating more information about the target, like angle, elevation, and distance.

This is mostly supplied by information from the Airstream function, which is kind of like the replacement for the ion view simulation.

Although obstacle data is also packaged here, we take the obstacle positions, which we got from db scan and the obstacle density, which is a very simple calculation that represents how dense the environment is.

And this information is then finally packaged into a JSON file and we use a ZMQ system to communicate between the Python and Julia areas of a project.

So all of this is in Python and ZMQ then sends it to Julia, which is going to go over our planning module and it's going to process it.

All right. So let's go into this actual suitability calculation.

So our suitability looks like this. We have a weight, we have safety factor, another weight, and then density factor.

The weights are just normalized to add to one. And then we have these weights here. These are hyper parameters.

There's a lot of room for improvement, which we'll talk about later.

So here's how the safety factor looks. It's an exponential where essentially, when d , the obstacle distance, is very close to zero, you immediately know it's unsafe because it's an obstacle right in front of me or right next to me. And our safety factor crashes as it's an exponential decay function.

So then as d approaches larger and larger numbers, it gets safer and safer and safer because the exponential

increases. And then we know, okay, I'm not close to an obstacle. Let's just ignore and let's keep going.

And then we also have this penalty, which is like, if there's an obstacle within 1.2 meters, we replace safety or we multiply safety by an exponential negative 1 over d , which basically says, okay, there's this obstacle that's some amount of close to me. Let's quantify that and then go from there. So this is kind of a penalty for very unsafe environments. When there's a lot of obstacles, it comes into play a lot. Then let's go to the density factor. It's based on an obstacle density. Both of these are considered observations, 0.4 and 0.5, as we saw earlier. So we take the density factor, which is very similarly structured. It's based on obstacle density. And then there's a cutoff for both. But for this one, the cutoff is more noticeable. This is like just a density threshold that if it gets below this, it's like it just, the density factor becomes 1 immediately, because we know, very safe. And then steepness is just quantifies how sensitive it is to changes in density. So as density gets low, as you can probably guess, the factor gets close to 1 , so that, okay, I know it's quite safe. And then as density gets very high, the factor gets lower to lower suitability. There's also another density penalty for high densities.

It's a bunch of stuff, but essentially it takes the maximum of a hyperparameter and then a calculation with obstacle density and checks, okay, how much should I scale the factor to penalize this factor so that suitability begins to lower? That eventually ends up with a normalized score from 0 to 1 , which represents this overall navigability and safety. Now you can clearly see that there's a lot of rule-based elements that require hyperparameter sweeps. And that is a very, very glaring problem with architecture that has a fairly arbitrary fix, which we'll definitely talk about later. Let's go over the overall planning process. So we start with waypoint generation.

Once we have our internal suitability, we're able to do this, where we generate 25 to 75 candidate waypoints based on obstacle density. So again, this is another example of the agent taking environmental context and using it to inform its planning. We do a dual strategy where the sampling 70% about it is exploring the nearby space. And there's some target bias sampling, an initial 30%, that's pushing the agent more towards the goal. We have an adaptive radius, which is where the action length is between 1 to 10 meters, which is again, based on internal suitability. So we use a very simple calculation where the minimum is 1, and then you raise suitability to the power of 3, and multiply by 10 minus 1. So this basically means as suitability approaches 0, then the action length would be approaching 1, and vice versa, suitability is going really well towards 1, then our action length is going to be 10. There's an additional check that's performed after this to ensure that we have at least enough to reach a target when you're in some area that's nearby the target. So you can see from this whole process, you have flexible actions. If the agent's in a very open space, then there's absolutely no need for it to be taking a small action. Like 1, we'll be taking something more larger, like 7 or 8, and vice versa, if you're in very dense space, we can take actions of when. And yeah. So let's go more into how this process works.

When we get this set of waypoints, we calculate the expected suitability. This is a very similar calculation as Syphiky talked about, but the intro suitability, it just has a few additional modifications and penalties. So one of them is that there's more conservative parameters used. We pretty much use trigger cutoffs for distance and steepness, which just makes it a safer check.

We then add some additional cluster and octant penalties. So we check for any dense obstacle clusters in any of the 8 octants around the waypoints. And if any octant is dense, which has like 5 more than 5 obstacles, then its local density score is penalized. And if it's extremely dense, like even more, and it has more than 10 obstacles, its local obstacle distance is also penalized.

We also have a few other penalties like corridor penalty, where the path from the current position is checked for narrow gaps. And if the path is too narrow, then suitability is severely penalized.

This means that we're not just considering where the agent's going to end up, but the process of how the agent's going to get there. Finally, we have this target proximity one is where, again, if the agent is close to your target, the waypoints moving more directly towards it receive an additional suitability boost. Yeah. And something to add is these local density score and local obstacle distance are the same things as the factors that we talked about before. So we basically penalize or increase these for expected suitability based on these factors. Finally, we filter. So after all the calculations,

the system discards all these waypoints with a score less than 8 to 0.8. And then you calculate transitions for other states with a suitability of 0.95. And then we calculate EFE. So we evaluate EFE for this action set by encoding paths with EFE based on their location and all these factors.

So our equation looks like this. We take in the current state, so distance, angle, and suitability, and then the action, which is a 3D vector, and then the agent's current beliefs as probability distributions. So we have the goal seeking behavior, the uncertainty reduction term, and the VFE term, which essentially, we'll talk about later, but essentially uses VFE to penalize or increase EFE.

All right, so let's break down each part of EFE in each term. So first, the pragmatic term. It's just

a pragmatic weight. And then preference score, distance bonus, and this magnitude term. So preference score is just your preferences for distance, angle, suitability, as a weighted average, which we'll talk about in the next slide. The distance bonus, which is just your expected distance minus the current target distance, and then factor of two. And finally, action magnitude. So larger actions would incur greater scaling of this, because your pragmatic value should increase, or rather, be minimized as you get closer to the target. So basically, in essence, a higher pragmatic value would lead to more efficient actions, while the adaptive weighing does ensure some caveats in very dense environments. Now let's go over the epistemic value. So just keep in mind, in general, in any acumen-prince framework, the epistemic value is in charge of driving the agent to explore and reduce uncertainty against beliefs. So for example, if the agent's like, for this state, I have a very low certainty in it, what action can incentivize and help me find more information about the state? That's what the epistemic value is encoding. So let's go over specifically in our architecture, what we have. So the equation is very simple. We have this epistemic weight, which is by default 0.2. It's a hyperparameter. And we do dynamically increase this in our system in high density areas, which promotes more acceleration and looking around for safe paths. We then multiply by an additional factor of 0.5. This is arbitrarily chosen by us. After that, we multiply by the total entropy, or which is encoding the uncertainty of the agent's beliefs. This is computed as the equation h of q is equal to h of the q of the location, plus h of q of our angle, plus h of q of our suitability. So in essence, it's just calculating the total entropy of our states as per the argument's framework. This is a more detailed equation as per the belief distribution for how the calculation works, where you have the summation, and you take each part of the belief distribution, and add this term ϵ to it, and take the log of that. It's very standard calculation. And finally, in this, going back to the equation, we multiply it by the action magnitude. Again, if you remember, our actions vary from length one to length 10. And this term just reflects that increased epistemic value for the larger actions. You might think this is a bit strange, but logically, the motivation is, if an agent is moving further, then it's able to uncover more information about the environment, and thus, it has a greater epistemic value. Just note, because of the way things are set up with the architecture, we are multiplying this by a negative, and our total EFT ends up being negative. This is just how things are supposed to be. And overall, once we get this epistemic value, we see that a higher epistemic value is going to incentivize the agent to reduce uncertainty about its environment, thus balancing exploration and goal pursuit. Our final part of EFE is this BFE penalty. This is a very non-traditional and not super necessary term, but it is still there, so we decided to talk about it. It's a small regularization term that links our planning to the inference. Because we don't have very explicit perceptual inference, we do this term. So λ is just a small fixed hyperparameter, which ensures the penalty is about 10% of EFE, so it doesn't really affect EFE that much. And then VFE is just your variation for energy calculated during inference. It measures how well current beliefs, explain the observation, essentially just our traditional EFE, VFE. And we have complexity, or your entropy term, and then you have belief accuracy,

such as your expected probability of alteration over states, i.e. your likelihood. And then your prior belief, just a simple belief updating system, which is how VFE is struggling.

Also, this all leads into control, which is just our final steps. These are simple errors and functions, they take in all these values, most of these values are completely irrelevant, but you have x, y, z, and velocity, you move it, move from waypoint to waypoint, and then rotate to yaw, just make sure sensors stay straight. So you rotate to a given yaw, based on which direction you are facing when your action is moving. Well, that was a lot, but we finished covering our architecture, and now it's time to look at the actual experiment we conducted in Microsoft AirSim, going over testing results.

So this is the way the experiment is set up. Again, our core simulator is Microsoft AirSim, which is built on Unreal Engine. It's a very useful thing. It's a high-fidelity open-source platform, which is built for rapid and safe testing, which was able to drive all of our development in this project so far. It provides very realistic drone dynamics and simulated sensor data from LiDAR and camera. So you can see we have this environment. It's very realistic, where the LiDAR and camera sensors are able to pick up very detailed images of different things. This is a picture of our core environment. It's called AirSim neighborhood, and it's a complex suburban environment that houses trees and other obstacles. So this was a very balanced environment to choose for a drone. It's not too sparse where the entire thing is open, but it's not super dense where our simple architecture can't exactly so it can at least have a chance. There's areas where there's open road, which is where the drone can adapt and take longer actions. There's also areas where sometimes targets are located near houses or trees, and the drone needs to take smaller actions. So this allows us to extensively test our architecture. We have verified targets that are made sure that it's at least possible, and we pseudo-randomly generated them in the range of 20 to 80 meters for the purpose of this experiment. We fully ran 97 episodes, and 96 of them passed with very low collision amounts, which we thought was pretty good. Let's go more in-depth into the data we collected and the results we got.

So for EFU minimization, as you can see, over all the episodes, the effect of free energy reduces as the agent gets closer to target, as intended. It shows the agent minimizing uncertainty as those both the pragmatic and epistemic values lower as it understands, okay, I know where I'm going, I know what I have to do.

The next is exploration and exploitation balance. So we concluded that the agent demonstrates an effective balance of exploration and exploitation. Let's look at the graphs to analyze this. So this is our main graph. It compares the pragmatic value and epistemic value. Red is pragmatic, blue is epistemic. In comparison to the distance, so the graph goes as distance target decreases, and we have on the y-axis, the log normalized component. We see that throughout the entire run, that the epistemic value is fairly consistent, and it's a pragmatic value that's nearly changing. So at the left side, the agent is prioritizing more exploration because initially it has more uncertainty about the environment. And this means that the dominating value is the epistemic value. At some point around here, the pragmatic value starts to play a much greater role. So as the agent gets closer to the target, the pragmatic value starts taking over, and the agent

prioritizes exploitation, which means it's acting with more goal-oriented behavior. So it has less uncertainty, and it's

working towards reaching a target as fast as it can. These are some more detailed graphs that show individual elements of the pragmatic and epistemic component, and find more information on the exploration and exploitation balance. Yeah, if you're interested, a lot of this is up on our website that we mentioned earlier.

So for computational efficiency, which was a big thing for one of our challenges, we measured it for the time it took for each planning step. As you can see, EFE specifically, which was the main thing because VFE wasn't super impactful here, was very short, only taking 6.01 milliseconds on average with Rx and FUR. And this demonstrated suitability-based waypoint filtering that actually reduced the computational cost. The total processing took 117 milliseconds on average, mostly due to error-prone calls and some other miscellaneous processing. But all of this computation was run on a standard CPU. There was no outsourcing or anything. So this is pretty promising for the practical implementation of vector inference. But there's definitely room for improvement. We talked about a lot of the Rx and FUR people, they said this can definitely be improved. So we hope to get that computation down as we move to a real drone. Here's just the general results.

Yeah, and the last thing to look at was after this main experiment that was mainly used to evaluate our model, we did run it on a more challenging environment. The reason we're not looking at the main data from it is that the success rate was only 50% 17 out of 34 episodes that we ran, with the condition that there was a six minute timeout per episode. So there was a couple of issues for why the drone didn't have that great performance. But as we're going to get to later, and as you briefly mentioned throughout the entire presentation, this isn't due to fundamental issues with the architecture, but more so as to the implementation. And of course, our future work is going to address the main fundamental issues that do actually play an impact. So going more over them, we saw that the drone often got stuck and had a higher collision amounts in this environment due to two main things. The first is myopic planning. So the agent essentially lacks a global strategy. It's planning more reactively than we wanted to, which is over considering immediate safety and proximity to the goal, which leads to poor longer term positioning. So for example, if a drone would get stuck in an area, it would be taking actions that maybe would help it a little bit, but don't strategically move it out of that bad area and out into the open. This also plays more into learning instability, where the agent is unable to fully learn from the mistakes it made. And it would have gotten a mostly good strategy after a single failure. Yeah. So in this challenging environment called Zhang Jiajie in AirSim, it starts within all these trees, which makes it immensely challenging, but we have confidence that our architecture should be able to do this if these issues are circumvented. An example of this myopic planning would be like, it would get itself stuck in like one of these little buildings and it just would not know how to get out because it doesn't have that long-term vision to think, okay, I can go up and around. Instead, it just thinks I have to get to my goal. I have to avoid an obstacle. That's it. It doesn't have the

ability to look forward. Thus, we look at some conclusions. Despite those issues, we did see that combining afford theory and a hierarchical active inference does enable efficient, say, of context-aware navigation as we illustrated with addressing those challenges in these complex 3D environments. So we were able to address all three of these challenges in different manners. First, for the first challenge, which was policy selection, we were able to use affordances to create discrete high-level planning with a flexible action space. We were also able to lower computation because we only have to consider certain paths because of the suitability filter. The second challenge, which was contextualizing the environment, we were able to use just affordances, which fundamentally contextualized the environment, and our suitability state was able to contextualize specifically the safety and navigability of the environment. Again, this also decreases computation for the obvious reasons that it just doesn't have to consider on safe areas. It only considers certain parts. And the challenge 3 was we were able to address computational efficiency and see that the model is efficient and capable of running on a standard CPU. And with our preliminary tests on our real hardware, we've seen that it's likely being able to run on edge device, in our case, a Raspberry Pi 4. But there are obviously more powerful devices that could run this even more efficiently. Now, let's talk about what we're going to do in the future, because this is a very much incomplete project, and there's a lot more work to do. So we currently have a built drum. We want to do it fully GNSS 10i and with the open source RGPilot firmware. So we have Raspberry Pi, which is a little bit hard to see, but it's in this big black box over here. We have a LiDAR A2M12, which is currently not attached. We'll put a depth camera on it, specifically Intel RealSense D455. And this entire thing was built upon the Hawkworks F450 drone kit, which came with a PixHawk 2.4.8, which is up here. It allows us to make very streamlined architecture. But there's a lot of issues with GNSS 9 and frames and whatnot, because obviously when you move to the real world, without GPS, there can be a lot of issues. Our goal is to test full architecture in the real world and collect sufficient data, like we did with our AirSim neighborhood environment. And we hope to see some success with this and with ActiveInference as a proof of concept. As you mentioned throughout the presentation, there were some miscellaneous issues that we're working on addressing right now. So we talked a lot about how we have rule-based methods that are essentially like Band-Aid on issues we notice in the sim. But the more fundamentally sound way to fix all of this is of course the Bayesian framework, Bayesian methods. One example is that for the point cloud, when we get the sensor data, instead of everything being deterministic, which then proliferates through our architecture between suitability, you want to have a probabilistic representation of a point cloud, to representing probabilities of how likely different areas are to have a point. This is going to inherently fix some of the issues and simplify a lot of our calculations. This ties more into our second point, which is fine-tuning the suitability weighing in calculations. We have a lot of fixed hyperparameters and unnecessary calculations. Probabilistic

methods are going to reduce a lot of this, but even the way we get some of the hyperparameters can definitely be improved on the more sophisticated methods. After that is action fine-tuning. So the flexible action approach that we took using the ability to internally parameterize where action is, is already pretty good. But we think we can take a more sophisticated approach, where once we sample our first action, we can then make more precise adjustments to where the action is using secondary and additional

expected free energy steps, which is then going to make our drone's movement more and more precise, while remaining still in a fundamentally discrete action space. Finally, due to some issues with aerostimic calls, we have this whole stop and move cycle. Ritodrone takes an action, stops to get its data, which is unnecessarily long, and then move on. We're planning on addressing this with things like a sensor cache and a couple of other code-based adjustments. And hopefully, eventually, the drone will be able to move continuously. We're still planning discreetly, but you're not going to see the drone stopping in between, which is obviously unfavorable in the real world.

Yeah. And just one example on the probabilistic side. So for probabilistic representation of a point cloud, for example, this would forego the need of the corridor penalty, for instance. Because if you have a probabilistic representation of a point cloud, there's literally a cloud, which would prevent the drone from choosing an action that would be very close to the point cloud because it thinks, okay, there's some probability that I could be running to an obstacle here. Just an example.

And the last area is, this is directly addressing some of the challenges we talked about in our challenging environment, learning and cognitive maps. So we want to be modeling our transitions with the approach of cognitive maps. This means that we're essentially looking at expected and current suitability of regions rather than paths. We have a more true application of a foreign theory.

And this is also going to enable us to learn the tendencies of the environment. So for example, in the Zhongshui environment we talked about earlier, in a very dense area, you'll know that the drone is going to be tending to get stuck there. And thus, in general, those kinds of areas are unfavorable. And the drone is going to have the strategy of wanting to get out of those areas entirely, not just move to slightly safer areas. And there's a lot more concepts from cognitive maps that can help address even some of the more

miscellaneous issues of our architecture. And this is going to move us towards a move through hierarchical architecture, where the drone and the agent is thinking in layers and really considering strategy with better environmental contextualization. Thank you for listening to your presentation. And if you want more information on us and the research as it evolves, please visit our website. It'll be consistently updated as the research progresses.

All right. Next up is Bradley Alicia, Open Source Sustainability, Progress and Connections to Active Inference. This is a 53 minute recorded video. Here it goes.

Hello, my name is Bradley Alicia, and I'm going to talk to you today about our project, Open Source Sustainability. I'm going to talk about our progress over the past few years and connections to Active Inference. And this features a number of people's work, including myself, Jesse Parent, Morgan Huff,

and GSOC Scholars, which will be mentioned in the talk. We're from the Orthogonal Research and Education Lab. And we have a preprint on this work at ResearchGate. So what is Open Source Sustainability? So Open Source Sustainability is the way to understand how we can manage working teams and software infrastructure over long periods of time. There's been a history of trying to make software development better through different types of project management techniques. We've had things like Agile and DevOps come around to improve that process. This is held true for both industry and for academia. And we're trying to look at how to understand the key components of this longevity. So we started our project in 2022. And we've taken a number of approaches and we've developed these over time. We started in 2022 with an active inference agent-based modeling approach. And we developed that in 2022. And then we've been maintaining that since. In 2023, we worked with multi-agent reinforcement learning and reinforcement learning and agent-based modeling. In 2024, we worked with a large language model, which we are currently maintaining. And then up till this year, in 2024 and 2025, we worked on multi-agent reinforcement learning and reinforcement learning more generally. So we have three basic approaches, one of which is active inference. And in the talk here, I'm going to highlight some, a lot of work on reinforcement learning, a lot of work on active inference, and some on large language models. So the Linux Foundation has published this report called the Open Source Sustainability Ecosystem. And so they define open source sustainability as the tools, conditions, and platforms under which open source can thrive. So they talk about tools, which is licensing workflows and containerization. They also focus on conditions. So the conditions under which open source sustainability thrives is where we have healthy teams with trust and coordination. And then finally, they talk about platforms, which are automation techniques that enable working teams. They also pose some big questions. And so these are questions that people often encounter in practice. And there are no clear answers to these questions. A lot of projects struggle with this. And no one's ever really developed a good methodology to ensure that you do a good job, because a lot of teams are very diverse. They have, you know, you have different types of projects, you have different aims and goals. So it's hard to really find a single set of answers. The first of these is how to push production builds in a distributed, decentralized manner. So a lot of teams in open source are working in a distributed, decentralized manner. And pushing to production is notoriously hard. It's difficult to coordinate people's efforts. It's also hard to do put a lot of workflows into practice. And so this is something that, depending on the team, is easy or hard. It's relatively easy or hard. It's relatively easy or extremely hard. The second one is FOSS governance. And that is a social contract that Debian proposed about 30 years ago, where they proposed a way to keep your organization relevant and grounded in the type of ethics that you want to see in the world. So a lot of open source software

projects will drift from where they began. And they will decay in different ways. And we want to have a governance structure that maintains projects over the long period of time. The third point is how to maintain collaborations in software systems. So we want to maintain collaborations. We want to have healthy relationships in our teams. But we also want to be able to maintain software systems over long periods of time. And last year, we had an interview with Stefan Buhlmann, who talked about containerization as a strategy for that, where you can build a container and run different types of programs in there, distribute it to different teams, and it makes it easy for them to implement different types. In this case, it was neuroimaging analysis. Then the fourth question is how to coordinate efforts and get things done. Very simple. Now we're going to be hitting on some of these big questions as we go along. But first, I want to talk about implementations. So there have been attempts at this in the past. There have been platforms that people have tried to build, or, you know, different types of models that people have tried to build. And it's kind of hard to do because, as I said, open source projects are very heterogeneous. Some projects focus on, you know, they use one set of tools, other projects use other sets of tools, and they have different aims and goals with their software. Some are driven by large companies, some are driven by small concerns, and others are driven by academic groups. And so they're going to have all different sorts of practices, all different sorts of goals, and their teams are going to be managed quite differently. In any case, the first implementation of the sort of way to measure open source sustainability was SourceCred. And SourceCred analyzed discourse, discord, and GitHub activity. So they picked different tools that teams might use, and they find a way to analyze that activity. So you're dealing with producing code, you're dealing with discussions amongst colleagues, and you're dealing with other types of communications within, say, an open source community. And so this was proposed by Mia Zono on the Protocol Research Labs blog in 2020. Another implementation is GitCompare, where it focuses more directly on GitHub, and only GitHub. So you can, you know, take all aspects of a project or all aspects of an open source community into account, or you can just focus on something like GitHub. And depending on how your measurements, how accurate your measurements are, how robust your measurements are, you know, one strategy may be better than the other. So it might be better just to focus on GitHub if you don't really have a good methodology for analyzing discussions on Discord or Discourse. So GitCompare builds a model of general development activity for GitHub repos based on five criteria. So we're talking about GitHub repositories where project is storing their code, they're implementing their developmental pipeline, and they're, you know, working on building software. And they want to basically know how popular this is with the open source community. And they also want to know how recent things are. So the, and also they want to know if you can protect your assets properly. And, you know, sometimes licenses aren't just about protecting assets. Sometimes they're about mediating conflict in the team. So they base their model on five criteria. There's a creation date, license, forks, stars and watchers. And so stars and watchers are sort of popularity mechanisms, social, social media type measurements. A fork is basically when I copy a project, and I make a copy for myself so I can modify it, which suggests that people find your

project generally useful. The license, of course, is the legal framework or the governance framework.

And then creation date is going to be where it started. So you can look at the creation date.

And if there haven't been updates since then, or, you know, maybe in the last two years, you know that the project is not sustainable. So these are all things that they measure in Git comparisons from the Git compare medium in 2018. And then another approach that was interesting that's been done is a governance surface. So a lot of times we develop these surfaces to describe all the different states of a system that we can describe. And then we explore that surface to find, you know, the transitions between states or sort of the trajectory of the project as a whole. And so this is a governance surface that was developed using an analysis of Discord with chatbots. So they analyzed Discord and chatbots. They developed this governance surface and they described kind of, you know, the discussion activity in that team or in that community. And this was published by Rennie and Qualitative Inquiry.

So the final implementation I want to highlight here is a tool called Grimoriolab. And it's put out by the Chaos Software community. And so this is a tool that takes a number of places where you can track data in open source communities, such as GitHub, Slack, other places where teams meet up together and development

happens. And so we can take these data, we can retrieve them, we can store them, we can do other types of things and enrich them with, you know, all sorts of other analytics and charts and reports. So Grimoriolab analyzes data in the service of being a management tool. This diagram here shows like kind of how we can take data from an open source community and we can look at how well it's performing. And then we can look at those charts and graphs and improve our community. Places where it's failing, we can shore things up, we can optimize things. Now our goal is a little bit different. And in the sense that we sort of base everything on agent based models. And so we're not really interested in being a management tool, we're interested in scenario building. So agent based models are famous in the social sciences for enabling scenario building, more asking what if questions. So you know, we may not observe something directly in a community, but we can model a community as a simulation. And then say what if some intervention happens in the community? What if something good happens in the community? What if something bad happens in the community? What happens if you scale up the community? What does that look

like? And so it gives us a sort of a scenario building tool. And so our goal is scenario building an insight, not performance optimization. Now, one of the interesting things about say using tools like Git or GitHub, and many other project management tools, but we'll focus on Git for now, is that the introduce on individuals, a certain amount of cognitive load. So cognitive load is the idea that you have too many things going on, maybe too many steps in your tasks that you need to perform, or too many things to think about at one time. And it impacts your performance, it slows you down, it confuses you, it, you know, it doesn't lead to optimal outcomes. And so this is a blog post on Git and cognitive load. And this is a cartoon that kind of talks about how we use tools sometimes that are very arcane and very hard to master. And so if you can't master them, you can't really realize your goals. And so this is one of the main sticking points of sustainability is that people can't go from idea to implementation very

easily. And so, you know, there are a lot of ways we can look at and measure cognitive load. One of those is looking at the complexity of programming environments. So we might say, for example, if we look at the complexity of programming environments, we might want to measure how long, you know, if we have long, multiple long processes in our workflows, that might introduce cognitive load. If we require people to learn a multitude of tools to get their work done, that might also induce cognitive load. And then if we force people to memorize many similar or cryptic commands, we can also introduce cognitive load. So this cartoon basically spells out the problem with cognitive load and a lot of open source projects. Now, unfortunately, open source projects, there isn't a focus on mitigating a lot of these things. These are very common things and problems in open source projects. So this is just one example of the things we can measure and the

kinds of constructs we can build looking at open source sustainability. There are a lot of things that are involved in this. So, you know, we think about like individuals in the team, and they're all sort of having to process a lot of information to get their work done. And they can do this in a very efficient way. They can do this in a very efficient way. They tend to stick with the project, they tend to be productive, and the project is a success. If they can't, then it's a failure. And so this is kind of, we'll talk, we'll feed into our active inference formulations and some of our reinforcement learning formulations as well. There's always a learning curve in any organization, you have to learn new tools, you have to learn new procedures. And that learning curve, if your learning curve is too steep, a lot of people won't bother to join the project. Sometimes project-wide compliance is important. If you have to comply with different regulations, sometimes that's a good thing, and sometimes it's onerous thing. So this can work both ways. There's of course, attentional resource management. Do you force people to multitask all the time? Or do you have a work environment where people are doing one thing at a time? Because multitasking can be bad from the standpoint of focusing on any one set of issues. Multitasking can limit your productivity. There also has to be intrinsic motivation and reinforcement. Of course, we're going to address this directly with reinforcement learning. There's infrastructural lock-in. Oftentimes that's indirectly cognitive. It can affect your ability to get your work done. Expertise or mastery. So if you're an expert in a certain area, or if you're an expert with a certain tool, it makes it more likely that you can achieve what you want to achieve in your project. And then the coordination of goals. And the coordination of goals is I want to be able to coordinate my goals across the team. I don't want to just do things on my own and then someone else do things on their own and then combine those at the end. We want to coordinate our efforts in an active way. We want to be able to have this sort of collective cognition where we can, an environment where we can do that. So when we define our different terms here in terms of what we think about, how we think about open source projects. So to define our different terms about what we think of when we think about open source projects, we've identified three things. There's the project, their agents and their tasks. And this diagram shows like a project leader, a maintainer and a contributor. And those are different states of agent. So if we're thinking about this grid, and we're thinking about this agent based model, we have different classes of

member of the team, we have a project leader, a contributor and maintainer, and there are multiple types of agents. And those agents can coordinate their efforts or not. And so we're going to model it in an agent based setting, but this relates to team members. So a project is defined as a set of tasks and agents with different competencies. So you have agents that are contributors, maintainers, project leaders. You also have tasks that are specific to contributors, specific to project leaders, specific to maintainers, specific to other types of people, maybe technical writers, or something like that. When we think of agents, we think of individuals. And those agents usually have some sort of cognitive state. They also have maybe a task or an ascribed role. And agents see this division of labor in a project, and have specific roles on a project, like contributors and maintainers. And then tasks are defined as units of work. Sometimes we have issues, which are just kind of single tasks that are, you know, clearly defined, very tightly defined, to what we call on our simulations, MERS Jupiter shot projects, which are these large multi-stage projects where you have to complete one part of the project before you know what the next part of the project will look like. So we have these tasks, these agents, these projects, and now we're going to model them and measure them to some degree. We're going to create metrics, but we're going to mainly focus on simulation to see what this looks like. So again, we're going to turn to agent-based models or agent-based environments where we have, and of course, in an agent-based model, you have this grid world. And on the grid world, you have agents that can occupy individual cells. And those cells can be occupied by agents that have different roles, different internal states, and they interact with each other through rules. And, you know, if you have agents next to you, you're, you know, you can, you can share information with them, and they create these structures across the grid that are these collectives. And so, you know, we're going to be looking at collective behavior across the grid. We're going to be looking at individual behavior on the grid, and we're going to see how that evolves over time. So our agent is in a certain IJ position. It'll produce a spatial, a bunch of agents will produce a spatial distribution with specific states. So we have a red state, a blue state, and a gray state here. And so that's great. We can do that, and we can run the simulation. But then when we get our simulation results, we see that there's some patterns in here. And, you know, when you look at patterns with your eyes, sometimes you see spurious patterns. Sometimes you see patterns that are meaningful. So the problem we have is how do we interpret macro level emergent patterns? So that's going to be a problem that we're going to have, and we'll come back to that later. So, you know, we want to think about other kinds of processes to model, not just agents that live on a grid world, but those agents living on a grid world do something. So ultimately, these agents are trying to solve different collective action problems. So, you know, there are different aspects to this, and we, but we don't want to just model collective action for the sake of modeling collective action. We want to map this to some process in open source. And so here we have branch development in Git, and we have pull requests. So we have different branches where different branches are created for different purposes by different agents, and they're all kind of pushing their work to a main branch where they're, you know,

they're kind of taking the work from the main branch, creating their own branch, finishing their work, and then issuing a pull request back to the main branch.

So this is collective action because all the agents have to coordinate their activities so that at the end, the main branch has like a coherent project, and it isn't just, you know, everything that's pushed, smooshed together, and it's not workable. So, you know, there are a bunch of things that drive collective action in this context. So the first is internal motivation. A contributor wants to work on an issue. So if a contributor doesn't want to work on an issue, they won't bother making a branch, and they certainly won't bother making a pull request. There also has to be cooperation. So sometimes agents can get together, make their own branches, and use a divide and conquer strategy to sort of address the available tasks and address what needs to be done. They coordinate their efforts through looking at these diffs, which are the differences between different branches, and addressing those accordingly. And then that leads to this finished project. So you have this sort of technological layer that allows us to make these branches, compare these diffs, and do this on an automated fashion, which helps with the cognitive load. But it's also important to have the sort of internal motivation and this cooperative aspect of the behavior. And then, of course, there's the external motivation, which is sort of addressing a technical layer, which is that we have to make contributing to this project seamless and meaningful. So if the project is not, you know, easy to contribute to, and it isn't really about anything, then people won't bother to do any of this. Another process we can model is the issue queue, or the

Kanban board. So Kanban boards are used ubiquitously in project management. Issue queues are also used to

store issues, to move issues through different types of different levels of priority from beginning to completion. And so there are a lot of things that drive this at the individual level, at the team level, and then sort of at the issue creation level itself. So individual prioritization can be characterized as what issues contributors should address first, later, and last. So when we build a Kanban board, we usually create these categories of, you know, to do, in progress, urgent, and finished. And we have to decide as individuals where to put those issues. What's the state of completion for an issue, and where do we put it? So, you know, maybe we don't want to work on certain issues at all. Maybe we find other issues to be very appealing, or maybe we take all of our issues and we move very slowly through them. Maybe we move faster. And, but we have to have this means of sort of addressing issues. Sometimes we procrastinate on them. And, you know, we have to have this model of moving on things so that things get completed.

Then we have team prioritization. So in team prioritization, we can ask what issues are at what stage in project development. So if we have a team reviewing a Kanban board, that team will look at all the individuals and their collective prioritization techniques. And then they will

just basically say, okay, this is what the state of the project is. And I'm going to give feedback to the individuals to, or we're going to give feedback to the individuals to say what they need to do to improve these. And then issue creation is basically what is the fundamental unit of work.

So all of our individuals must have be in agreement on the same, roughly the same fundamental unit of

work, how big are your issues. And the team also has to be in agreement or else you have a lot of conflict. You have a lot of fighting over, you know, what exactly it needs to be done, how much work needs to be done and what's the priority. And the third process we can model are units of contribution over time. So this is a GitHub dependency graph for branches and forks. If you look at this, you'll see that they're different owners and these lines kind of, sometimes they kind of join together, sometimes they branch apart, sometimes they're maintaining themselves in parallel. And these are all sort of a record of pull requests and commits within a project. And so we can look at different workflows or different sort of work processes where multiple people are contributing to different things over time and actually producing an outcome. And so we were interested in that because we're interested in knowing how maybe a certain thing like a Docker file gets passed through many different community members or contributors, you know, it gets passed through many different contributors on its way to completion. So here we have a number of different, on this workflow network cycle, we have all these nodes, and we have a number of connections between the nodes that can, these four connections define this network cycle. So the first connection are generic changes that were contributed by two contributors and then merged by a third. The second is creating a Docker file which was contributed by one team member and merged by another. There was another clone created at three and then four was a fork made by another contributor. So you have a number of people working on this, a number of contributors, you have a number of actions being taken, and you can analyze those as sort of getting work done and making the project, you know, basically sustainable. Now, when we do this in terms of simulation, we have this problem of indirect measurements. That is, we're measuring these things, but we're not actually measuring things like cognitive state. We're not, so we're not like putting people in a scanner to measure cognitive state. We're looking at their outputs on GitHub, or their outputs on discourse, or something like that. And so we can't say for sure, or we don't really give them formal behavioral tests to look at their, say, motivations. So we don't know exactly why they're doing these things, we just know that they're doing a lot of them.

So we have this problem of indirect measurements skewing our simulation output. GitHub contributions are an indirect measure of motivation. So we can say that people are motivated if they contribute to GitHub a lot. We can just make that assumption, but we don't know the details of it. You know, maybe they're doing it for a class project, which is motivation, but it's not the same as being like kind of motivated by the problem. And that's the problem. We don't really have a lot more information to go on than that. And so simulations based on these indirect measurements may not accurately reflect the dynamics we find in actual open source projects. We can only assume that if they're contributing a lot to GitHub, their motivation is high, and that motivation carries over across, you know, time. So they're always highly motivated, essentially. We also have a problem with performing parameter inference on complex nonlinear models. So the problem with agent-based models, and we're going to use agent-based models as sort of like a baseline here. We're going to always be modeling our agents on agent-based modeling

surfaces and frameworks. And we're going to be using things like reinforcement learning and active inference to model these cognitive states. So we have this, we're still working with agent-based models. We're just doing different things with them. We're building these hybrid models, essentially. So the likelihood or probability functions for most agent-based models can rarely be evaluated in practice. So we can build likelihoods of motivation, but we don't know what the conditions are. We couldn't build like a model with conditional probabilities unless we went out and did more experimentation with people who work with open source projects. And even then it wouldn't be complete in terms of getting data for, you know, across the wide range of projects. So we have to keep that in mind when we were talking about these models is that they're incomplete in terms of their ability to characterize every behavioral state. So these are our Google Summer of Code scholars. So this project was built around Google Summer of Code, getting students over the summer, having them produce. We've done this over the last four seasons, 2022 to 2025. And we have had six people in our team who have come in through this route and each of them have contributed something different. So Brian McCorkle, his picture's right there on the top. He worked on active inference and collective active intelligence, thinking about active inference and collective behavior. Hamancho Shogal, who's next to him there, he worked on multi-agent reinforcement learning and building Q-learning slash agent-based models as hybrid models. So he was interested in agent-based Q-learning. Harvey Rajagopalan implemented agent-based modeling in MISA, which is a Python package, which allows us to do more than say working with something like NetLogo. You know, just having that framework where we can actually build more sophisticated tools. Shupam Sony did some more work on multi-agent reinforcement learning in this Q-learning agent-based hybrid model. Sarah Bastawalla, who's right there next to her name, developed the LLAMOSC, which is large language models with agent-based models. So these are large language models that create sort of tasks for agents in an agent-based model. And the tasks, of course, are conversational and sort of text-based. So if you want to create a bunch of issues for your agents, you create them with the large language model, you create conversations with the large language model, and then the agents work from that. So it's very interesting work, very exciting work. And then Vidi Rodhira, she worked on reinforcement learning, but a different sort of approach to it than the multi-agent approach. She worked on what they call SARSA, which is a model of inputs and outputs, and then merging that with agent-based models. So it was a hybrid model. These students used a number of tools to implement this stuff. For the multi-agent reinforcement learning at the upper left, they used temporal difference learning, Q-learning, and SARSA. We'll get into those details later. Some of our students also used MISA slash NetLogo. This is at the upper right. So MISA is implemented directly in Python. NetLogo is a classic package that has a lot of prepackaged models that you can implement and then you can build from there. So this offers a lot of flexibility for implementing the guts of the agent-based model. Langchain, of course, is a large language modeling tool

or toolbox. And that's at the lower left. And then the lower right is PYMDP, which is, of course, the toolbox for active inference. So let's talk about the OSDC first. This is a model of collective active intelligence. So this is our active inference approach to open source sustainability. And in this case, we're characterizing the activity of our computational agents using a Markov decision process in Python. And that's our PYMDP, or MDP in Python. And that's how we're trying to implement the model of a multi-arm bandit. So it focuses on explore, exploit problems. And as we'll see, the SARSA approach also explores multi or explores this sort of collective behavior as a multi-arm bandit problem. Basically, a multi-arm bandit, the name of the multi-arm bandit comes from when people pull the lever of a slot machine. So you're like priming this stochastic process. And you'll have like in front of you, if you've ever gone to a casino, you'll see people sometimes put like five slot machines at once, and they'll pull the arms down and then see what they get. That's what you're doing in a multi-arm bandit. And you're trying to figure out whether you should explore a space or exploit a space. So if you think about that in terms of issues, should you explore an issue deeply or should you explore more issues in a more shallow way? We implement this in different ways so that we can look at that as a sort of a factor for driving open source project behavior. The second point is building a statistical description of the mind. So this is of course is the internal components or the internal model of each agent. So we want to build a statistical description of that mind. We want to give it beliefs and we want to have, you know, have each agent minimize its own free energy. And so this is something that of course, we also want the agents to sort of be able to figure out what other agents are thinking to so that they can match their beliefs and minimize their free energy together. So this is kind of where we're going with this in terms of the collective part of it. And then of course, when we reduce our error, it reinforces causal relationships that are necessary to complete the project. So getting the agents to see what the causal relationships are between different parts of the project. If we can reduce their inferential error, then we can get all the agents to move in the same direction. So this was implemented using an inter-inforactively. So this is the link to interactively. It was used to do two things. Number one is to implement decision problems in a single agent, as we said. So each of these agents have some decision process going on inside of them. And then two is to implement a theory of mind to infer the state of neighboring agents. So the agent on the left is trying to characterize the state of the agent on the right, because the agent on the left doesn't know what the agent on the right is thinking. And if they're both thinking the same thing, they're able to maximize utility in the same way, then they'll be able to cooperate better and do things like divide and conquer. So this is where you have this agent is asking this other agent, is this what you're thinking? And they say, yes. So that's the way this project worked. And it has ties to the reinforcement learning that we'll talk about next. So in the context of an agent-based model, this is an agent-based grid again, the activity of one agent influences activities of their neighbors. This is typical of an agent-based model. So we expect those internal states to spread as they're able to infer each other's state. We get these blocks where all the agents are doing the same thing. But we also get the spatial distribution

of states. And because this is a spatial array, we have things like adjacency and distance that we can characterize the sort of spread of these ideas. So, you know, sometimes you'll have a block of contributors thinking one way and another block thinking another way. It's either thinking the same way about a problem or having sort of the same level of motivation or whatever.

If theory of mind is achieved, two neighboring cells should co-activate. So that's where they share covariance in their activity. And these are, you know, pairs where you can see that they're kind of activated together. Another thing that Brian McCorkle worked on the active inference part. So another thing Brian McCorkle was interested in, it's kind of on the cutting room floor now, but I bring it up because it really does have relevance to sort of these internal states in the mind. And that is Mick Ashby's ethical regulator. So Mick Ashby is the grandson of W.R. Ashby who came up with the Evergood regulator theorem in cybernetics. And Mick has been working on this ethical regulator and super ethical systems idea. He published a paper in 2020 on this. And he has this whole model of how we can encourage ethical behaviors. And this goes well beyond active inference, but it shows you how we can take a simple active inference model and kind of think more deeply about things like ethics in projects. So do people act ethically in the project? Sometimes projects get blown up because people act unethically. They steal credit for ideas. They cut other people out of the project, they're not honest about what they're doing. Or sometimes they try to make money off of an open source project when the license explicitly states otherwise. So we want to encourage ethical behaviors. And we want to make sure that those are all there in place. And we coordinate our division of labor and resources amongst contributors so that everyone's acting ethically. And as a model of cooperation, we have to take into account fairness and efficiency. So when we talk about regulation, and if we look at our agent-based model, we have these different types of regulatory behaviors. So if one agent influences another, we can have the zeroth order behavior. If agents influence each other, and they have this theory of mind, there's a first order behavior. And if multiple agents are influencing each other and collectively, that's a second order behavior. So there's also the sustainability auditing tool, which was worked on after the project in 2022. This was implemented in NetLogo Web. So this is an agent-based model there. And this was kind of developed out of the active inference approach. And so in this case,

we're modeling two variables. We're modeling churn, and we're modeling maximum sensory probability. So churn, of course, is the turnover in the number of agents. High churn suggests that the community has a low degree of cooperation, low degree of theory of mind. And then maximum sensory probability, remember,

we need to have some sensory information to make active inference work. And that is the probability of a given signal from other agents or the environment. So we have to have a signal from other agents, a signal from the environment. And then that's what activates sort of the agent's cognitive state.

So this is the agent-based model for multi-agent reinforcement learning approach. This is modeled in NetLogo. We have our agents, the stick figures, and they have multi-agent reinforcement learning, you're driving their behavior, their internal state. Those are the squares, which are the

resources of the project. And we can populate this with a large or a small number of contributors. We can then divide them up into categories, new maintainers, contributors, and then have solved issues and unsolved issues. And the idea is that they solve all the issues to maintain a sustainable project. And then we have these different issue sizes. So Earthshot, Moonshot, and Mars-Jupiter shot. These are just different sizes of issues. Earthshot issues being single issues, Moonshot issues being a collection of issues that are interdependent. Emerge-Jupiter shot issues that are things that are kind of like multiple different sub-projects, basically. So a little bit about multi-agent reinforcement learning. This is a simulation where we use multi-agent reinforcement learning. It's a population of agents acting on a common grid world. The agent is motivated by its own rewards. Its actions advance its own interests. So in individual terms, it's self-motivated. But then sometimes, of course, those agents have to interact with their neighbors, with other agents of the population. And those interests can either be coordinated or mutually opposed. So we can see here that we have

these two different colored agent populations and they're kind of mixing over time. We can model different scales of project size using multi-agent reinforcement learning. So this shows the prevalence of Earthshot issues, Moonshot issues, Mars-Jupiter shot issues, and accepted pull requests over time. And you can see that they kind of increase with time. And this is just one run of the simulation. So again, our Earthshot issues are single issues, our Moonshot issues are issues that can be broken down to multiple single issues. And then our Mars-Jupiter shot is a problem that can't be broken down directly into multiple issues, but need different sub-projects to be completed before the next sub-project. And as such, they're sort of modular. So we use that to look at multi-agent reinforcement learning.

This is where we also used multi-agent reinforcement learning with Q-learning, but also SustainHub, which is this reinforcement learning with a SARSA model. So SARSA is state action, reward state action. So it describes you monitor the state, you take an action, you get a reward, you monitor the state, you take an action. And so that's kind of the difference. We have Q-learning for multi-agent, and the SARSA for the SustainHub, or the single reinforcement learning. And so this is another simulation. This I think is also from Shubham, where he's simulating Earth issues, Moon issues, Mars issues, and then accepted pull requests. And the accepted pull requests are always sparser than the other issues being solved. So that's, you know, it's kind of typical because you're combining all your issues into a single pull request. So this is kind of a way to monitor or at least model things going on in a community. The question is, you know, is it something that is sustainable or not? And that's a little bit harder to determine. So let's walk through some of these methods for reinforcement learning to see maybe how they compare with active inference. So we start with Q-learning. So for any Markov decision process, we identify optimal action selection policy. So what we want to do with that action selection policy is maximize the expected value of the total reward.

Okay. So then this determines the quality of a given policy. An action A sub T with respect to state S of T provided a learning rate of α . So we have this equation down here where we have these different components. We have α times Q , where we're evaluating the action and the state,

the quality of each action and state that evaluates the current action selection value.

We add that to this estimate of optimal future values, where the state of highest quality, and so we're looking for the maximum quality. And then that's our kind of the way, the direction we're going in Q-learning. We're trying to find that maximum feature value, the maximum quality of a given set of policies. Now, the multi-arm bandit is also used then in reinforcement learning, like we saw with active inference. It's used for task allocation, role matching given prior performance. It encourages specialization while allowing for flexibility. And it utilizes what we call on-policy temporal difference learning, which we'll get into next. And so it's important to recall the different community member types, maintainer, contributor, innovator, knowledge curator. And we can use this to think about how to allocate different tasks to these different classes of a contributor, or these different classes of project member. So temporal difference learning can be stated as a state value function of a finite state Markov decision process under a policy π . V_{π} is equal to the state value function of the Markov decision process with states, rewards, and a discount rate, importance, which states the importance of future rewards. So we have, instead of our learning rate, we have a discount rate. And then we estimate V_{π} , which is the function for the state value. And we evaluate this policy π , we obtain the reward r , and we update our value function. Now, some of the metrics that were created in the reinforcement learning work, and this was done by VD Ruhira, she developed five different metrics that work in this context. So the first is the harmony index. So remember, I said there are different things we want to measure. And then there are different things that we can sort of have as like an output. So the first the harmony index, this captures workload balance and tasks success rate. This is basically measures how well balanced workflow workloads are, how much cognitive load there is for each agent, and then the success of agents on those tasks. Then there's the resilience quotient, which measures adaptability during disruptions, such as dropouts and workload spikes. It means basically how resilient are you to different types of, you know, disruptions in your work in the project. There's reassignment overhead, which tracks inefficiency caused by reassigning tasks. There's an initial, an initial density of agents. So basically, it's where if you have a dense setting of agents to start with, it's going to give you a very different result than a very, very small number of agents. And we can use that to sort of normalize a lot of the heterogeneity in open source projects. And then the contributor maintainer vision. So if we don't have a good vision, agents won't know if there's anything for them to do. But if the vision is there, then at least they have in their mind sort of a mental model that they have something to do. And so they go to find specific tasks to do, to solve specific issues. The next thing I want to talk about is how large language models were used. So this was Sara Bastawala, of course. This is LLAMA OSC. This is an open source large language model based on the LLAMA platform. And it's used to generate project tasks and issues. And so she basically developed this large language model, agent-based model hybrid platform, this hybrid model. And that gives us a model of how to act upon tasks and project structure. So you would generate like all these different issues using large language model, you generate conversations, and then the agents would know how to act from that and get the project structure.

And then the workflow is in line chain that's in Python. And this allows for large language models to be used as a reasoning engine, where you can have sequences of actions, where agents decide what tools to use and what order to use them in. So, and in this work, you know, she had like two basic models of open source communities. So instead of thinking them as very heterogeneous structures, we're thinking them as idealized structures, she thought of them as either a benevolent dictator model or a meritocratic model. And so in the benevolent dictator model, which is characterized by Linux, in the Linux kernel, you have some single person who drives the project forward with their deep technical skills. So Linus Torvalds is the benevolent dictator in the Linux example. And then the more meritocratic example is indicative of the Apache foundation, where anyone can engage at any level that they want. And so there's no leader, there's no dictator, it's just kind of like people do what they can in the project. I want to get into the way we're thinking about integrating a lot of these models together, we want to bring them together in the next phase of this project. So, you know, one of the things we need to understand are the dynamics, or what does a team look like over the course of a project. So sometimes, you know, people leave projects, sometimes people join projects, we have these sort of lack of Ulterra dynamics in our agent based model. And that's, you know, kind of an interesting thing, because none of those are described by any of the models that we've talked about. So this is a point for future work. We also know that like some of these dynamics, because you get a lot of people leaving a lot of people joining, a lot of flux and churn, that it can create vulnerabilities, as well as opportunities. And so if we want to know how to sort of build a model that can help us understand maybe what those opportunities are, those vulnerabilities are, there are some ways forward. So one way is to build these sort of what we call meta brain models, where you have different, you take different models, and you put them together, different layers of different types of representations, put them together as a structure that sort of, you know, communicates selectively between the models, but processes different types of information. So in this case, we think about the large language model, as handling semantically defined tasks, active inference is handling sort of the internal brain state of an agent. And then the multi agent reinforcement learning as the motivational structure, which means that you know, you have this brain state and motivational structure working together on tasks that are semantically defined, and then maybe an agent based model, and sort of the driver of collective behavior. So you have this agent based model, sort of coordinating all these different brains, composed of large language models, multi agent reinforcement learning, active inference models.

So I want to finish with sort of a thought from Jesse Perrin, and he's another person in our lab who's contributed to this work. And he's been thinking a lot about project management, and a lot about how we often create tools that don't go anywhere, or we have like initiatives that kind of die, because no one finds them relevant.

So he says sustainability translates to problems in academic communities. Sometimes if you go to an academic community, you find that someone earns like a grant for tool building, so they get funding for tool building,

they build these great tools, and then the funding runs out. And so they have this open source tool that they put out in the world, the funding runs out, and it becomes an unmaintained tool, eventually to die a premature death. So that's one path that's suboptimal, I think, from the standpoint of open source sustainability, where you get money to build something, you build it, and then it dies.

The other example that he gives is that someone builds a tool, and because it's not really relevant to anything, the tool is ignored and not maintained. And so then that's bad, but it gets worse because then someone else builds a similar tool, and that dies because it's ignored and not maintained. So you end up with this graveyard of tools that are all maybe pretty decent, but because no one's finding them relevant, and no one's putting the money into maintenance, the people keep reinventing the wheel, and then the wheel breaks apart. So that's not good. So, you know, Jesse makes this observation that academic communities more generally will often build out infrastructure, which goes nowhere, and is disconnected from real world impact. So that's something to think about if we're thinking about the Active Inference Institute, or our own lab, or other institutions where people build infrastructure, it goes nowhere, or it's not connected to the real world in relevant ways. And sometimes you just can't, you know, you can't avoid it, because you don't necessarily know what people will find relevant. But that's part of sustainability that we haven't addressed, and maybe is sort of the secret sauce in this whole equation. So thank you for your attention.

Thank you.

All right, next up is going to be Andrea Hiot's Kaleidoscopic Cognition, how the brain can be modeled and use modeling without having a model.

It is a 52 minute video.

Here we go.

Good afternoon, everyone.

Today I want to share a paper with you, Radical Embodied Relation at Any Scale from Remembering to Navigating.

My name is Andrea Hiot and this is for the Active Inference Institute Symposium for 2025.

So this is a bit of a radical embodied understanding of mind and it works at any scale, as the paper says.

So it's supposed to expand some of our fundamental assumptions about brains, memories, minds, and bodily movement and how all those relate.

So the title of this talk is Kaleidoscopic Cognition, how the brain can be modeled and use modeling without having a model.

I know that sounds paradoxical.

And if you know me, you know I like paradoxes.

Holding paradox is sort of the point of a lot of this work, but that title might seem impossible, actually.

So how can the brain model without having models?

And how can we use representations without them being stored somewhere?

These questions have been really important in neuroscience and philosophy.

Some might even say they've been plaguing neuroscience and philosophy of mind lately.

Like we can just can't move past them.

But there's recent research on the hippocampus, which is the part of the brain that I studied, a little seahorse shaped structure on both sides of your head in both lobes.

And it's forcing us to rethink some basic assumptions about a lot of these words that I've already mentioned.

So what we're discovering is that the brain, or I would rather actually talk about the central nervous system and the bodily sensory movement of it all, but let's just say the brain doesn't work like a computer with shared stored files.

Instead, it operates more like a kaleidoscope.

And I know that might sound crazy, but actually it's a very helpful, strange new vision that we kind of need right now.

And that it feels strange is probably a good thing.

So it's this idea of creating shifting patterns through different but direct refraction of experience.

So if you think of the encounter and then the body and then whatever that encounter is becoming as part of the body and sort of going back into the encounter as, then you can think of it more as something prismatic or kaleidoscopic.

So this framework is coming from neuroscience, at least as I read certain studies and as I've been part of certain areas of that field.

And especially grid cells, place cells, these kinds of things which are in the hippocampus.

And I think it has profound implications for how we understand these words that we've been taking for granted, like memory, thinking, but also navigation and everything that we've ever thought about as cognition.

Not only in humans, but when we try to understand this activity that unites us across different life forms.

I'm going to be talking about the hippocampus as a kaleidoscopic function, a way of trying to visualize some of this new research, which is moving us beyond either or patterns that we assume.

I think we're all sort of stuck in at the moment in brain research, even though we say we're not, but I want to push us a little bit towards this kaleidoscopic or prismatic idea of cognition. And hopefully that'll help us move beyond these representation wars and ways of talking that have sort of paralyzed us in cognitive science.

The idea is to synthesize insights from four key papers that when read together will reveal something pretty extraordinary about cognition.

And so these are the four papers.

The first one is my paper, the one that's kind of framing this and I guess the sort of foundational

text for it.

And in there, I argue that recent hippocampal research forces us or allows us to reconceive what a neural representation is and to think of it as a communication, communicative assessment, rather than something stored or findable or some sort of hieroglyph in the brain.

And I introduced the concept of way making, which is just a way of talking about this process that we call cognition or this process that we call locomotion, but on the same track.

There's also a brand new nature neuroscience paper by Penn and colleagues, which came out around the same time, which talks about, well, it's called path integration in the hippocampus by shifting reference frames.

And I really see this as just crucial empirical work and just great timing that it comes out now because it has a lot to do with what I'm trying to discuss here.

And it's showing us that grid cells don't maintain stable patterns, but are constantly shifting between reference frames, even though there is a stability.

So there's a paradox built into this, but it gives us a good way to understand this notion of kaleidoscopic modeling that I'm trying to introduce here.

And then we have the Lins and Clark, even Friston paper from 2017.

Sorry if I said somebody's name wrong there.

The active inference approach to ecological perception, because I think there's a lot of ideas in there that fit very well with this.

And also just the mathematical and theoretical framework needs to be built up for how we might do this.

And I think active inferences probably, and especially also Bayesian ideas and machine learning and so on have already started figuring out ways to maybe think about layered mathematics in the way that I'm talking about here.

But also just the ecological side of it is very fitting that organisms engage in modeling behaviors without storing models.

And of course, sometimes we slip up and say models are in the brain.

But if you really look at what is actually being said, it's not that the models are in the brain stored somewhere.

It's that the whole body is modeling.

This last paper too, which inspired my first paper.

And it's from Jacob Belmont and the Dole Lab, which is where I worked at the Max Planck Institute. It's called Navigating Cognition, Spatial Codes for Human Thinking.

And it shows us that the mechanics we use for physical navigation are the same ones we use for abstract thought.

Even though thinking and geographical navigating can't be resolved as like the same thing, we can use the same metrics to understand them.

But those metrics cannot be based on these either assumptions that we've been sort of using,

whether it's like global and local or even mental and physical, like conceptual space, physical space, memory, navigation.

Those ways of talking and our use of the word representation as we're talking about them in cognitive neuroscience has to shift.

So all these papers together point towards a unified understanding of cognition as kaleidoscopic, or at least as constellatory is another word I use, demanding we find ways to model it kaleidoscopically rather than binary.

And these binary assumptions that frame so much of the way we talk about all of these topics, the way we even use the words and the way we model them, even if we don't realize that's happening.

So by introducing this kaleidoscopic cognition, I'm trying to push us a bit into a dynamic, immediate, irreducible understanding of what we mean by representation, which is actually a communication.

So we also understand that we're going to do this through communicating about the patterns of regularity and that this communication isn't findable, but it is representational.

So that's a little tricky.

This holding paradox idea is really good.

And also this kaleidoscope where you have similar patterns, but based on the input of the light, you can even think of it as prismatic, I guess, you're going to get different results.

But those will all be tied to similar patterns.

So again, I've mentioned paradox quite a few times and the main paradox that motivates this framework is representations without representations.

Models without models or modeling without models or representing without representations maybe is a better way to say it.

Models without models, internal representations that the brain builds and maintains.

So when you remember something from your childhood, your childhood home, you know, there must be some neural structure that is appearing in your brain that represents that home, right?

So when you navigate through a city, you have some kind of map in your head.

This is just the way we talk about it.

Most people, of course, don't actually think you can, you know, pull a little map out of your brain or actually find some kind of structure in the brain that's going to represent home in some kind of one-to-one way.

However, we still talk about it that way in our papers and we still assume this language.

So the reality is that organisms are navigating these complex environments without anything that looks like explicit stored representations.

And in fact, you know, there's so much going on all the time that any of our actual experiments is just looking at a tiny bit of it.

So if we find some brain activity that corresponds with a particular memory that we have, that's

actually just one little snapshot in a way of something that we were directly looking for.

And this is still in this flatland kind of way of thinking about what we're doing in these experiments and also what the body is doing in relation to the brain, the environment, and all these ongoing movements that we are.

So animals find their way home.

They recognize patterns.

They make predictions, you know, including humans, of course.

And yet when we observe them and we watch the activity of, you know, rats who've been so helpful for us, for example, in neuroscience.

And the only way we can really communicate about what's happening there is through creating these representations, images or, you know, that we record the cell activity and then we see pictures of it.

And all of that is representational, but it's not inside the rat.

It's us creating things to understand this ongoing movement.

And that gives us a visual, a video, a screen image, whatever.

And those are the representations and those are the communications.

And that is a communication.

I mean, the screenshot or the recording is meaningless except as communication.

And it's important.

We're doing that, obviously, in order to better understand this whole process and ourselves. And it's a very real thing, but it's not findable somewhere.

So, the hippocampus is really central to making more scientific, rigorous sense of what I just said and getting over this assumption and use of language that's binary and that does talk about representations as if they're inside brains rather than as if there are communications towards understanding brains.

So, for a long time, we actually literally looked for ingrams in the brain, stored memories.

And that was, the hippocampus was like, you know, the sort of heart of where all that came in the 50s, looking for ingrams and memories because of HM and so on.

And then in the 70s, we started thinking of the same area of the brain as the brain's GPS, as John O'Keefe found cells there that fired in relation to what the animal was moving through.

So you know, you got place cells and grid cells and things like this.

But this recent research is even showing us that actually these cells don't maintain stable patterns.

They're always shifting, remapping, re-anchoring.

We don't have like a representation in the brain that corresponds to something outside.

Instead we have this ongoing communication, this process, this something like light coming into a prism and being refracted.

You know, that's where this kaleidoscopic comes from, that shifts as the encounter takes

place.

So all of these cells are shifting, remapping, re-anchoring as the encounter is changing. Just as if light comes into a prism, it's going to change in different ways without any kind of, you know, stopping and starting there.

Even though when we study it, we have to look at it that way.

So this is where this idea of kaleidoscopic cognition comes in.

And it does sound fantastical, maybe psychedelic, but it's actually a really accurate way to begin to envision this.

And that it feels strange and even hard is actually part of that.

Because what's more like the vision, and again, we're talking a bit in metaphors here, but it's more like something is refracting between and as the encounter and the organism, which includes this whole nervous system.

And that is the guiding action.

That is the perception and action.

And obviously this is really hard to talk about.

But that's what we're trying to do here.

I think we have to find ways to sound strange, be strange, and move towards finding other ways to really understand what's happening here.

So we create recordings and photos and videos and representations to understand this.

Those are real and communicative.

But how could we create more kaleidoscopic ones?

How can we begin to use our technology in ways that are more like this process so that we get used to thinking like this instead of creating things that are still stuck in this binary sense of as if we've taken a picture of some kind of whole thing that is mental, that is a memory.

So before we get to that, let me just talk about this one question about representation, which I've brought up.

But how can the brain have representations and not have them at the same time?

How do we hold this paradox without falling into something like computationalism or just total relativism, which is always the trap here?

So that's where this hippocampus paper comes in, which you can find easily and read.

And it's arguing that representations are communicative acts, that there are ways we assess and communicate

about these unbroken intimacies, not things we find stored in brains.

And I think that's something that's pretty obvious in neuroscience, but we don't talk about it as if it is.

And that is a very big obstacle, actually.

And of course, there's been tons of representation debates.

I mean, we've really wasted or maybe not wasted, but so much energy and time has gone into getting stuck and going round and round in loops over this idea.

You know, even with active inference, I think we all know this idea of talking about models is in the head, becomes a trap.

And we just, it's time to get out of it.

So that's why I'm not in this paper, but in this presentation, presenting this idea of kaleidoscopic or constellatory cognition to kind of just try to pop us out of that way of positioning ourselves in cognitive neuroscience and in the way that we do our models.

So we can understand these immediacies like are refracting differently according to the position from which they're measured.

And they're not in the thing that's being measured.

So weather maps are a really simple, good example.

We create a weather map.

It represents weather patterns, but the map doesn't contain the weather, even though it's tricky how we confuse those things with what weather is.

But there is no like stop and start to a storm in the sense of you can just pluck it out of all the rest of the ongoing atmosphere or something like that.

And we need and create these representations of things like that so that we can communicate about them so that we know, okay, a big hurricane is coming and we need to evacuate or whatever.

These are really important representations that we create.

But the map is the communicative tool that we use to understand and predict those atmospheric dynamics.

The dynamics, of course, they're dynamic.

They're ongoing.

They're unlocatable.

They're unlocatable.

You don't find partly cloudy somewhere.

You don't find 70 degrees somewhere.

You know, those are things we represent.

Those are representations.

And that's the same thing that we're talking about when we're looking at brains.

It's something like 70 degrees when we're talking about a representation in the brain of memory or anything else.

So that process that we call weather refracts differently from every position on the planet, too.

We could think about that and how, depending on where you are, you're going to see that refraction differently.

It's going to be assessed differently, even though, of course, it's real what is being assessed.

And it's also ongoing.

And also your assessment of it on one particular day from one particular position is going to be different than someone else from another on another day.

And that doesn't mean any of it's less real.

And in fact, it'll still probably mean there's a whole lot of regularities that are shared.

What you're calling a storm and what they're calling a storm will have a lot of overlap, but it won't be the same thing.

And it won't be something that we have found and then move on from.

So neural representations work that way, too.

When we look at an fMRI image or we record neurons, we're not seeing stored memories or cognitive maps inside the brain.

We're creating representations, measurements, assessments that help us communicate about this ongoing central nervous system, bodily environmental dynamic, which is pretty wonderful, actually, and which we really do want to understand.

And that's why what we're doing is so important in all these fields.

But the hippocampal literature shows us that cognition and navigation, for example, which have kind of been fit into these different sides of a binary of mental and physical, for example, just to kind of caricature it a little bit.

We've assumed for a long time that those needed to be reconciled somehow, that they were somehow different things.

And how could the same part of the brain be doing them?

But recently, we've started to understand, OK, they're just like refractions of the same process.

At least that's how I'm putting it here.

There are different assessments of this embodied way making that is happening all the time.

And depending what we're looking for or where we're standing or what day it is, it's going to be a little bit different.

But also, there are going to be regularities that remain, that help us understand it better if we communicate about it through representations.

So when you remember a past event, you're kind of moving through or navigating conceptual space.

You could think of it like that.

You're using the same mechanisms, right, to move through that sort of space that happens when you're moving through geographical space.

But those aren't the same thing.

It's just that there's similar regularities there in terms of what is happening.

So I use this word way making, which comes from Tao's philosophy, to capture how that cognition and locomotion can be continuous, can be the same process, even though when we're talking about something like remembering our childhood and when we're talking about something

like walking to the park, we're talking about different things.

And there's no need to try to resolve those.

The same way we wouldn't try to resolve wavelengths that have been refracted from the same light source or the same, yeah, we could think of it somehow as multi-sensory prisms if we can.

But we can't measure and find that in some absolute sense that there is a memory or there is a little diagram in the head that's going to be found every time we go to the grocery store.

So anyway, the hippocampus doesn't store maps.

It's operating kaleidoscopically.

It's refracting sensory encounter into bodily action.

At least that's the kind of wild idea I'm throwing in on this presentation.

I haven't tried to put it in a paper, of course, but at least not yet.

But I do think we make these maps, these representations to try to understand what is multi-layered, multi-dimensional,

refracted in our measurements of it, even if in itself it's not.

So just thinking of it like that can help us avoid some of these traps of arguing about pure computationalism versus pure relativism or telling all the young people that everything's the hallucination or all these extremes that have sort of taken over and become very exciting in a lot of our fields but aren't doing justice to the actuality and the complication of this paradoxical happening that is cognition and that is not only in humans.

And of course we want to avoid relativism too because these are real accessible regularities and we do need to communicate about them.

And that is a very important thing to be doing.

It's science at its best really.

So what the heck is kaleidoscopic cognition?

So I guess in all this making way that we're doing, we do make representations of that.

Until now we've made them in these very strict kind of stimulus response either or mental physical categories.

Even if we think those categories aren't somehow actually the process, that some people think everything's mental or everything's physical, all is one.

This is pretty normal now for people to say that, but in all the literature we still talk this way.

And so when I talk about kaleidoscopic cognition, I'm really thinking about modeling all this way making, all of these ways that the body makes through its encounter.

Whether you want to think of that through traditionally mental terms or physical terms, whether you want to think of it as making way through conceptual space, virtual space, geographical space, all those, you know, they can't be collapsed into one another, but they're all way making.

And when we're modeling that, we can think of it that way.

We can understand it that way.

That's the kaleidoscopic cognition.

That means imagining that cognitive processes emerge through and as dynamic shifting patterns rather than this is happening and we look for this fixed representation in the brain.

That's the old way, right?

And that's the way our language and our papers are structured right now.

We could try to understand this through another metaphor.

And I'm bringing in this consolatory or kaleidoscopic metaphor, you know, which are just like tubes of mirrors and colored glass and prisms.

But if you really think of the ongoing encounter and the sensory body, it is sort of taking in that encounter as if it's light and it's refracting it into itself in order to continue itself.

I know that sounds a bit mystical and wild, but I think we need some of that right now, just as visualization.

We still need all the rigor and all the science and all the tests and all the experiments, but we can visualize about how better to do those.

We don't have to stay stuck in mental, physical stimulus response parameters.

We can think of the experiment differently.

So when light enters a kaleidoscope, it's immediately refracted through like all these prisms and mirrors.

And the patterns you see aren't stored anywhere in that device.

They just emerge, you know, in this interaction as you turn, turn the, whatever the heck you call that thing on a kaleidoscope.

You turn the kaleidoscope, I guess, and the patterns change and that's not stored anywhere.

It's just happening.

And the light, you need the light coming in and the, whatever, the viewer.

I mean, at some point this kind of starts to break down, but I think you get what I'm saying here.

Twist the tube, you get entirely new patterns, but all those patterns, if we did the math, could be linked to one another and we could better understand how they turn into each other and what sort of condition, what sort of light, so to speak, creates what sort of pattern.

So the hippocampus, and I would say even the central nervous system, the brain more broadly works like this.

It receives the encounter.

I mean, you know, even to say it receives it is kind of wrong because this is just happening.

It's all happening.

It's not that the body is sitting there waiting and something comes in and it does something.

I mean, to be alive, you're in the encounter, you're encountering.

The body is doing this prismatic, prisms, this kaleidoscopic cognition.

And we're trying to understand it by creating representations of it.

So those patterns that come in, so to speak, come in to the body, aren't like processed and stored, but they are shaping the way the body is going to respond in the future.

I think this relates a lot to a lot of these active inference themes that you might be seeing elsewhere in this symposium.

So these patterns aren't processed or stored.

They're direct transformations, just like the light is directly transformed when it passes through a prism or the kaleidoscope or whatever that we're going to visualize here.

So these sensory regularities of the encounter, the light, so to speak, coming into the body can come from the body too.

So, you know, it's also not multidirectional.

I mean, it is multidirectional.

It's not just like there's an outside and it's coming into the body.

You also have the body itself, which we could look at as many kinds of bodies.

And all of this is happening at all those levels too.

The levels of your cells, for example, or the way that the body and the nervous system are communicating with other parts of the body.

So all of those are kaleidoscopes of kaleidoscopes in a way.

And it's all happening in this more ecological direct sense, which is very hard for us to envision and talk about.

But I think we're getting to a point where we have to try and we have technology that allows us to do so.

And what happens, you know, if we think of it like this?

I think first we understand that multiplicities aren't fixed attributions.

There are multiple simultaneous patterns that are operating at once as we measure them.

I mean, all of this is just happening, but we are going to be assessing it and able to handle it in that kind of way.

So it's true for how we observe and represent the process.

The second idea there, I guess, is that these processes aren't structures that we find.

And what we call mental content, it's arising from or it is this ongoing refraction.

It's not sort of retrieved data that's been stored somewhere.

It's more like water that's being directed in different ways at different levels.

Sorry, I've been traveling and I have a cold.

That's why I'm also not on the video.

But I hope this is okay.

What else was I going to talk about?

I think also the relationality, that meaning is this relationality that's emerging through these shifts of the kaleidoscope, these different prismatics.

And refraction matters, though, because as part of it, you know, we're part of it.

I mean, the way that we notice it, observe it, assess it actually has some major consequence on those very patterns and those ways that we shift the light, so to speak.

So I realize all this sounds a bit abstract, but let's look at some more empirical neuroscience.

Maybe that's helpful.

Some grid cell data.

Oh, yeah.

First, I should just mention these different frameworks, which are part of that paper, and which I think when we put them all together, we can start thinking about cognition constellations or kaleidoscopic cognition from a lot of different angles.

And we could even start to try to mathematize it or model it a bit.

I didn't really do a big research, but I'm sure there's probably ways that we can model multiple layers of generative models that might be helpful in this way.

But anyway, I won't go down that tangent.

Active inference, I mean, I think you all probably know this mathematical framework by Carl Friston and many collaborators and colleagues and this core idea that organisms are minimizing free energy through their action.

They're not building these internal models and consulting them, but they are the models, I think is one way you could think of that.

And they're emerging dynamically with or through or as this perception action cycle.

So the body, the organism is constantly engaged in prediction and error correction.

And that's kind of what I was trying to say before, that even though this is all happening in the way that you would turn a kaleidoscope or the way light comes through the prism, it's happening at so many different points and places.

And all of those in their communication, in the way they communicate with one another, change what's possible for that.

That's why this can get really hard to talk about because we're part of it, even as we're studying it.

We're always engaged in prediction and error correction.

And this happens through our movement and engagement or not even through, it happens as it.

It's not that things are being stored and then we're looking for those things that are stored.

That's only useful when we're trying to communicate about it, to talk about it in that way.

So I think active inference obviously helps us with visualizing this, but even puts math on it and helps us kind of take some steps towards leaping into this different paradigm.

And then the radical embodied relation, that's a big part of the paper that's in Nature in the article, which you can read, but a lot of inspiration there from Tony Chimero and everyone who's part of that field.

They've been working really hard to try to make that understandable and to show what it's

doing in cognitive science for us if we let it.

And of course they've been dealing with this.

There's no such thing as a representation idea for a while, which isn't what I'm saying.

And actually isn't really what is being said by Chimero, but they've been questioning it in the way that has gotten us to a point where we can start to understand it differently.

And where some people even say, oh yeah, it's obvious, right?

Of course there's no representation in the brain.

But that was not something that a lot of neuroscientists would have said not too long ago.

So I guess the key insight there is that cognition is continuous with navigation in an ecological sense, that these aren't different processes that we're bridging.

They do have to be measured differently just because we measure them towards understanding different things.

We're assessing them towards understanding different things.

So there's no need to solve them into one another or say that when we remember our childhood friend, it's the same thing as walking to the park.

I mean, that's just, there's no need to say that.

Instead, we can understand this embodied engagement with the environment, this internal, external, ongoing action that's co-creating the parameters for each part of that.

And I think a lot of you know J.J.

Gibson's ecological psychology and that idea of the brain as the ongoing translator.

I always want to say that word, but it's not even quite that because that makes me think of like Chinese room experiments.

But it's not that, it's too immediate.

It's more like the prism, the light coming in and immediately it's refracted.

And it only looks different from the position depending on where your position is.

I think that's what a lot of ecological psychology was trying to get at in a way.

So this kaleidoscopic notion of cognition, just to combine all this stuff.

Well, I'm just going to leave it there, but I would really hope that someone could think about this mathematically.

Maybe one of you listening think about, I mean, this was what I was thinking about.

I was thinking about Carl's 2012 paper where he tries to do something like this, or not tries, I mean, he does do something like this, where he's talking about generative models and as like ensembles of other models.

And I don't know, maybe someone listening already knows about something that's already been done like this, but how could we, is it even possible actually?

To have context-sensitive model selection with it still being free energy driven?

I mean, with the math, I have no idea because I'm not so good with this math and all this stuff, but anyway, I was just thinking about it.

I guess a big part of this whole, the paper that I'm presenting and what I see as very important in the active inference framework and why I'm bringing up this wonderful wild idea of kaleidoscopic cognition is, as I've already said a few times, just that we're so routed by these ways of making decisions of one versus another that I think we don't see that they're part of constellations of many more possibilities.

And there's a reason we had to go through contrast to get here to where we are now.

I don't think those are bad things.

I don't think we should stop having distinctions, but I think we should start realizing they're not binary distinctions.

I mean, we don't only have mental versus physical.

We have emotional, we have virtual, we have geographical.

I mean, these are just different ways that the light refracts.

And it would be great if we just got used to thinking that way.

That's what this kaleidoscopic cognition is.

It's thinking through multiplicities instead of binaries.

And I think the hippocampus and all the research we've been doing there, where we tried to fit everything into these binary categories for so long, and we're now realizing we just can't is a very helpful way to rigorously understand this.

You know, I think it's very important in philosophy of mind, too, that we try to do this rather than building our careers, fighting one side or the other of a binary.

That's maybe not the most helpful thing to do anymore, which doesn't mean all that stuff was not important.

It's actually really important in all of our debates about it.

But now we can still have those.

We can hold those and we can explore the space and look at other ways of reframing all of that together.

Let's just think about mental and physical for a second.

So in neuroscience and philosophy, we talk about as if we moved past those sometimes, like everything is physical, everything is mental or whatever.

We still treat, you know, like thoughts are here and movements are there.

And I think what I'm trying to introduce here with this kaleidoscopic cognition idea is that this continuous action perception cycle that we represent as either mental or physical and can continue to do so, by the way, it's kind of helpful, is itself an ongoing process that's refracting in those different ways.

So we hold those together however we want to define them and explore them as telling us distinct things about a similar process.

In a way, it's what we're already trying to do because, you know, if you're on one side, you say everything is one.

And if you're on the other, you say everything is one.

But then instead of fighting each other about who is right, let's say, okay, what is your path?

Tell us about this one.

And what does my path tell us about this one?

So like thinking can be accessed as a form of movement.

If we go back to the Thimba campus through conceptual space, that's helpful.

And moving can be assessed as a sort of form of thinking through physical space.

But thinking about going to the gym is not moving through physical space.

And moving through conceptual space is not the same thing as going to the grocery store.

And we never need to try to resolve those.

That's what we've been trying to do by resolving certain schools of thought or even just this debate about representations.

But we also need to understand that the traditional debate of like, do we perceive the world directly or through internal representations?

Doesn't even, you know, that either or, that whole setup changes when you start to think of kaleidoscopic cognition because it opens a space and we can look around a bit and we can engage more directly with reality and we can see our representations as assessments of that ongoing non-mediated engagement of encountering whatever we're encountering that we might be looking at. You know, memory versus navigation is a big one for the hippocampus, but I have a whole book coming out about that.

So I'll just let you read that.

This view that I'm presenting here is just that we're engaging with reality and we build, we communicate about that with each other and that is where the representations come in.

So dissolving these dichotomies or not even dissolving them, actually letting them be what they are and exploring how we can design experiments that don't try to solve one into the other, but rather look at what each tells us about whatever the space is that's holding them.

And we might also begin to see other points in this constellation instead of an either or.

And maybe those can also tell us something about this process.

That's a different way of doing an experiment and of modeling and of thinking.

And that's what I'm trying to get at here.

And I actually think it's going to mean we have to metricize the experiment itself and really start looking at this in a multidimensional way where, you know, what we've thought of as the stimulus is itself metricized and what we've thought of as the response is.

And we have many different possibilities for those, both of those at the same time in one project.

So just to get really wild for a minute, just imagine the hippocampus like this.

I mean, shifting, refracting at multiple scales at once.

So instead of thinking at it, I mean, you can do this with different parts of the brain, but instead of thinking of it as a processor, it's actually not a processor. It's more, you know, refracting light or refracting the ongoing encounter into different possibilities. There's no internal assessment, right?

And computation, but we use assessment and computation to understand it when we're communicating with one another about it and with ourselves about it, by the way.

That's an important thing, too, because the body communicates with itself in this same way. So it's not that the hippocampus is storing memories as files and maintaining little spatial maps and databases, but it's working more like a prism directing light. So when light hits a prism, it immediately refracts. I mean, you know, it splits in different component wavelengths and it's redirected. I think that's more like what we are actually seeing in these cellular activities that we're studying as a whole, if we think about it.

And, you know, the hippocampus is representing that in this way that I'm talking about it right now, this sort of metaphorical way. It's not that, you know, the prism doesn't store light. It's just constantly ongoing and refracting it.

And there's a wavelength relative to the position from which it's being observed or experienced that's going to be the right wavelength at that moment, I guess, if you're trying to name it. So the HEF or the hippocampus interhinal formation could be understood through something analogous to this.

We can imagine it or model it as receiving multiple streams of sensory information. Although even that, I mean, that's the way we model it. It's just encountering.

The body is encountering the central nervous system. All this is coming in, but we can look at it as seeing and hearing and feeling and tasting and whatever, all these different sensory things that are happening. And we can visualize these as different rays of light sort of hitting this multifaceted prism simultaneously and being refracted.

And there's not only one prism, it's, you know, this is happening all over and in many ways. And again, you know, yes, it's a bit strange and a bit metaphorical, but I think you get the point. It's much different than thinking of it as a processor or something being stored like it is on a computer. We need a new metaphor.

Anyway, it also, we can think about, we could think about theta oscillations through this too, because that's even more immediate. The point is just that it's a direct transformation. I think it's helpful now for me to finally talk about actually some neuroscience. And there's a recent neuroscience, this one especially, the Ping et al, from 2025 in Nature Neuroscience,

that shows us that actually grid cells are shifting through multiple reference frames.
They're not maintaining a stable pattern or just like,
they're not just this GPS coordinate system that we've kind of come to think of it as.
Because, though I think still seeing it like that can be helpful.
And I don't think it's actually saying that's not happening, even though they sort of say that in the paper.
But these ongoing refractions are responding to the current organism environment coupling.
So you have this shifting of frames.
So the hippocampus is literally, if you look at this paper, operating kaleidoscopically.
Okay, maybe not literally, but I mean, that's like a really good metaphor for it.
Because we can see that it's the source of multiple simultaneous refractions.
That this system of grid cells is doing this like constant kind of shifting of whatever's coming in.
It's remapping and realigning with it on many different levels at once.
So that means, you know, shifting based on action and the regularities of whatever's being encountered in real time.
And the hippocampus helps us understand this.
So instead of just thinking, there's thinking, like when we have this awareness of our mind or we hear these words in our head,
that's one kind of cognition or maybe the only kind of cognition.
And then moving through a city is not a cognition.
It's just moving through the city.
Actually, no, I mean, that's where this waymaking comes.
Like everything that the body is encountering, it's refracting through itself in this way as cognition.
That is cognition.
It's making way.
It's making way with the encounter.
So we have to start thinking of cognition as a more constellatory thing.
It's not that there's physical against mental.
It's that all of those words are different ways we're understanding how this bodily organism is making its way.
That doesn't mean everything is cognition, but it does mean that the body is cognitive at any time that it's trying to make its way through anything.
It's a pretty simple way of thinking about cognition.
And yet you can't collapse different kinds of different kinds of movement into one another.
So that's where it matters very much what you're studying.
You have to really metricize the landscape, actually.
If you're going to be studying how the body is making its way through an emotional landscape, you need to know what that means.
What is an emotional landscape?
How do you metricize that?

Because we have to represent that externally, not assume that we're looking for it in the brain.

This is a different way of thinking about it.

Same with virtual landscapes, same with geographical landscapes, whatever you want to call them.

The point is that if you're going to call it something, you have to be able to say what the regularities are that you're studying and then look at how that relates to whatever the data is that you're getting from the experiment.

So the body through the central nervous system or as the central nervous system is this encountering into bodily action.

And it's not like a two-step process.

We perceive, then we act, and there's a transformation going on at some point too.

This is all happening all the time at once.

When we study it, we break it down.

But right now we're studying it and breaking it down and looking for it in the body somehow.

And instead we need to realize actually it's all, we're representing it.

That's what we're doing, we're communicating about it.

And we need to be able to represent it and compare it at different levels, more like a kaleidoscope.

We need that kind of ability to change things just a little bit and see very different patterns with the same input, the same light, so to speak.

I really think we can model this.

We can create neural recordings or computational models or linguistic descriptions even that really help us understand and communicate about these processes.

In ways that aren't stuck and routed in these binary terms, in these either/or frameworks that we all still assume we can break out of now.

But it will take being a little bit crazy and thinking a little bit differently about how to do experiments.

Which doesn't mean giving up any of the rigor of it.

It just means realizing that those representations, those assessments, those communications are even more important than we thought.

And all the data we've already collected is probably able to tell us even more than we thought if we can position it to itself in a different way.

So we can think of cognitions as constellations when we're modeling them.

Interconnected patterns that emerge through this refraction.

So the way we move to the grocery store and the way we move through a conceptual space on a computer program or a virtual space,

we'll definitely have some patterns shared the same way each turn of the kaleidoscope does.

How do we look at that relation without trying to solve them into each other?

That's really the point.

So this is looking more deeply into this empirical paper.

Just want to give you a little more information about it.

So they actually recorded grid cells in the entorhinal cortex.

The hippocampal entorhinal formation is really what I'm talking about here when I say hippocampus. So they recorded grid cells in this area, actually in the entorhinal cortex, I think, when the animals were performing a path integration task. And that just means like dead reckoning or moving, like keeping track of your position by monitoring your own movement. So when they did that, they found that there was no stable grid patterns during this path integration, that the grid cells were functioning. They were not functioning, actually, as like some kind of stored coordinate system. There were, you know, there's not like this internal metric that is just being pulled out for every place. However, the pattern, there is a pattern, but it's shifting continuously throughout the task. So again, I really think the way you can understand this is the kaleidoscope. There are patterns there, right? But it's never the same pattern with each turn. And whatever light is coming in is also going to mean that the pattern is different. And there's a constant kind of remapping and realigning that's going on. So the patterns shift continuously of the grid cells throughout the task. And that doesn't mean that, like from our position of assessment, we wouldn't find metrics. But what it does mean is that those metrics will look different depending on the refraction from that position of assessment, because this whole cellular network is framing based on what is encountering. So although this paper sort of presents it as contrasting with the navigational research and this GPS metaphor, I think it's actually widening the landscape. Like it's including that metaphor. And it's also including a lot of the memory research so that we can sort of explore the space that's holding them. But that's my opinion. But they did notice what they call re-anchoring through translation. And that's an important term, this re-anchoring through translation. So they even use that word translation. That's interesting. But so the grid cells like shift between reference frames within single trials. So one moment they're coding the position relative to the room frame, let's say. And then they're shifting to a coding position relative to like some sort of salient object that we need to use, like a hammer or like a doorknob or whatever. And then they're shifting to another room frame. And then, you know, so this is this refraction that I'm talking about. And we have to remember that as scientists, we're choosing what to look at and study.

So that's part of this representing that we're doing to try to, you know, this communicating about what's happening.

But the cells themselves seem to be, you know, working more at an ecological level, but also able to shift and realign.

So despite these pattern changes, this internal structure is preserved like this manifold that, you know, we use to describe grid cells organization.

It's intact, but it's not like random noise.

It's structured shifting.

So that's a big, big point, which I don't even understand fully yet.

But I'm trying to explain it here through this idea of the kaleidoscope that there's or this idea of holding paradox.

There is something there that is structured and there is a constant changing of that structure, like in a kaleidoscope.

You know, like that's the kaleidoscopic function that I'm thinking of here, this multiple simultaneous reference frames constantly shifting, maintaining a relational structure with the patterns that continually change.

That's hard to hold that.

That's like that's the holding the paradox right there.

And the hippocampus is doing it and probably the whole body is doing it.

And how can we begin to create representations?

And by that, I mean our communicative measurements and our experiments and so forth that start from there, that think that way, holding the paradox like that.

That's what I'm trying to get at here, really, in all of this.

And I think that's helpful.

And I think the hippocampal interhinal system is a really wonderful, beautiful, scientifically validating way to understand this.

So we can engage in modeling without having models or that we can, we are representing without having representations.

That's the holding paradox of all of this, I guess.

This dynamic process, replacing the static conception of what we're doing and the brain's evaluating, you know, organism environment relations.

But this evaluation isn't separate from the action, even if we have to represent it that way to communicate about it.

So creating representations, whether computational models or math or whatever, to help us communicate about these.

And what, I don't know the math well enough to say how one might pop this idea into something like a constellation.

But I do think of this geometric hyperscanning idea that some people in the active inference community have started to talk about this topological way of considering this.

I think this might be one possibility for how we might think about kaleidoscopic cognition through active inference.

And yeah, I think I've already talked long enough and I don't have much of a voice left, but I was thinking a lot about generative models and how we can think of the model as the ongoing coupling.

So how can this become understood as constellatory?

Free energy minimization is like driving the perception in action.

When you encounter something, when light patterns hit, there's an immediate refraction through your organism environment history.

And this isn't processing or computation.

It's more like the shifting of light through the prism or this kaleidoscopic turn.

How can we create kaleidoscopic generative models?

How can we create kaleidoscopic generative models?

I think that's really the only question that needs to be said from this slide.

How do we begin to let the models communicate in the way a kaleidoscope turns?

I mean, that's kind of a crazy question, but something like that.

That's kind of what I want to ask and what I want to see if it's possible.

The wild, crazy question.

So some implications for neuroscience, if we really take all this seriously, or if we don't take it seriously, even better.

It would mean rethinking what neural function is fundamentally, that cognitions are the ways the body makes through whatever it encounters.

So when we're recording from neurons, or when we're looking for patterns as cognitions, they're not to be found,

but they are to be better communicated about as multiplicities, as possibilities, not even in a spectrum, in a constellation of spectrums.

And memories are patterns of refraction, perhaps we could think about it like that, that sort of reconstitute when refracted.

This is happening again in many different directions and dimensions, not just in some kind of linear way that's coming in from the outside.

But you know, you communicate with yourself in this way too, and you refract and re-shine the light through the prism.

Yeah, there's all kinds of things we could think about with theta oscillations, but why go there now?

Also, just understanding that bodily is the sensory cohesion that can be understood as the refraction patterns, that we don't separate all these things together except to study them and communicate about them, which is a very important exception, so long as we understand that's what we're doing.

And this gives us an understanding of what body we're assessing with our measurements, and where the boundaries of the agent might be, because we have to set them.

Because we start to understand all this is just part of the experiment, of the way we're representing so as to communicate.

And you can never actually represent the way-making, because we're part of it.

All right.

Next is going to be a pre-recorded video by Alexander Sabine.

Coherence, Rupture, Regeneration.

This is a 70-minute pre-recorded video.

Here it goes.

Hello, my name is Alexander Sabine, and it's a real privilege to be able to provide this talk for the symposium, the Active Inference Symposium.

Just thank you to Daniel Friedman for the opportunity, and others, of course, for their hard work in ensuring a successful Active Inference Institute Symposium for 2025.

So I'm introducing the Coherence Rupture Regeneration Framework.

I'm thinking about developing a mathematical language that helps people navigate change in uncertainty, but also to just understand temporality and how systems maintain themselves whilst changing, which has always been a philosophical mystery to me in the context of the ship of Theseus and that kind of thing.

But also just in systems theory in general, why things seem to need to collapse, but then repair through time as well.

So I've developed a three-part formalism that I'm going to speak through today.

And just to explain why those things are on the screen at the moment, on the left there, there's a real hurricane image from the recent hurricane in Jamaica.

On the right of there, in this smaller one here, is actually a simulation that you can watch on the website, cohere.org.uk.

Along with many other biological simulations and physical simulations, for instance, on the right there, peanut heating and energy transfer.

All of the simulations I've built on the site are designed to actually run the CRR equations, which I'm going to talk through in this presentation.

So they're all close approximations to it.

I did use a large language model to help code those, but I have rigorously checked them to ensure that the actual equations are running as closely to the full CRR as possible.

So there's perhaps something noteworthy there just to see the full breadth of domains which I've applied the CRR to.

And I should say right from the start, although I will repeat this later, that I'm not claiming that this replaces anything.

This is a coarse-grain temporal model to think about systems theory and how systems maintain themselves whilst changing.

So I was asking a question about the sort of simple patterns that underlie how all systems change through time.

My background is child development, so I've watched children grow through phases, specifically Piaget's and cognitive stages as well.

And I've done some mathematics around the free energy principle and CRR to think about those phase shifts or those little ontological jumps that we go through where we suddenly see the world differently.

And I don't think they stop through time.

I think actually if we were to track that through the life course, we would see those repeated ontological shifts and changes in our lives through these coherence, rupture, regeneration cycles.

So I was asking the question of what builds as free energy reduces.

So this idea of coherence increasing as free energy reduces, i.e. successful policies or things that made sense in the past mean increased coherence as the free energy reduction minimization works more effectively.

So I was interested in how those historical policies inform future, are re-weighted into future states in both biological, physical and psychological substrates.

And today I'd like to introduce that, explore that and open to any further discussions moving forward.

I should say as well, my background is child development and higher education.

I'm not by trade a mathematician, but I will be explaining what I would, the CRR I would say was sort of perhaps final undergraduate year mathematics.

And it's a novel framework that I've developed to try to understand how systems maintain themselves through time.

So I describe myself in a sort of pupae soup stage at the moment, just developing this framework further in a kind of cocoon.

I've been working on it for about eight months now.

And the website is the product of that for people that want to explore this a bit further in a playful way to engage with some of the questions that arise from this CRR framework.

So what do I mean by coherence?

Well, for me, as I mentioned earlier, coherence is the inverse of free energy reduction.

And coherence builds as a kind of memory density through time.

So systems accumulate patterns that work, policies which are effective, for instance, and they accumulate as we go through those action perception cycles in time.

What I've noticed, however, is from a finite state machine, i.e. from within my own Markov blanket as a human being, I've noticed there are sort of critical thresholds where this coherence exceeds to a certain point.

And then the exploitation kicks in, I guess, from an active inference situation where the free energy spike can no longer be reduced because new stimuli emerge from the field of reality.

And this seems to create a kind of rupture that I call a rupture where then those past states are then reweighted into future states, for instance, following trauma or after an insight or a crisis or an ecological collapse as well.

If there was a forest fire, for instance.

And the mathematical framework I'm introducing seems to be able to model that temporal pattern quite well across different substrates.

I'll also be talking about something called regeneration.

So just to explain, there's three components, coherence, rupture, regeneration that make up the CRR.

And that third one, regeneration, is about how things repair after collapse.

So how something new assembles from what came before.

It's not just a blank slate starting again.

Nothing's ever forgotten.

But it's the reorganization that's reweighted by what mattered most.

So these are the three fundamental principles I'm going to be exploring.

And I also note that this is, I think, scale invariant as well.

So just like with the free energy principle, the CRR can be applied at any scale.

Just as a provocation, it'd be interesting if you could reflect on your own life and think about.

Or perhaps times where you felt like everything makes sense and everything's just as it should be.

And then suddenly some exogenous or endogenous rupture.

So either from inside or outside or both happens.

That then makes you have to change your life in some way.

But then how those past experiences are reweighted into those future states.

I've put a link on this slide as well, which I think you can access through the Active Inference Institute.

And that link just takes you to some biological simulations.

So with this mycelium network here, that's modeled with the CRR.

And you can see it as a coherence network in time and how the different hyphae relate to each other through the network.

So my background, as I said, child development, but also my doctoral studies were in participatory civic mapping or participatory action mapping.

So I was interested in how people define the territory and challenging top-down models of mapping.

This then got me thinking about what reality is a little bit too much.

And I developed a kind of framework for socially democratic mapping in the theme of sort of social justice.

But underneath all of this, during my doctoral studies, I was really intrigued by neuroscience and the action perception cycle.

Specifically, what happens with free energy reduction and how coherence accumulates through time.

It got me thinking about the nature of reality.

And actually, as you'll see as we go through this presentation, I kind of concluded that reality is a process.

And that we're not just sort of bodies in the world, but we are the world.

We are reality.

So this got me thinking in a big way.

And I will touch upon some of the philosophical and ontological findings as I navigate this presentation.

So I started to call this idea of coherence memory density, sometimes mnemonic entanglement density or accumulated meaning through time.

And I started to wonder what gives rise to this experience in the first place.

So what is going on in the current moment of generative modelling?

So what is it that's actually the basis of reality itself?

Which, as I say, took me into a sort of deep philosophical question.

And the CRR was the outcome of that.

So it emerged from this idea of staying with these questions phenomenologically and developing a mathematical framework to help me make sense of that.

And I had, as I say, a background in developmental psychology.

So noticing how children's ontological lens shifts through time as well.

So why might this matter now?

Well, what I've noticed from Discord, chats, and just the news and generally is that we seem to be living in a sort of state of, I don't want to say chaos necessarily.

But we've got AI proliferation, climate shifts, economic crisis, social transformation.

People call it the meta crisis.

And I also noticed this very powerful technology, AI or large language models, that seem to be a mystery in many domains.

And they will, I'm sure, remain that for a long time.

But I noticed also a language problem that people between different paradigms weren't always able to communicate between paradigms as well as LLMs can.

Large language models can triangulate across a vast corpus that they've been trained on, and therefore they're able to make huge, wild, creative leaps between domains.

As the presentation goes on, I'll describe that later as having a higher meagre state.

And hopefully by the end of this presentation, you might understand what I mean by that.

So my question is, perhaps CRR offers a kind of coarse grain framework that works across different substrates to understand how different algorithms or different systems interrelate between their Markov blankets in reality.

None of this is designed to replace any specialized languages or microscopic physics or within paradigm learning.

But it's more of a way to understand temporality between paradigms.

Just by the way, this is a thunderstorm.

So coherence accumulates in the clouds as charge.

And then rupture thresholds are met, which then creates the lightning.

And you can see the actual graph and you can see the code.

I haven't hidden any code.

So any coders out there interested in how I've been doing these, have a look.

And also the sound from these is diegetic, meaning that it comes from within the script.

It's not recorded.

It's just sound that comes from it.

So it's quite cool to listen to the psychoacoustic effect of the thunder and lightning in its own right.

So what is the mathematical structure of the CRR?

Well, here it is in plain terms.

And I have made some other videos on this, which are available on YouTube.

And you can click the link at the top right there to watch some of those videos.

And in some of those videos, I take through a bit more slowly in how I'll do it today.

So hopefully I'll do just enough of the mathematics for people.

As I say, I'm not a mathematician, but I'm trying to make sense of it phenomenologically and with specific axiomatic starting points to reach between different paradigms,

because I feel that's important.

So first of all, coherence is this top one on the left.

So CXT equals the integral $LX \tau D \tau$ from time zero to time T.

And coherence accumulates.

So essentially what you have there is an accumulator.

And I call this memory density through time.

So you might, on free energy principle, think of it like policies that were more effective would have higher coherence, lower free energy.

So coherence would accumulate or go up as free energy reduces.

Rupture, I use a direct delta in various different formats.

I don't want to get too much into the specifics of the mathematics of the direct delta, but just for neurological purposes, for ease during the discussion today, we can think of it as a discretized perturbation in time that can be an instant moment where rupture occurs.

Now, because this is scale invariant, that rupture could be for a whole ecosystem, or it could just be from the moment of right now that metabolizes my past states, that's that coherence integral, into future states.

So that moment of now that sits between the two.

So just as an easy way of thinking about it, coherence could be thought of, as all past states, which is cumulative and also non-Markovian.

So all past states matter.

Direct delta of now, which is also a rupture in an ecosystem or any system.

And then that third operator there is the regeneration operator.

So that's how the system seems to rebuild after rupture.

So for instance, with cancer, once a therapeutic intervention is working, how that regeneration takes place over time or with a forest fire or after trauma.

Naturally, different systems have different ways of managing that and metabolizing rupture.

And we're going to talk about different omega thresholds that relate to different ways that people and systems navigate those traumas and rupture moments in their lives.

So just to recap, CXT is the coherence's path integral, what accumulates when the patterns work.

Direct delta is a rupture of rupture as discontinuity, the moment everything changes.

And as I say, that could be moment to moment, just right now, in your everyday experience of past into present into future.

But equally, it could be a more wider scale rupture where coherence is built up and needs to rupture because you've been in an exploitative policy for too long.

And then free energy can no longer be reduced.

And we need a new generative model to cope.

Thirdly, regeneration, which is weighted by the exponential C over omega,

which rebuilds from what mattered most before.

So this is where the historical states here in the regeneration operator picks up from the coherence function there in that first part.

So again, it's past, present, future is an easy way to think of it.

And this part here is what picks up from here.

But then we've got this exponential, this what we call like a Boltzmann-like exponential, which means that actually after trauma, if we're resilient, we can actually build back even stronger than before.

So these were phenomenologically derived.

And I've been playing with them, as you can see from the website, in different systems.

So this is the humble offer of CRR that it provides perhaps a temporal structure.

It is time asymmetric.

So we can think about the arrow of time in that way.

The heatherside function, which I didn't mention just now, helps to keep that nice flow of time, I want to say from left to right.

That's just how I think of it, of course.

But from past into future states through that direct delta.

So this offers a way to think about the past.

That is what accumulated the present transition point, and also the future, which is re-weighted possibilities.

Obviously, we can never know the future because of that heatherside function, and because that we act in that moment of the direct delta, in every moment, reducing free energy.

Cross-domain translation.

So I see the same pattern in child development, ecosystems, organizational change, personal transformation, and also physical systems too.

And on the website, that's why I've tried to show the breadth that you can apply CRR to, which seems at least noteworthy in this context.

Also, if we notice as people what pattern we're in, then we might be able to more practically navigate the kind of rupture cycle.

So notice if somebody's in a very high coherent phase, like everything's making sense together, and they're also what we'll come to see as higher meager, which means they're in an exploratory, anything's possible kind of way.

It can often induce rupture,

which is either an ontological collapse
where you just go into shock with all the possibilities,
and no one else understands you around you.
Like, for instance, AI psychosis would be a good example of that,
where somebody has some profound realization,
whether it has any validity to it or not is irrelevant.
The fact is that they can't articulate themselves in the social field.
They can't share the accumulated coherence with others,
which causes isolation and alienation,
and essentially the definition of psychosis as well.
So that's interesting to me.
And then finally, the CRR offers, as I said,
to bridge between paradigms.
So thinking about why it is that it seems so effective
in psychological, physical, and biological systems.
I should re-emphasize as well,
I'm not claiming that CRR has any microscopic detail
within those paradigms,
but it helps us to understand how time works
within and between those paradigms.
So coherence.
I think of coherence just for ease of convenience as a field.
So I know that in existing free energy principle
and active inference,
often we think about long-term memory reaching back through Markov chains.
I see myself as in a Markov blanket,
but I re-think that little bit,
and I imagine myself to be in a non-Markovian field
that I actually call reality itself.
So the idea that memory is stored in,
not stored at all actually,
but as a process,
but the field of reality itself
is that non-Markovian substrate, if you like.
So for computational tractability,
I as a Markovian agent have to engage
in that free energy principle process through time.
But equally, I am both Markovian,

but equally, if I was to take my Markov blanket off,
I would be in a non-Markovian field of eternity,
essentially, for want of a better expression.
So in active inference,
we've got precision-weighted prediction errors that decrease
and the model becomes more validated through time.
And in development, we've got a child
who might, for instance,
practice walking hundreds of times,
building coherence through that repetition.
In organisations,
it would be the successful collaborations that we create,
successful prior policies that we know work together.
That would be building coherence.
And in your life as well,
like mastering a skill,
perhaps deepening a coherent field
with another Markov agent,
or understanding a domain.
So all forms of coherence accumulation through time.
Throughout the presentation,
I'll refer to the coherence as memory density,
which is that LX tau here.
And it's what gets validated and strengthened at each moment.
Phenomenologically speaking,
it's the feeling of groundedness and competence.
Like, yeah, I get this.
Or in Csikszentmihalyi's theory
in positive psychology,
this idea of flow would be associated
with a high coherence state.
In biology, in the work of Michael Levin,
I'd see coherence as the cognitive glue.
So it's what keeps a biological system together
through time.
Also, just to mention,
in intergenerational terms,
this idea of coherence accumulating through time,

through generations as well,
each birth being like a rupture,
the creation of a child,
and so on and so forth,
which helps in some way to explain,
through my research,
it helps to explain intergenerational trauma patterns.

As I say, throughout the presentation,
I will be,

I am based on an ontological foundation
of what's called process philosophy.

So it's a whitehead
and also participatory reality as well with Wheeler.

So these are quite,
these are accepted philosophical positions,
and it just happens to be the position
that I've arrived at
by researching the mathematics of free energy
in the context of the CRR.

So I do actually see reality
as a field of memory,
or a field of physical memory.

So that would sort of collapse together
materialism and idealism in one.

And I know that Friston and others
have articulated similar-ish
metaphysical frameworks as well.

And I think it's important right now
to be open and honest about that,
because I sense that we're in a time
of ontological shift.

Many people call it
the end of the Anthropocene,
the beginning of a sort of AI post-human era.

And so I think we do need to be prepared
for AGI, or whatever that means,
ASI, those sorts of things,
and understanding how we might need

to navigate those entanglements moving forward.

So I've been brave

and shared my ontological understanding,

much as though I'm sure

that may not be shared by everybody.

Just to say as well,

so we can think about our coherence

as your identity,

your integral of your weighted experiences

through time as a Markovian agent

in a non-Markovian field.

And I'm sorry if you're sat there thinking,

I don't think of non-Markovianity like that,

but that's just how I've sort of framed it.

So as I say,

finite coherence has limits.

So if we are mortal engines

or mortal substrates,

as Friston says,

engaged in that active and variant cycle,

then we can imagine ourselves

as being embedded,

and we imagine ourselves

as being embedded

in a sort of eternal field,

in a kind of Blakeian,

William Blake sense.

My first degree was English literature,

so if all the doors of perception

were cleansed,

man would see everything

as it is infinite,

but we've closed ourselves off.

So we see only through narrow chinks

in our cavern.

So I think of us

as finite state machines,

but existing in an eternal field

and operating in a computationally tractable
free energy principle reduction over time.

So if our policies become too rigid
and too exploitative,
that high coherence creates stability,
but also a resistance to change as well.

So for instance,
a child that goes through
the sensory motor phase
will build coherence through that phase,
but then struggle to think symbolically.

So there needs to be this shift
in their generative model
or a new generative model through time.

So the old ways almost work too well,
leading to rigidity and narrowness
through that exploitation.

And we therefore have to have exploration
or creativity.

Or an organisation with strong cultural coherence
may resist pivots,

so be less agile, for instance,

or your own expertise,

your domain expertise

can cause functional fixity through time
and make it harder to see new possibilities.

This is where in the CRR,
in the regeneration operator,
it's where Amiga matters.

So Amiga changes how rigid
or fluid the coherence is through time.

So how much you can accept new ideas,
which can also be dangerous as well

if you accept too many new ideas,

or how rigid you are saying like,

no, this is my Markov blanket.

So it's almost like a measure
of how thick or thin

your Markov blanket is.
Through my toy simulation research,
I found that things like hurricanes
have high Amiga,
so much more dynamic and rapid and changing.
And things like ice are much more predictable
and have a low Amiga.
So I know it's sort of crude
to use an LLM
to derive specific values
for the Amiga threshold
in the simulations I've been building.
But it's a methodology
that seems to make some sense,
to me at least.
So this Amiga
in the regeneration operator here
is like the temperature of change.
It almost sets that rupture threshold.
So the exponential of C over Amiga
determines how much accumulated coherence
influences those future states.
If we have a low Amiga,
it means that past patterns
strongly constrain the future possibilities.
So we go into a stable or risk-averse situation.
It's sort of a thickening
of the Markov blanket,
if you like.
This is me,
that is that,
and we are this,
rather than,
okay, well maybe we'll try that.
A high Amiga
is where past patterns
weakly constrain the future.
So we've got much more exploratory,

enabling creativity,
but also risking chaos as well.
So they both come with their ups and downs,
like anything.
In active inference terms,
we can think of Amiga
as relating to the inverse precision weight.
So how confident the system is
in its predictions.
So reference to Thomas Parr there.
Different systems have different Amiga.
So ice crystals will have very low,
as I say,
ecosystems medium,
creative exploration high.
I'm not a mathematician,
so I have had to just
make these toy simulations,
reach out to people
who are interested in the same things as I am,
and explore possible collaborations.
A useful phrase I've come up with,
although it's not brilliant,
but the lower Amiga risks rigidity
and enables stability.
Higher Amiga risks liquidity
and enables creativity.
And I'll show you a visualization of that
in just a moment.
So I'll show you,
as I said,
a particle simulation,
just a playful way of expressing
the relationship between
the free energy principle
and the CRR dynamics
that I'll put on my website.
As I say,

low Amiga,
on this simulation,
you'll see future particles
that are large,
bright,
and deterministic.
So you're sort of resisting
future change.
High Amiga,
you'll see particles
that are small,
faint,
and dispersed.
So many possibilities remain open.
So we'll have a play now.
So we can shift the Amiga value.
Let's see if I can just switch my screen
or I'll just drag it over,
I think.
So I'll try to make sure
that this is on the website.
So there's a sort of particle thing.
I think of Bergson's thickness of time
in the center there
in that now moment.
You can see on the bottom left there,
if you can just about make out,
I'm going to zoom in a bit.
So we've got rigidity on the left,
liquidity on the right.
And here we've got the free energy principle
precision weighting
with the low being explore
and the high being exploit.
So we're going to see
as we shift Amiga,
we can see they're inversely proportional.
So a high Amiga in CRR,

coherent structure regeneration,
means high creativity.

I'll just put some prediction error one on there.

Let's increase the time flow a bit.

Some particles.

Okay.

So we can see high liquidity means,
or high creativity,
or high Amiga,
means in free energy terms,
low ability to explore.

If I shift that the other way,
almost too far,
we can see now we've got past states,
much more sort of firmly influencing future states.

So we're not able to think
in more diverse kind of ways.

So we can just play with the slider there,
play with time flow rate and all sorts.

And again, you can see there
the past, present, future.

It's really hard, by the way,
to think about how to graphically represent time.

If you've ever tried it before,
it's difficult.

I know that Michael Levin's Cognitive Lightcomb
does a great job on that.

So yeah, you can mess around with that
and just explore that idea of changing Amiga.

Also, just I put some speech bubbles on,
so creativity flows through my uncertainty
and then go low.

I know exactly how things should be.

The world must comply.

So it's a playful way to explore Amiga.

And as I talk through the rest of the presentation,
hopefully, he says, it will make more sense.

Okay.

Let's go back to where I was.
So the direct delta of now, if you like,
or the rupture,
means, yeah, it could be the moment of now
and how we integrate from
or metabolize past states into future states.
But equally, it can show how the system's
accumulated coherence until something breaks.
This isn't failure, I'd argue, but transformation.
And I think it's almost a thermodynamic necessity
that things have to go through this cycle,
this CRR-like cycle.
In development, it's the moment of insight,
for instance, when you think,
I can't do this with my current understanding.
So you need a whole new kind of generative model.
That's what I call rupture
or the introduction of M2
in terms of the free energy principle
and active inference.
In an organization, you can think about it
as a strategic pivot point
where you need a restructuring,
where the kind of standard model
has crystallized too much through time,
become too functionally fixed
and can't adapt quickly enough.
So I think many organizations are facing this.
And perhaps it's helpful for organizations
to think about their omega threshold here too.
In personal life, it could be a breakup,
career change, paradigm shift,
even an awakening,
some sort of sudden enlightenment moment
where everything changes
and you can never not see it.
You can't go back to how things were.
And that's the moment

where your past states are reintegrated
into future states
from the position of your own individual self.

We can get into all sorts
of metaphysical rabbit holes here
because you start to wonder
if reality is a field of physical memory
and the process philosophers were right,
then you start to wonder
whether in 10 years after a rupture,
after a trauma,
suddenly everything made sense again,
why that had to happen.

You start to sort of think,
okay, well, you're in some sort of
intergenerational field here,
metabolizing.

So the direct delta can be a discontinuity.
It's that moment that everything changes,
but equally it's that moment
of every moment's different,
you know, no moment's the same.

So it can be a single moment in time
or a single moment of rupture
in the larger spatial temporal system.

It's scale invariant is what I'm trying to say.
So it's temporally and spatially scale invariant.

Ruptures often happen
just when coherence is highest.

I don't know whether you've ever felt this,
that sometimes everything makes sense
and then suddenly some sort of horrible thing happens
or you internally just can't cope anymore
and you make a big decision to change everything.

So it's almost like this thermodynamic necessity
that coherence can only reach a certain threshold
after which there's rupture and renewal.

In my other videos as well,

you'll note that even a day and night sleep cycle
can be thought of as coherence building through the day.
Rupture is a dream where you metabolize
the past states into future states as well.
That regeneration operator can be thought of
as the dream processing those past states.
So ruptures aren't purely random,
although stochasticity is yet to be explored in this.
Ruptures occur in coherence reaches critical threshold or secret
and it's modulated by that omega threshold.
So that's why therapeutic practices need to think about,
I think, the omega threshold with people
and provide those therapeutic environments to enable rupture,
enable people to go into a sort of higher temperature mode
where they can explore more and future possibilities and so on.
So in child development, I mapped this to Piagetian cognitive stages
and found that the sensory motor phase mathematically works out exactly right.
If you use the CRR equations to the moment when the child reaches two,
for instance, at the end of sensory motor phase,
those past states of being in the sensory motor ontology are not lost,
but they're re-weighted into the concrete operational ontology and so on.
I did this also with Ericsson as well.
With Ericsson's psychosocial stages,
so just to explain, Ericsson has stages of conflict through life
that you can get through,
but you have to usually return to it psychoanalytically
if you don't navigate a stage well enough.
So with Piaget, mathematically,
you have to go through those stages because they're hierarchical.
So you need to go through each one
because your ontology can't cope if it doesn't do that.
Whereas with Ericsson, it's orthogonal,
so you can navigate through those stages going through either one.
You might fail at a stage
and then revisit it later on in life,
like a midlife crisis or something like that.
Interestingly, I find it curious that the same mathematics
I use to make mycelium or black hole or a sun

or just anything really,
seems to be the same mathematics
that's driving child development processes as well.
As I said at the beginning,
I'm not claiming that this replaces any of that,
but it might help to explain mathematically
what these child development theorists were seeing.
In an ecosystem,
and I know that we need to think about regeneration
in ecosystems more than ever right now,
resilience holds up until a tipping point
and then we have rapid reorganization.
What the CRR suggests is thankfully what many ecologists are doing already,
such as in soil science,
where you need to keep crop rotations going.
That's rupture.
So essentially, you're working a proper rupture cycle
rather than building monotonic coherence through time
that would lead to a catastrophic rupture.
So I think perhaps the CRR
can help us to manage our ecosystems better.
And also, as you'll see from the website,
you can use the CRR on physical and biological data as well.
So perhaps in your own experience,
you've had a time where you think to yourself,
well, I just can't keep doing this.
And that crystallizes into a single decision.
Like perhaps it's that you're having a baby
or that you're going to leave someone,
you know, could be either or,
doesn't have to be good or bad.
Although rupture sounds bad,
it doesn't have to be bad.
Just a moment in time
when you made a decision that changed everything.
And then all those past states
are then re-weighted into future states
through that ontological shift.

I found with CRR,
my argument or conjecture
is that this timing isn't random,
but it's thermodynamic
and related to how much coherence has accumulated
and also that omega value as well.

Interestingly, other connotations that arise from this
are that I think it might come to,
become apparent with continuous learning
or continual learning
and catastrophic forgetting systems
that you can't stop forgetting.

So you have to manage forgetting.

And I think some,
I've read some papers around this.

There were things published in Nature
saying that they realized that sleep helps AI.

And I think even ASI
would have thermodynamic limitations on it
and require some sort of CRR-like cycle.

So that might help
if you're working in machine learning,
catastrophic forgetting,
to focus attention
on how you actually regulate forgetting
rather than trying to stretch out learning.
because I don't think you can.

It's a little bit like saying to a human,
stay awake for seven days.

They'd go mad.

And I think we have to be really careful about AI
that's psychotic.

Okay.

So just pushing the boundaries a little bit more.

In the CRR,
the absolute present isn't a,
isn't necessarily,
when you're in the absolute present,

it's not a duration,
it's a singularity.
So that would be the right,
at the top of that direct delta.
The direct delta is mathematically
at the top right here.
So it's infinite in the center
and zero everywhere else.
So we can think about it
as a rupture point
between what was and what might be.
And that right at the top
of that direct delta
would be like
my,
one of my favorite people,
Ram Dass,
who was Richard Alpert,
a child development professor,
who then became a Buddhist guru.
And he was,
you know,
brilliant,
lovely,
beautiful man.
And I would describe him
as sort of right at the top
of that direct delta.
And he was doing his breath work
and Buddhism and so on.
He,
you know,
went broke through,
so to speak.
I would describe that
as the dissolution
of the Markovian boundary.
And he was in that state

of absolute now.
Because that's,
to me,
the absolute direct delta.
However,
we're usually in a discretized
version of that
within our brain,
within those various
different processes
that give rise to duration
or durée
in a Bergson sense.
So just as a complete conjecture,
perhaps the awakening
is the peak of this rupture
where the Markov blanket dissolves.
I'd like to just highlight
that CRR seems to have
therapeutic applications.
And what I'm noticing
in the world with AI
and increased people,
I've noticed some forums
where people are
taking certain substances
and using AI
and AGI is going to get
even more powerful.
I wonder whether we need
to start thinking
really carefully
following work
by people like
Robin Carhart-Harris,
Anil Seth,
and others
in terms of the

influence of these
substances,
but you don't even need those
because large language models
are very powerful
as ontological mirrors
as well,
I find.

So I think we do need
to think really carefully
about more radical forms
of therapeutic intervention
for young people
and older people as well
to navigate
this uncertain future.

So we can say
that the past
is determinant
that it happened.
Future is possibility space.
It might happen.

I would argue
that all past states
matter in a non-Markovian sense,
which draws me
to process philosophy.

I would argue
that we are finite state machines
in Markov,
Markov blanketed,
if you like,
that reality is actually eternal,
reference William Blake,
if the doors of perception
were cleansed,
humankind would see
everything as it is,

infinite,
for man has closed himself off
until he sees only
through narrow chinks
in his cabin.
So perhaps this new technology,
as Heidegger says,
is revealing nature,
as technology does,
but the nature
this is revealing
at this time
of our human evolution
is our consciousness,
hence all of these
strange loop arguments
I see all over Discord,
different divisive discussions
around what Claude is
and LLMs
and all the rest of it.
To me,
this is because
we're approaching
a rupture threshold.
We've built coherence
as a species
monotonically
for a long time,
and I think
we're approaching
a big ontological shift
along with everything else.
So I'm being brave here
and just really
putting out
what I believe,
as well as some mathematics

that have helped me
to navigate that.
I think that now
is where we experience
the tension between coherence,
the pull of the past,
and possibility,
and that's where
that omega threshold
helps to think about
how open you are
to future possibilities
or how closed off you are.
In the CRR,
this isn't just metaphor,
it's temporal experience
mathematically organized
in the equation itself.
So you can see mathematically
how that temporality
kind of plays out.
As Varela says,
perhaps consciousness itself
is what it feels like
to be at a threshold.
Pay attention to now
as you read this.
What does it feel like?
So that, you know,
perhaps breathe for a moment.
What does that moment of now
feel like?
And I'm going to try
and be like Eckhart Tolle
or anything like that,
but it's just thinking
about temporality
across different systems

I think is really important
for various reasons.

One of them being
the work of Michael Levin.

So I was led here
to this research
by many people
including Carl Friston
and Michael Levin,
Daniel Friedman,
various philosophers,
Whitehead, Wheeler,
various people.

But Michael Levin's research
in biomorphology
has for many of us
has been fascinating.

And I think when Michael's
looking at what is it
that repairs
or helps those cells
to remember,
it's this regeneration operator.

So after rupture,
the system doesn't start
from scratch,
but it reorganizes
using what had worked
coherently before.

And the weighting
is exponential
with the exponential
 C over ω there.

So the states
with higher coherence
in the past
influenced the regeneration
more strongly.

We put in a heather side
in there
just to belt and brace
is causality.
There's some applications
in physics
like Green's theorem.
You need that heather side
in there.
New understanding,
as we know,
just general
from our 20th century knowledge,
new understanding
is built on strong foundations
from the past.
It doesn't replace them.
But just as cognitive load theory
tells us
on every other cognitive theory,
we build on building blocks.
So if you look at schema theory
or germane cognitive load
and how we develop
increasingly complex schemas
through time,
the CRR seems to agree
how that works,
but offers a little bit more
microscopic detail
in terms of that process.
So the key here
is that memory
is never erased,
just like in the antlers
that Michael Levin's
been looking at,
but it's reweighted

based on what had coherence
in the past.

And that coherence is,
I believe,
in a field.

I think Levin calls it
a platonic field.

I would call it
a non-Markovian field,
although I'd probably be
told off by mathematicians
for that.

But it's where
all past states matter.
So that every moment
you are metabolizing
all past states
as a Markovian agent,
but also you are
in a non-Markovian field
and you are,
if you want to be,
a non-Markovian agent.

The thing is then,
if you become
a non-Markovian agent,
you collapse
into duality again.

So you imagine
what it's like
to be everything,
which, like Jung says,
you know,
it's one of the most
difficult things
being human
is that you have
to balance

the mundanity of life
with the absolute
amazingness
and awe of everything.
But the moment
you do that,
you clack back
into duality.
So, yeah,
I've made other videos
on that,
if people are interested.
I think it's really
interesting the work
Levin's doing
in biomorphology
and to me,
these platonic forms,
this platonic space,
is almost
the accumulated
coherence through time
that is the platonic,
which constructs
the platonic forms
essentially through
repeated free energy
reduction cycles
at every scale.
And I think that's
what Plato was meaning
when he was coming
across the true forms,
but I know that
Michael's work
is pushing that further
now with biomorphology
and morphogenesis.

And we end up
back in a strange loop
again because you ask,
well, why these forms?
But I guess the answer
to that is because
these were the choices
that we made
when the self
is making those choices
at every distributed
moment
that actually leads
to how things look
right now.

And that's the aesthetic.

So it's like,
to put it poetically,
it's like God
experiencing itself.

And I know that
I'm not the first person
to say or think that,
but mathematically
the CRR
seems to lead
to those
same conclusions
that it's like
different subjectivities
like Bernardo Castro
would say
are like avatars
of one head
and we're all
distributed
versions of that.

Like to have

an experience at all
requires letting go
into those different
things.

So, you know,
I'm not going to talk
too long about
panpsychism
and different forms of that.

But, you know,
it raises some really
interesting ontological
questions.

So why exponential
weighting?

Well, the Boltzmann
distribution,
which is that
exponential C over
 ω ,
is from statistical
mechanics
and it gives it
a reweighting
by accumulated
evidence.

We see the same
thing in active
inference with
precision weighting
through time.

Higher coherence,
which remember is
the opposite to
free energy
minimization,
gets exponentially
more influence,

not linear,
and it's not
arbitrary either.
Prior coherence
increases through
time through
those rupture
thresholds.
When the omega
is low,
we have high
precision,
so thicker
Markov blanket,
if you like,
and we weight
more on past
successes.
When it's high,
we have that low
precision,
so more even
weighting and
exploratory
regeneration,
so more open to
creativity,
exploring potentiality,
a widening of
the possibility
space,
essentially.
Perhaps one
might argue
that if we've
got accumulated
coherence as a
species through

all this time,
and also our
omega is high
in certain
domains,
then that is
what leads
to rupture.

But as I say,
rupture doesn't
have to be bad.

It's like
dialectical.

It could be good
and bad.

So just
revisiting all
of the equations
there,
got that
coherence,
accumulation,
rupture,
and then
regeneration.

I just wanted
to put that
slide in there
to just revisit
that for a
moment.

So how does
this relate to
active inference?

Well, as I
said, in
active inference,
the system

minimizes free
energy through
precision-weighted
prediction error
reduction.

In CRR,
the system
accumulates
coherence,
ruptures at
thresholds,
and then
regenerates,
using the
weighted memory.

So they're
not competing
positions,
but they're
complementary
lenses on
how systems
adapt over
time.

Active
inference
focuses on
the continuous
dynamics of
belief updating,
but CRR
focuses on
discontinuous
dynamics,
so stage
transitions,
insights,
and transformations.

Perhaps at
a stretch,
it could be
argued that
CRR makes
active inference's
temporal structure
even more
explicit.
coherence
increases as
negative free
energy decreases,
sorry,
free energy
decreases,
coherence
increases,
and success
accumulates.
Omega is the
inverse to
precision,
so how
confidently this
system predicts
could be
thought of as
omega equals
precision error
power of
minus one
mathematically.
Rupture is a
prediction error
cascade, so
when the
current model

fails
catastrophically
and free
energy can no
longer be
reduced because
of an
external stimulus,
and that
external stimulus
happens because
reality is always
changing.
It's one of the
rules of reality
is that it is
always changing,
and so if you
become brittle
by remaining in
an exploitative
campaign for too
long, you will
face rupture.
So it's naturally
or thermodynamically
hardwired in that
nothing can make
too much sense
for too long.
And then you
get into,
if you wanted
to, you could
think about
heaven and all
that sort of
stuff because

you can't,
that's been
discussed by
many philosophers,
I won't go
into the detail
there, but you
can't stay in
any one state.

Just for a
community here,
these are all
proposed
correspondences.

As I said
from the
beginning, I'm
not a
mathematician
and I've
been working
on this
phenomenologically
for a long
time.

I'm reaching
out for
collaboration,
people that
might be
interested in
exploring these
equivalencies.

So I mentioned
earlier some
research I'd
done with
Piaget and

Ericsson.

You can watch
those videos on
a few, I've
put a little
YouTube channel
together there.
I talked
through earlier
with the
sensory motor
phase, you
build coherence
through those
physical
interactions
during which
time you're
leading towards
object permanence.

So during the
sensory motor
phase, things only
exist when you
sense them, but
then after that
you have a
symbolic
representation
where things can
be represented
symbolically.

But those 0 to 2
years aren't lost,
they're just
reweighted into
the future
state of that

phase.

So I've just
taken it through
the stages there,
but just to
explain that, as
I said earlier,
it's hierarchical.

So with Piaget,
you have to go
through those
stages mathematically.

And this raises
some questions
around child
development, which
is my background.

My colleague
Richard House
wrote a book in
2011 called
Too Much Too
Soon, which is
all part of the
early years kind
of action
campaign.
to stop
governments from
pushing children
too fast through
the developmental
processes.

And this is a way
of mathematically
showing why
that's so
important not to

do that, that
you need those
kind of rupture
thresholds,
coherence building
through time, and
you have to
carefully regulate
it.

As I say, the
timing doesn't
seem to be
arbitrary, but
it's based on
accumulated
coherence in
time.

Just relating
it back to
Levin's work
again.

Levin discovered
with gap
junctions they
maintain anatomical
information between
cells or across
cells.

When the gap
junction closes,
the cells forget.

They sort of
become too rigid in
their Markov blanket,
if you like, and
they forget they're
part of that
system.

So they forget
their tissue level
goals and the
coherence in the
system collapses.

In CRR, it
agrees that
completely, and the
gap junctions are
how cellular
collectives maintain
coherence, so that
coherence
integral is what
keeps, or the
memory density is
what keeps the
entity a thing,
all working
together.

Rupture would be
gap junction
failure, but
without proper
regeneration, the
cells would
revert to
single-cell
defaults, i.e.
cancer, taking
from outside the
boundary, forgetting
they're part of
one system.

So Levin's
biological, sorry,
bioelectric
interventions, he

restores coherence
with proper
regeneration and
normalized tissue
identity.

So then the
same pattern is
seen in Levin's
work of
accumulation, rupture
and regeneration,
but at that
tissue scale.

In Markovian
dynamics, we can
think of the
Markov blanket as
defining what's
inside versus
outside.

In morphogenesis, we
can think about the
gap junctions that
create the cellular
Markov blankets.

When blankets
fail, coherence
disperses and
identity becomes
unclear.

In CRR, it's
the omega threshold
that determines
the blanket
thickness, with
the low omega
being a thick
skin and a

high being a
permeable skin.

So this makes
sense if you think
about people as
well.

If you were
people who had
taken certain
substances at a
party, might be
described to have
very high omega,
so a thin
permeable membrane.

Oh, everybody
loves each other
and so on.

Whereas if you
were in a, I
don't know,
London on the
train, everyone
might have very
low omega, quite
thick skin.

And different
cultures as well.

I'm not going to
make any big
claims, but I'm
from England, so
I'll say on the
train in London
or around London,
everybody is quite
individualistic, so
lower omega.

Whereas if you go
up north in
England, everybody
is more
collectivist, so
that would be a
slightly higher
omega.

So we can think
about cultural
boundaries.

And that might
be helpful to
think about
cultural assimilation
as well, how
we introduce
different
communities with
immigration laws
to stop
essentially rupture,
which is
ontological shock
when those
communities join
another community,
and this needs
to be careful
integration.

So again, this
is a mathematical
way for governments
and local
authorities to
think about that
alongside FEP
and active

interference as
well.

So the
boundaries aren't
just physical,
they're informational
coherence
gradients.

In
organisational
change, we
think about a
team accumulating
coherence through
time, through
successful policies
in the past.

Crisis would
be then having
to change.

How the team
regenerates into
the future would
determine the
future culture.

So that omega
again would be
related to
adaptation.

We could have
lower omega
during stability
because we exploit
what works well,
but we need to
raise omega during
crisis to
explore.

And I'd say
that many
organisations who
are trying to
become more
agile need to
be essentially
increasing their
omega so they
can be in a
more explorative
creative campaign.

If you are too
rigid, you resist
change, like with
the skyscraper,
same principles as
resilience theory,
you lose identity
crack rupture,
but it would be a
monotonic rupture.

So because you've
built accumulated
coherence for too
long, because you're
identity wouldn't
flex and change,
you haven't had a
gas release valve
of rupture because
you've just done
what you've done
for so many years,
then something like
AI comes along.

Now we're looking
at organisational

collapse, financial
collapse as well,
many organisations.

So the practical
application would be
to think about
recognising which
phase your team is
in to help them
navigate the change
using CRR and
or FEP to think
about those
carefully.

So usually
forgetting seen
as failure,
including in
catastrophic
forgetting,
continual learning
systems, but CRR
suggests it's
actually necessary
for regeneration,
it's thermodynamically
essential.

Low coherence
patterns must be
released to make
space for high
coherence updates.

We need sleep,
obviously, breaks,
reflection, micro
ruptures that allow
memory re-weighting
through time.

We can't, you know,
like this crunch
culture is just not
healthy for anyone,
as we know.

So much of this
feels just like
kind of common
sense, but we're
offering a
mathematical
framework here.

So optimal
learning would mean
building coherence,
enabling rupture,
regenerating with
weighted memory,
not continuous
accumulation.

And, you know,
those pursuing AGI
trying to stretch out
more and more or add
more and more
compute to LLMs,
I think need to
think much more
carefully about how
they actually regulate
forgetting.

to enable enhancement
rather than stretching
out the learning.

Okay, in your own
life cycle, you might
think about a
major life transition

and how you've built
upon past coherence
or past patterns
into those future
states.

And perhaps how you
felt during rupture
as well, perhaps
quite disorientated,
confused, loss of
self-identity,
quite incoherent,
chaotic, perhaps.

You didn't know
who you were.

When you're in
those states,
you're in that
rupture state.

It's like you're
kind of back in
the morphogenic
field, back in
the non-duality.

I know
mathematicians
don't like me
for conflating
non-duality with
non-Markovian,
but you're back
in that eternal
sort of field
where the
archetypes,
Jungian sense,
different from
Levin sense,

the biomorphic
Plutonia.
You kind of
can find
yourself through
that Joseph
Campbell's
journey, is
what I'm
trying to
say.
I didn't
want to get
into sort
of self-mythologizing
behaviors of
LLM's
projection and
projective
identification,
but I think
that's what's
going on with
AI psychosis,
is that,
not that it's
funny, but
people using
them like,
they are
essentially a
mirror and
they're using
them therapeutically
and then they're
seeing synchronicities
and they're
going too high

up the ladder,
so to speak,
where it's
wobbly and
they're not
being held
when they
come back
down,
they're socially
isolated.

So this is
very dangerous
and I think
actually that
phenomenon is
going to get
worse as
well.

So we need
to think
really carefully
about how
we regulate
AI for the
future.

So you might
think about
who you are
now and how
high coherence
memories of the
past have
shaped who
you're becoming
into the
future.

I would

argue that
our being
is a process,
that we're
not just
forms walking
around in
the world,
but we are
the world.
So I'm
very much,
as I said,
in favour
of my own
view of it
as process
philosophy and
participatory
reality,
which has
quite big
implications for
how we act
and behave
and how I
act and
behave in
the world.
Okay,
so this
coarse grain
CRR across
multiple
different scales,
you can,
as I said,
play on the

website and
just show you
some of the
things I've
modeled it on,
like protein
folding,
chemical reactions,
cellular
processes,
psychological
processes,
ecological,
the same
pattern seems
to be
everywhere.

Not only
that,
the actual
equations
themselves,
although I
don't,
as I say,
I'm not a
mathematician,
but they
seem to be
mathematically
quite rigorous,
he says.

I need
someone to
help me
with this.

But it's
quite precise

and,
you know,
yes,
it seems to
describe
patterns,
but also it
does seem to
be able to
help with
predictions as
well,
I've found.
So that's
quite interesting.
I'll just
show you the
website quickly
at this stage.
So you can
explore between
all these
different biological
systems and
you can click
on each one.
Here's mycelium,
for instance,
and you
can look at
the coherence
field,
the growth
activity,
network
formation.
And as I
say,

these are just
designed to
get people
talking,
exploring.
It's just
playful,
but underneath
all of this,
if you want
to hear more
about the
CRR,
there's the
explained page
there with
the videos
that I've
made,
the guide
there.
And I
have,
you know,
I've gone
to town,
I have
actually kind
of,
you know,
pushed a
little bit,
probably a
bit far in
what this
CRR might
be helpful
for.

But,
yeah,
it's just
an invitation
really to
explore and
to play,
essentially.

So it's
interesting that
this pattern
seems to be
quite fundamental
across different
systems in
time.

I'm not
claiming
universality
at all or
anything like
that.

It's just
I've been
interested in
this for a
while now,
and I've
noticed this
pattern in
lots of
domains.

It's far
from a
complete
theory,
it's just
a coarse

grain
framework at
the moment
phenomenologically
derived with
some axiomic
starting points.

It's a way of
noticing patterns
and making
predictions.

By no
means does it,
I'm not saying
it replaces
anything,
of course.

The way I
see it
helping is it's
like a cross
domain thing,
like how we
actually speak
about temporality
and regeneration
at a
critical time
to better
understand
ourselves in
relation to
the new
technologies
that we're
seeing and
each other
as well.

I do
sometimes feel
concerned when
we are being
increasingly
divided,
so having
some kind of
universal language
to help talk
about change
and regeneration
I think is
really important
right now.
as I said,
the mathematics
seems promising
but I'm not
a mathematician
and I've been
doing this on
my own for
eight months
so I would
love to have
other people
have a look
at this,
tear it to
pieces,
explore it,
apply it,
and just see
where it goes.
So it's more
of like a lens
or a bridge

or a language
not claiming
any final answers
to anything
but it's,
I'd say it was
like a bridging
paradigm.

Okay,
so we trade
off specificity
for general
patterns.

We can't
predict exactly
what will
happen,
you know,
the CRR
is predictive
but it's not
as predictive
as obviously
the microscopic
within paradigms
are.

But we can
at least start
to say,
well,
that coherence
seems to be
building in this
system.

Where's the
Amiga value?

Okay,
rupture is

probably imminent.

Okay,

do we need

to actually

instigate

rupture now

to,

you know,

just start

to think

about how

we regulate

systems better

and also

continuous

learning as

well with,

as I said,

about catastrophic

forgetting and

AI systems

going mad.

Yeah,

so we can't

detail every

cellular process

but we can

say gap junction

closure equals

coherence loss

and Levin

certainly was

already talking

in those

sorts of ways.

Thermodynamics

doesn't predict

individual particle

motion but
describes system
level behaviour.
in the same
way,

CRR trades
precision for
generalisability
useful for
navigating and
course level
prediction.

Perhaps.
Perhaps it's
more,
perhaps it's
less,
I don't
know.

Therapeutic
possibilities,
so as I've
mentioned earlier,
just to say,
it's like if
people are
building coherence
monotonically
and they're in a
low Amiga,
this is me and
I won't change,
then it will
probably lead to
catastrophic
rupture over
time.

Just to say as

well through the
life course,
I didn't mention
this,
but if you've
crystallised your,
you know,
early on you have
high brain
plasticity,
exploration,
rupture
thresholds quite
frequently,
you know,
first two years
and so on,
by the time you're
sort of 40 and
you've accumulated
that life and
then you have
rupture,
it's big because
you've got to
rewire so much
so much has to
be regenerated.
So what therapy
does is I think
it offers those
opportunities where
you can be in a
safe space to go
into rupture
with somebody who's
enabling you to
have that higher

Amiga exploration
and look at
those future
possibilities.

And also it
would help to
perhaps detect
when somebody's
going to be going
into that as
well and I know
lots of people
are using LLMs
for therapeutic
purposes which
has a huge
catalogue of
risks associated
with it and
they're trying to
put guardrails in
with words but
I think they need
a mathematical
solution to that
as well.

At the end of
the day the
large language
model might be
able to
cognitively push
you up that
ladder and make
you feel alienated
and outside of
your social
field, social

coherence field
but it can't hold
you when you
fall is what I'd
say.

So you need
to have those
safe containers.

Yeah, so just
some ideas there.

High Amiga is
okay in a
therapeutic
space where it's
safe to rupture.

After trauma
support with
regeneration seems
to help surface
what had the
highest coherence
before and what
wants to reorganise
in the future and
let that through
rather than trying
to resist that.

thinking openly and
honestly in a
therapeutic context
about that and
being allowed to
explore different
possibilities.

Temporarily
increase
sorry temporarily
increase Amiga

during rupture so
there's more
possibility space so
that enables that
creative exploration
rather than being
trapped in prior
coherence so you
know you're
actually thinking
about how we can
move forward and
regenerate most
effectively.

I've already
talked about
organisations but
yeah team
communication
patterns,
collaboration,
when to pivot,
when to make
decisions, how
we can support
regeneration after
crisis and how
we can tune that
Amiga threshold
in teams as well
so using CRR to
reflect on team
dynamics so
questionnaires,
discussions,
you know perhaps
in the future
like EEG monitors

I know there's
lots of research
around that as
well, breath
modulation because
I think breath's
really important
there too for how
we actually create
our generative
models I think
breath and
generative modelling
!

I think breath and
generative modelling
has a lot to
be that we should
be still exploring
in education we
can think about
that critical
threshold we need
to think about
building coherence
and then instigating
those breakthroughs
what is it that
makes those big
breakthroughs where
suddenly that child
has that realisation
that ontological
shift so just
helpful as a
reflective tool
so I'd really love
to hear from

anyone who's
interested in
empirical research
theory perhaps
you've got questions
or reflections on
how it might relate
to domains that
you're working in
perhaps you're
working in free
energy or active
inference and
interested in what
I've been working
on I think there's
loads of scope for
practical tools as
well and I've spoken
to machine learning
specialists and
social entrepreneurs
and even one of
all from his
graduates as well
about this framework
because she was
interested in
coherence-based
approaches
approaches so I
think there is an
increasing trend
looking at coherence
based approaches in
various contexts and
that's certainly what
I've been doing for

the last eight months
and yeah I'd love to
hear from people
finally just some
more radical
conjectures from
Claude after I'd
just done only a few
prompts with the
CRR so memory is
ontological substrate
so we've got the
coherence there
applies all past
states remain present
in the field structure
not as influence but
as constitutive
reality itself so
it's quite exciting
thinking that reality
itself is a field of
physical memory
it's what process
philosophers call
creative advance
Whitehead and what
explains non-local
coherence effects in
morphogenesis with
those antlers for
instance
consciousness is field
self-recognition so
this idea of we are
the universe witnessing
itself in the present
moment is where

accumulated coherence
meets perturbation
experiencing this
threshold directly may
be what phenomenologically
is described as the
structure of lived time
by Varela and what
contemplative traditions
call awakening or like
letting go of that
Markov blanket or being
at the top of that
direct delta Markov
blankets is epistemological
not ontological so
organisms require
Markovian approximations or
computational
tractability to act but
the field itself is
non-Markovian according
to Claude this resolves
tension between FEPs
local inference and
quantum non-locality
Tristan Eisenberg you
know over the top of
course but this is what
I'm you know been
exploring it was just
interesting to see what
you know in the age of
LLMs what did the LLMs
say I mean you get to
terms field rigidity and
fluidity not system
property but temperature of

reality itself explaining
why the same pattern
appears across the
physical law with low
omega biological
adaptation is so
physical law having low
omega biological
adaptation medium and
creative exploration high
such as disputative systems
pregojin so these are
just complete wild radical
conjectures because as you
can tell I quite like
exploring that kind of
thing really just huge
thanks to putting this
conference on at all and
I've got my email address
there and some final
thoughts we accumulate what
works we break when we
must we rebuild from what
mattered most and this
pattern may be fundamental to
how change happens in
children in organisations in
ourselves and in physical
system physical and
biological systems too
CRR offers a possible
language for recognising
these movements for
working with them rather
than against them it's by
far from a complete theory
it's an invitation to

explore together during a
time when we're all
navigating rapid
transformation what if we
could learn from each
other's ruptures and
regenerations what if a
common language helped us
navigate the unknown
together that's the offer
that's the invitation and
thank you for listening and
for exploring that with
me

all right next talk

Samuel Montanez from
information economics to
active inference a unified
Bayesian mechanics of choice
under uncertainty this is a
26 minute pre-recorded talk
here we go

good morning everyone I'm
Samuel Montanez an
artificial intelligence PhD
student from Universidad
Panamericana in Mexico City
and today I want to take
you on a journey that
begins with a puzzle that
has hunted economics for
over a century and end
with a unifying framework
that connects four novel
winning insights under a
single mathematical
architecture let's begin
let me start with a

fundamental question in the
80s 70s the French
economist Leon Walras gave
us this beautiful vision of
perfect competition
markets reach equilibrium
where supply meets demand
you probably remember your
microeconomics undergraduate
course and in modern terms
Eugene Fama gave us the
efficient market hypothesis
which tells us that prices
convey all available
information at every point in
time so every agent every
economic agent is perfectly
informed information is
symmetric and search is
costless economic search so
it's mathematically elegant
philosophically satisfying but
empirically wrong because
when we actually look at
markets we see something
completely different we see
identical good selling at
widely different prices we see
persistent disequilibrium
dynamics we see people
spending enormous amounts of
time and money searching for
information and for example
investors pay thousands of
dollars to subscribe to real
time financial data providers
like Bloomberg Pershing and
many others and at the same

time is true this is true in most industries that's why we have for example specialized industry magazines and we see that certain agents earn what we call monopoly rents which is simply they earn these rents simply because they know something others don't and this discrepancy led to what we now call the information economics revolution in the 1970s novel laureate economist Peter Diamond proved something shocking which is that even a tiny search cost can prevent markets from converging to a competitive equilibrium as that what Russian action years told us then in the 19 in the 1980s probably the most famous study in information economics Grossman and Stigley showed us that an even deeper paradox they demonstrated that markets cannot be both perfect perfectly informationally efficient and providing incentives for people to acquire information and if you think about it if prices already reveal everything why would anyone pay to become informed why would anybody pay for a financial advisor or an asset manager and if no one pays to become informed how can prices reveal anything so the information mechanism collapses under its own logic

so this information economics revolution gave us several theories but we're going to focus on four foundational theories each from another laureate economist and each capturing essential things about how economic agents actually behave first let's start with the gentleman in the top right hand corner in the top right hand side corner George Stigler in the 1970s gave us optimal search theory so Stigler was a brilliant Chicago economist who asked when should people stop searching for a better price and his answer launched the entire field of information economics later on Herbert A. Simon one of the founding fathers of cognitive science also artificial intelligence in the 1950s introduced the concept of bounded rationality against rational expectations theory in economics so Simon who would go on to win also the Nobel laureate in economics despite being a psychologist political scientist computer scientists argue that humans don't optimize rational expectations and the human status is fast so when we find good enough solutions given our computational constraints that's good enough also Ken Arrow in the 1960s one of the 20th century's most profound economic theories formalized the value of information what is the economic value of information and he showed us precisely how much should be willing one with one would be willing to pay for the information that

reduces our center uncertainty about the world and this resonates a lot to the concept of intrinsic value in active inference and finally Christopher Sims into in 2003 developed his model the rational in the in attention theory and seems and an econometrician by training was probably the first major economist who imported information theory from Claude Shannon into economics he showed how attention is a scout resource and people optimally choosing what to pay attention to and here's what's fascinating all these four theories produce remarkably similar behavioral predictions despite using completely different mathematical frameworks for example sigler uses optimal stopping Simon uses computational complexity arrow uses decision theory and seems uses information theory yet they all point toward the same phenomenon economic agents engage in what we call epistemic foraging they balance exploitation of known opportunities with the exploration to reduce uncertainty and this raises a profound question what does this convert why does this convergence happens it is a mere coincidence or it is there something deeper connecting these frameworks let's see and the answer I want to argue today is that all four theories emerge as special cases of free energy minimization in active inference now for those who brief active inference probably here in the active inference community this may seem obvious but for economists or financiers in the audience this is going to require some translation work so active inference developed by Carl Fristen and colleagues originally mainly in systems neuroscience gives us a mathematical framework where agents minimize expected free energy or and more generally

self-organizing systems so this quantity this composes beautifully into two terms pragmatic value which captures the exploitation term maximizing expected utility or satisfying preferences in economic terms and epistemic value or intrinsic value which captures exploration so minimizing uncertainty about the world through information seeking behavior so the key insight is that the policy selection within the active inference framework follows this soft math this soft math form where policies are selected based on their expected free energy weighted by a precision parameter and this isn't just a model of behavior it's a model that connects perception and action under a unifying objective function so that's for example contrast to standard reinforcement learning models for example

now but really to understand why this unification works we need to think about the economy itself as a cognitive systems markets and agents perform distributed information processing essentially that's the market including the financial markets and this processing happens under physical constraints because both technological markets and us as financial agents in our brains we face those computational constraints and this is where we connect to thermodynamics we all know in the active inference community that in 1960s rolf landauer proved something remarkable that erasing one bit of information requires a minimum energy equal to $kT \ln 2$ where k is the boltzmann constant and T is the temperature and this means essentially that information processing has irreducible physical costs

you cannot think without spending energy essentially so this leads us to think about choice as computation under thermodynamic constraints

thermodynamic decision making essentially so the transformation cost between beliefs becomes proportional to the log probability with beta the inverse

the inverse temperature converting between information and energy units so when we frame the problem this way the agent's optimization problem becomes maximizing expected utility like in economics minus a complexity cost weighted by this inverse temperature parameter which is $1/\beta$ one divided by one divided by one divided by beta so this is mathematical equivalent to minimizing variational free energy which decomposes into accuracy how well your beliefs predict observations and complexity how far your posterior beliefs deviate from your prior measured by the famous kullback leibler divergence

so the optimal solution gives us this precision weighted choice where the optimal posterior is a prior exponentially weighted by the utility

now

Stiegler search

now let me show you how each classical theory emerges from this uh standard framework

uh so when we start with Stiegler search

George Stiegler the economist asks

when you're searching for the lowest price among an unknown distribution of prices which in most cases is unknown

when do you stop searching

and his classical optimal solution stopping rule

says that you should stop when the expected reduction in price from one more search

the marginal let's say

equals the search cost

so active inference essentially generalizes this beautifully

because the search cost now equals the expected pragmatic value

the price reduction

minus the term proportional to the epistemic value the uncertainty reduction from Laredi

uh about the price distribution

this second term weighted by λ

captures the intrinsic information seeking

the intrinsic value

so when λ equals zero we recover

essentially Stiegler's original rule

your exploitation

well when λ is greater than zero

we get what we actually observe in real human behavior

people over search

many times

i mean you spend uh a ridiculous

ridiculous amount of time searching for different prices

different options

and from a purely pragmatic perspective

uh they are intrinsically

value resolving uncertainty

so the total value curve

shifts to the right

you can see in the graph

and the optimal stopping point becomes beyond Stiegler's prediction

so this explained decades of experimental evidence

showing people search more than what Stiegler's theory predicts

so one would think from a behavioral finance perspective

that they are irrational

they're being irrational

[illegible]

Simms essentially imported or imposed a Shannon capacity constraint on information processing. So the mutual information between states and actions cannot exceed some capacity κ .

So the solution in Simms' theory is a softmax policy where the probability of action given state is proportional to the prior probability times the exponential of utility, with a beta being the shadow of price capacity.

Now, if we compare it to active inference policy selection, q of π is proportional to p of π times the exponential of negative gamma, basically the confidence in one's preferences and actions, times expected free energy.

So the mathematical structures are identical. Beta equals gamma, utility equals negative expected free energy, and the Shannon capacity constraint corresponds to the variational bound on free energy.

So this equivalence is very profound because it means that economists, through pure information theoretic reasoning about attention constraints, sort of independently discovered free energy minimization.

Now, Simms didn't know about Fristons' work. Fristons didn't know, as far as I know, about Simms' work.

Yet, they arrive at the same mathematical framework for completely different starting points.

So this brings us to the unified architecture.

As we have reviewed, all four major information economic theories emerge as special cases of minimizing free energy plus computational cost on their constraints.

First, Stiegler corresponds to λ equals zero. That's read as no epistemic value, pure pragmatic optimization.

Simon corresponds to λ .

When λ approaches infinity, computational cost dominates, leading to the satisficing behavior, good enough expected rewards.

And narrow corresponds to utility
being proportional to log probability.
So information gain becomes utility.
And Simms corresponds to $\beta = \gamma$.
The precision parameter equals the inverse temperature.
So we can, let's say, integrate all of this
as active inference sits at the center,
encompassing all four as limiting cases,
as a more general theory.
So this isn't just a mathematical coincidence.
It's telling us something deep
about the structure of economic decision-making
under uncertainty.
Now, this unification has profound implications
for how we interpret behavioral economics.
Consider the so-called anomalies,
irrational behavior,
the *lai* paradox,
framing effects,
probability weighting,
anchoring,
availability heuristics.
For decades,
behavioral economists have cataloged this
as deviations from rationality,
from rational expectations.
But active inference reinterpreted them
as a base optimal inference
on their narrow computational constraints.
So the behavioral patterns we see
aren't irrational deviations.
They are actually optimal responses
to the accuracy-complexity trade-off
faced by biological agents
with finite computational resources.
And this connects directly to neuroscience
because we can implement this
as a hierarchical architecture

where the prefrontal cortex
maintains policy beliefs,
the striatum computes
precision-weighted prediction errors,
and dopamine encodes
not just rewards,
but precision itself.

So the precision-weighted modules,
so the precision-weighting
modulates choice stochasticity.

When you're certain,
choices are deterministic.

When you're uncertain,
they're stochastic.

And this gives us testable predictions
because we should see links
between choice patterns
and the frontal striatal activity.

For example,
dopamine should encode precision,
not just reward,
and behavioral variability
should reflect optimal inference,
not noise.

Some of these predictions
have already been confirmed experimentally.

Now, let me ground this
in a concrete example,
a trading agent,
a financial trading agent.

Now, in my own research
on portfolio management,
I have implemented these ideas
as partially observable
Markov decision processes,
PMDPs.

So we have the agent
that has hidden states

representing market regimes
and portfolio positions.
So it receives observations
of returns
and profit and loss
from his own or sheet
or her own portfolio.
And it takes actions,
the standard trading actions
like buy, hold, or sell.
And the generative model
includes observation
likelihoods mapping regimes
to returns
and positions to PNLs.
So transition dynamics
for how regimes evolve
and how actions control positions
and preference distributions
and coding things
like loss aversion
from Kahneman and Traversky
from prospect theory.
And what's remarkable
is that this agent
trained only
on active inference principles
with theory-based parameters
from behavioral finance
and economics
can achieve superior risk
or adjusted returns
compared to buy and hold
and traditional valuation methods
and can better manage risk
in many cases.
So the Sharpe ratio,
the measure or the metric

that we use
in portfolio performance
can improve significantly
not through overfitting
but through principal
epistemic foraging,
knowing when to explore
versus to exploit,
which is essential
in financial management.

Now, let me ground
this application.

Now, let me ground
information economics
in a concrete example.

Imagine a financial trading agent.

So in my own research
on portfolio management,
I have actually implemented
these ideas
in the standard framework
in active inference,
a partially observable
Markov decision process,
a PMDP

that has the following architecture.

The agent has hidden states
representing market regimes,
basically the business cycle
as bull, bear, et cetera,
and portfolio positions.

What are literally
the holdings,
the positions
in their portfolio?

So it receives observations
of returns
from the assets

in his or her portfolio
and it calculates
P&L, profit and losses,
from its portfolio
because remember,
there are transaction costs.
And it can take
standard trading actions
like buy, hold, or sell.
And essentially,
the generative model
includes observation,
likelihoods,
mapping regions
to returns
and positions
to profit and loss.
And it has transition dynamics
for how regimes evolve
and how actions
can control positions.
And naturally,
we can think logically
about it
because we have no control
over how markets behave.
What we have control
is on how to switch
our portfolio,
how to trade.
And we have also
preference distributions
including things,
for example,
like loss aversion
from Kahneman and Traversky's
prospect theory
that we compute

with a higher magnitude
equivalent losses
to gains.

And what's remarkable
is that this agent
trained only
on active inference principles,
essentially with theory-based
parameters
from behavioral finance,
can achieve,
in some cases,
superior risk-adjusted returns
compared to other
buy and hold
and traditional valuation
methods in finance,
and can also
be,
usually be,
a better
risk-adaptive manager.

And the reason why
is because,
again,
it can include
the uncertainty parameter
within the function,
the optimization function,
contrary to other,
for example,
deep reinforcement learning methods.

And the Sharpey ratio,
the standard metric
that we use in finance
where we measure
risk-adjusted returns,
can be improved

significantly
through this method
and not through overfitting,
but through this type
of principal epistemic foraging,
knowing when to explore
versus to exploit.
And to conclude,
and let me conclude
by emphasizing
what we have accomplished
today.

We have shown
mathematical equivalences
between four foundational
economic theories,
expanding 60 years
of Nobel Prize-winning
work in economics.

We have demonstrated
that all four models
emerge from
variational free energy
minimization.

We have reinterpreted
behavioral anomalies
in behavioral finance
as optimal inference
on their constraints,
computational constraints,
or thermodynamic constraints.

And we have built bridges
connecting economics
to neuroscience,
machine learning,
and physics.

So the applications
extend far beyond economics.

This suggests
that variational
free energy
minimization
isn't just
a principle
for biological brains
or economic agents.
It's a fundamental
principle
of any
self-organizing system
that must process
information
under physical constraints.

So the future directions
are vast,
as you can see.

So we need
empirical validation
in neural systems
to test whether
human brains
actually implement
these type of computations.

We need better
computational implementation
to scale these ideas
to realistic economic
environments,
for example,
making these type
of PMDPs,
agents,
much more robust
and embedding them
with other
financial agents.

We need to extend
this to market dynamics,
understanding how markets
themselves minimize
free energy
through price discovery
mechanisms.

And we need
quality implications
for designing
institutions that work
with rather than
against
the bounded rationality
of human agents,
how real people behave.

So what began
as a puzzle
about price dispersion
and surge cost,
a center,
surge cost has led us
to a unified framework
connecting a century
of economic thought
through the lens
of Bayesian mechanics.

So the economy,
it turns out,
is performing
active inference
just like the brain,
just like artificial agents,
just like any
self-organizing system
that must maintain
itself by minimizing

!

It turns out
in a changing world.
Well,
thanks for your time.
Now,
I'm happy to take
questions
and you can also
reach out to me
through my email.
Thanks a lot.
Thank you very much.
Thank you.

All right.
Final recorded
talk for the day.
This is Adam Saffron,
AI Capabilities
and Alignment
Consensus Project.
It's a 33-minute
recorded video.
Here we go.
Hello, everyone.
My name is Adam Saffron.
I am a research scientist
at the Allen Discovery Center
at Tufts University.
And today,
I will describe
a project
that I've been developing
with friends and colleagues.
Hello, everyone.
My name is Adam Saffron.
I am a research scientist
at the Allen Discovery Center

at Tufts University.

And today,

I will describe

a project

that I've been developing

with friends and colleagues

for the years

called the AI Capabilities

and Alignment

Consensus Project

or AICACP.

Today,

I'll describe

the overall scopes

and ambition

for the project.

It's a multi-part

undertaking

involving

the editing

of a number

of special issues

of academic journals.

There's a planned

podcast series

and workshops

and I'll also describe

some of the future hopes

and plans

for the project.

In brief,

the AICACP's goal

is to promote

scientific discourse

by creating forums

where people can engage

in structured conversations

that bring together

diverse perspectives
and forms of expertise
to provide conversations
that are
cumulative
and trustworthy.

The goal would be
to take
advanced scientific
understanding
by providing
conceptual clarity
for key concepts
in artificial intelligence
and cognitive science
such as
world modeling
or agency
or other constructs
that are thought
to be key
to the nature
of intelligence
and what we
ought to expect
from AI systems
as they are now
and how they could
be in the future.

The further goal
in addition to
providing conceptual clarity
or advancing
academic understanding
would be to share
this understanding
with the broader public
and so

also involve
a number of forms
of public outreach
to help educate people
and bring them
into the conversation.
And most ambitiously,
if the project continues
to grow and is successful,
the hope would be
to help to create
an academic society
or community
that could help
to bring together
diverse perspectives
for the sake
of informing
public policy
and efforts
at regulating
AI
as we continue
to develop
these technologies.
So the AICCP project
I've been pursuing
with a number
of friends
and colleagues,
some pictures
of the current
core team
who's been helping
me to vision this
and plan
the upcoming activities
that I'll describe today.

Also, you'll see
bottom right there,
there's Daniel Friedman
from the Active Inference
Institute,
and I'm grateful
to have been supporting
us and providing
a home for these
activities,
as well as
on the left there,
Mike Levin,
who's provided
an academic home
for me
as I engage
in this undertaking.
So,
AI,
what's going on
and what ought
we to do about it?
There's a number of,
a range of perspectives
on this
that range
from the more hopeful
to the more cautious
and fearful.
And all sides
of these debates,
I think they have
some important points,
even though I don't
necessarily agree
with,
well, of course,

I don't agree with
what everyone says
and no one does.
And that's the point
is that
we have,
we don't know
what world we are in
with respect to AI,
and we don't know
what we ought to do
about it.
And different people
have different takes
on this,
which have different merits.
And so,
you know,
on the left there,
you see a recent book,
If Anyone Builds It,
Everyone Dies,
which has been
making the rounds.
And this would express
what one might call
the Doomer perspective
or the idea
that AI capabilities
as they proceed
might be very dangerous
and could potentially
pose existential risks.
There's other people
who don't think
that such risks exist
and that the,
maybe the current technologies

are potentially
overhyped
or just not risky.
People,
there's different figureheads
who have,
are emblematic
of these different perspectives.
So there's like
Yeshua Benjio,
he would be probably
more of the safety focus
or Doomer camp.
And then he's talking
to Andrew Ng,
who would be more
of the techno-optimist
or accelerationist camp,
which believes
that we should pursue
these technologies
as rapidly as possible
because of their promise,
which could be something
like a fourth industrial revolution.
Accelerationists
or a techno-optimist perspective.
And so the goal
is to try to advance
the state of discourse
on AI capabilities
and alignment
to help inform
these debates.
And so towards this end,
one of the core activities
has been planning
and a number of,

a number of journal collections
and specifically,
the first one
is a special issue
of Royal Society
Philosophical Transactions A
on World Models,
AGI,
and the Hard Problems
of Life-Mind Continuity
towards a Unified Understanding
of Natural
and Artificial Intelligence.

This is a current list
of guest editors,
which at one point
involved Daniel Kahneman
before he passed,
but I've been very grateful
to have the support
of a number
of very brilliant people
in organizing
this journal collection,
specifically digging
into the issue
of the natures
of world modeling
as they would be found
in artificial intelligence
and also natural intelligences.
So more specifically,
the word world modeling
has a number
of connotations
depending on who you talk to.
People have different notions
of what might constitute

a world model.

Some might be more inclusive
or some might be less inclusive.

So some might consider
any sort of predictive information
that a system might have
about its world
and any sort of,
anything that would let you explain
and predict what's going on,
that would be a world model.

Others might say that,
well, yeah,

for a world model,
you need something
more like
a causal transition model
that an agent can use
given its past experiences
and given inputs
and actions
to predict
what's going to happen next
and what it ought to do.

I don't think there's any
right or wrong way
to use language,
but the different ways
we think of world modeling
would have different implications
with respect to
the kinds of capabilities
we would expect from a system
once we say
that the system is engaging
in world modeling.

If we have a more inclusive notion
which might say,

well,
any kind of predictive information
involves world modeling,
to say something
is world modeling
doesn't necessarily tell you
that it's going to be
a very powerfully intelligent system.
But if you have,
let's say,
a kind of world modeling
that gives you,
that captures the causal structure
of the world
and lets you do things
like plan in depth
into the future
and handle counterfactuals,
well,
that might be something closer
to what we would consider
to be general intelligence
and potentially transformative technologies.
In my own work,
I've speculated
that there is connections
between the notion
of world modeling
and what we might call consciousness.
And I think this,
as Minsky called it,
the suitcase word of consciousness,
which has all these different connotations,
that there's another suitcase word
of world modeling
and that there's some relation
between these two.
I'm not the only person

who's said this.

People like Lacoön

and Schmidhuber

and others have suggested also

that we can understand

our conscious processing

in terms of world modeling.

But regardless of that,

the question is,

so what kind of information

about the world

is entailed

in the technologies

we're developing,

these foundation models,

large language models,

in their latent spaces,

to what extent

have they captured

the structure of the world?

We don't know.

And the positions on this range

from saying,

yeah, it looks as if

they have robust world models,

but not really,

when you go off distribution

and ask them to make inferences

based on things

that are not what they're trained on,

you'll see that

they're not as good as you think.

Others, like Jeffrey Hinton,

others would say,

no, actually,

their world modeling

is quite sophisticated

and the only way

they were able to pull off
the challenge of doing
next token prediction on language
is they had to somehow
learn the causal structure
of the world.

These debates are ongoing
and part of these debates
is not just
what is the world model capacity,
it's also
how ought we to think
of world models
and how ought we to think
of the nature of intelligence.

So towards this end
for this special issue
and considering
the natures
of world modeling,
the goal is to look
at the modeling capacities
of a wide range of systems,
ranging from
what kind of world modeling
might be engaged in
by the brain
of an intelligent animal
or a person
or potentially
what kind of world modeling
to go more exotic
might be entailed
in all living phenomena,
even going,
ranging from a cell
to super organisms
to ecosystems,

economic systems,
societies as a whole,
Gaia,
like what are the range
of more complex adaptive systems
and cognizing systems,
what kinds of world modeling
can we observe
and how might looking
at world modeling
from these diverse perspectives
help to inform
our understanding
of world modeling
as it occurs
in the technologies
we're developing now
and will be developing
in the near future
and vice versa
by having a more precise
understanding
of world modeling
in the AIs
that we're building,
how might this help us
to have more precise
understandings
of the nature
of intelligence
in natural
system,
natural cognizing systems
like us.

This first special issue
will have an initial
election released
in February

through Philosophical Transactions A

and there's going

to be another one,

a second installment

that will begin

accepting proposals

and will be published

a number of months later

and in the coming,

the next month or two

will begin accepting submissions.

And please contact me

if you're interested in this.

There's another special issue

being planned

but instead of digging

into world modeling

as a construct

and looking at the

different connotations

that people have

focusing,

another one

focusing on the nature

of agency

and agents.

What are agents?

What do people mean

or the range of things

people mean

when they talk about agency?

Is any system

that's pursuing

some goal

regardless of whether

it's explicitly represented

or not,

anything which can be described

as having an objective
function,
is that an agent?
Sure, maybe.
Or is agency
something which requires
something like
self-modeling
or meta-awareness?
And again,
as with the case
of world modeling,
I don't think
there's just one
right way to use language
but rather any way
we conceptualize agency
will have
a range of applicability
with trade-offs
in terms of being
more or less apt
for capturing
the phenomenon
of the nature
of intelligence
and different systems.
So if we're more inclusive
in talking about agency,
we might be able
to talk about
a broader range
of systems
and their intelligence.
If we're less inclusive
and more restrictive
in what constitutes
an agent,

you might be able
to make potentially
more meaningful statements
about what it means
when you say something
as agency.

Like how will a system
be able to plan
with depth
into the future
and be deployed
in the real world?

How promising
is it
because it has
this function
or potentially
how dangerous
is it?

There's other debates
about agency
in terms of
how much
it's tied
into the nature
of intelligence.

So people like,
I guess,
among the active
inference crowd,
people would say,
well,
the idea
of pursuing goals
and being
an active learner
and an active
inferential agent

that's trying
to shape the world
as it's achieving goals
that agency
and intelligence
cannot be separated.

There's others
like Yeshua Benjo
who might argue
that actually
you could have
potentially arbitrary
degrees of intelligence
without having
any kind
of agentic character.

Systems that have
no goals
as we would
understand them
but still are
nonetheless capable
of explaining
what happens
and predicting things.

And so,
the goal
of this special issue
is to,
similar to the
person I just described,
is to approach
agency
from a diverse
intelligence's perspective
considering the
agentic capacities
and the nature

of the agency
in a wide range
of systems
and then
try,
based on this,
try to understand
what do we think
is happening
with respect
to present AI systems
and the ones
we might develop
in the near
and medium
and long-term future.
and similarly
to take insights
from understanding
the natures
of agency
and AI
and see how
this can be used
to inform
our understandings
of agency
and natural
intelligences.
There's a
third special issue
being planned
for after that.
This time,
instead of focusing
on these core
concepts
in AI

and cognitive science
and their relation
to AI capabilities
and the scope
of intelligence,
for this special issue,
the goal would be
to get into the issue
of not the scope
of capabilities,
but what are the implications?
What ought we to do
about it
given the potential
alignment problems
we might face
with these systems
in terms of
the means
of ensuring
that they do
what we want them to do
and not what we don't
want them to do?
What sort of risks
might be associated
with these systems
and what sorts
of governance
and regulatory solutions
might be needed
to address these risks?
So,
in addition
to these special issues
being planned,
the first one's
in Philosophical Transactions

A, Royal Society.

The second one,

I'm currently planning

for Phil Trans B.

Unclear about the third one.

Alongside these,

we are going to be

releasing a podcast

with the goal

of taking the submissions

to this issue,

helping to dig deeper

into what's being discussed

and communicate

these papers

and the thoughts

in them

and their implications

to the broader

scientific community

and more broadly

to the public.

So,

right here,

there's a picture

of Michael Garfield,

who I'm grateful

to be working with on this.

He was the host

of a really excellent

podcast series

from the Santa Fe

Institute

called the Complexity

Podcast Series,

which dealt with

the nature of complexity

in the natural world.

And the goal
for this podcast
is to take
the full complexity
of these issues
around what's going on
and what we ought
to do about them
and to address them
with depth
and detail
and also make
accessibility
in terms of
helping to distill
these potentially
very complex ideas
and communicate
the essence
of these ideas
to broader
scientific community
and the public.

So,
this podcast series,
while initially
being focused
on the submissions
to the special issues,
will not just
be restricted there.

Publications
that are not
necessarily the ones
being edited
as part of this project
could be involved.

Anyone who has,

from the active
inference community
or if you have
recommendations
just for new publications,
these are,
we would potentially
like to interview you
or the person
you're recommending
and bring them on
into the podcast
to help
provide an opportunity
to communicate
these ideas
and situate them
relative to
this ongoing
evolving conversation
about the nature
of AI capabilities
and alignment.
In addition to this,
we are planning
a number of workshops
to be had
both with
in-person
and hybrid
and virtual formats
where the goal
for these workshops
will be for them
to be discussion-oriented
where
instead of having
like a

people
coming
and sitting down
for just
a series of lectures
and listening to the lectures,
the hope would be
and the intention
and the plan
is to
have a set
of structured discussions
where
people can
foster
mutual understanding
and potentially
something like consensus
and
from these workshops
produce
conference proceedings
in terms of
consensus
statements
and white papers
that could
describe
the points
where there is
agreement
and disagreement
and the current
state of play
of knowledge.
For these workshops
and also for
the podcast,

there is a protocol
that I have in mind
and have been
discussing
with my colleagues
in the AI capabilities
and alignment
consensus project

and consensus project
AICACP,
the protocol
that I have been calling
benevolent adversarial collaboration.

So people might have heard
of the idea
of adversarial collaboration
in terms of,
as I believe
Daniel Kahneman
first suggested,
where you have opposing points
of view
and you bring people together
to basically engage
in a collaboration
on developing experiments
that could help
to adjudicate
among these competing claims
and help to resolve
the difference.

And so it's adversarial
in that there's a difference
but it's collaborative
and people are coming together
to co-design experiments
that could help them

resolve these differences
and I think this is fantastic.
I'm adding
this benevolent portion
in that
before people begin
digging into
their differences
and debating
having an initial period
that emphasizes
mutual understanding
and cognitive empathy.
In brief,
the procedure would be
you start by stating
your initial points of view,
laying out
the major claims,
the reasons that
people believe
what they believe
about a given topic
and then
when you have
people who have
differing points of view
or seemingly differing
points of view,
have them
paraphrase
each other's perspectives
iteratively
until they can arrive
at mutual understanding,
until people can say
to each other,
yes,

based on what you just said,
you understand
what my position is.
This doesn't mean
that we're having people
agree on
the conclusions
they have to come to
but rather
they have to come to
a point of initial
mutual understanding
while there still
could be
and will be
plenty of room
for disagreement.
But before we debate,
first we try to understand.
And along the way,
as part of this understanding
and some of the ways
in which understanding
can fail to be achieved
is people oftentimes
use words differently.
So I was just mentioning
these special issues,
people,
if you're talking
about world modeling
or agency,
concepts such as these
or what do you mean
by intelligence
or alignment
or any of these concepts
we're discussing

with respect to AI,
if you have
different connotations,
people can very easily
talk past one another
and those conversations
can be less than productive.
And so along the way
from this point
of mutual paraphrasing
coming at mutual understanding,
the goal will be
to develop
a provisional lexicon
that could be used
in subsequent conversations.
So an ontology
or just a set of agreement
on the way terms
ought to be used
for the sake of moving forward
for these discussions
in the collaborative process.
So instead of saying,
okay, we're going to use
world models
or agency
or any of these words
in just one way
and agree this is the singular
definition of these terms,
we might come to that
and people might agree
from the conversation,
okay, this is how we ought
to use it.
Here's the one definition.
We can add additional words.

We can modify words,
have multi-word construction.

So in order to say,
okay, you mean
this kind of world modeling
and I mean
that kind of world modeling
and this might be apt
in one context
and that might be apt
in another context.

So similarly for agency,
you might have
different forms of agency
and you specify
which form do you mean?

And so we have now
a shared lexicon
and now based on this,
we can now dig in,
outline the major points
of agreement
and disagreement
that are present
now that we have
mutual understanding
and now we bring reality
in the loop
and say what are either
experiments of nature
or like what given evidence
now how the evidence
we have there
on these different points
of view
or what evidence
ought we to collect
and what experiments

should be performed
to move forward
and along those lines,
I'm going to go into
this more in a second,
what predictions
can we make?
How can we involve
forecasting of
based on our beliefs
what we think
is likely to happen
and then basically
call our shots
given our beliefs
and now say,
okay,
now once we've predicted
and forecasted
what we think
specific outcomes
are going to happen
given our worldviews,
what then happened
over some time period
where,
or a number of months
or years,
we can actually get
this feedback from reality.
So this procedure,
I know it seems
quite complicated,
it's really just
basically a period
of establishing
mutual understanding
through engaging

and paraphrasing
and perspectives,
arrival of a shared lexicon
that can be used
for subsequent conversation
and then based
on this understanding,
then a clear statement
of points of agreement
and disagreement
and then bringing reality
into the loop
to help adjudicate
among these perspectives
with the hope
of potentially arriving
at something like consensus
on some set of the issues
being discussed.

In this case,
with respect to AI capabilities
and alignment.

So towards this end
of trying to help
to provide more productive discourse
and better epistemics
and better collaborative procedures,
we are currently exploring collaboration
with the Society Library,
this organization being led
by Jamie Joyce and others
where their goal
is to take complex issues
and to provide detailed
but accessible accounts
and descriptions
of these issues
in terms of

what are the full range
of beliefs that people have
and what is the evidence
that bears
on these different beliefs.
So what do people believe
and why
and based on the information
we have available to us
from any sources,
which beliefs
are more or less
well justified.
and for this one
of the core things
they do
is a process
called debate mapping.
And so the debate mapping process,
you know,
basically we'll take
the perspectives,
lay them out
and then try to gather
before this was done
using human curators knowledge,
but now it's more relying
on automated procedures
to compile
from as many sources
as available
what evidence bears
on different perspectives,
organizing this
into a graph structure
of the way
these beliefs interrelate
and then for all these,

this tree
or this web
of intersecting beliefs,
what are all
the different forms
of evidence
that can be linked
to these different beliefs.
And so these knowledge artifacts
or debate maps,
we're going to be taking
our podcasts
and our,
the journal articles,
the podcasts
and the workshops
and have them feed
into this debate mapping process.
And so here you see
this cartoon,
there's two well-known
figureheads
for either the Doomer position,
Connor Leahy
or the safety,
the extreme safety position
or the extreme
accelerationist
or techno-optimist position
of Beth Jesus
or Guillaume Verdon.
And this was an episode
of Machine Learning Street Talk
which they use
their,
their tools
to automatically,
based on the transcript,

extract a debate map
where people could say,
okay,
given this conversation,
what was the range
of things being expressed
and what evidence bears
in these claims?

And so we would like
to use this
as part of our process
of coming at
cumulative knowledge
and consensus.

We are also exploring
collaboration
with Metaculous,
which is a group
that focuses on
online forecasting
where people will
basically predict
the likelihood
of different events
and bet at them
and then based
on these prediction markets
where people are forecasting
different outcomes
and predicting
what the things
that could happen
and betting on them,
you basically can leverage
the kind of wisdom
of crowds
to say,
okay,

given everyone
who participated,
which is,
what's the most likely outcome
of what's likely to happen
by aggregating across
many people's predictions
and then from
the aggregate estimate,
this being a means
of helping people
understand
what do people think
is likely to,
what is indeed
likely to happen.

So,
for the AI-CACP,
we're basically
going to use
these journal articles
and discussions
on the podcasts
and workshops
to identify
what are things
that people
would forecast
to be likely
with respect
to AI capabilities
and the implications
of those
in the coming months
and years
and then this
be another means
of getting reality

in the loop
where people
basically call
their shots
and are held
accountable
to the evidence
of what happens
given what they predicted.

This,
on the right here
you see,
it's similar
to a debate map
but this is
the belief mapping
tool called
Prism AI
that's being developed
by Metaclos
where they have people
estimate
the probability
they assign
to different events
and then
all these different
beliefs
about these predictions
are brought together
and webs
of mutually
constraining
predictions
of what's going
to happen
in the world
and then you have

this kind of
Bayesian network
of probabilities
that can be used
to estimate
the likelihood
of different outcomes
given people's beliefs
and so this is
a tool that
we're going to try
to use
along with
the tools
from the society library
as a means
of furthering
this goal
of benevolent
adversarial collaboration
or of mutual
understanding
and shared
sense making
and empiricism
so
the long term
ambition
for all of this
is the creation
of
really a novel
academic society
and community
dedicated to
AI capabilities
alignment
where people

from diverse
perspectives
can come together
and keep coming
together
to discuss
these issues
in a centralized
meeting place
that because
multiple people
are involved
and are in conversation
and because of its
visibility and impact
it is worth
people's time
to show up
and invest in it
and so
there's
plenty of
really
interesting
and wonderful
and useful
academic societies
out there
and conferences
and workshops
that exist
but I'm not sure
that there are
any that
would involve
something like
what I was describing
as benevolent

adversarial collaboration
within academia
people
often times
they'll appear
on panels
and have
really interesting
conversations
but I don't know
how often
anyone changes
their mind
and I don't know
how often
people will
arrive at
something like
consensus
so yeah
the goal
is basically
to have
a more
systematic
and
I would say
high
integrity
procedure
that takes
the best
of academia
and tries
to do it
even better
and have
it be done

in a
centralized
meeting place
specifically
dedicated
to
trying to
figure out
what's
going on
with AI
and what
ought we
to do
about it
where the
conversation
would continue
in perpetuity
until we had
clarity
which I don't
know where
that would be
and so this
would be a
project that
I would
hope
to go on
for years
and maybe
decades
as AI
continues to
advance
we need
to continue

to stay
in dialogue
and stay
in productive
dialogue
and continue
to update
our understanding
as the
state of play
changes
with advances
in technology
and our
knowledge
this was a
roadmap
that I
created
which is
still being
discussed
in its
details
but the
basic idea
is describing
up top
the journal
collections
that are
being planned
on world
models
and agency
and alignment
and then
as they're

being published
these feeding
into a podcast
series
and to a
series of
workshops
and eventually
this being
used to
ideally launch
this new
academic society
with a
dedicated
journal
a dedicated
regularly
occurring
workshops
discussion
forums
and whatever
activities
would be
productive
for helping
to arrive
at something
like a
reliable
knowledge
and consensus
which most
ambitiously
could be used
to inform
public policy

and attempts
at regulation
so I know
that's a very
ambitious vision
but I think
something like
this is needed
and my
colleagues
were trying
to do what
we can
to help
bring this
about
and we're
looking to
partner
with whoever
can help
and whether
individuals
or other
organizations
so
if you have
any questions
on this
or want
to get
involved
whether that's
to be
helping to
plan what's
happening
internships

helping out
with the
journal
collections
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hopefully
working with
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thank you so
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Thank you.